

Rewind project report draft

Practice Module for Certificate in Designing Modern Software Systems

Team number pending confirmation

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1. Introduction

1.1 Background

Rewind is a private shared-memory application for close friend groups whose members may live apart. Members contribute short ordinary moments to a time capsule. Submissions become sealed metadata; a server-driven cycle later compiles and reveals a chronological film, which remains in the group Archive. The submitted proposal frames delayed reveal as a way to reduce pressure to curate every contribution. That expected benefit has not been established by a measured user study in this draft.

1.2 Business Needs

The system coordinates trusted membership, bounded contributions, timed reveal and reliable media delivery. A group needs confidence that only members can access its conversation and released film, that quotas and corrections remain consistent across devices, and that an interrupted process can recover without duplicate contributions or lost accepted media. These needs drive explicit authorization, transactional rules, durable jobs and honest client states.

1.3 Stakeholders

The five proposal members are Gopal Guruprasath, Nguyen Kim Long, Sang Haoxiang, Phan Mai Tan Loi and Jiang Jiayu. Their individual model ownership, contributions and effort require factual confirmation. Other stakeholders are private-group owners and invited members, authorized operators and the module assessors. There is no recorded commercial sponsor or lecturer approval of the pilot exceptions.

1.4 Project Scope

Sprint 2 develops a usable private-group web journey. The current release clients are desktop web, iPhone Safari and the installed Home Screen web app. The codebase also contains Expo/native adapters; Android/APK and compiled iOS obligations remain visible but are deferred for this overnight goal. The final proposal still requires managed OIDC, PostgreSQL, recoverable managed backups and retro processing before upload in Sprint 3. A working password/SQLite/server-processing pilot does not satisfy those final obligations.

1.4.1 Functionality in Scope

UC01–UC03 cover account/session, private group and invitation access. UC04–UC06 cover video review/trim/submission, photo contributions and quota/correction. UC07–UC08 cover automatic cycle rollover, compilation/retry/publication and archival. UC09–UC11 cover owner prompt/timezone, opt-in reminders and private text/reply/reaction/reconnect. UC12–UC14 cover premiere/Archive playback, own-clip/group-film download and client install/update/invite delivery. Five packages organize these fourteen use cases; they do not cap the number of major-flow diagrams.

1.4.2 Functionality out of Scope

The first delivery excludes public feeds/profiles/discovery, followers, hashtags, licensed music, owner transfer, member removal, leaving/rejoining and account

deletion/recovery. Chat editing/deletion, read receipts, typing and attachments are excluded. Public-store approval and a commercial SLA are not promised. The membership exclusions are privacy/retention limitations, not complete production account management.

1.4.3 Quality Attributes

The proposal requires server-enforced privacy, least privilege, authenticated HTTPS, idempotent writes, source-audio preservation, valid portrait video, bounded retries, clear failure states and repeatable delivery. Its performance targets include API p95 at most two seconds and a 25-clip/150-second compilation workload within ten minutes. Measured results must name build, fixture, hardware and controls; green tests alone are not those measurements. Real storage semantics, provider receipt, physical camera and installed-client lifecycle evidence remain distinct.

2. Project Conduct

2.1 Project Plan

Sprint names are zero-based: Sprint 0 foundation; Sprint 1 13–26 September; Sprint 2 27 September–10 October; Sprint 3 11–24 October 2026. The owner execution plan estimates 211–365 focused hours for Sprint 2 and 76–132 for the deferred compliance tranche. These are planning estimates, not recorded effort. The WBS groups entry/access, capture/media, cycle/compilation, participation/reminders, authorized delivery, operations and five report packages. Actual member responsibility for at least one use case and one pattern problem is pending; agent task ownership is not human contribution evidence.

2.2 Project Status

Accepted dev contains the password/SQLite pilot, private groups/invites, capture review, contribution policy, persisted chat, cycle processing, authorized Archive delivery and the public-only PWA shell. PR #387 repairs the photo group label; #388 repairs chat API prefixes; #392 adds web portrait guidance. Two local HTTPS chat journeys and two group-switching journeys under #396 passed. #251/#252 retain broader acceptance. Safari/Home Screen #329/#350, physical orientation #327, hosted/S3 #344 and installed playback/download #345 remain open. #352 aggregates the journey; it is not another implementation closure. Newly recorded Sprint 2 issue #399 identifies an unsafe Demo/reset path that deletes shared media before restoreFixture when real-account data exists. The fail-closed fix is reviewed in batch bb3b089 and awaits dev integration. Worker validation reported 26 focused, 452 server, 612 frontend and 6 accessibility checks; these were not rerun for this report. At this source cut, no hosted reset/restore pass or final private-data boundary acceptance is claimed.

2.3 Project Metrics

The fresh baseline was fc2b8c811e9c13b176113fa0bdb2c015798990b4 with successful Quality and CodeQL runs. PR #387 tested head 64c6e40e56da31ab602bd6c82384b5d9241b7251 and merged as f1d74c510ecff566a25b1449daa8352befa9f00c with an identical Git tree. Its fast checks passed 148 root tests with one existing skip, 443 server tests and 605 frontend tests. These counts describe that exact run; they are not human productivity or time spent. Actual Sprint meeting decisions, burndown data and per-member hours remain pending where no verified record supplies them.

3. System Design

3.1 Software Architecture

Current pilot architecture

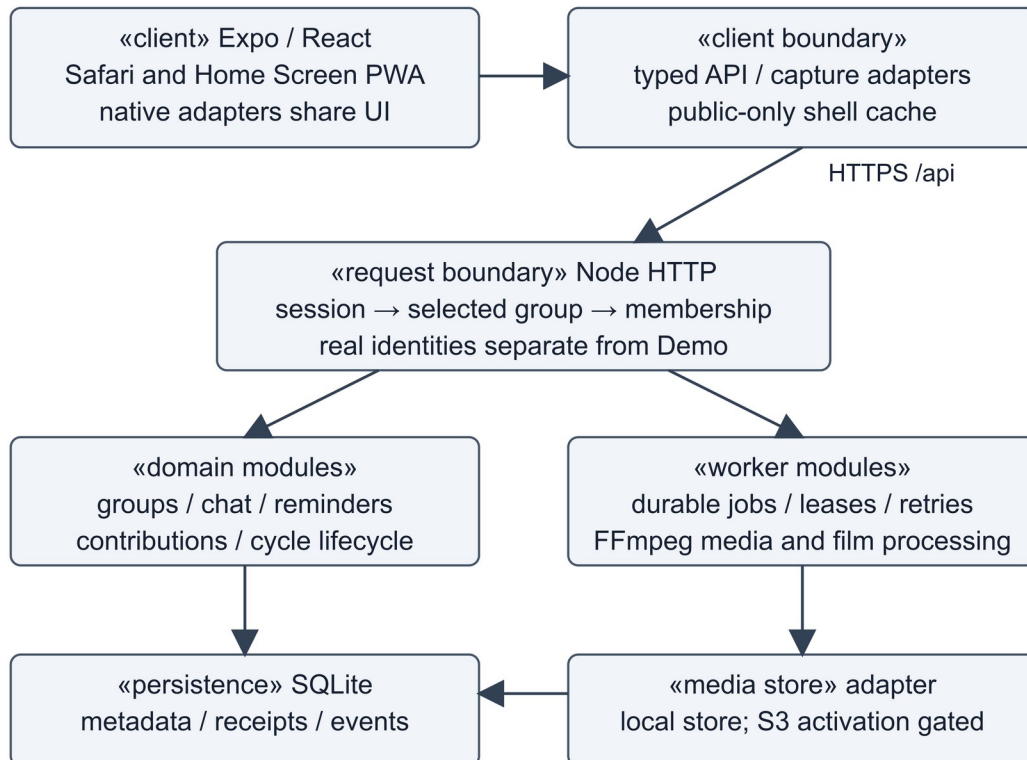


Figure 1 Current component and layer model. Source: accepted dev f1d74c5; src/runtime, server/src, server/migrations.

Future compliance target

Sprint 3 obligations; not current acceptance

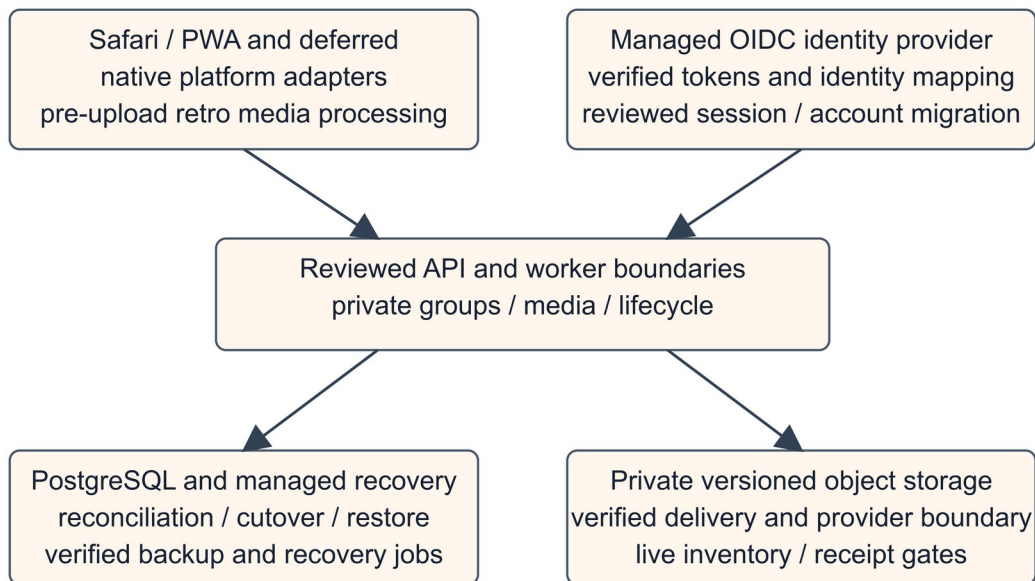


Figure 2 Final compliance target. OIDC, PostgreSQL, managed recovery and pre-upload retro remain Sprint 3 obligations.

The current implementation uses a TypeScript Expo/React client, React Native web/PWA output, a modular Node HTTP service, SQLite metadata and FFmpeg media processing. The web shell and API share an HTTPS origin with an /api proxy boundary. Browser authority uses an HttpOnly cookie; native adapters use a securely stored opaque token. Real and synthetic Demo identities are separate. Public shell caching never authorizes or caches private API/media responses.

Capture, group, chat, reminder and Archive UI modules call typed adapters. The server establishes the principal, selected group and membership before domain operations. SQLite transactions persist quota, invitation, contribution and lifecycle changes. Durable jobs/leases coordinate processing and retries; released-media capabilities recheck session, selected group, release and exact media integrity. Private S3/direct-upload code and isolated Terraform roots are prepared; actual hosted S3 semantics and live inventory remain separate gates. This source description does not assert every prepared adapter is activated in the hosted pilot.

The final target replaces pilot password verification with managed OIDC and SQLite with reviewed PostgreSQL, adds managed recovery, and processes retro treatment before upload. Provider notifications, final native packaging and live storage acceptance require their own verified boundaries. Target architecture is labelled separately from current implementation.

3.2 Transition from Analysis to Design

Six strategies guide the complete journey. T1 trusted identity maps conceptual Account/Member to session resolution, request-boundary adapters and transaction-

bound membership predicates. T2 platform capture maps a conceptual Recording to camera/file/store/review contracts and explicit ownership/disposal. T3 transactional contribution maps Budget/Correction to durable contribution/job/quota records and idempotency keys. T4 persisted lifecycle maps Cycle/Film to recorded state transitions, successors, leases and retryable jobs. T5 asynchronous participation maps Conversation/Reminder to ordered events, reconnect cursors, per-session destinations and durable outbox/provider adapters. T6 authorized delivery maps ReleasedMedia to fresh session/group-bound capabilities, client players/download adapters and public-only asset caching.

3.2.1 Static and dynamic transition guidance

For T1, keep principals distinct from client resource IDs; validate session before membership/owner query, then mutate within the authorized transaction. For T2, create a platform adapter and owned local media handle; review precedes submission and retake/success disposes content. For T3, bind a retry key to one contribution and apply both quota dimensions atomically. For T4, persist cycle closure/successor and durable work before execution; publication must observe ready output and maintain the 24-hour premiere. For T5, separate durable facts from live transport; reconnect uses a watermark and notification delivery is not inferred from scheduling. For T6, refresh temporary access at each play/download and clear stale protected media on group/session change. The package model pairs show these concrete static/dynamic mappings; the incorporated plates in §§3.4–3.5 retain the accepted model cuts and a refreshed source crosswalk.

3.3 Use Case Model

3.3.1 Overall use case diagram

Overall use case model

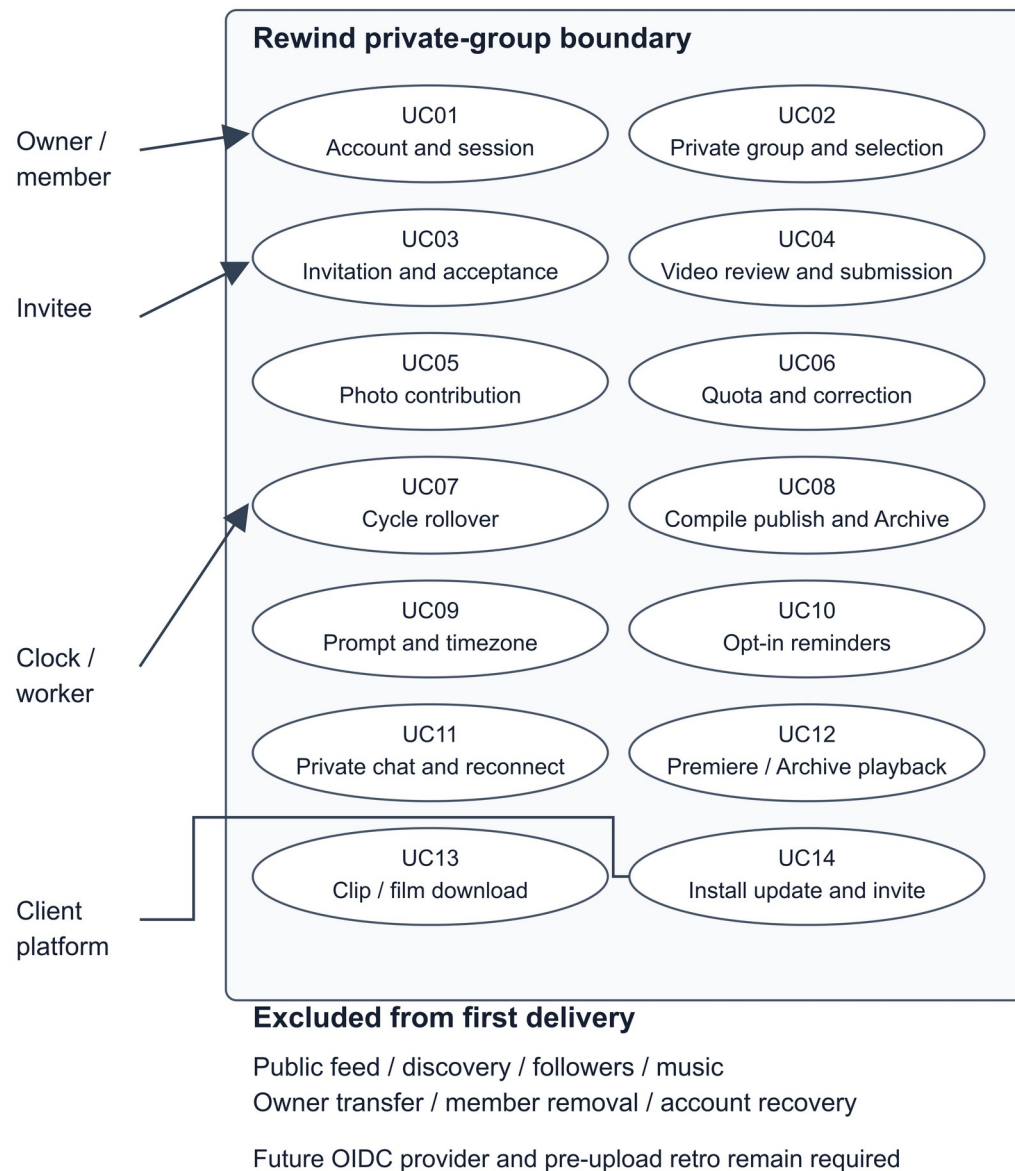


Figure 3 Overall use case catalogue. Actor associations are an overview; package flows refine the authorization roles.

The overall model groups UC01–UC14 under the private Rewind boundary, with owner/member, invitee, clock/worker and platform roles. Excluded public-social and account-management cases are outside the boundary. Managed OIDC is a future provider role. Package-specific diagrams refine the same boundary.

3.3.2 Use case descriptions

Each package's use-cases.md declares normal and relevant exceptional major flows: access rejection/expiry/replay; unsupported capture/permission, invalid media,

upload interruption; quota and correction boundaries; scheduler/processing failure and restart; chat reconnect/denial and reminder expiry; sealed/unavailable/offline media and client updates. A flow catalogue must enumerate every required paired sequence; a single happy-path sequence does not cover all major flows.

3.4 Analysis and Design Models

3.4.1 Package models and ownership

ACCESS UC01–UC03, CAPTURE UC04–UC06, CYCLE UC07–UC08, PARTICIPATION UC09–UC11 and ARCHIVE UC12–UC14 are separate source packages. Analysis class views describe domain responsibilities; design views name implemented modules/interfaces/persistence. Each major flow requires one analysis and one design sequence. This draft embeds all 137 accepted package models: 28 use-case class views, 88 sequence views for 44 major-flow pairs, 20 before/after pattern views and the ACCESS use-case overview. The figure register maps each plate to its unchanged SVG, package model text, issue and source cut. Real human ownership remains pending. No member is assigned a use case from commit counts.

Detailed model sources: ACCESS, CAPTURE, CYCLE, PARTICIPATION, ARCHIVE.

Complete-model plates preserve every accepted SVG and all relationships. The original A4 portrait body resumes after each model block. A3 landscape foldouts are used for most figures; 18 unusually tall or dense figures use 22 by 16.54 inch landscape foldouts. Principal 14 px labels print at least 8.4 pt; 12 px annotations print at least 7.2 pt. The smallest accepted 11 px notes print at least 6.6 pt. Captions are 9 pt. Native SVG and lossless PNG fallback preserve readable zoom without semantic redrawing.

ACCESS model/source register | issue #357

Accepted model cut fc2b8c811e9c13b176113fa0bdb2c015798990b4; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

src/auth/RealAccountProvider.tsx:55

src/auth/real-account-client.ts:103

server/src/auth/index.ts:122

server/src/http.ts:4141

server/src/groups/real.ts:15

server/src/groups/real.ts:40

server/src/groups/real.ts:45

server/src/http.ts:3516

server/src/groups/settings.ts:5

server/src/groups/settings.ts:44

server/src/groups/settings.ts:70

server/src/http.ts:3783

src/groups/RealAccountGroupExperience.tsx:659

src/reminders/RealGroupSettings.tsx:131

src/reminders/RealGroupSettings.tsx:184

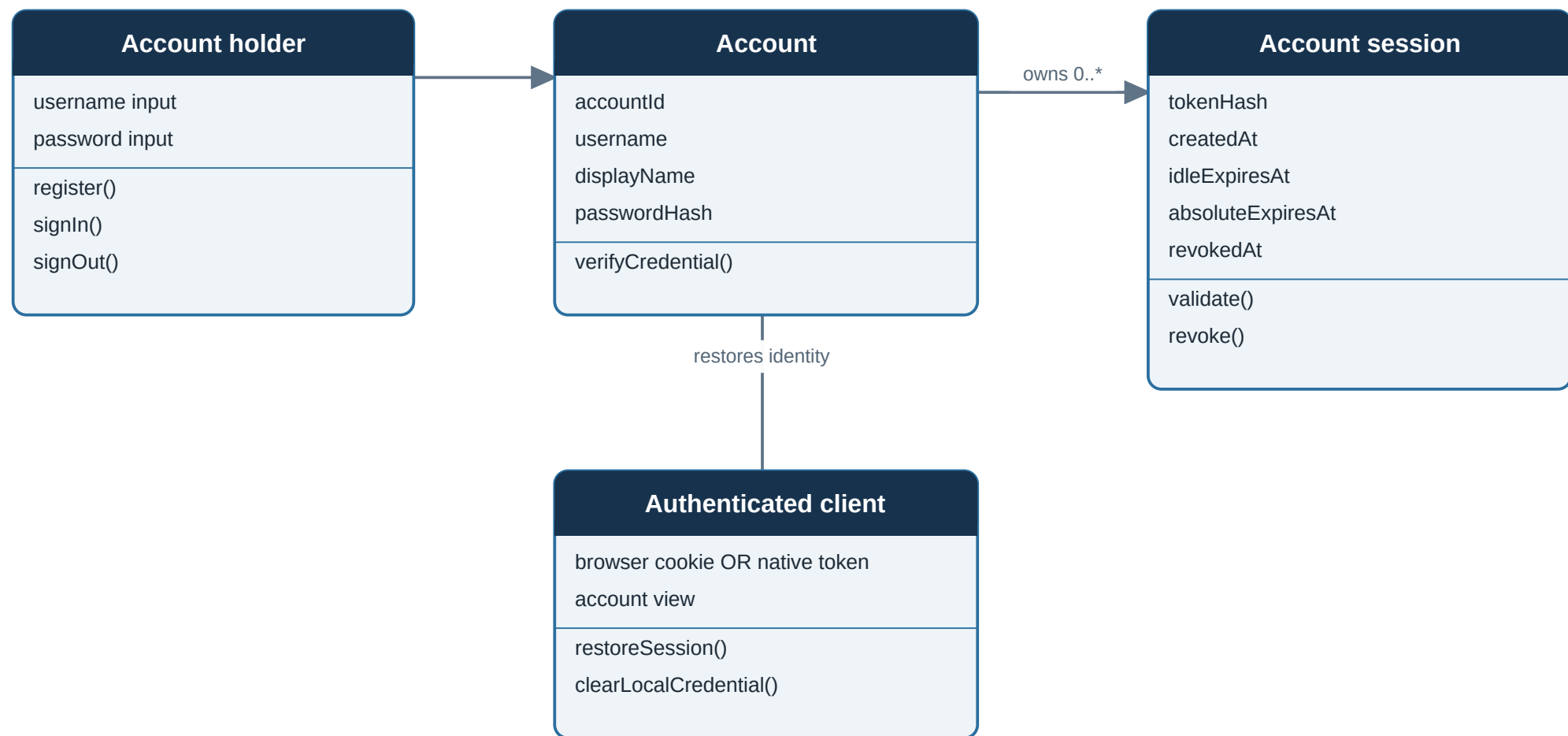
server/src/reminders/schedule.ts:3
server/src/reminders/schedule.ts:55
server/src/db.ts:219
server/migrations/025-real-group-reminders.sql:3
server/src/cycles/lifecycle.ts:276
server/src/groups/invites.ts:124
src/groups/RealAccountGroupExperience.tsx:688
src/invites/deep-links.ts:52
server/src/policy.ts:28
server/src/http.ts:312
server/src/groups/real.ts:121
server/src/http.ts:348

Figure M005 | ACCESS | uc01 analysis class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC01 — analysis classes

Conceptual model: account identity and server session; no external identity provider is active in this pilot.



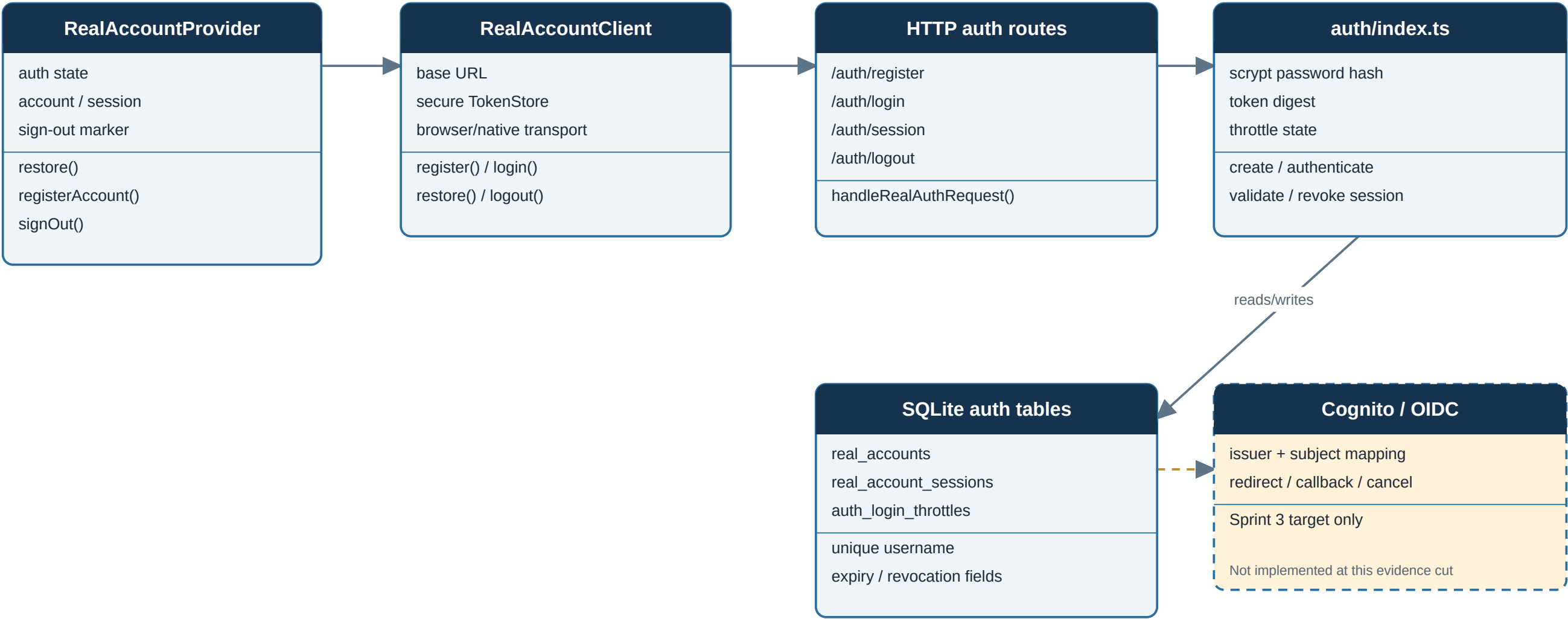
Source concepts: `real_accounts`, `real_account_sessions`, `RealAccount`, `RealSessionResult`. Analysis view does not equate a Demo member with an account.

packages/access/diagrams/uc01-analysis-class.svg

Figure M006 | ACCESS | uc01 design class
Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC01 — design classes

Current local-password implementation. Dashed Cognito box is the separate Sprint 3 target.



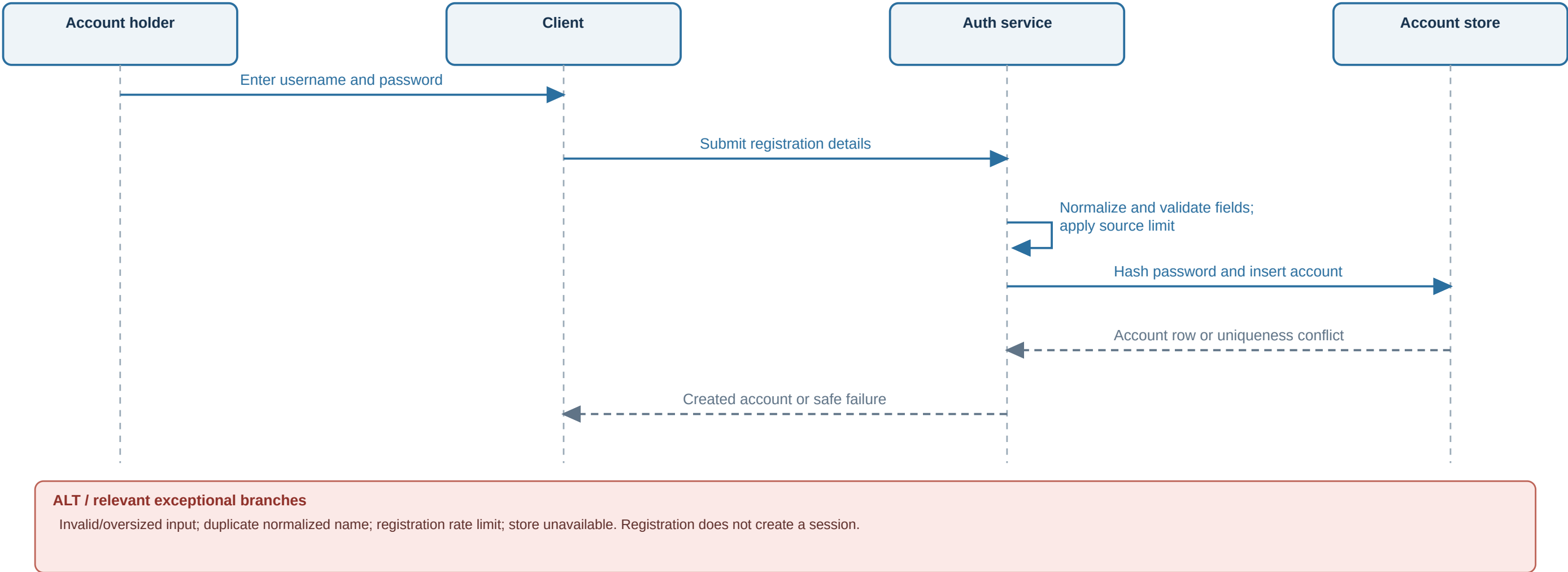
Code: src/auth/RealAccountProvider.tsx; src/auth/real-account-client.ts; server/src/http.ts; server/src/auth/index.ts.

Figure M007 | ACCESS | uc01 f1 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC01-F1 — Register an account — analysis sequence

Conceptual actor/service/entity interactions; success and relevant exceptional branches are both identified.

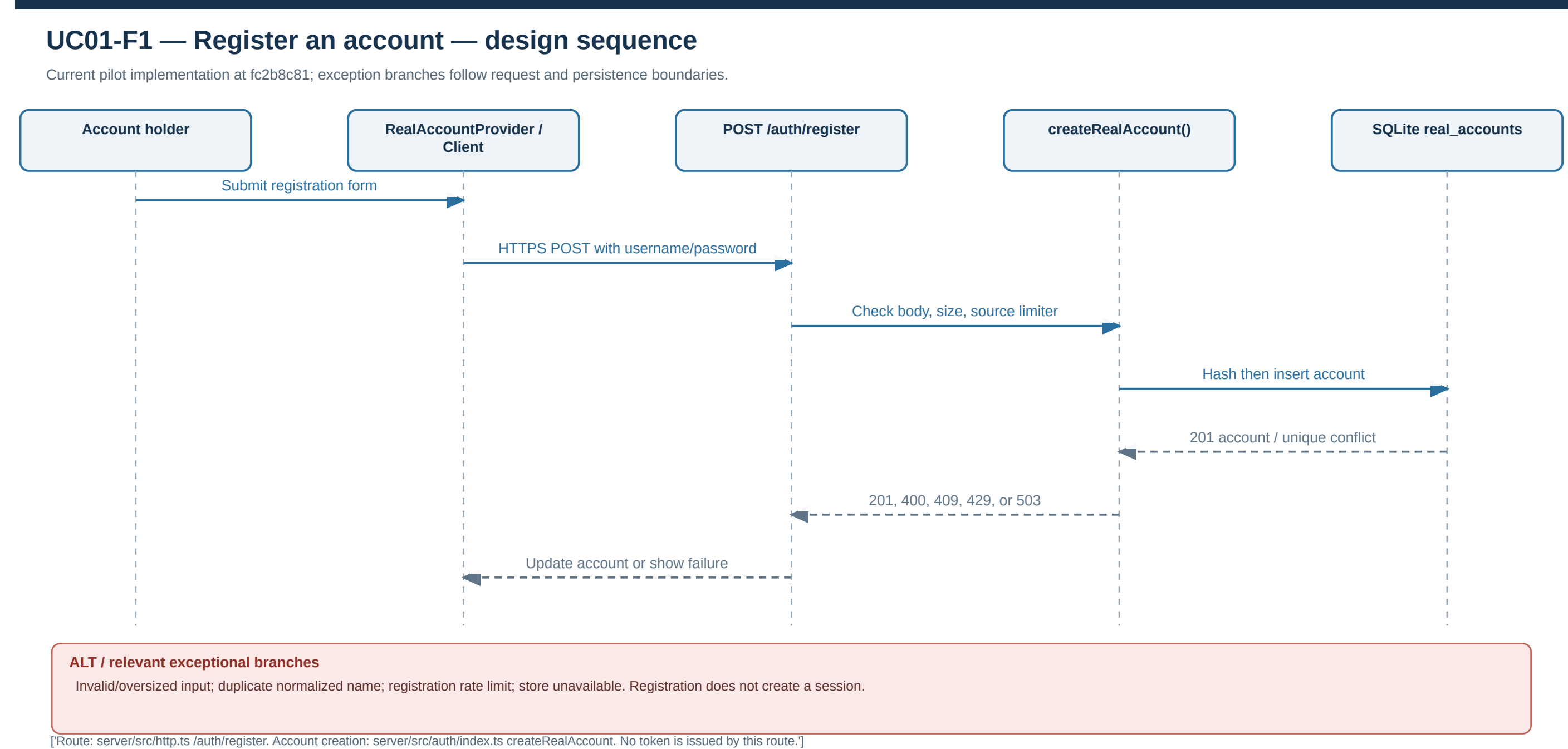


Analysis view; implementation names are deliberately omitted.

packages/access/diagrams/uc01-f1-analysis-sequence.svg

Figure M008 | ACCESS | uc01 f1 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



packages/access/diagrams/uc01-f1-design-sequence.svg

Figure M009 | ACCESS | uc01 f2 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

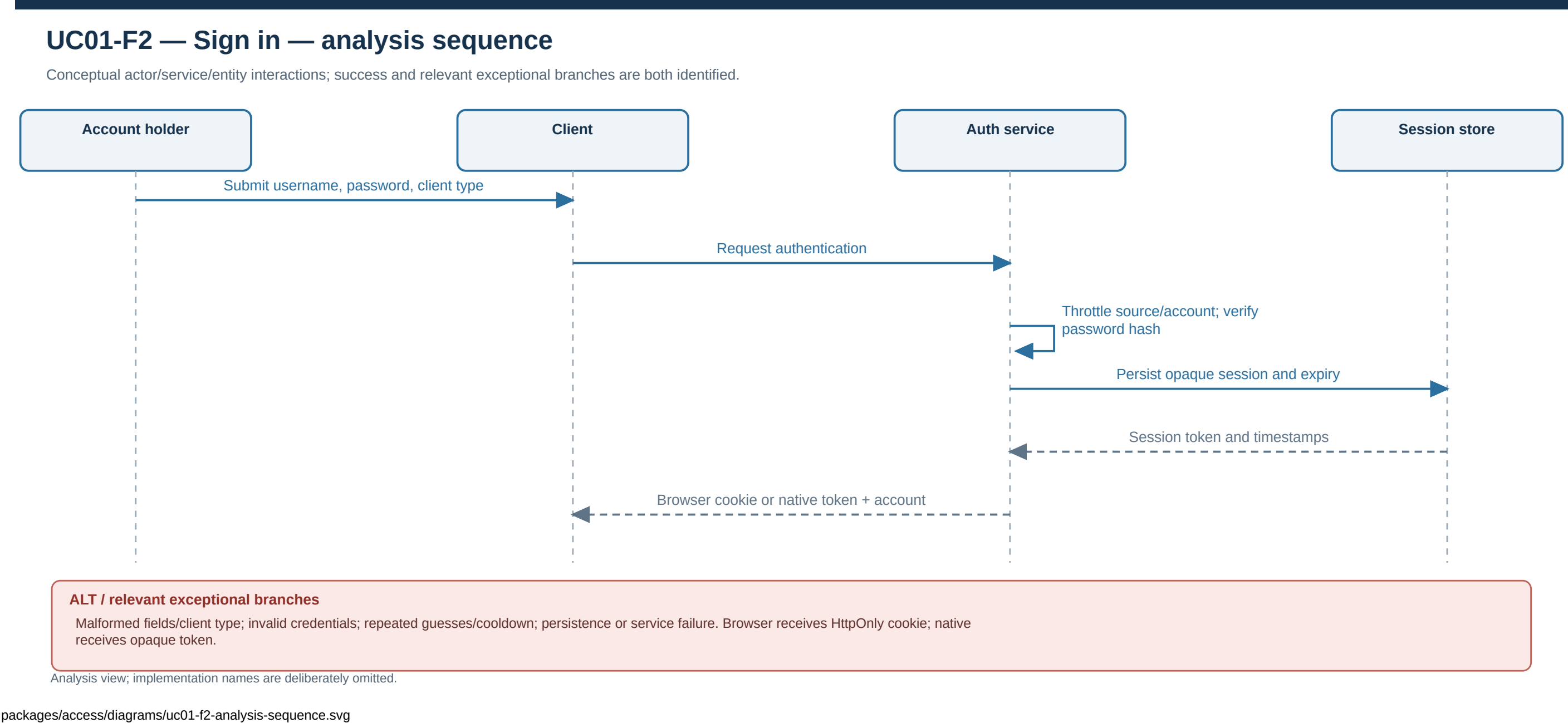
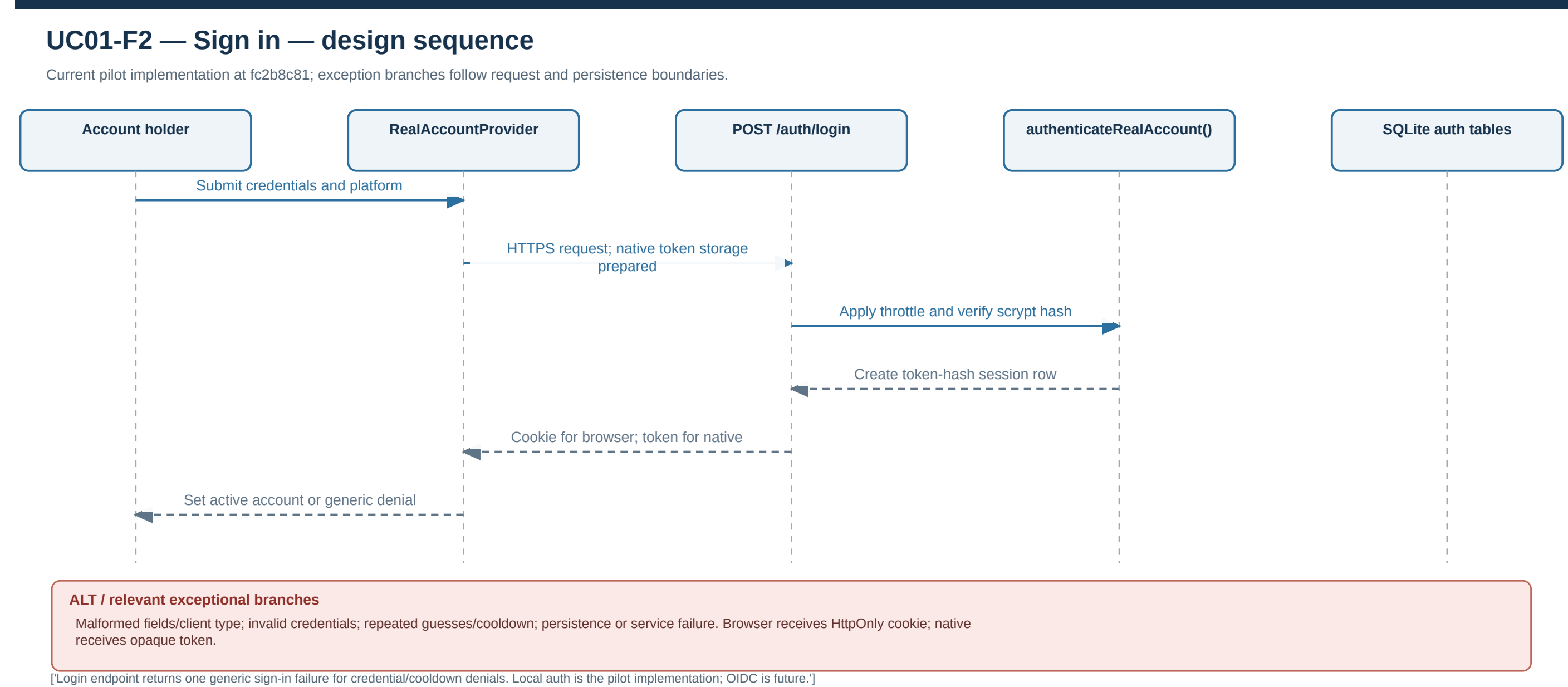


Figure M010 | ACCESS | uc01 f2 design sequence

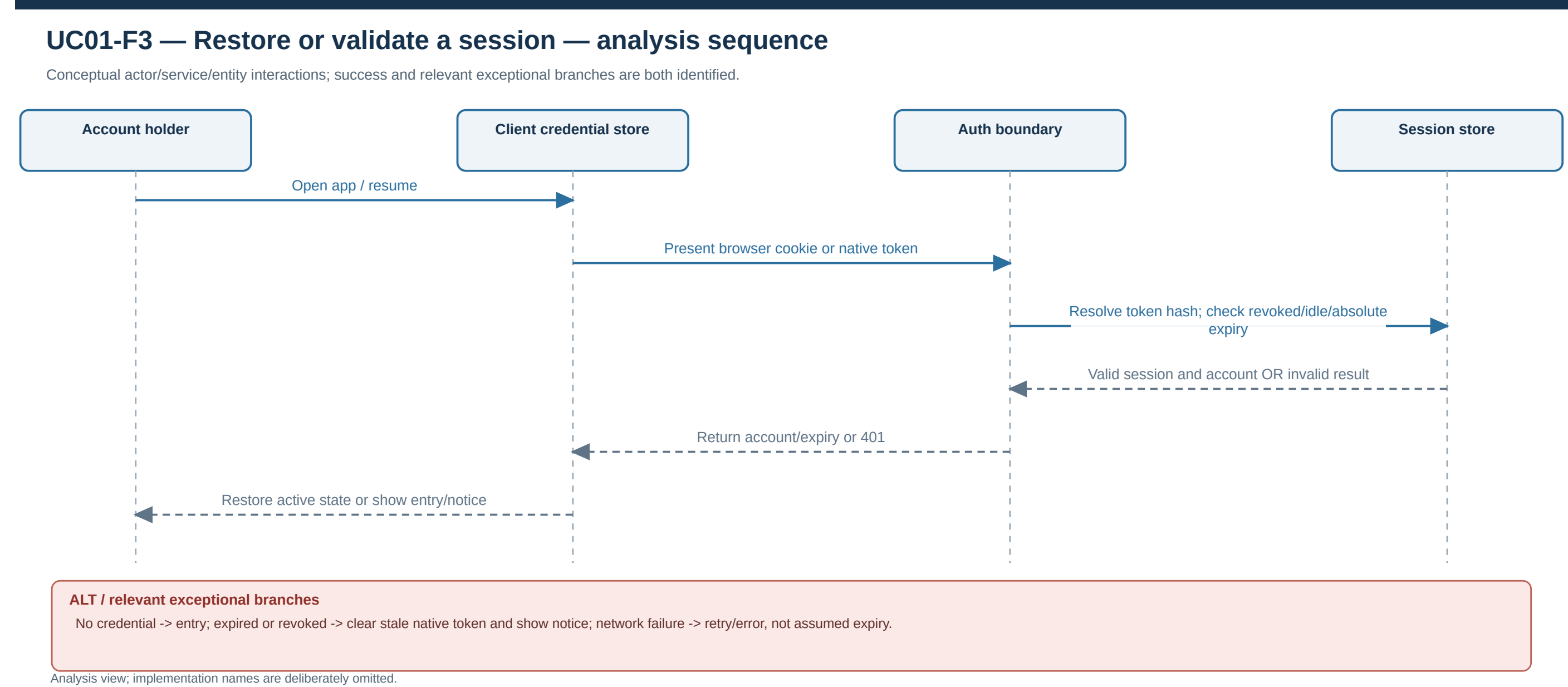
Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



packages/access/diagrams/uc01-f2-design-sequence.svg

Figure M011 | ACCESS | uc01 f3 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



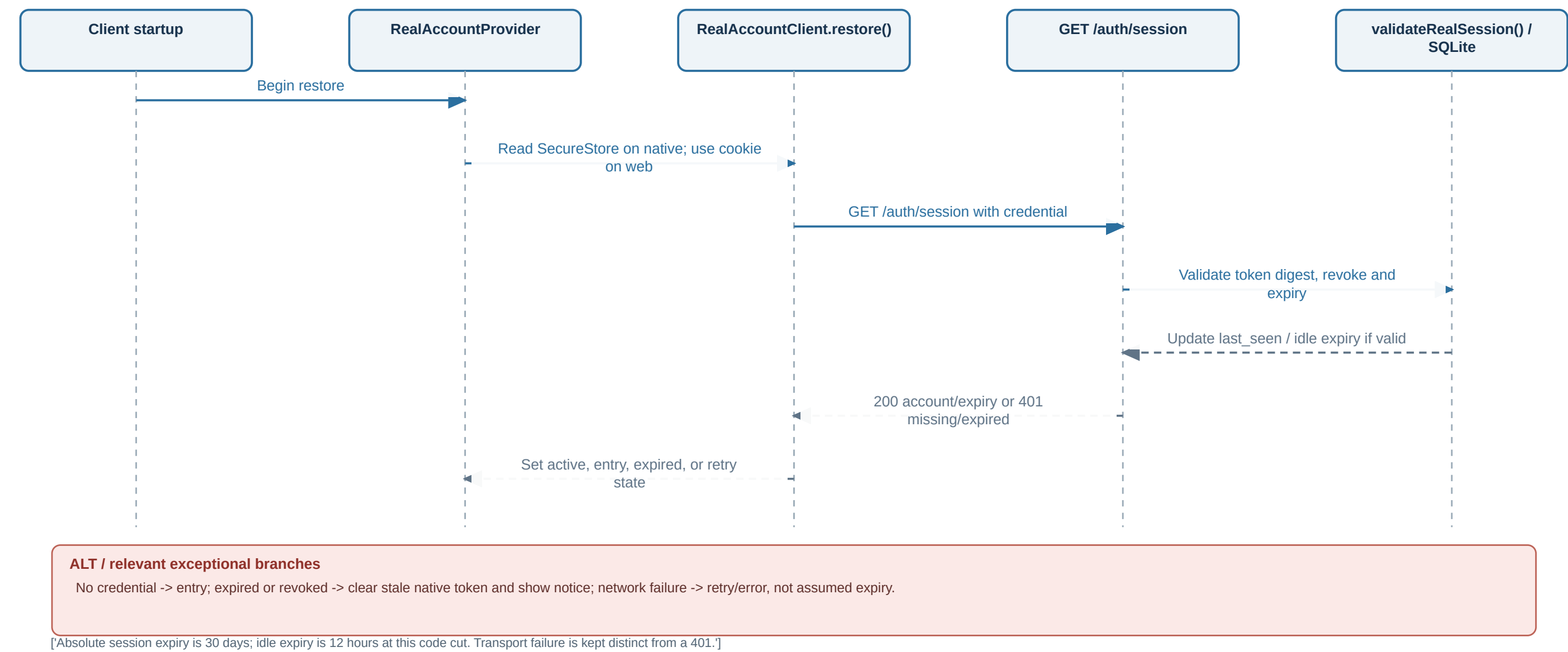
packages/access/diagrams/uc01-f3-analysis-sequence.svg

Figure M012 | ACCESS | uc01 f3 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC01-F3 — Restore or validate a session — design sequence

Current pilot implementation at fc2b8c81; exception branches follow request and persistence boundaries.



packages/access/diagrams/uc01-f3-design-sequence.svg

Figure M013 | ACCESS | uc01 f4 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

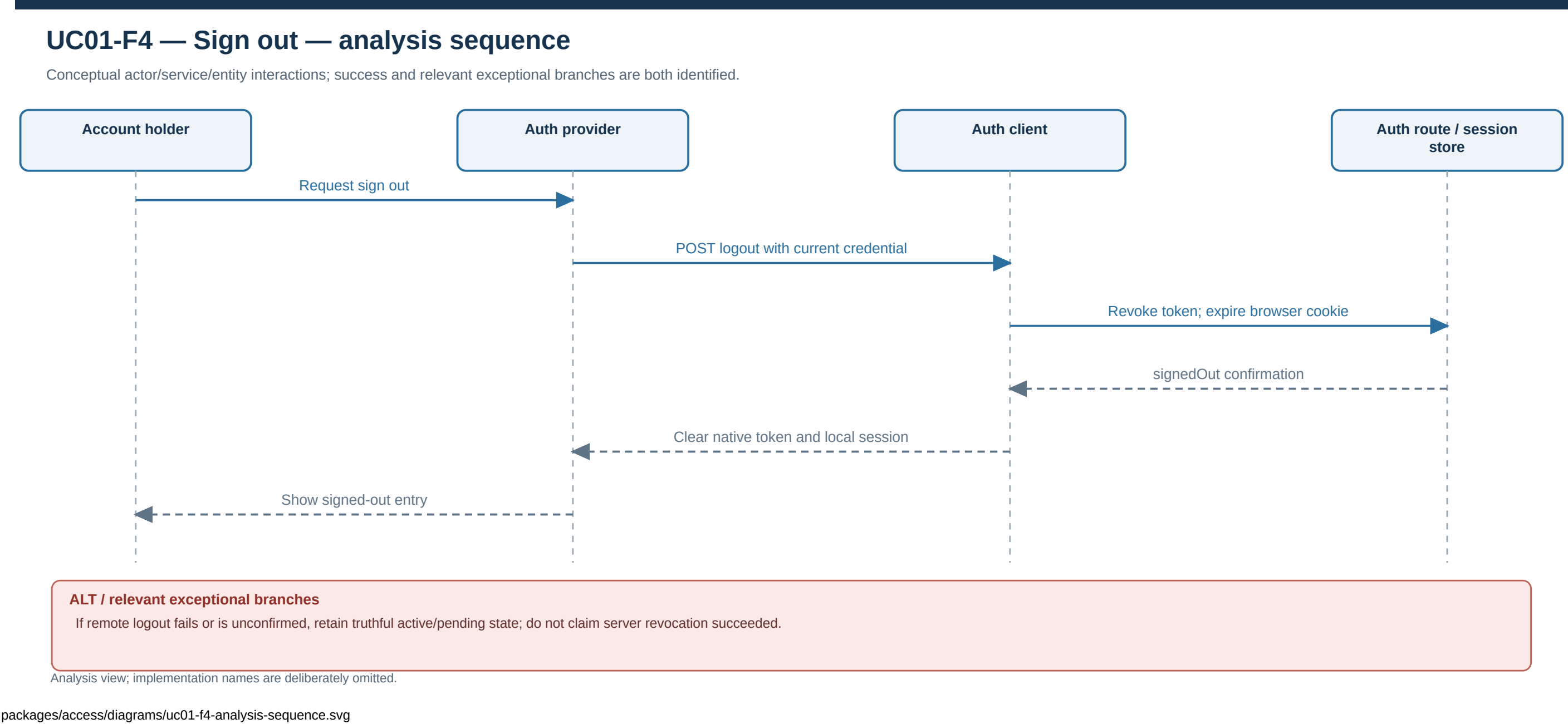
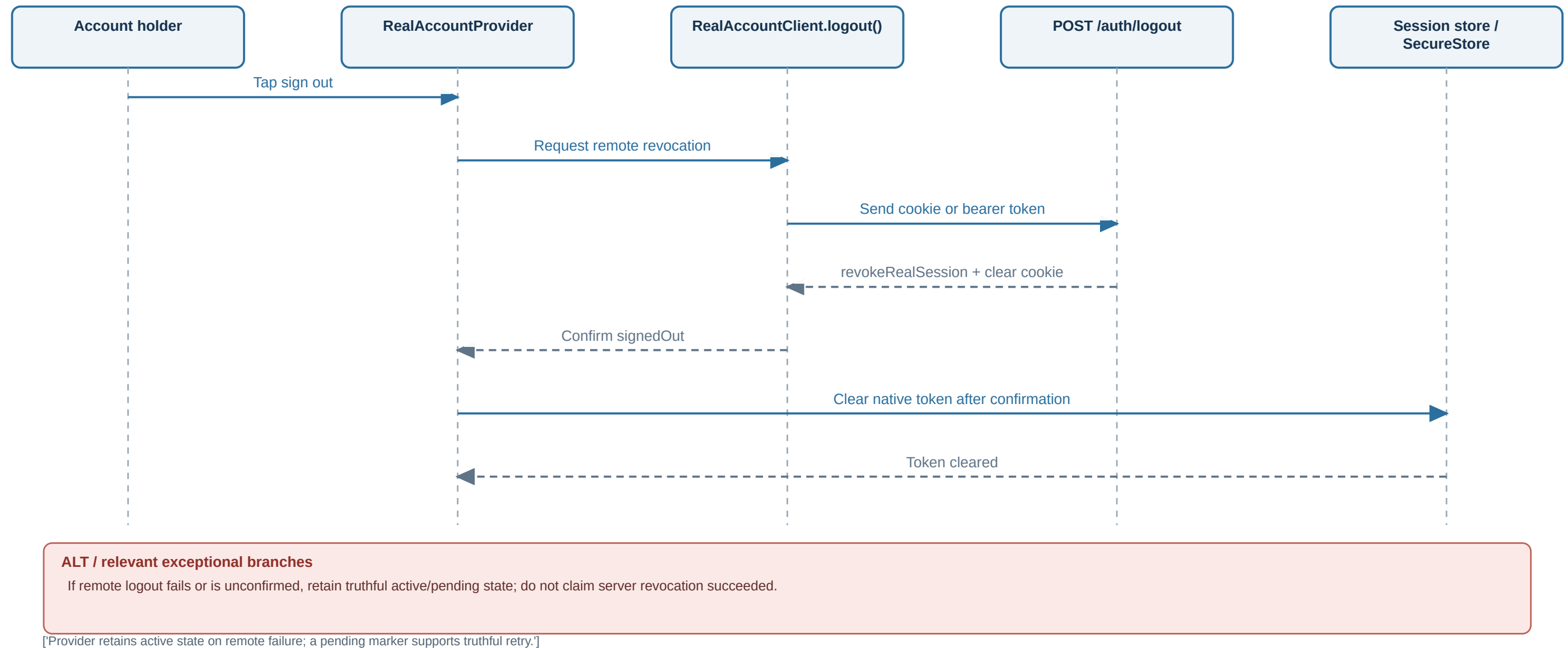


Figure M014 | ACCESS | uc01 f4 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC01-F4 — Sign out — design sequence

Current pilot implementation at fc2b8c81; exception branches follow request and persistence boundaries.



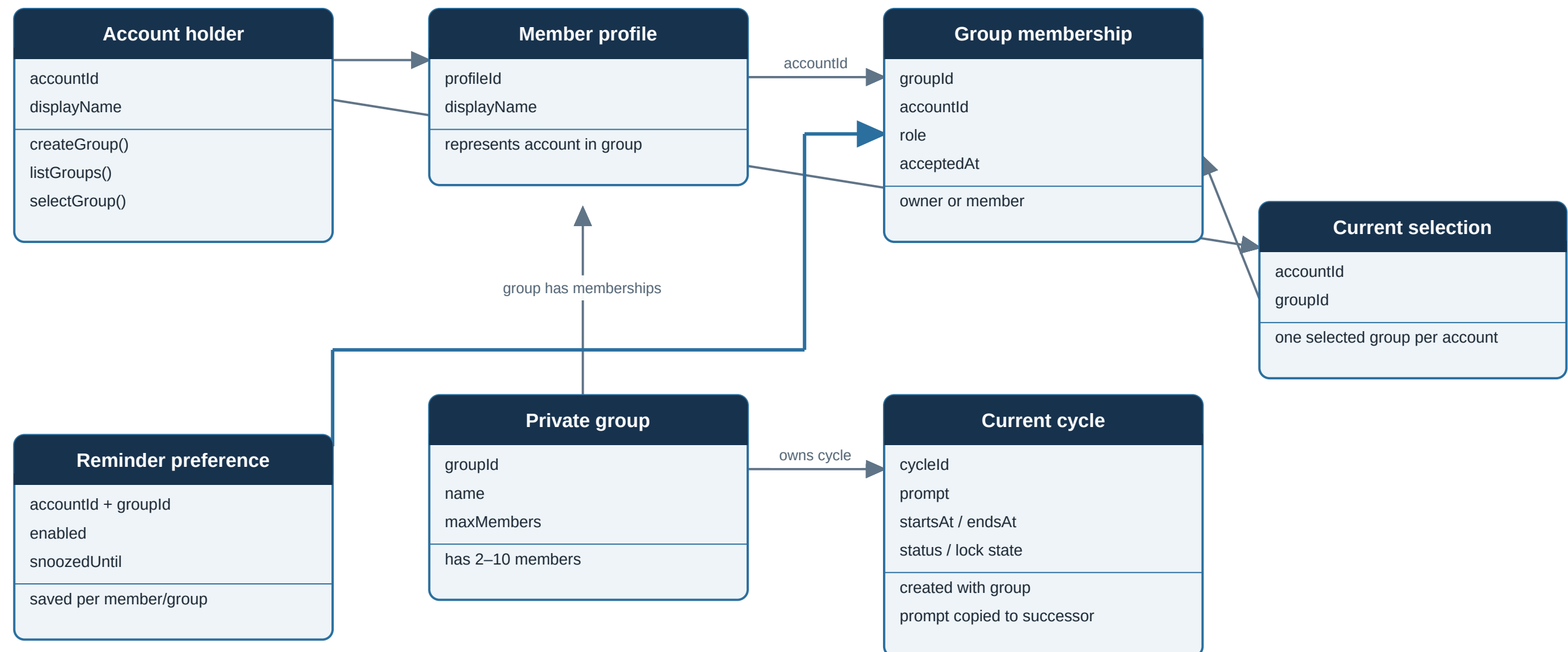
packages/access/diagrams/uc01-f4-design-sequence.svg

Figure M015 | ACCESS | uc02 analysis class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC02 — analysis classes

A real account owns membership; group settings belong to the owner, while reminder preference belongs to each member.



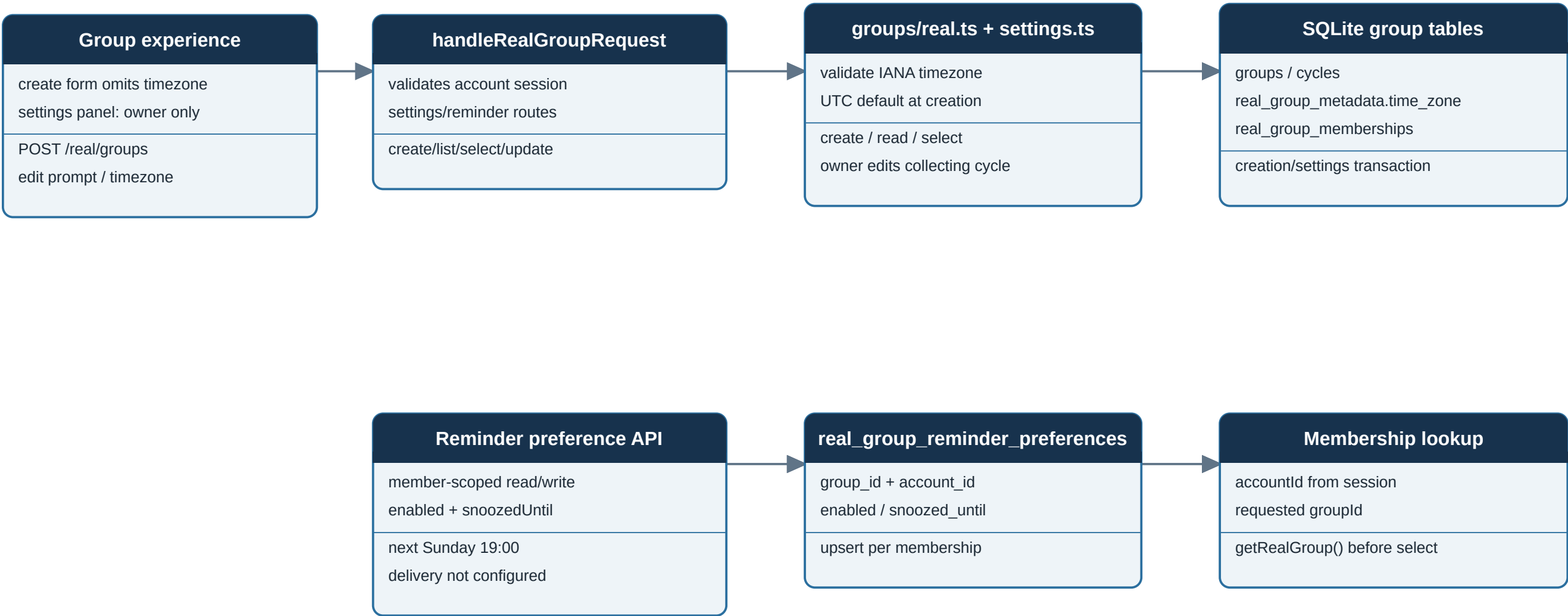
A group selector names the resource. Membership supplies authority; reminder preference is per account and group.

Figure M016 | ACCESS | uc02 design class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC02 — design classes

Group creation/settings are transactional; per-member reminder preferences are saved separately. Delivery is not configured.



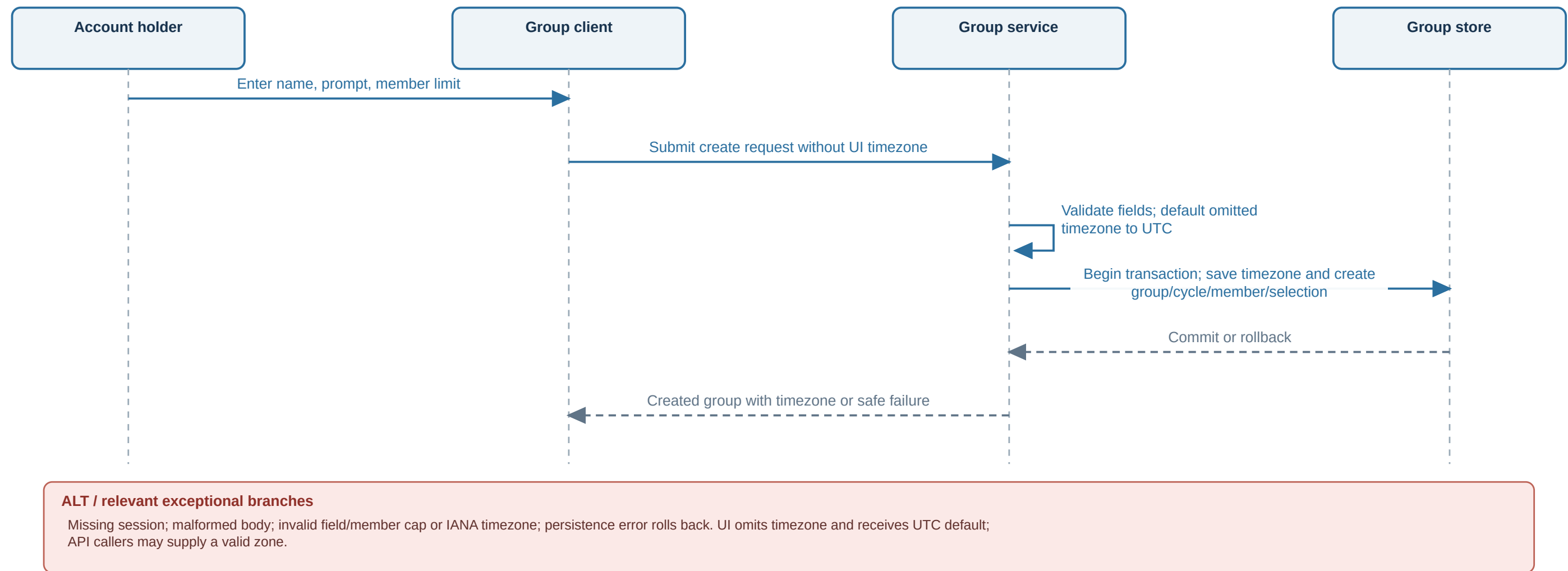
No membership removal, owner transfer, or account deletion flow is implied.

Figure M017 | ACCESS | uc02 f1 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC02-F1 — Create a group — analysis sequence

Conceptual actor/service/entity interactions; success and relevant exceptional branches are both identified.



Analysis view; implementation names are deliberately omitted.

packages/access/diagrams/uc02-f1-analysis-sequence.svg

Figure M018 | ACCESS | uc02 f1 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

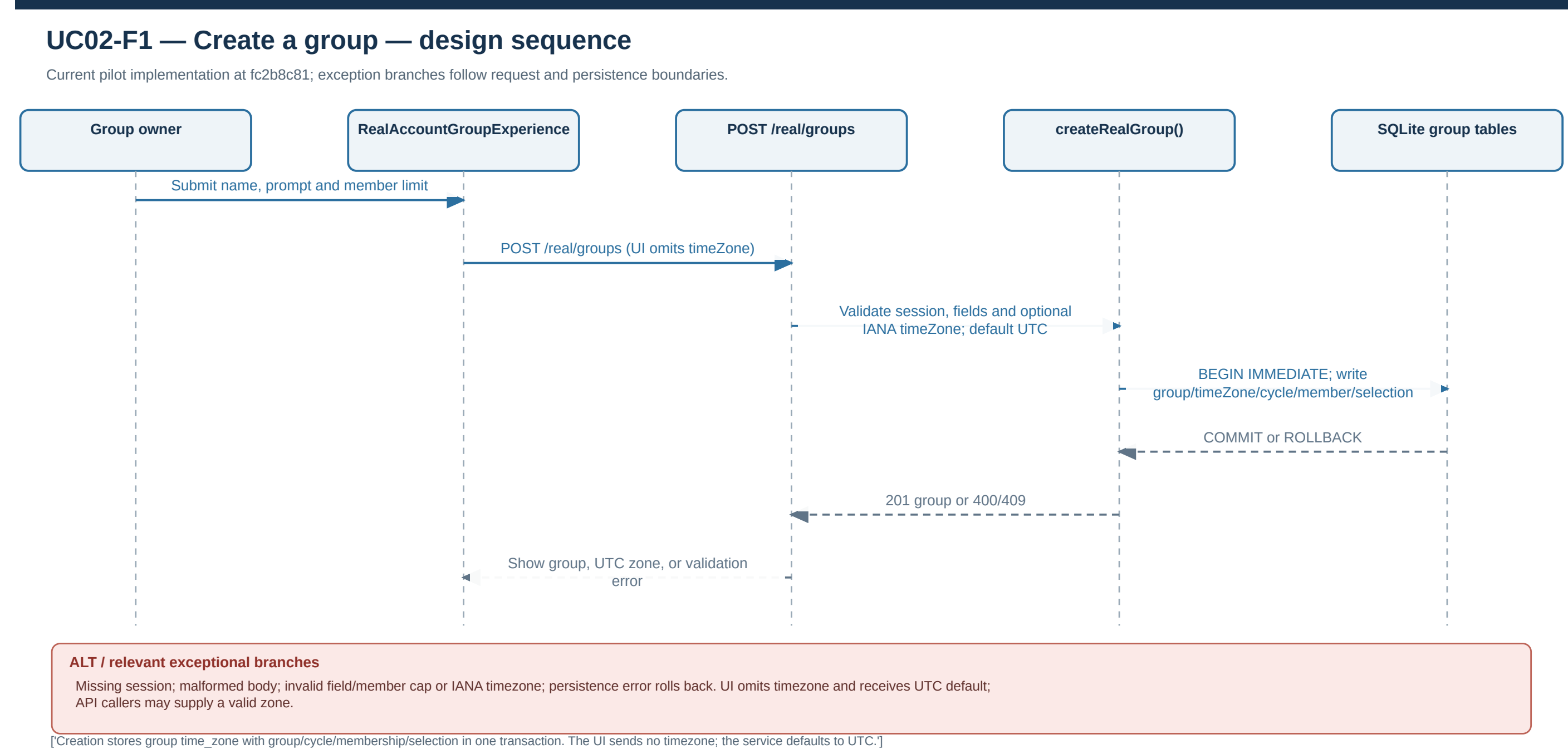
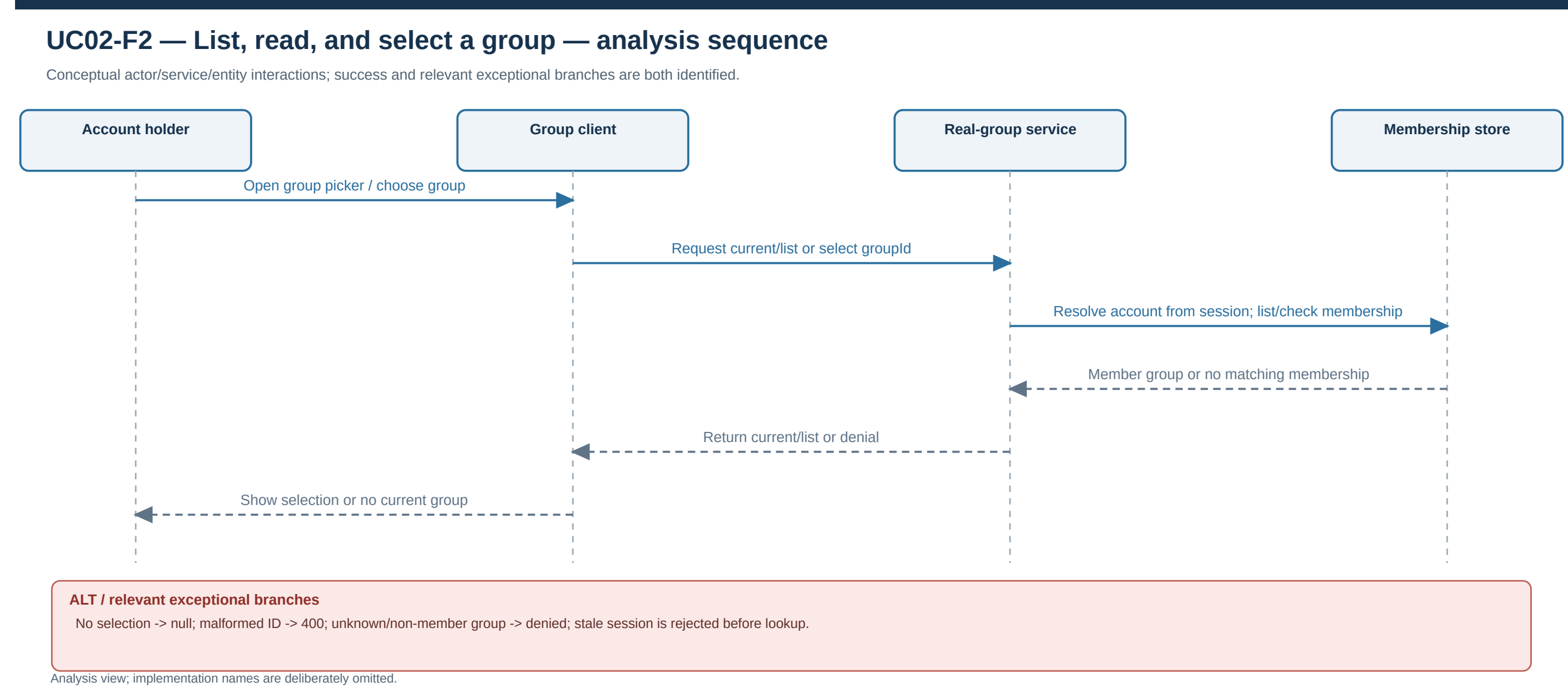


Figure M019 | ACCESS | uc02 f2 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



packages/access/diagrams/uc02-f2-analysis-sequence.svg

Figure M020 | ACCESS | uc02 f2 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

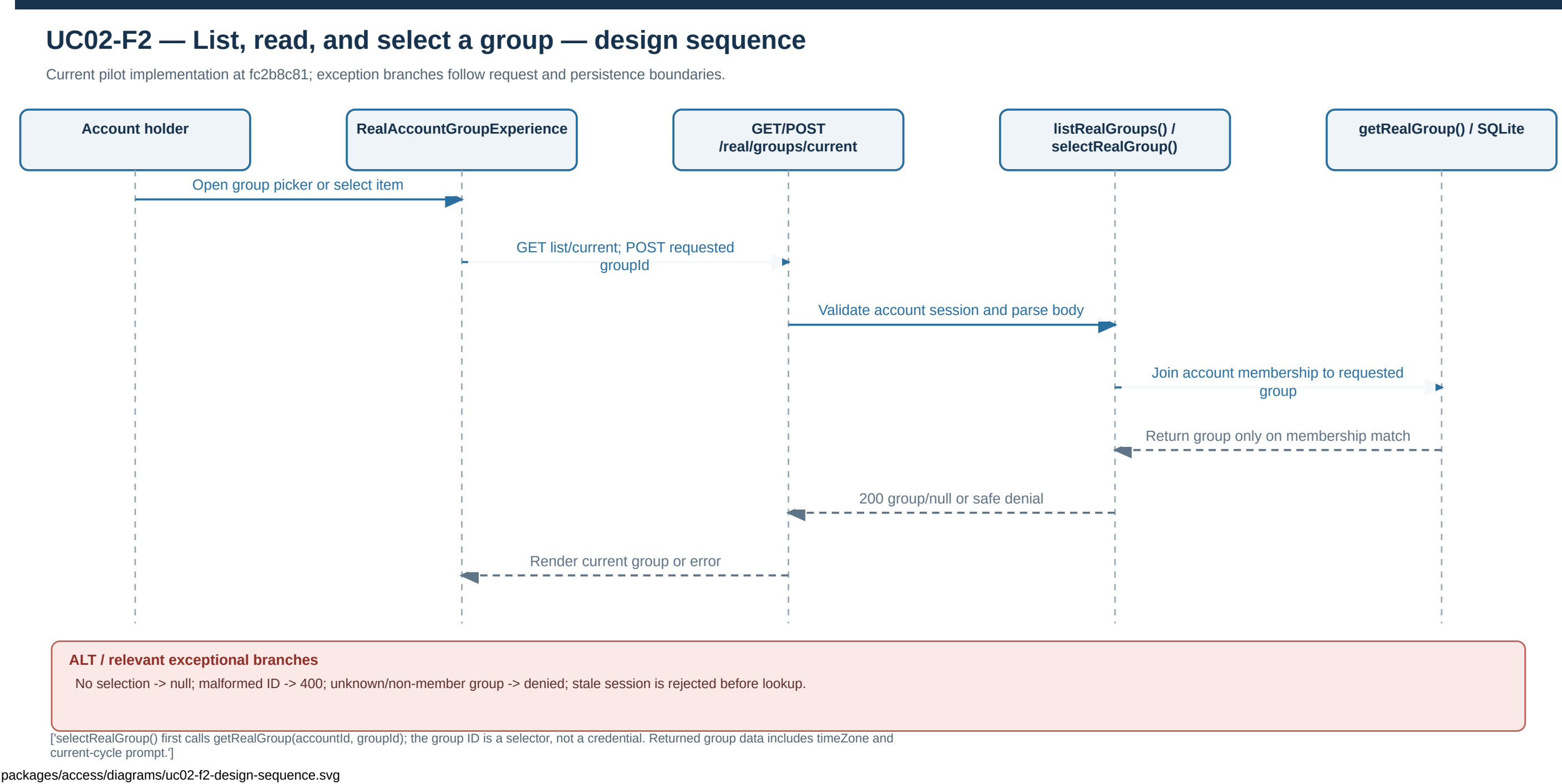


Figure M021 | ACCESS | uc02 f3 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

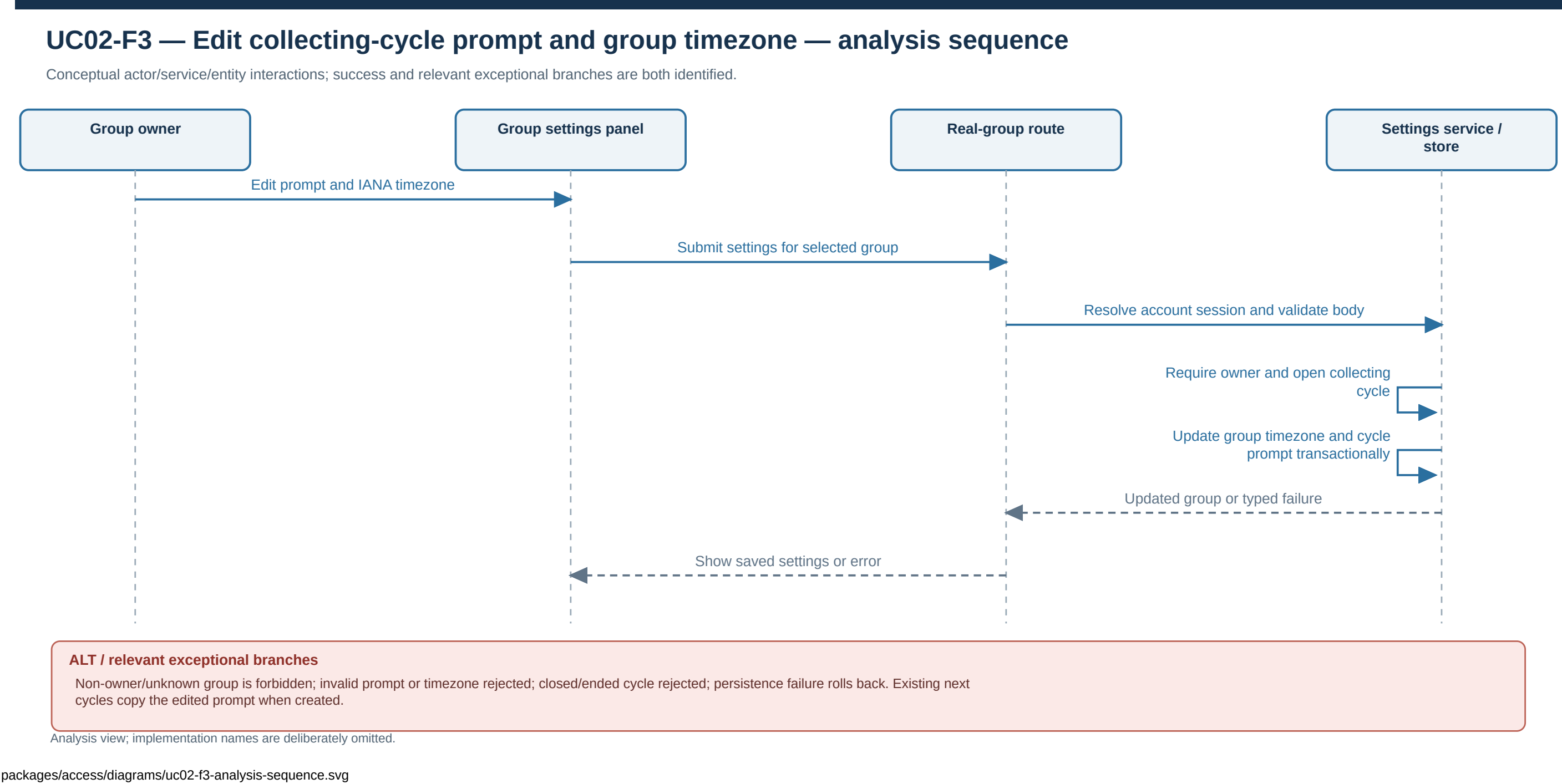
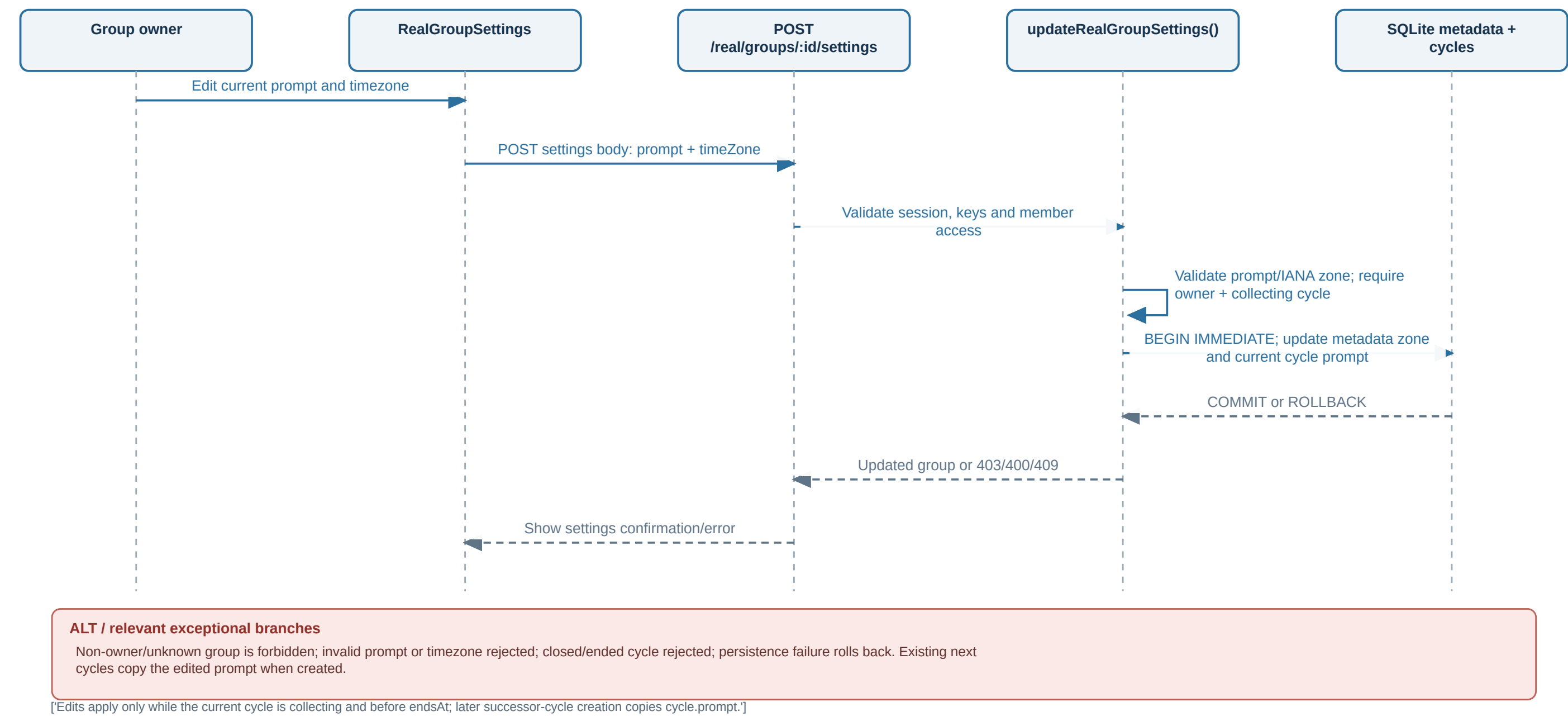


Figure M022 | ACCESS | uc02 f3 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC02-F3 — Edit collecting-cycle prompt and group timezone — design sequence

Current pilot implementation at fc2b8c81; exception branches follow request and persistence boundaries.



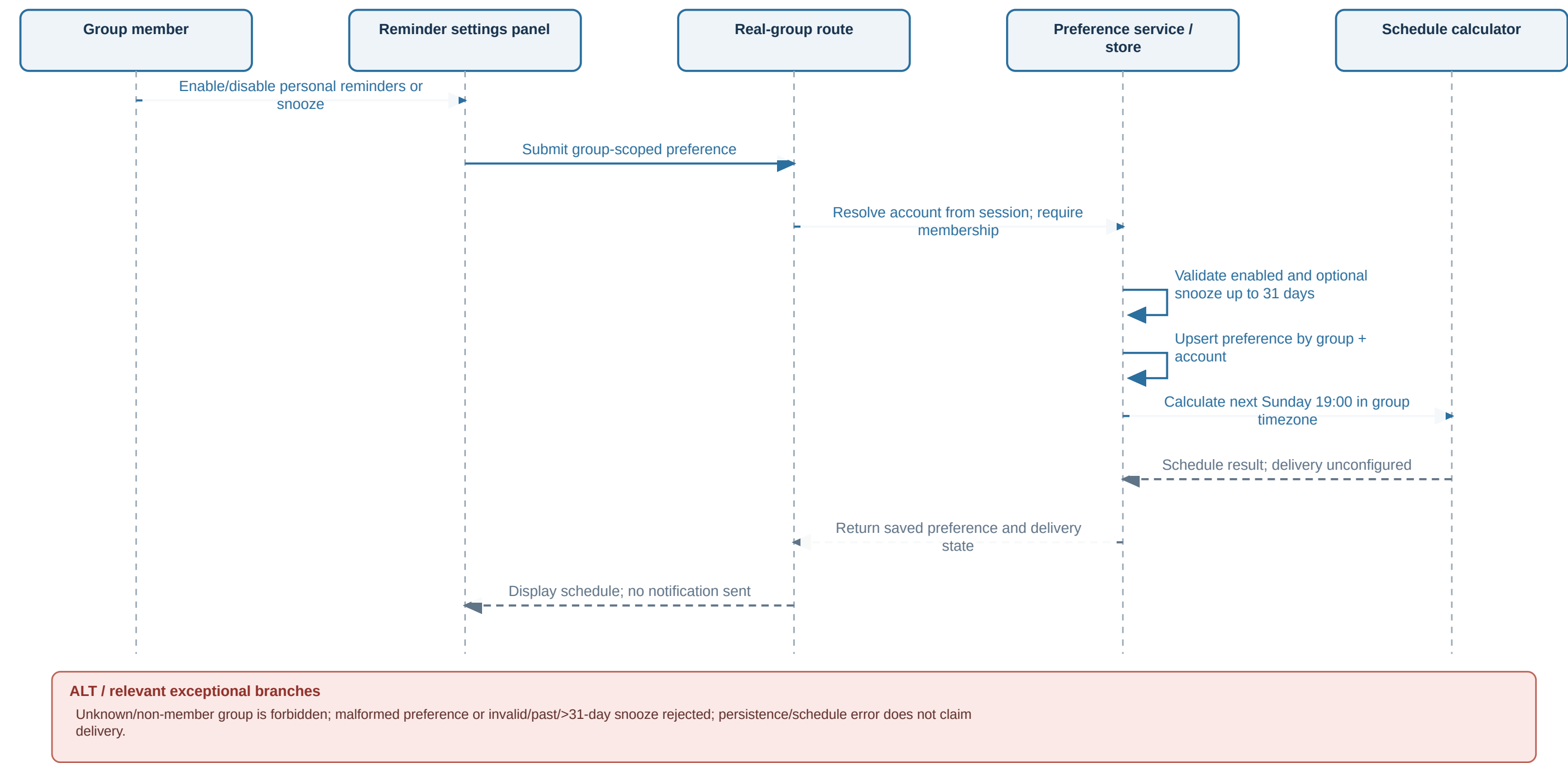
packages/access/diagrams/uc02-f3-design-sequence.svg

Figure M023 | ACCESS | uc02 f4 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC02-F4 — Save a member reminder preference — analysis sequence

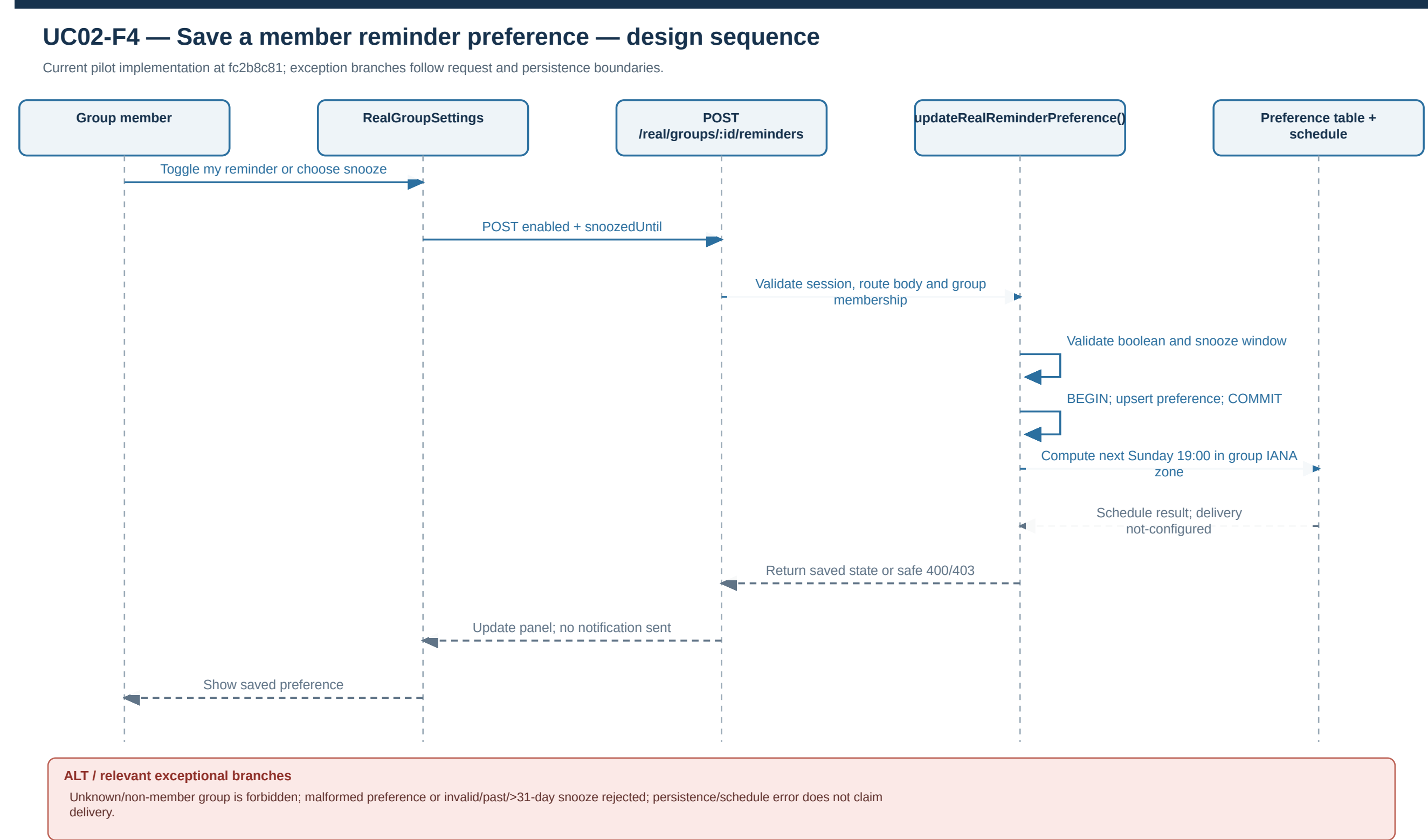
Conceptual actor/service/entity interactions; success and relevant exceptional branches are both identified.



Analysis view; implementation names are deliberately omitted.

Figure M024 | ACCESS | uc02 f4 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



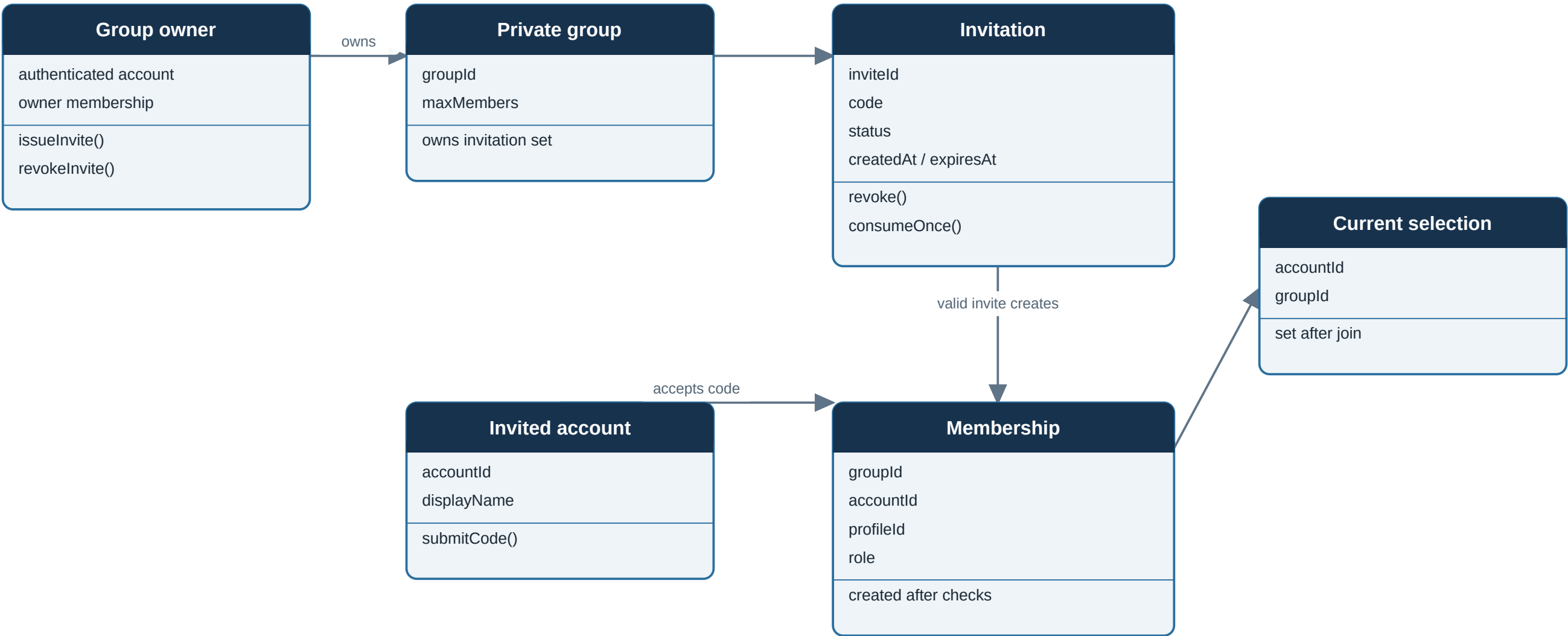
[Preference is member-specific; GET exposes schedule and delivery.state=not-configured. Saving a preference does not send a notification.]

Figure M025 | ACCESS | uc03 analysis class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03 — analysis classes

An invitation is a bounded, one-use capability that becomes a membership only after server validation.



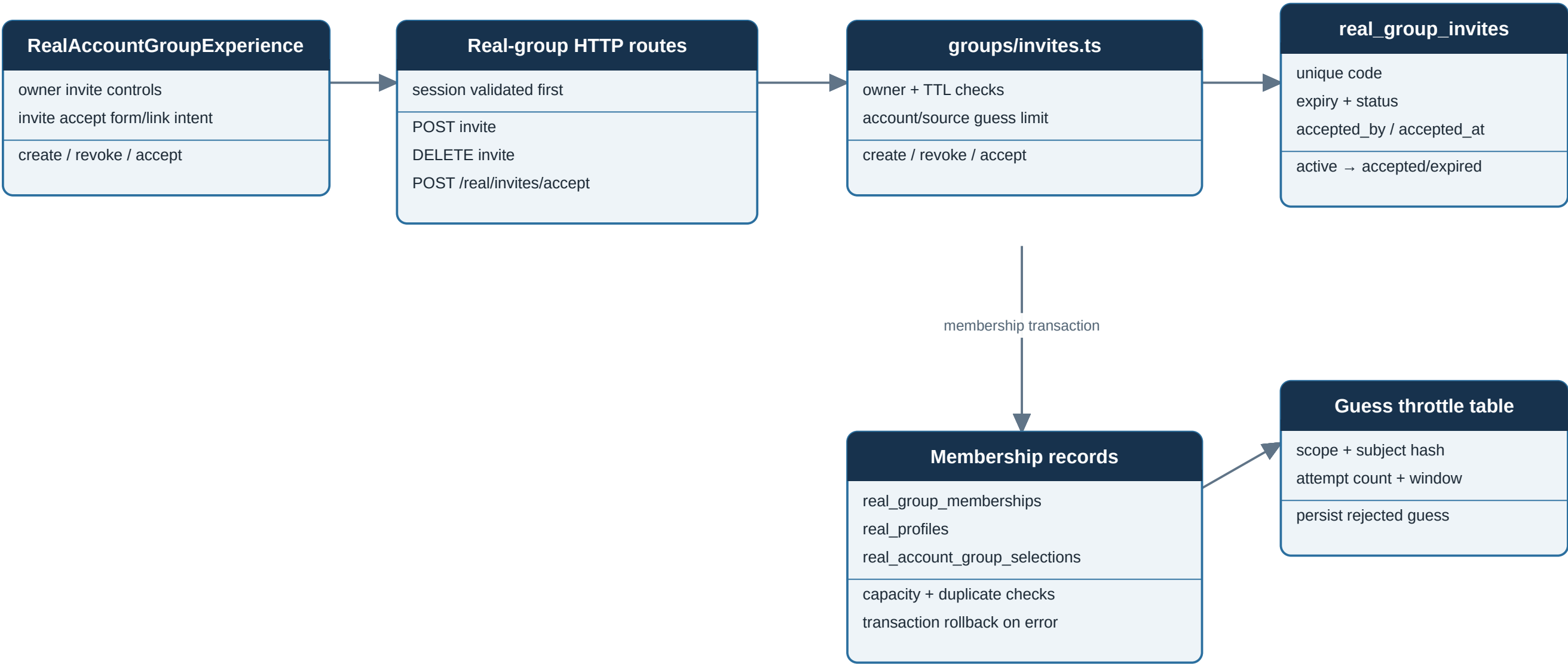
Capacity, expiry, replay, group consistency, and account/source guess rate are separate acceptance checks.

Figure M026 | ACCESS | uc03 design class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03 — design classes

The acceptance transaction binds invite consumption, membership, profile reuse/creation, and selected group.



Invitation codes are not account credentials. The client deep link carries a code and bounded expiry; the server remains authoritative.

Figure M027 | ACCESS | uc03 f1 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

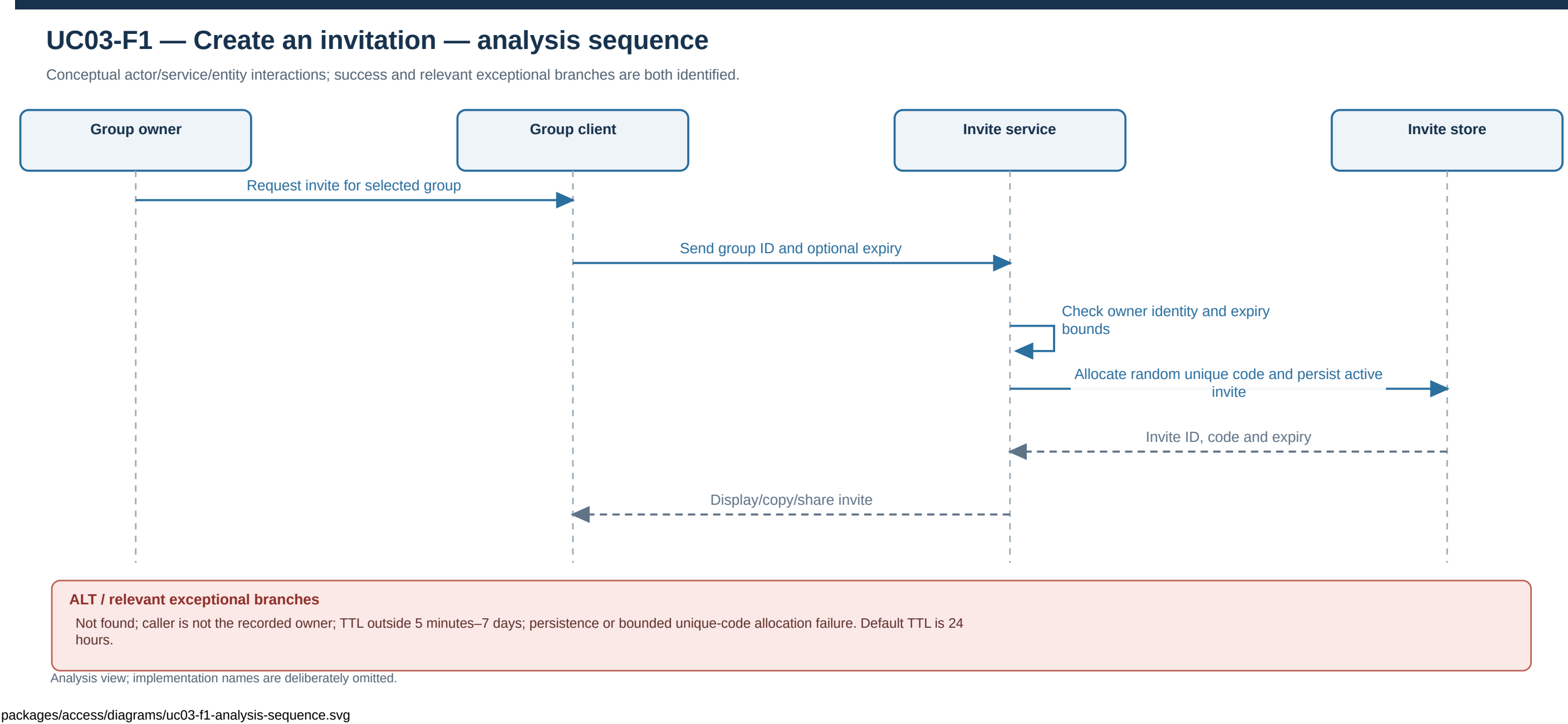
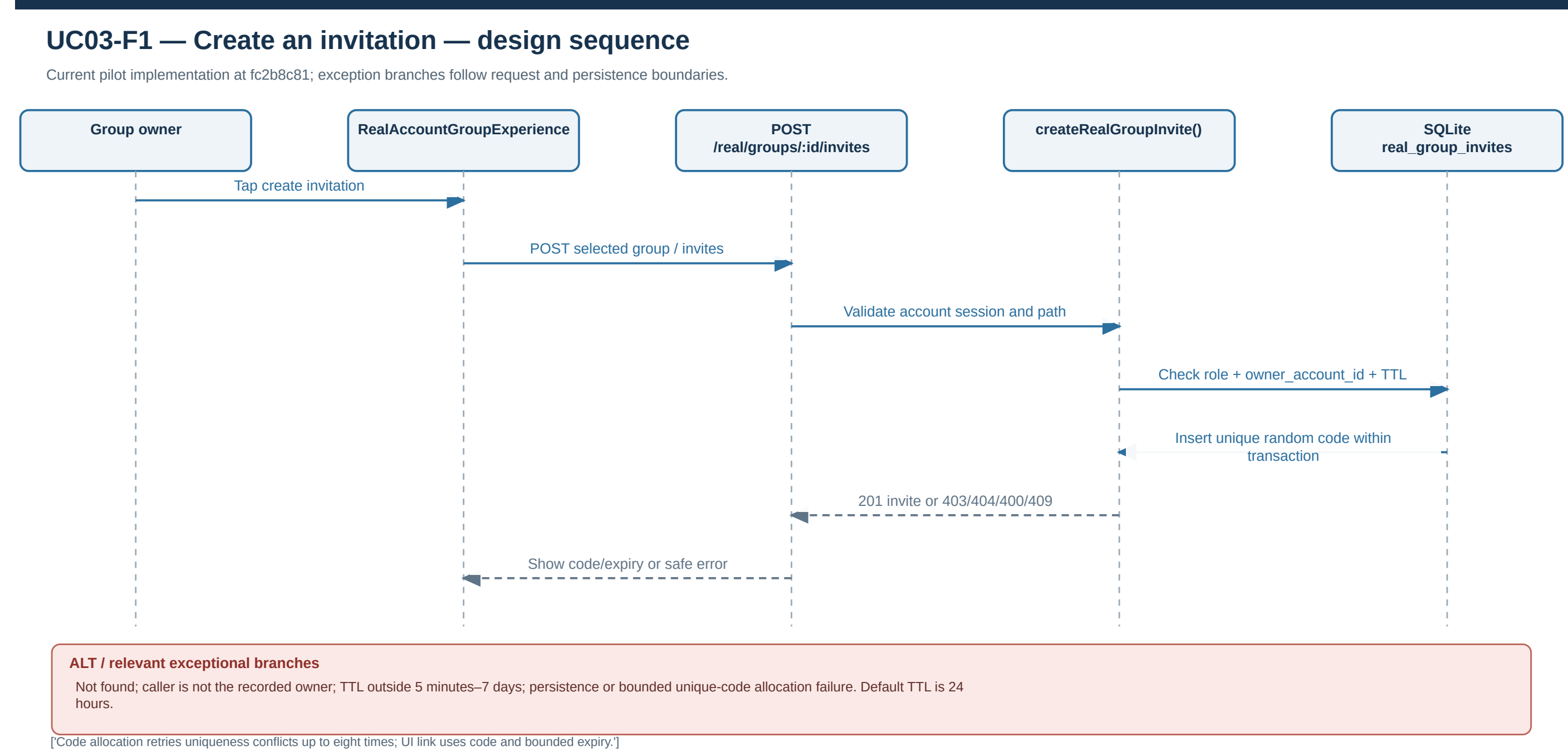


Figure M028 | ACCESS | uc03 f1 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



packages/access/diagrams/uc03-f1-design-sequence.svg

Figure M029 | ACCESS | uc03 f2 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

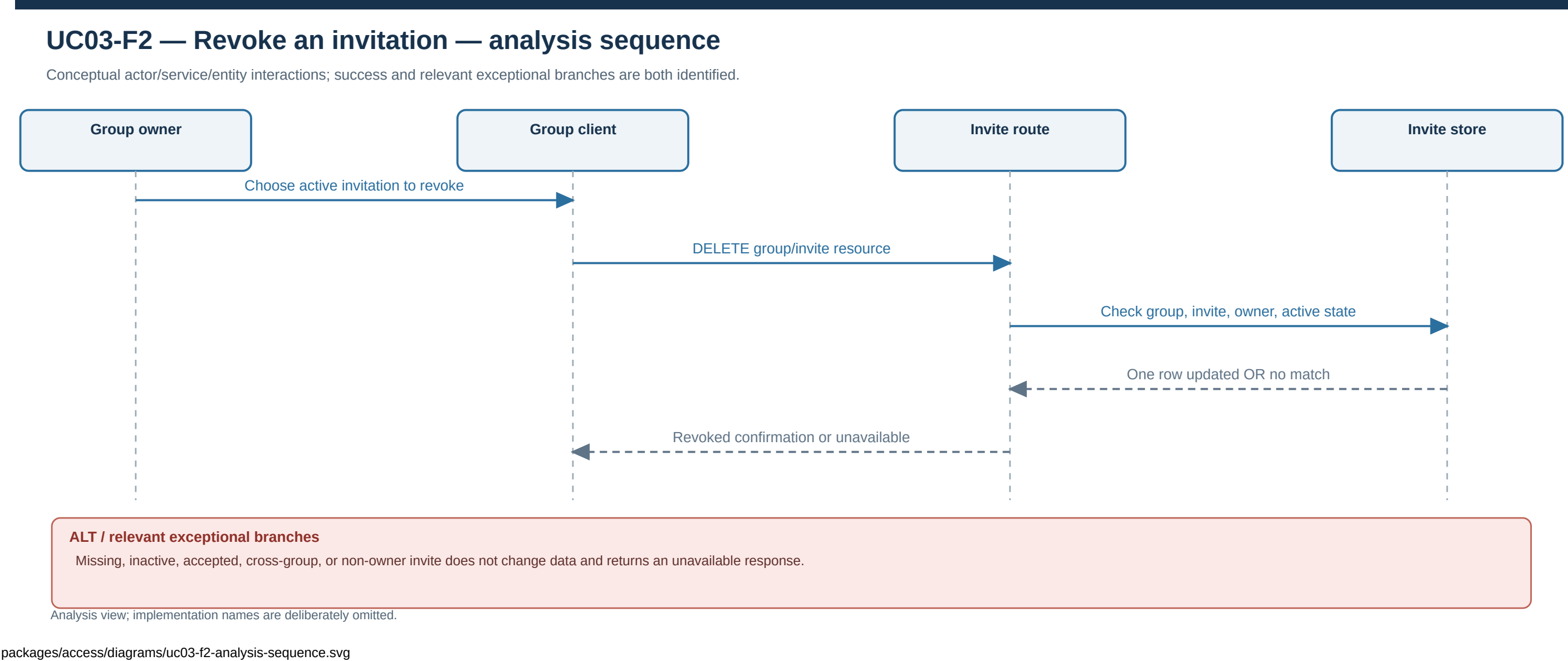


Figure M030 | ACCESS | uc03 f2 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03-F2 — Revoke an invitation — design sequence

Current pilot implementation at fc2b8c81; exception branches follow request and persistence boundaries.

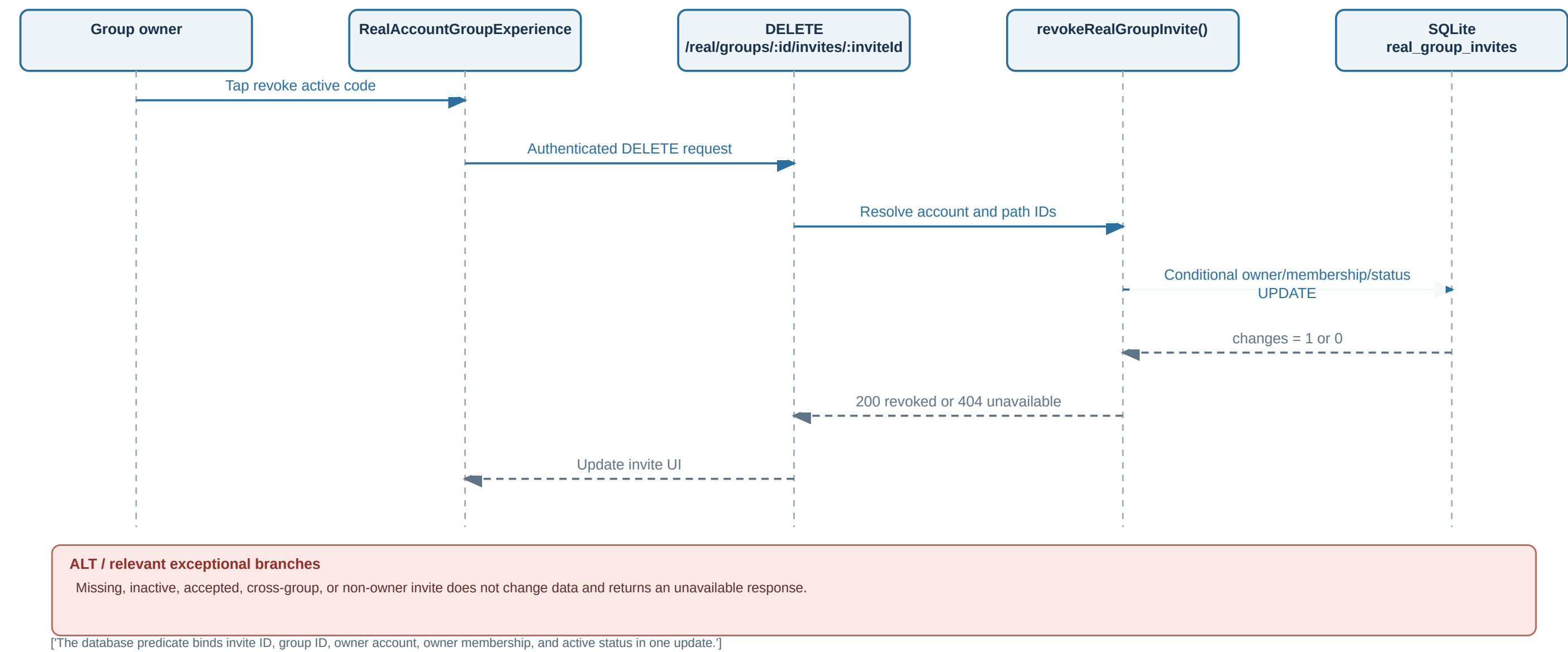
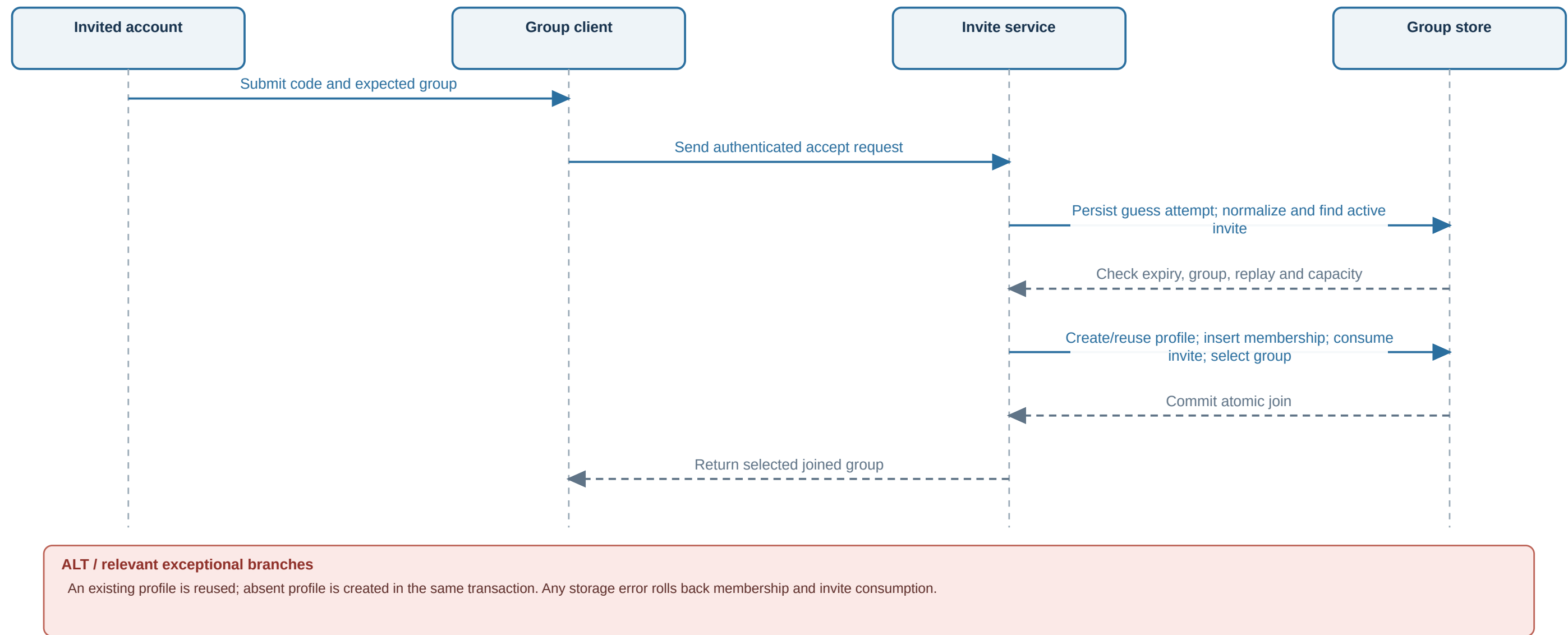


Figure M031 | ACCESS | uc03 f3 analysis sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03-F3 — Accept a valid invitation — analysis sequence

Conceptual actor/service/entity interactions; success and relevant exceptional branches are both identified.



Analysis view; implementation names are deliberately omitted.

packages/access/diagrams/uc03-f3-analysis-sequence.svg

Figure M032 | ACCESS | uc03 f3 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03-F3 — Accept a valid invitation — design sequence

Current pilot implementation at fc2b8c81; exception branches follow request and persistence boundaries.

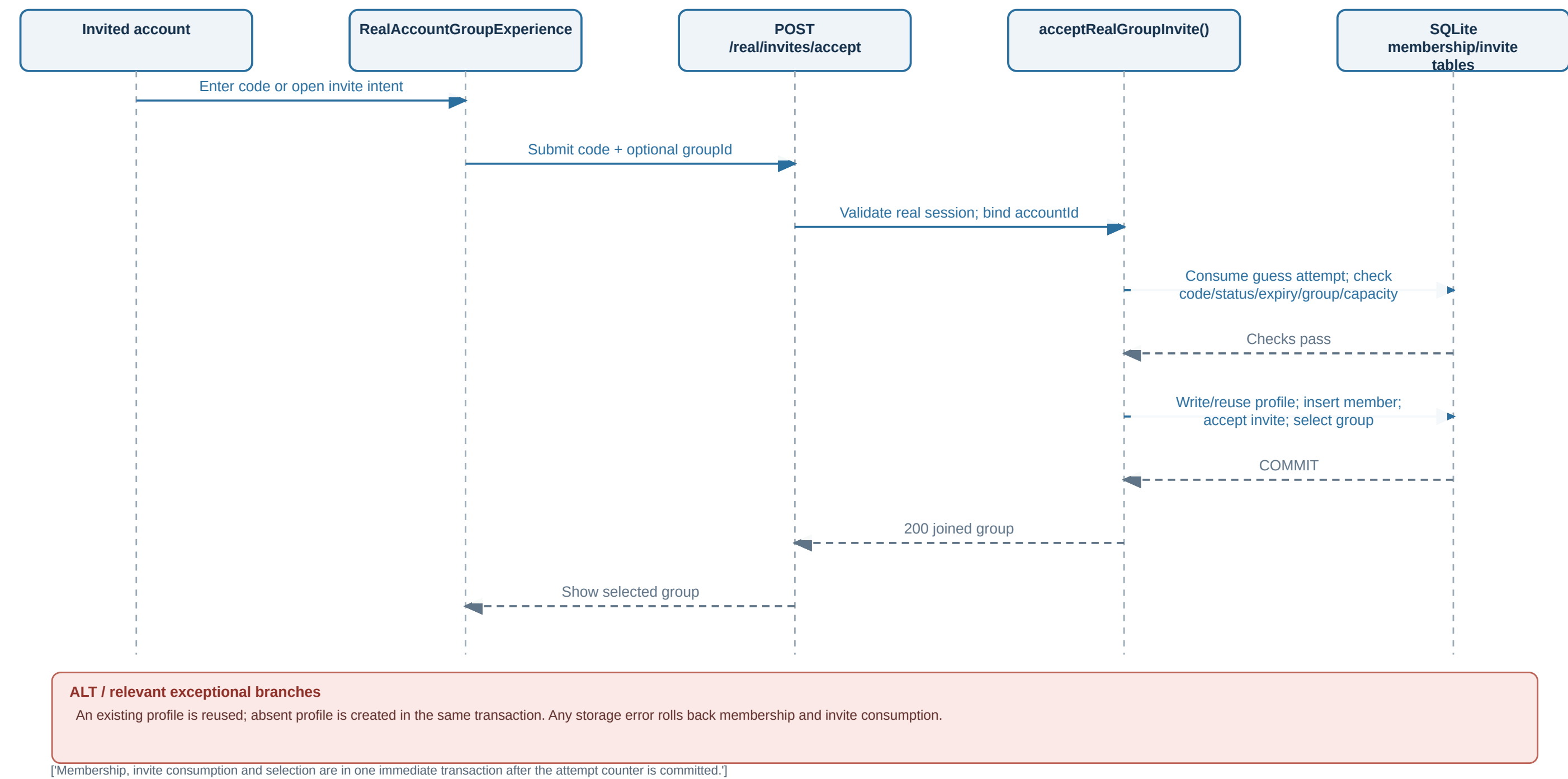
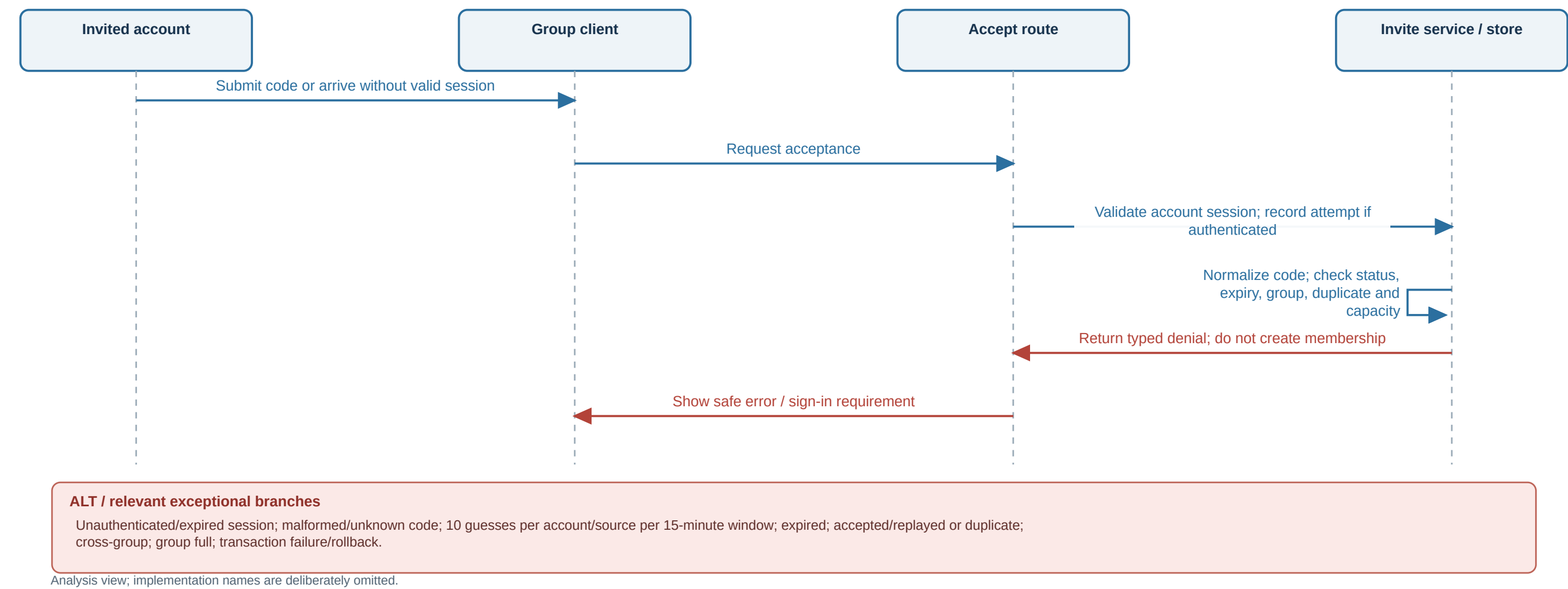


Figure M033 | ACCESS | uc03 f4 analysis sequence
 Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03-F4 — Reject an unusable invitation — analysis sequence

Conceptual actor/service/entity interactions; success and relevant exceptional branches are both identified.



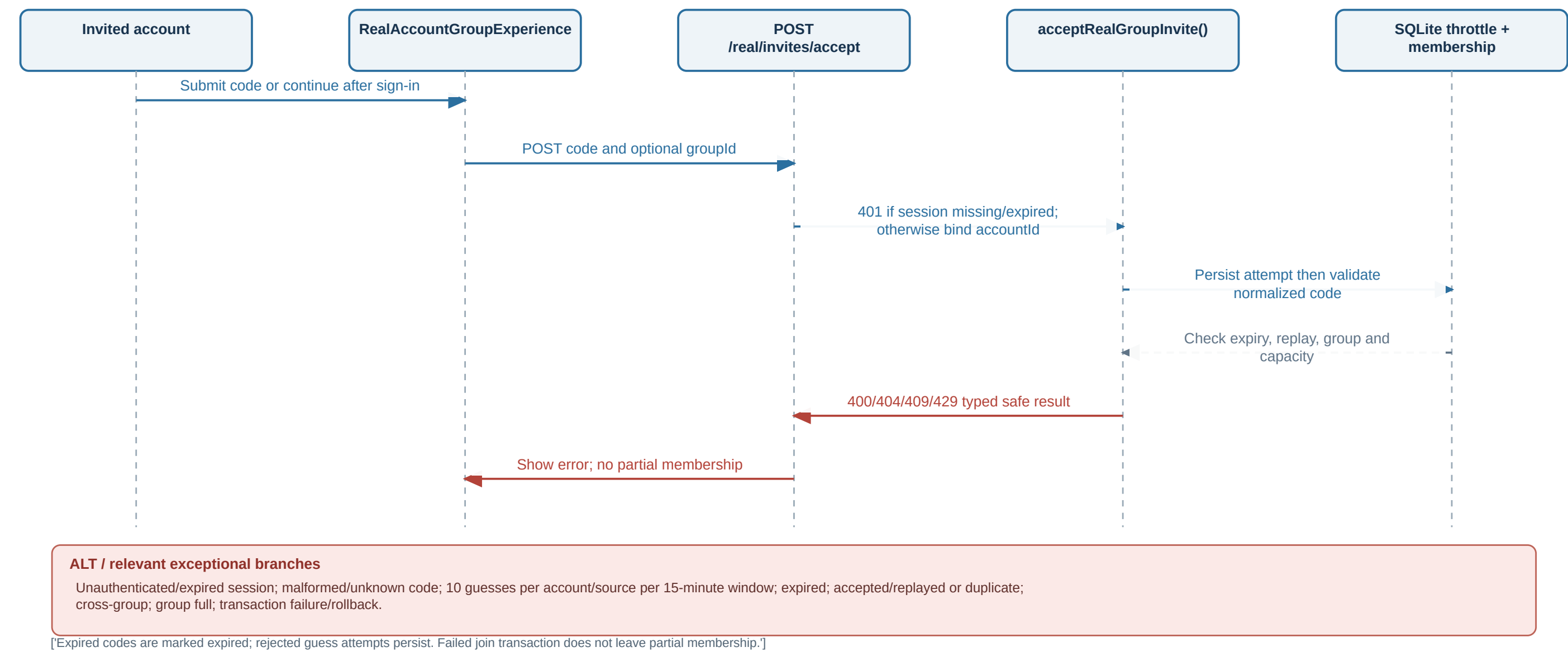
packages/access/diagrams/uc03-f4-analysis-sequence.svg

Figure M034 | ACCESS | uc03 f4 design sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

UC03-F4 — Reject an unusable invitation — design sequence

Current pilot implementation at fc2b8c81; exception branches follow request and persistence boundaries.



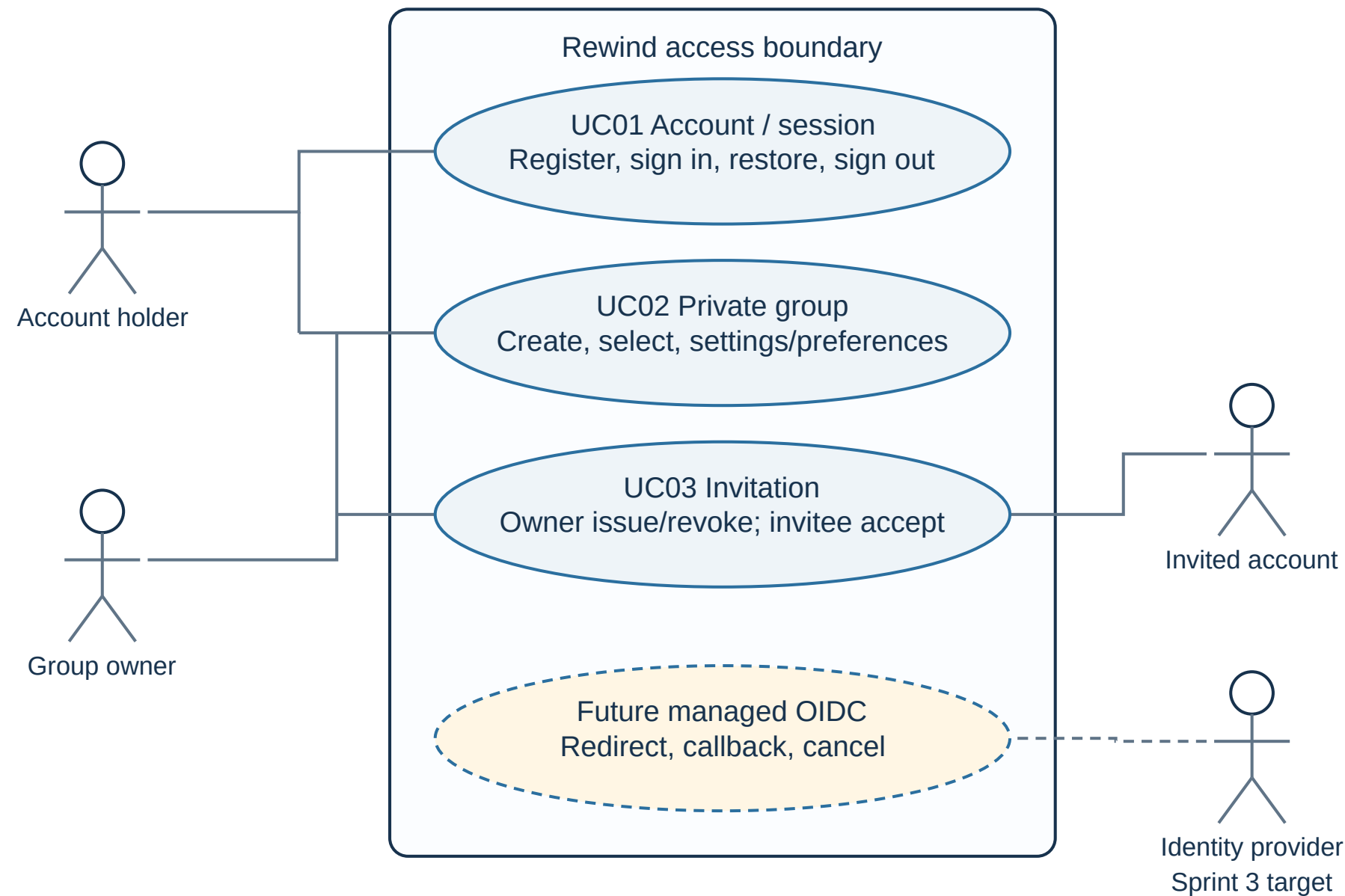
packages/access/diagrams/uc03-f4-design-sequence.svg

Figure M035 | ACCESS | use case overview

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

Access — pilot use cases and future identity boundary

UC01–UC03 flow details and exception paths are catalogued separately.



Excluded: public discovery, owner transfer, member removal, leave/rejoin, account deletion/recovery. Outsiders gain no private group authority.

CAPTURE model/source register | issue #358

Accepted model cut fc2b8c811e9c13b176113fa0bdb2c015798990b4; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

src/capture/VideoCaptureScreen.tsx:562

src/capture/video-review.ts:124

src/capture/contracts.ts:76

src/groups/RealAccountGroupExperience.tsx:360

src/capture/real-account-video-runtime.ts:46

src/capture/direct-transfer.ts:337

src/capture/CameraCaptureScreen.tsx:147

src/capture/still-image-session.ts:48

src/capture/file-store.ts:96

server/src/contributions/index.ts:77

server/src/contributions/index.ts:240

server/src/contributions/ledger.ts:575

server/src/ffmpeg.ts:206

src/capture/contracts.ts:101

src/capture/platform.ts:340

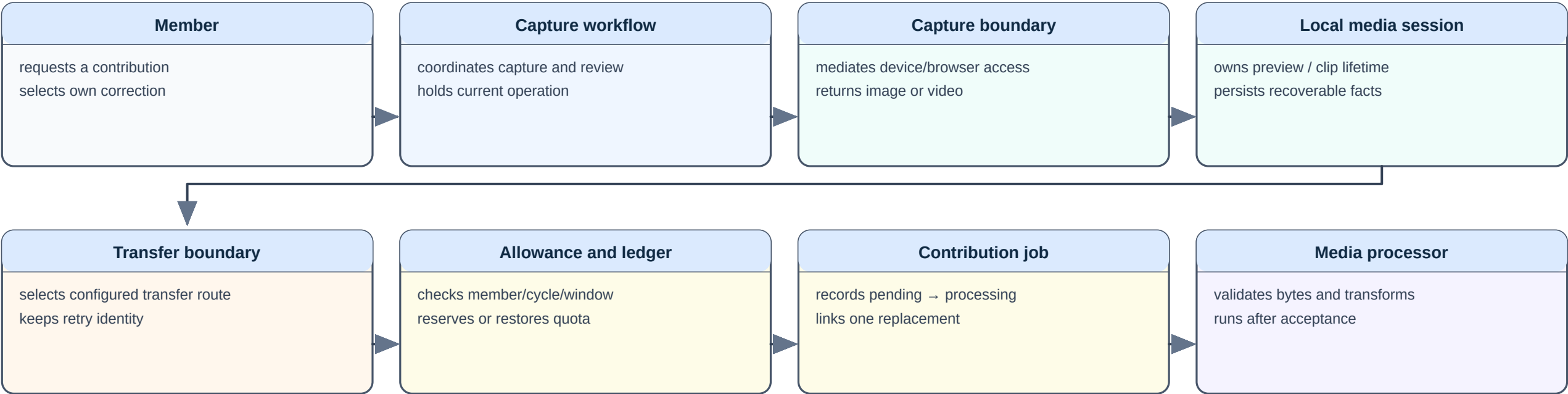
server/src/media/upload-intents.ts:386

server/src/media/store.ts:45

Figure M040 | CAPTURE | uc04 analysis class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

UC04 — analysis classes

Conceptual domain roles; implementation technology is intentionally omitted.



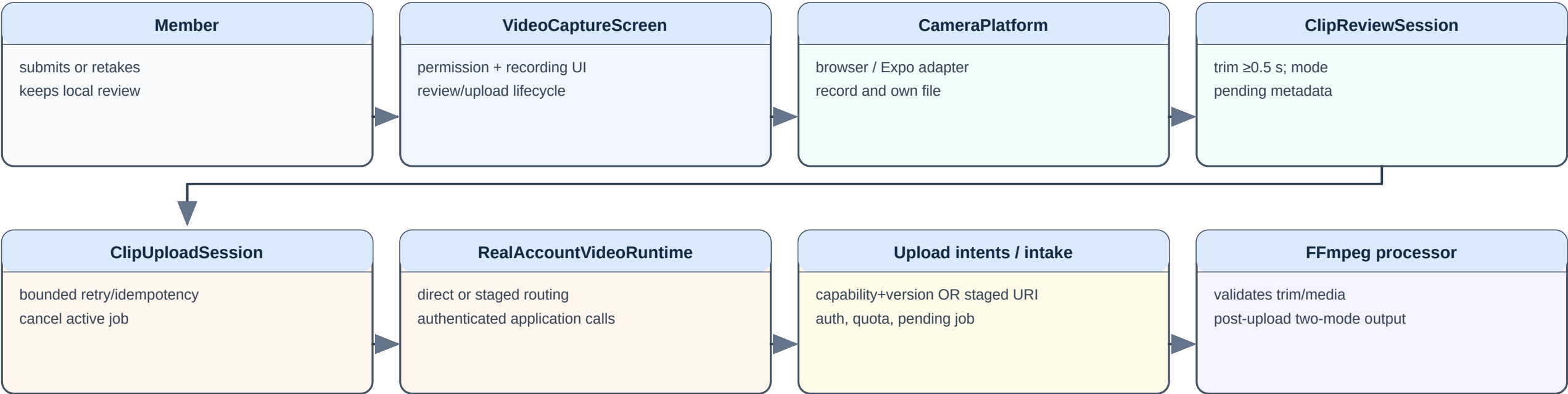
Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

packages/capture/diagrams/uc04-analysis-class.svg

Figure M041 | CAPTURE | uc04 design class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

UC04 — design classes

Code-backed responsibilities at the exact accepted commit.



Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

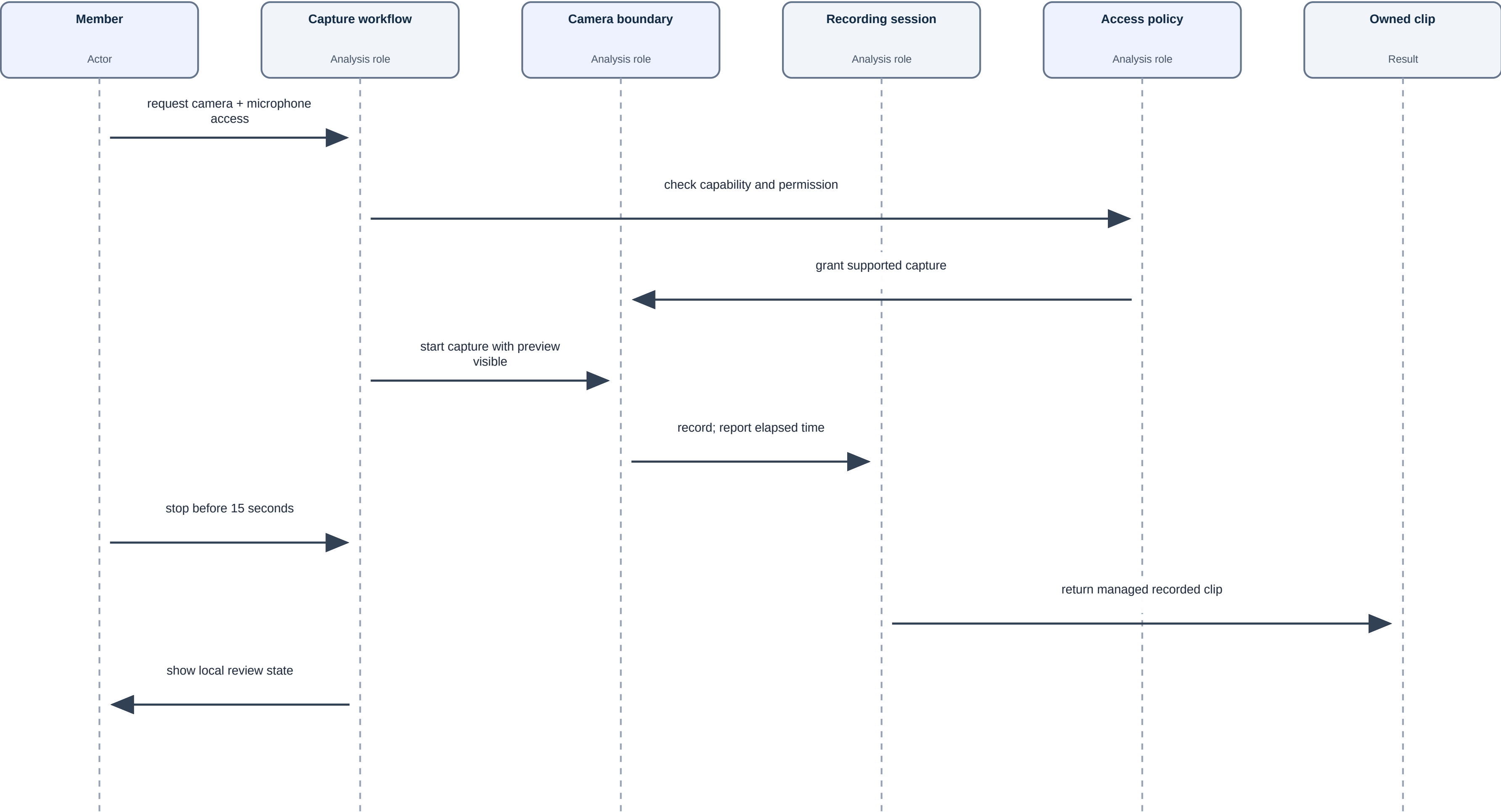
packages/capture/diagrams/uc04-design-class.svg

Figure M042 | CAPTURE | uc04 f1 analysis sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC04-F1 — Acquire access and record

Analysis sequence · permission, recording and owned clip



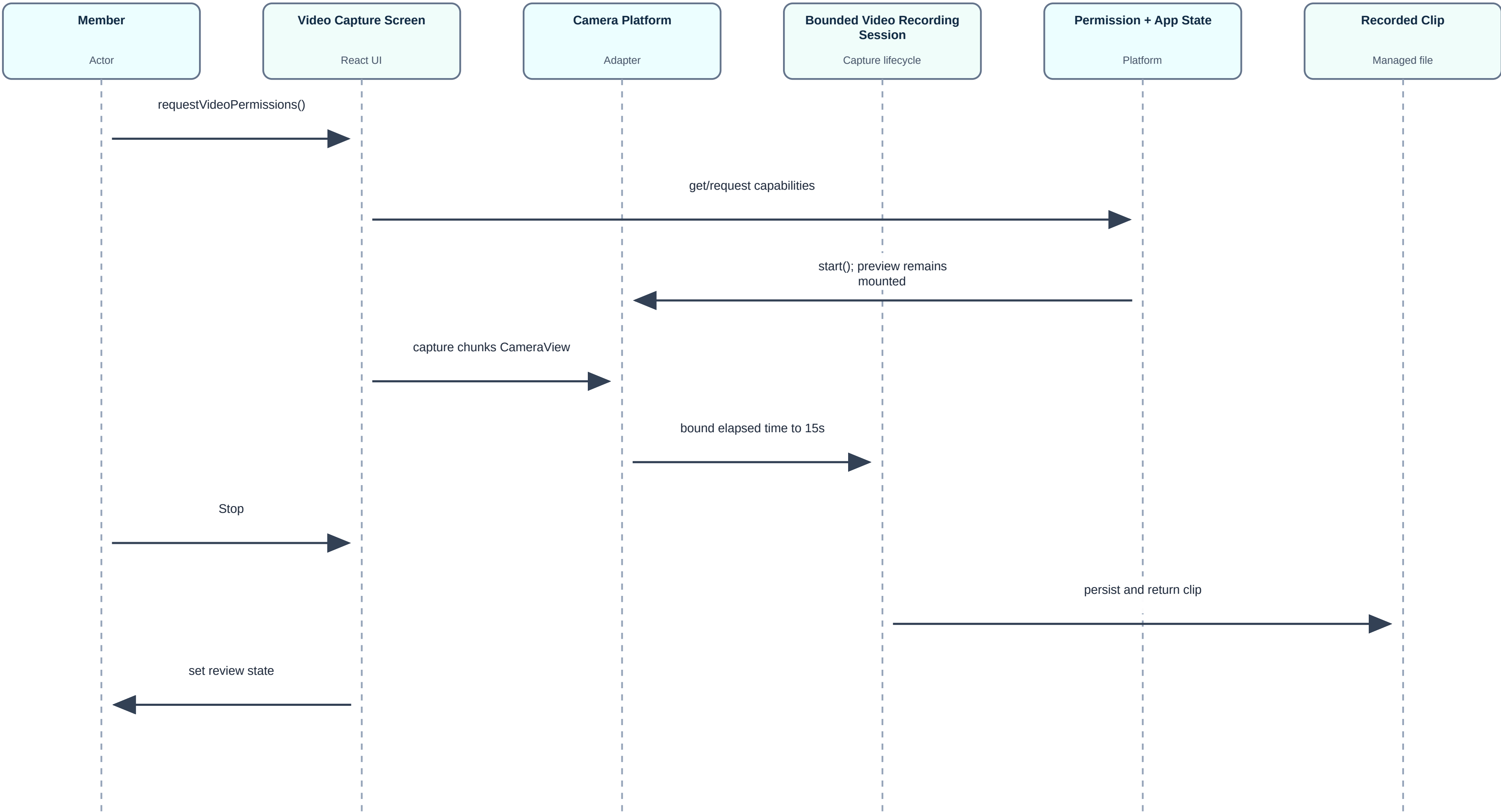
Relevant alternate / exception paths

Permission denied or blocked → recovery guidance. Unsupported live capture → labelled file fallback when available. Missing recorder, camera/write error → no review clip. Cancel, route exit, background, or stale completion → stop capture and release only media owned by that operation.

Figure M043 | CAPTURE | uc04 f1 design sequence
Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC04-F1 — Acquire access and record

Design sequence · permission, recording and owned clip



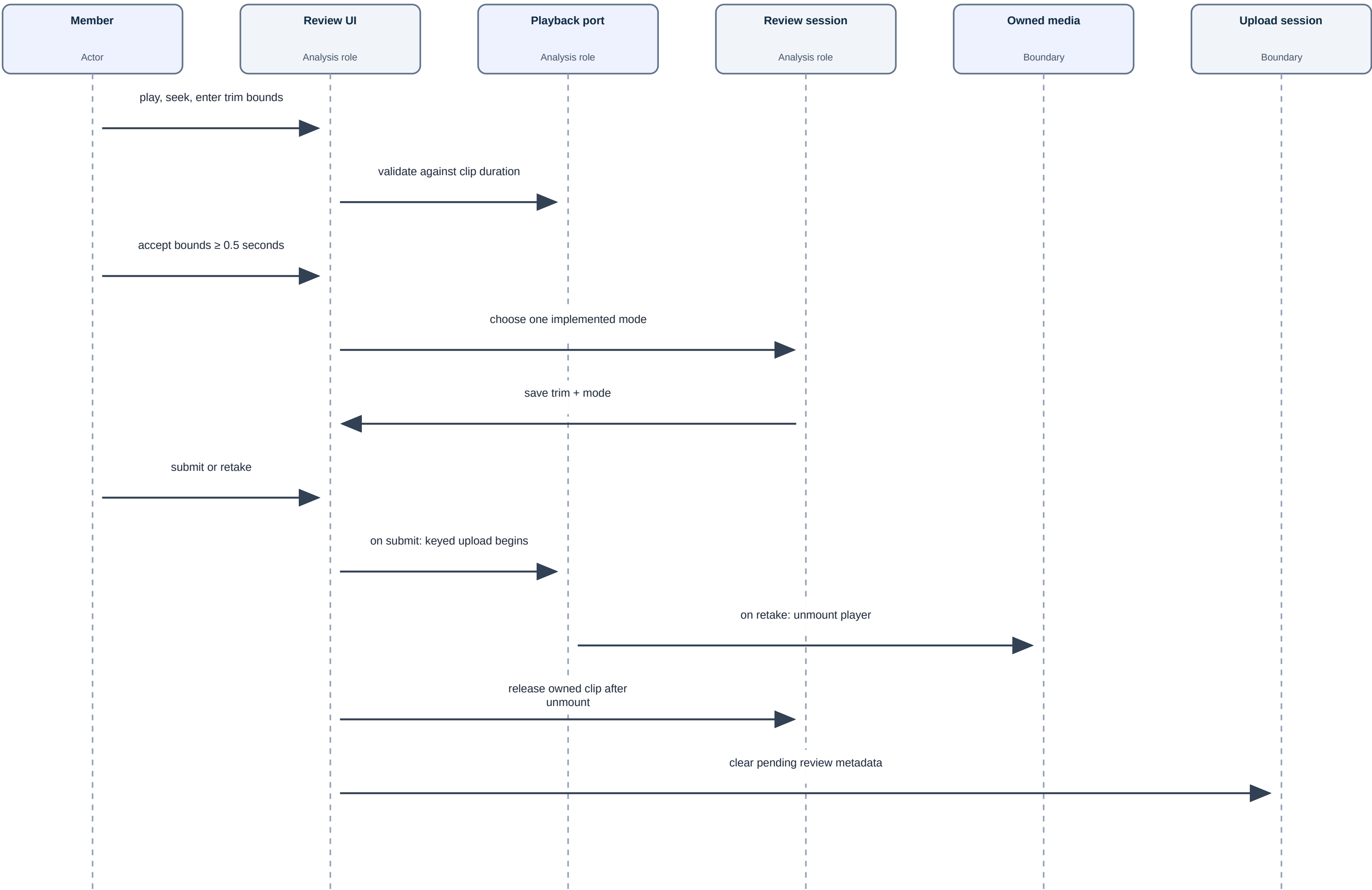
Relevant alternate / exception paths
requestVideoPermissions / refresh can fail or return denied; unsupported browser path offers file selection; start or file persistence failure sets an error.
Background interruption cancels recording; generation/ownership checks discard late completion.

Figure M044 | CAPTURE | uc04 f2 analysis sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC04-F2 — Review, trim, mode or retake

Analysis sequence · local review does not submit media



Relevant alternate / exception paths
Invalid/out-of-range/too-short trim remains in review. Landscape, non-MP4, missing audio or invalid duration fails validation. Player failure offers recovery/retake. Cleanup failure retains the clip and reports an error; backgrounded upload is cancelled and retry depends on the session budget.

Figure M045 | CAPTURE | uc04 f2 design sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC04-F2 — Review, trim, mode or retake

Design sequence · local review does not submit media

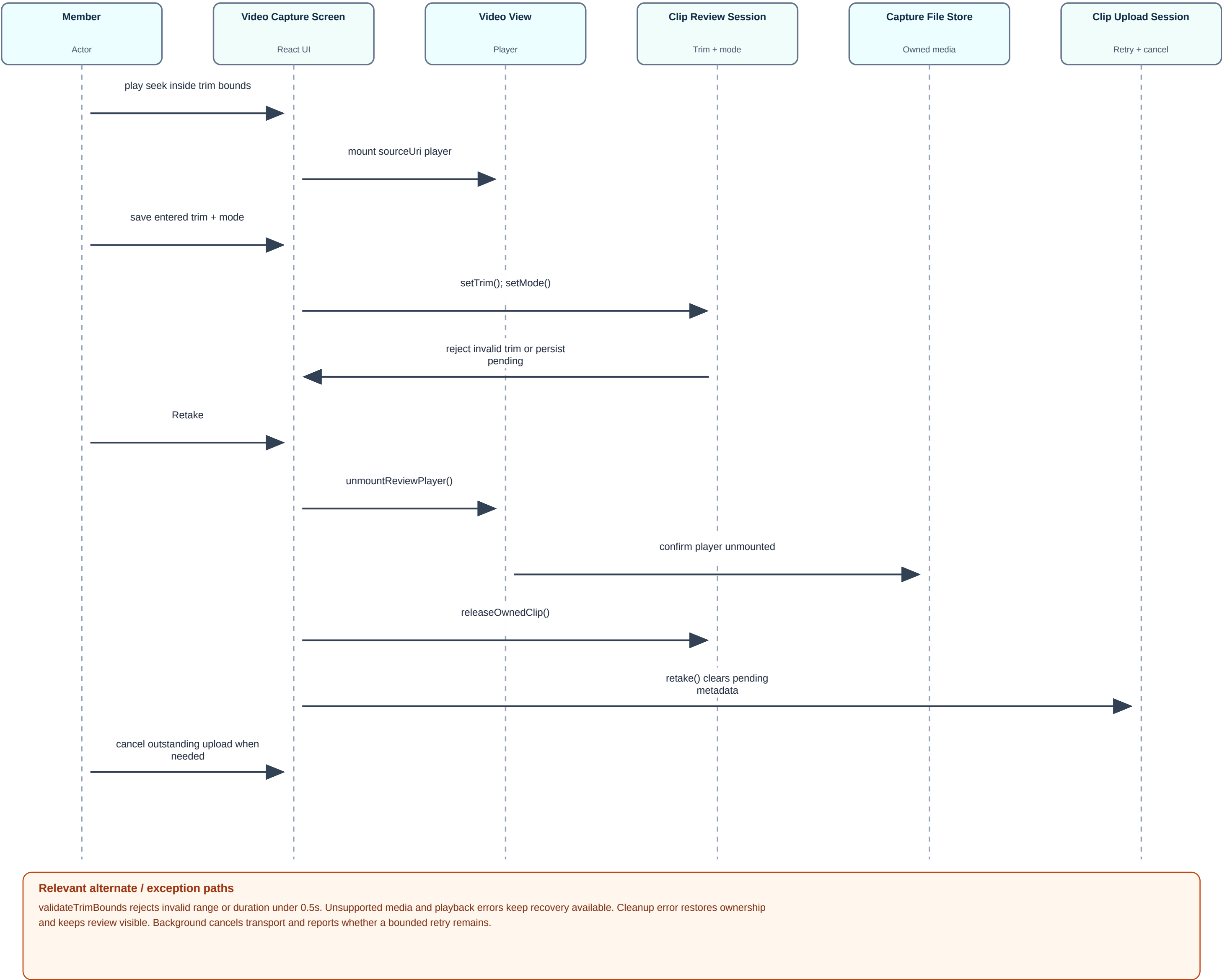
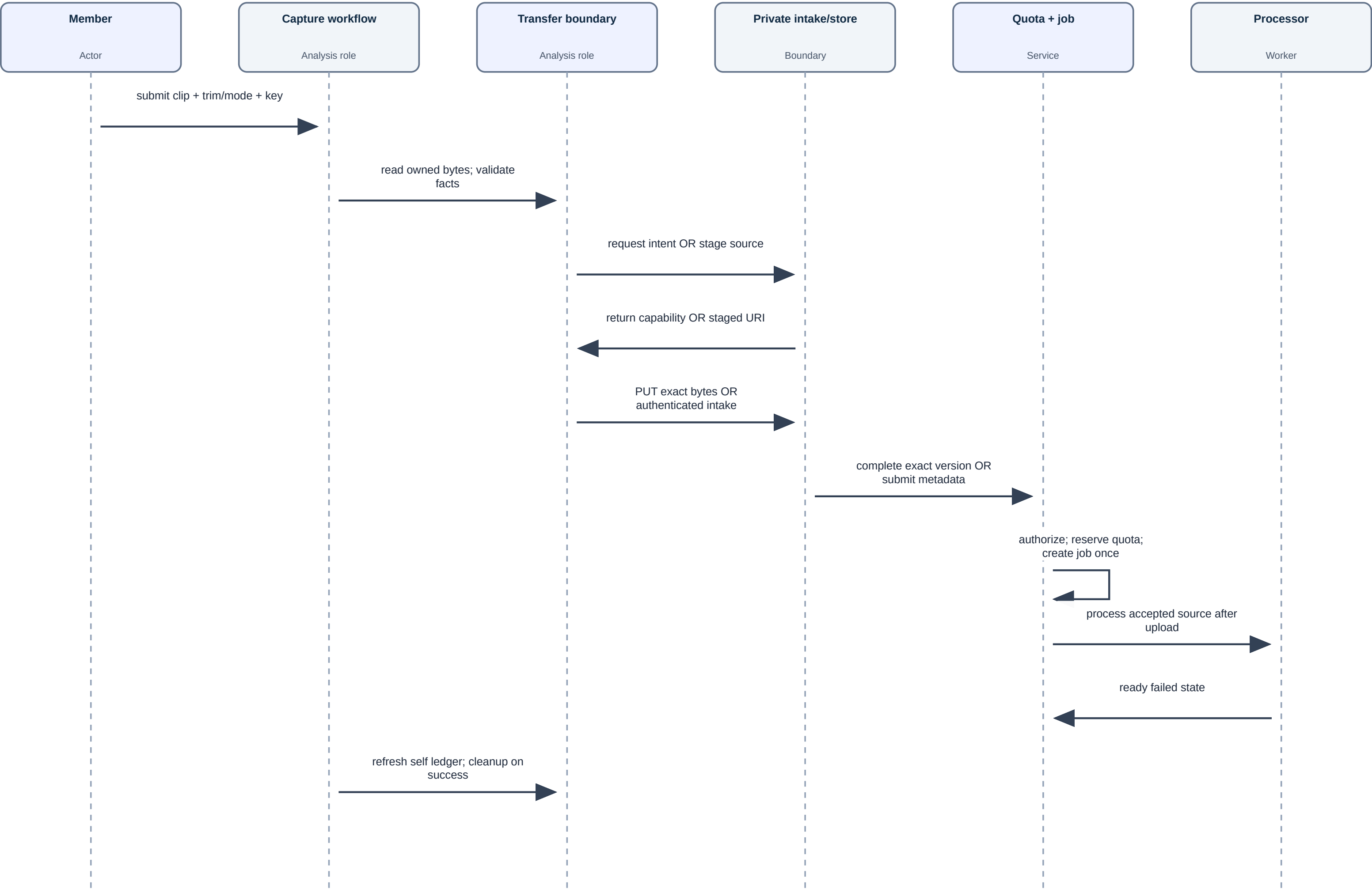


Figure M046 | CAPTURE | uc04 f3 analysis sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC04-F3 — Transfer, accept and process video

Analysis sequence · configured direct capability or authenticated staged intake



Relevant alternate / exception paths

Direct path: lost response → status/reconcile exact version; expiry, unknown/conflicting version, auth, cycle, quota or replacement rejection fails safely.

Staged path: source/media mismatch and idempotency conflict reject. Processing failure/status conflict returns retry state; cancellation is allowed only while owned job is cancellable. No cloud/provider acceptance is implied.

Figure M047 | CAPTURE | uc04 f3 design sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC04-F3 — Transfer, accept and process video

Design sequence · configured direct capability or authenticated staged intake

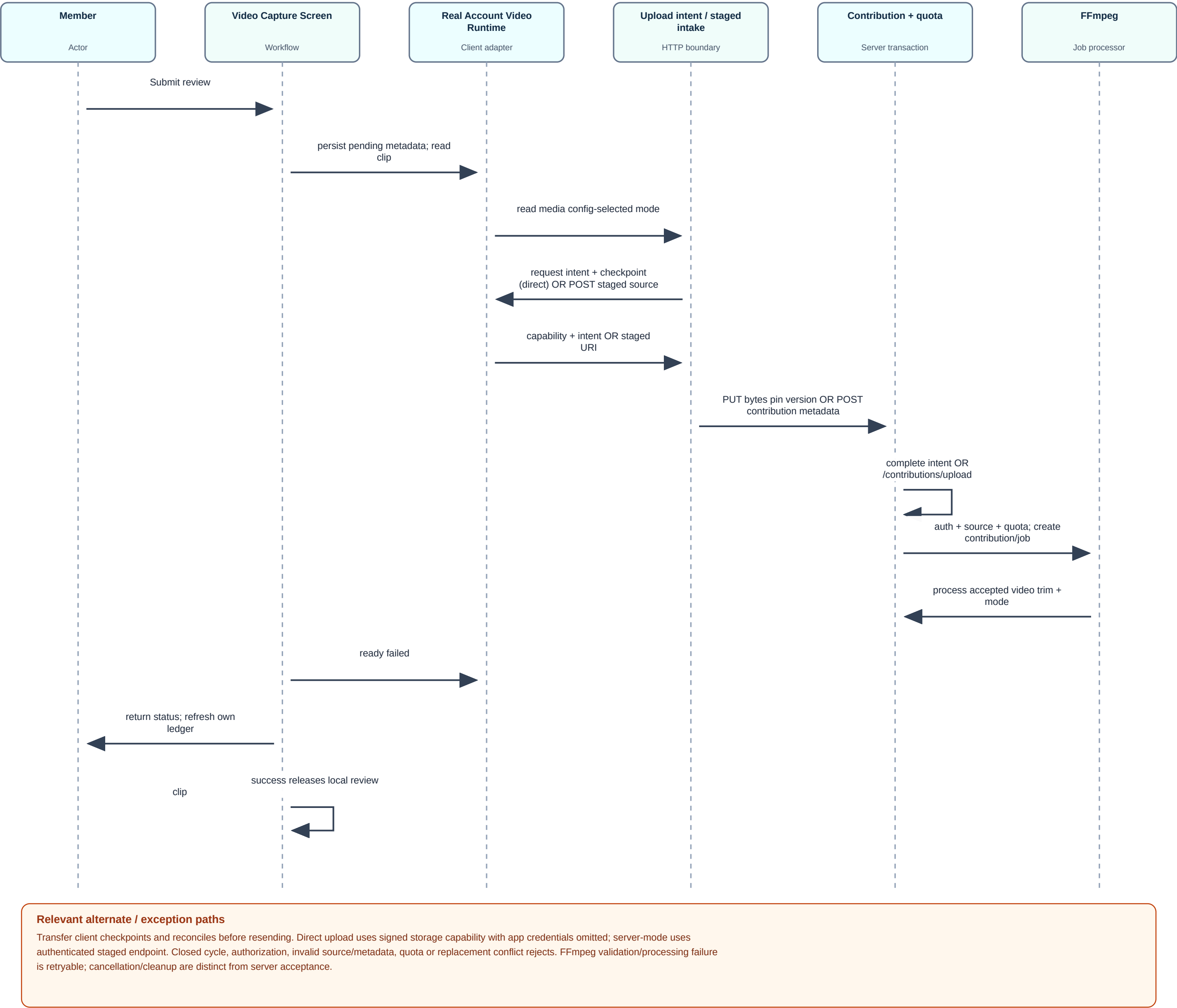
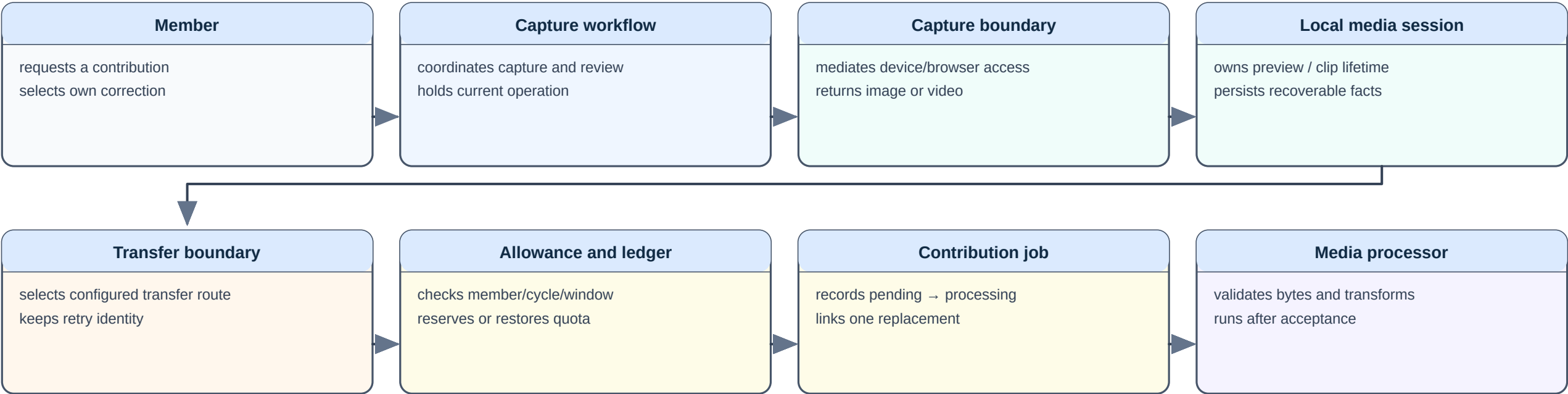


Figure M048 | CAPTURE | uc05 analysis class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

UC05 — analysis classes

Conceptual domain roles; implementation technology is intentionally omitted.



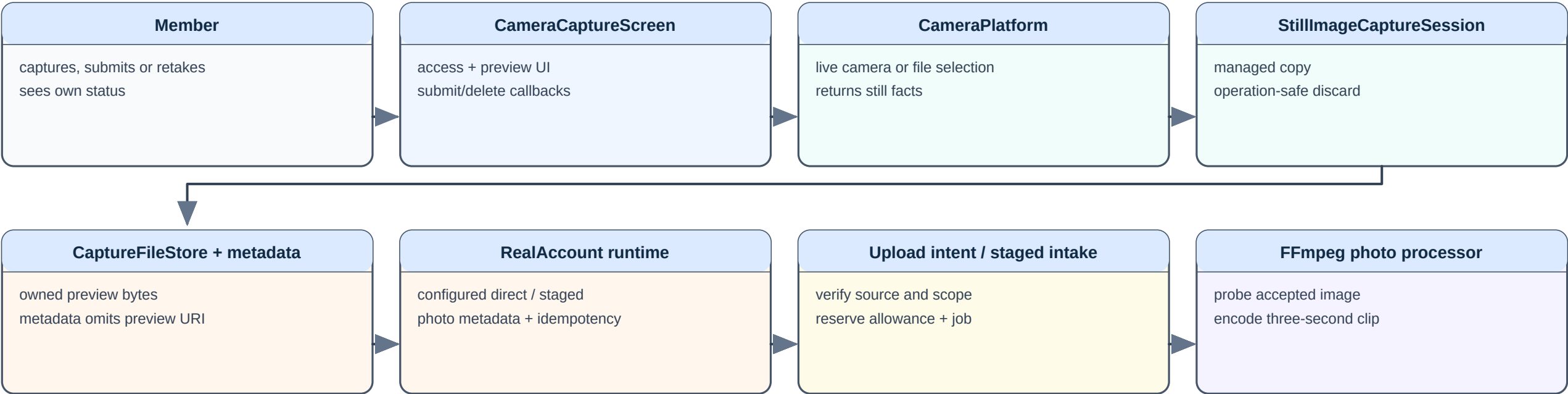
Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

packages/capture/diagrams/uc05-analysis-class.svg

Figure M049 | CAPTURE | uc05 design class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

UC05 — design classes

Code-backed responsibilities at the exact accepted commit.



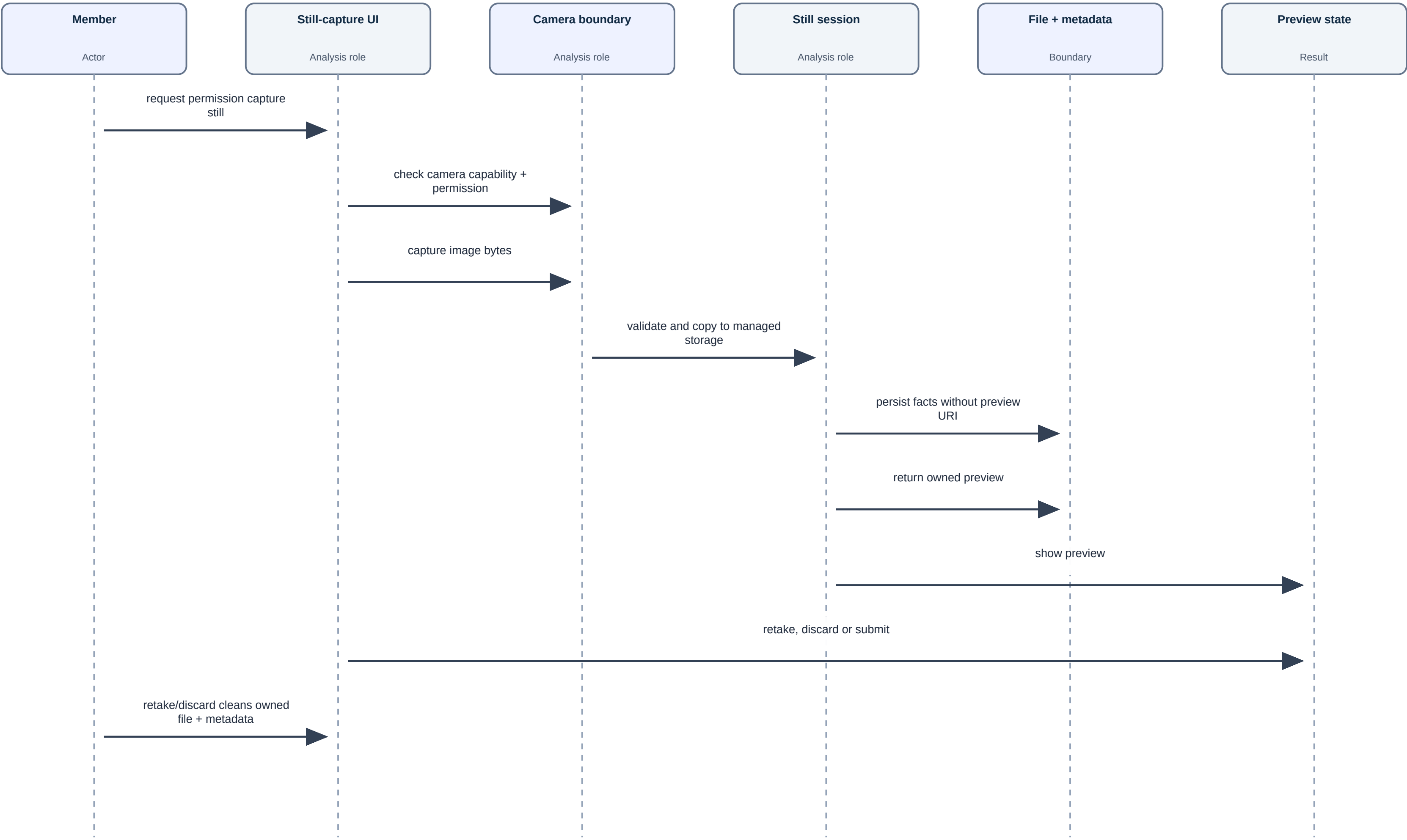
Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

packages/capture/diagrams/uc05-design-class.svg

Figure M050 | CAPTURE | uc05 f1 analysis sequence
Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC05-F1 — Capture and review a photo

Analysis sequence · managed preview and operation ownership



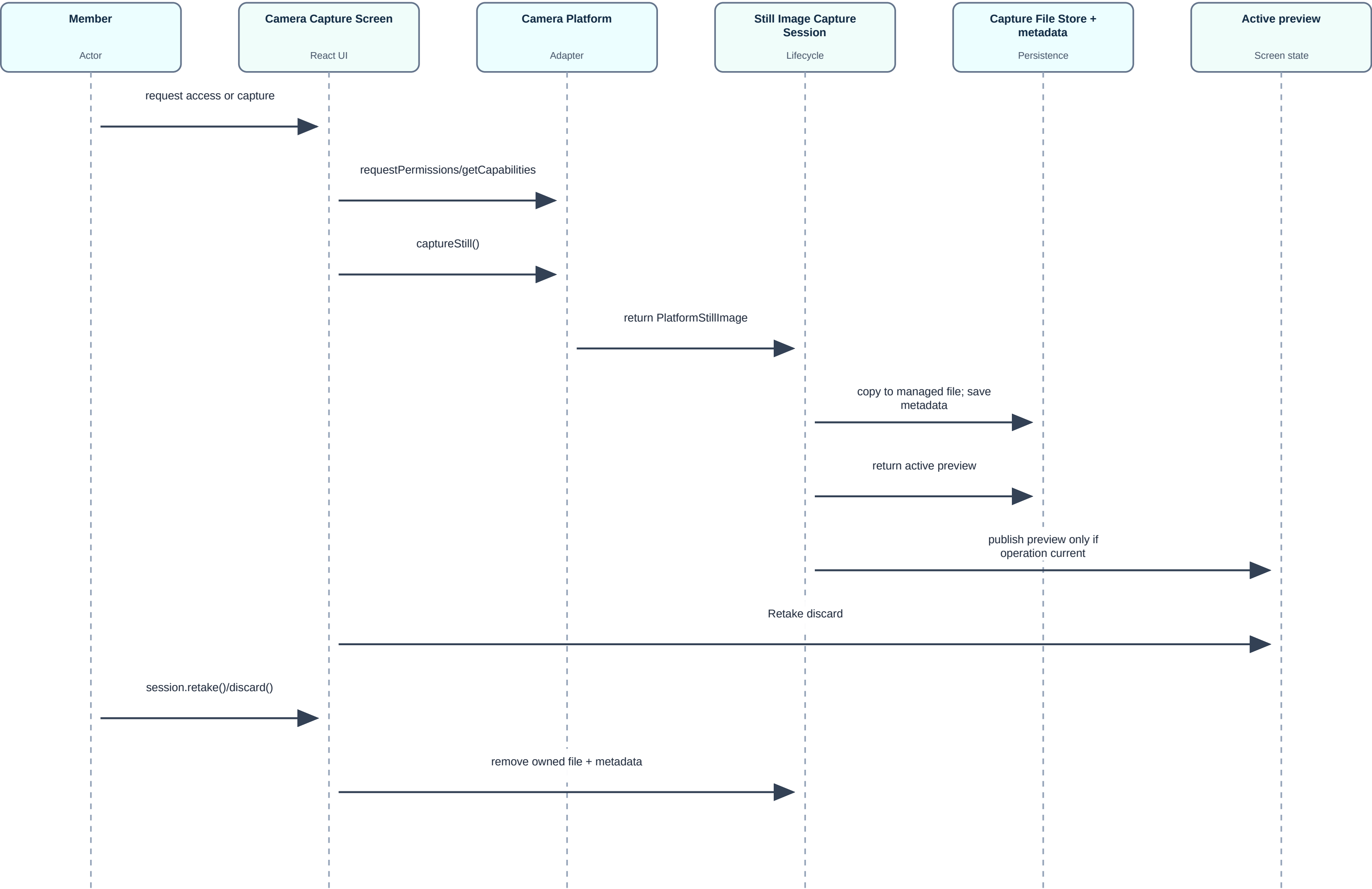
Relevant alternate / exception paths

Denied permission offers recovery; unsupported live capture may use labelled file fallback. Camera, copy, metadata or file-store failure never publishes ready preview. Late result after interruption is discarded only if still owned by the old operation. Background abandons in-flight capture and discards preview.

Figure M051 | CAPTURE | uc05 f1 design sequence
Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC05-F1 — Capture and review a photo

Design sequence · managed preview and operation ownership

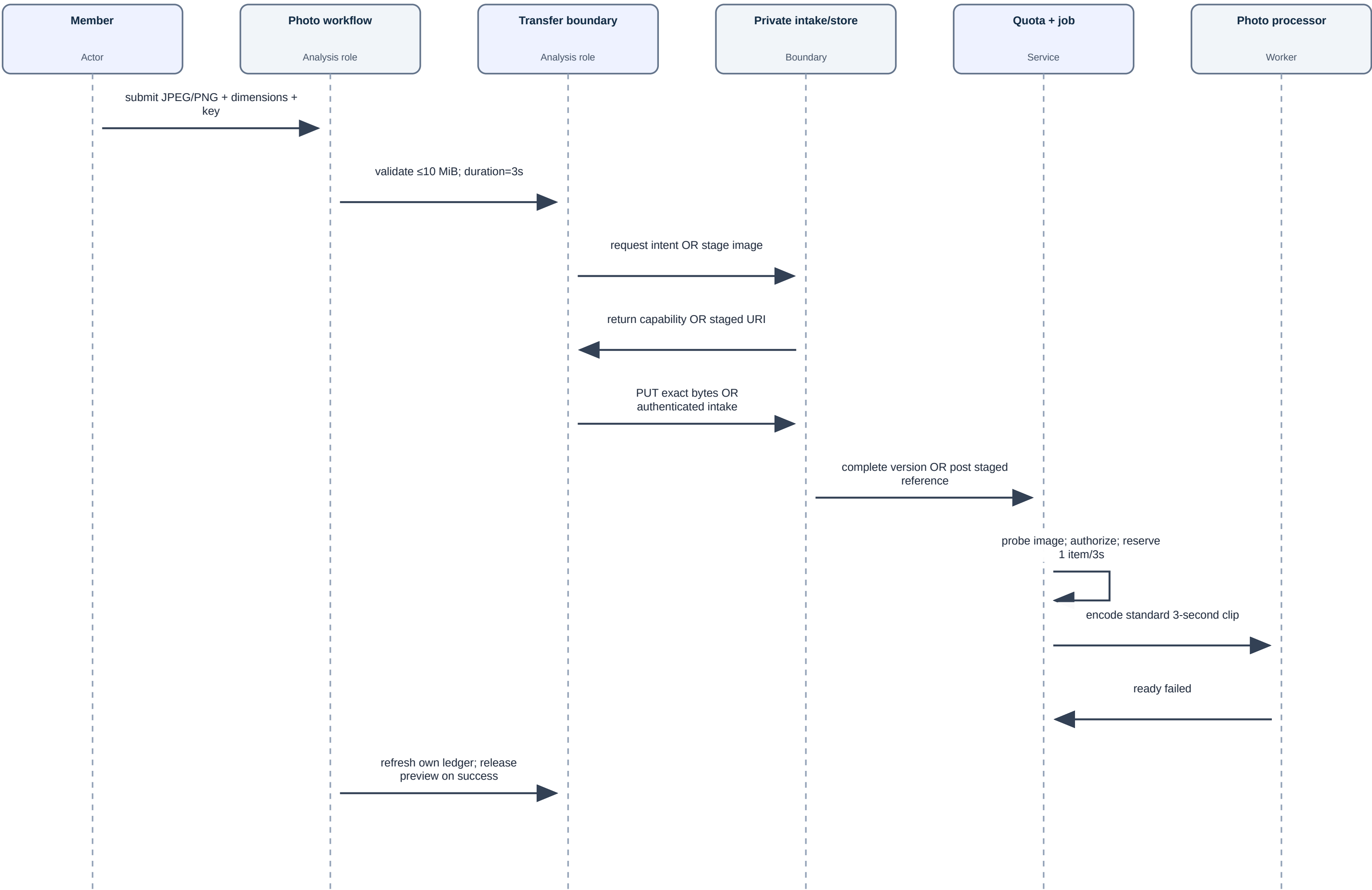


Relevant alternate / exception paths
Permission/capability denial, selected file fallback, capture/copy/metadata errors leave no ready preview. Sequence token invalidates stale async completion; only that capture file is removed. Background discards in-flight capture/preview; retained upload bytes survive route teardown.

Figure M052 | CAPTURE | uc05 f2 analysis sequence
Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC05-F2 — Transfer, accept and process photo

Analysis sequence · three-second contribution through configured transfer route



Relevant alternate / exception paths
Unsupported type, >10 MiB, invalid dimensions or corrupt server probe is rejected. Intent/version/auth/cycle/quota/idempotency/replacement errors cannot report success. Network/process failure retains retry state and bytes when available; cleanup failure is distinct from acceptance.

Figure M053 | CAPTURE | uc05 f2 design sequence
Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC05-F2 — Transfer, accept and process photo

Design sequence · three-second contribution through configured transfer route

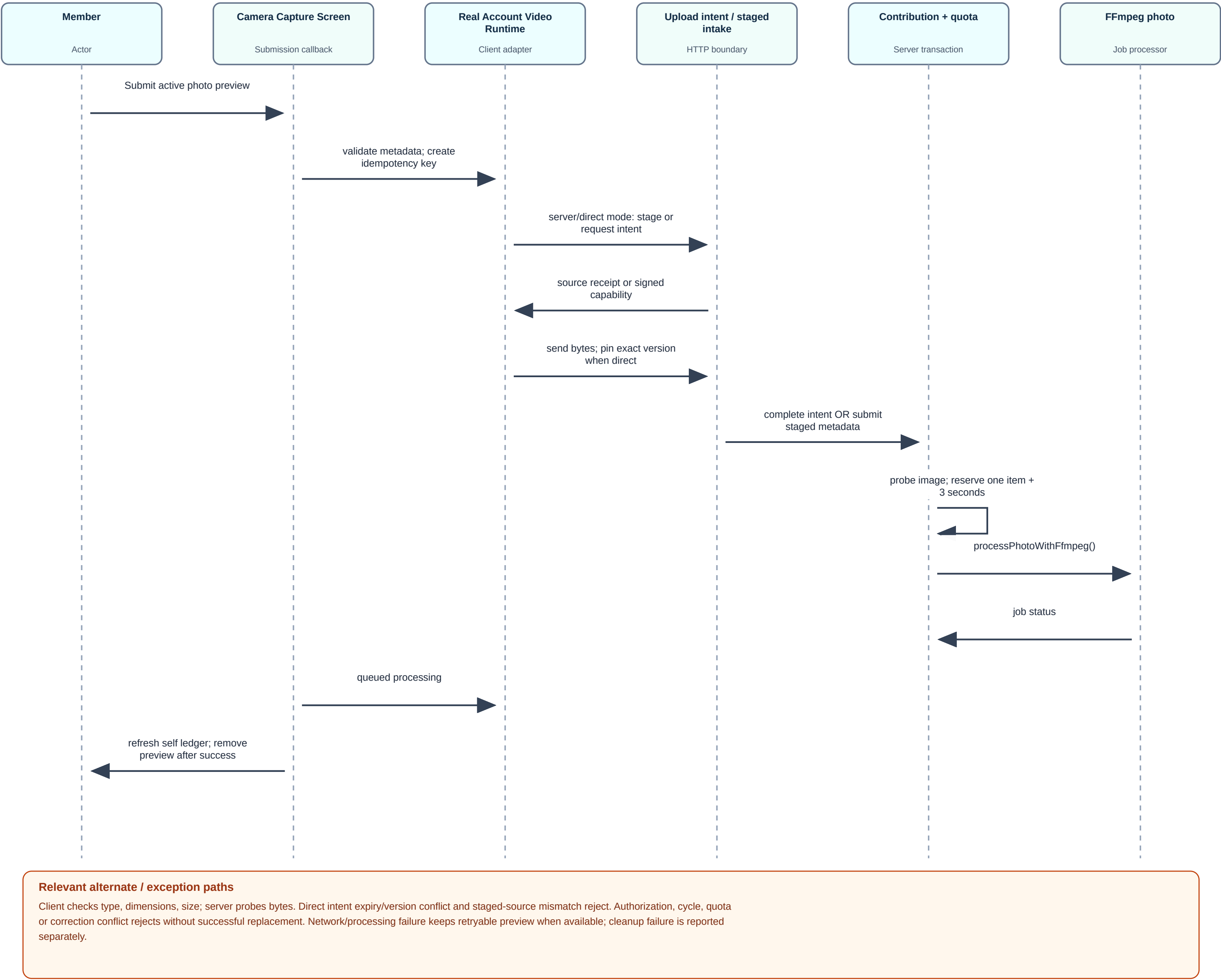
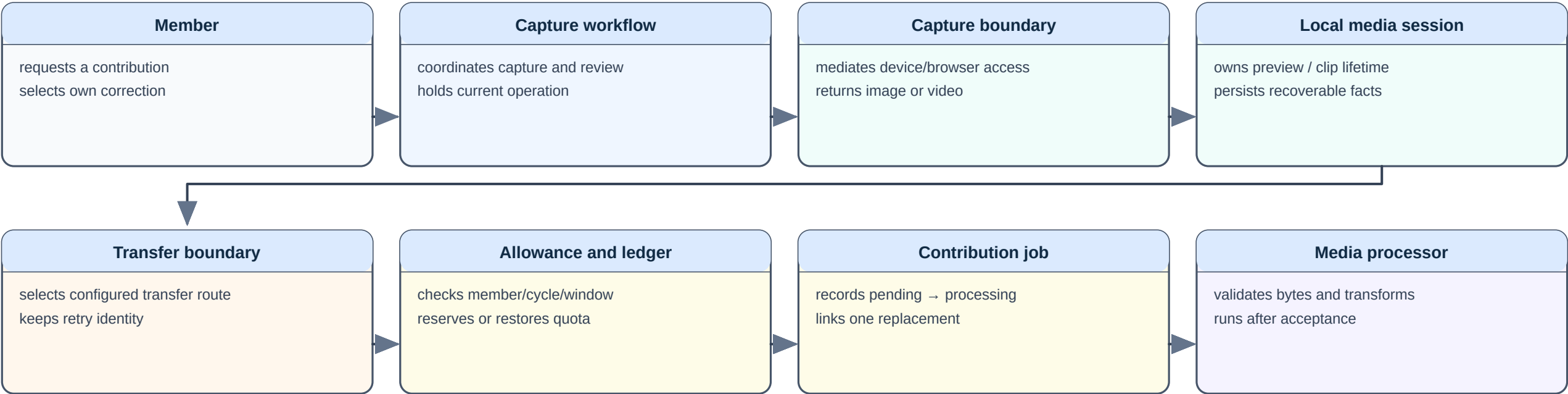


Figure M054 | CAPTURE | uc06 analysis class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

UC06 — analysis classes

Conceptual domain roles; implementation technology is intentionally omitted.



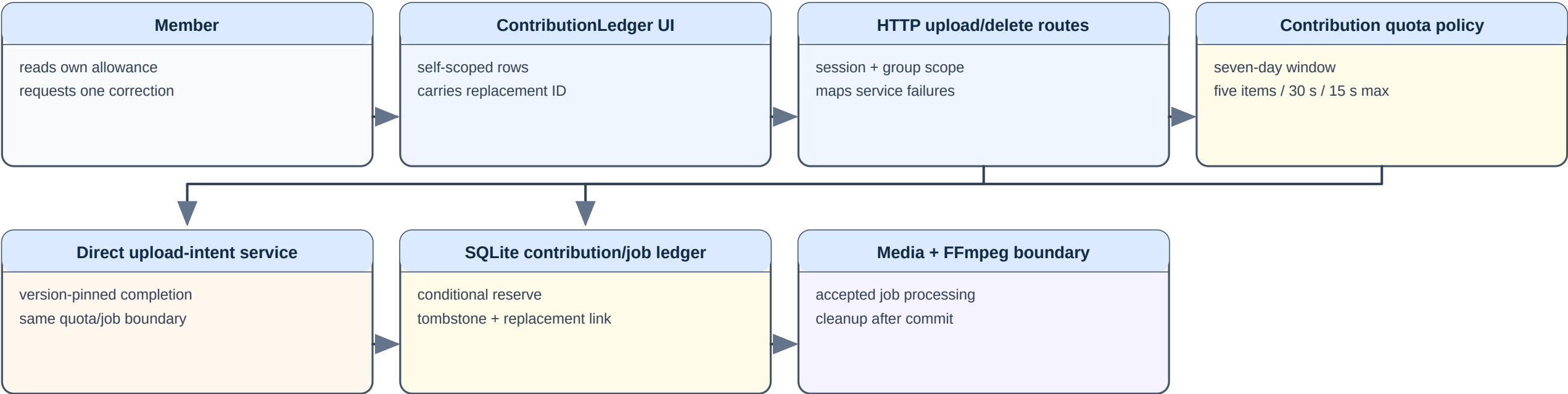
Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

packages/capture/diagrams/uc06-analysis-class.svg

Figure M055 | CAPTURE | uc06 design class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

UC06 — design classes

Code-backed responsibilities at the exact accepted commit.



Solid arrows show a responsibility/use relationship. The intent and staged routes converge on the authoritative quota and job records.

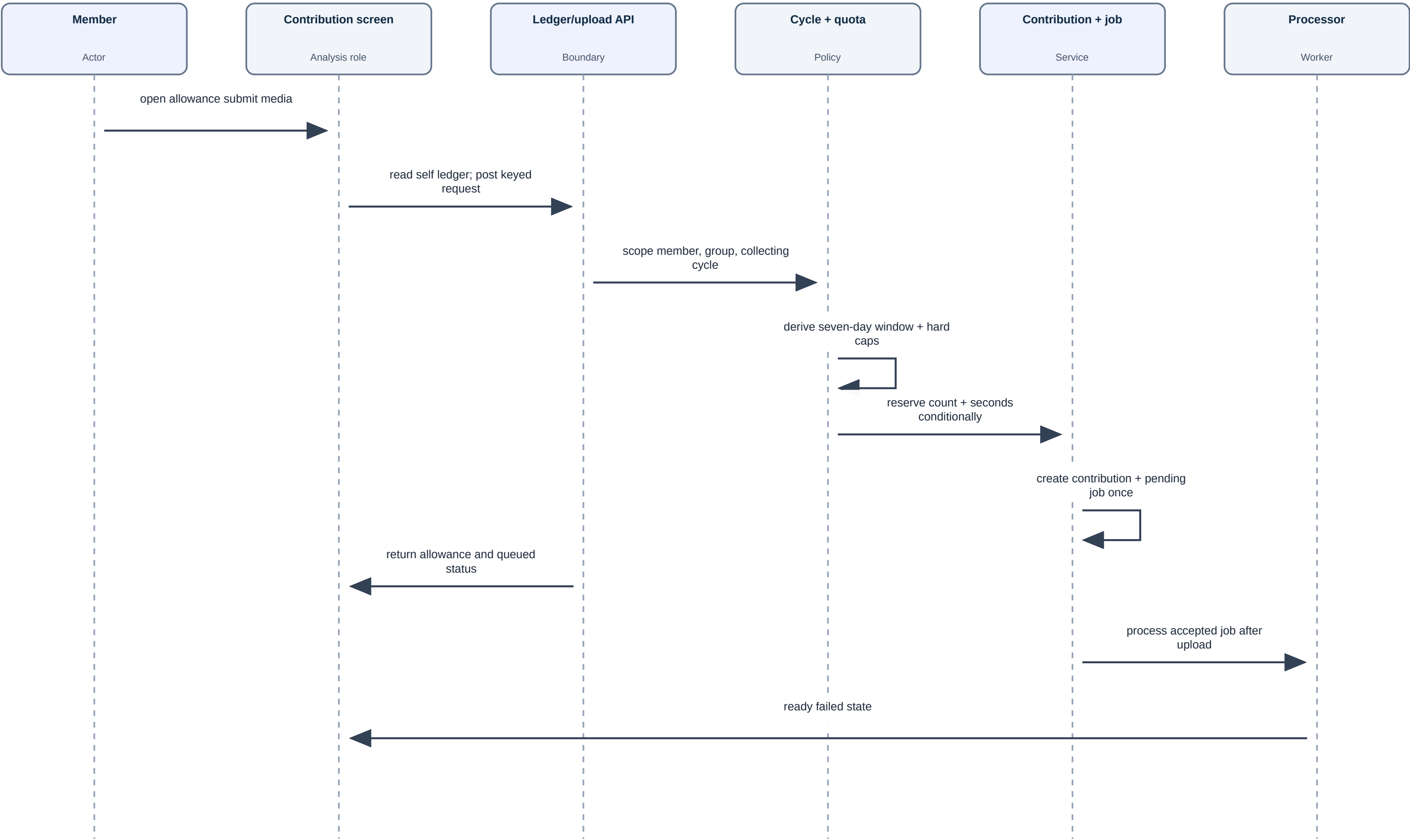
packages/capture/diagrams/uc06-design-class.svg

Figure M056 | CAPTURE | uc06 f1 analysis sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC06-F1 — Read allowance and reserve contribution

Analysis sequence · scoped read followed by one atomic reservation



Relevant alternate / exception paths

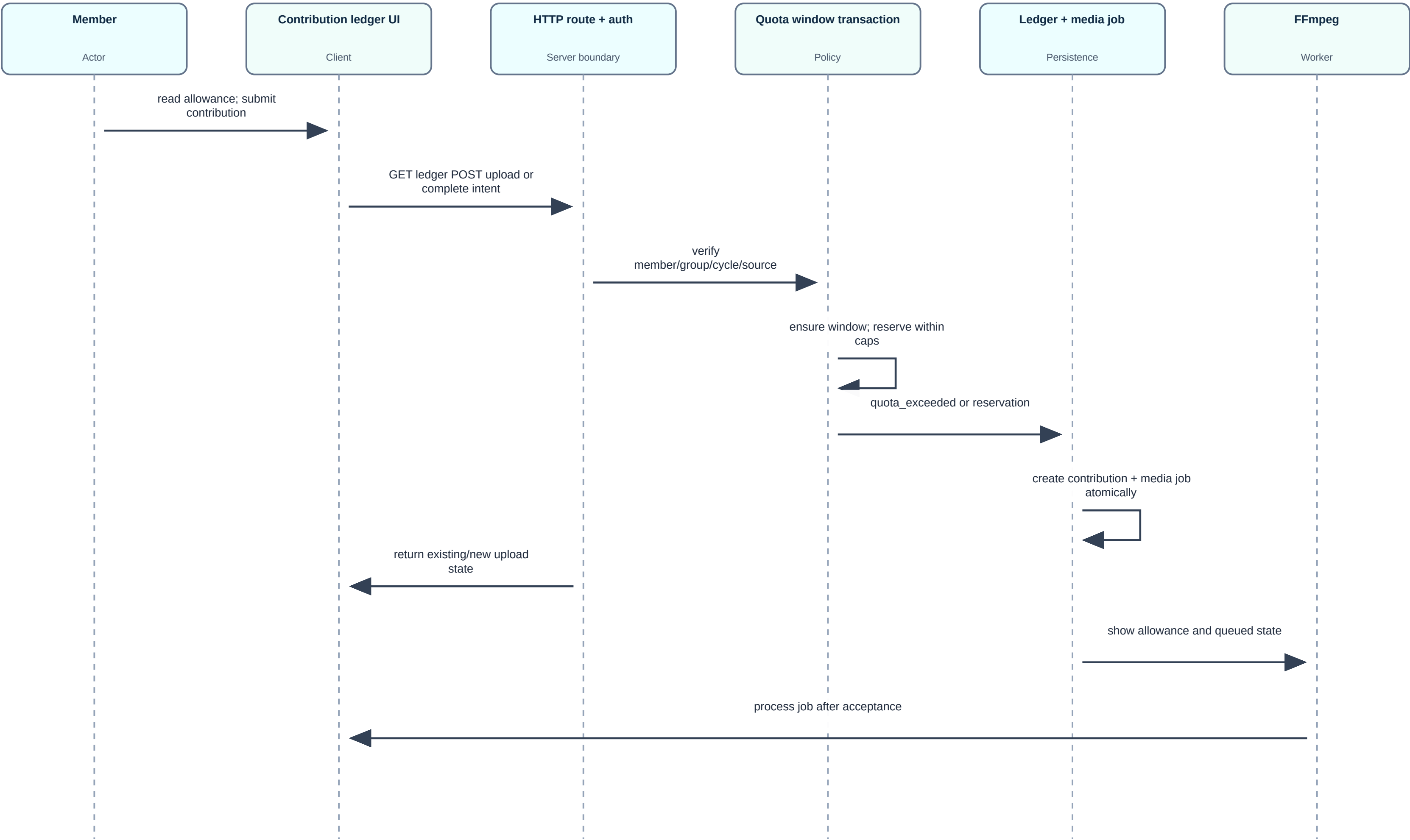
Invalid duration/media, missing or closed cycle, stale source/version, exhausted five-item/30-second allowance, conflicting key or transaction failure rejects/rolls back. Identical accepted key returns existing result without another reservation. Quota denial is not treated as offline retry.

Figure M057 | CAPTURE | uc06 f1 design sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC06-F1 — Read allowance and reserve contribution

Design sequence · scoped read followed by one atomic reservation



Relevant alternate / exception paths

Hard caps: five items, 30 seconds, max video 15 seconds; photo=one item/three seconds. Invalid media/cycle/source, duplicate key with different intent, quota denial or storage/DB conflict rejects. Transaction rollback prevents partial contribution/job state; identical idempotent replay does not charge twice.

Figure M058 | CAPTURE | uc06 f2 analysis sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC06-F2 — Delete and replace once

Analysis sequence · same-window eligibility, allowance restoration and linked replacement

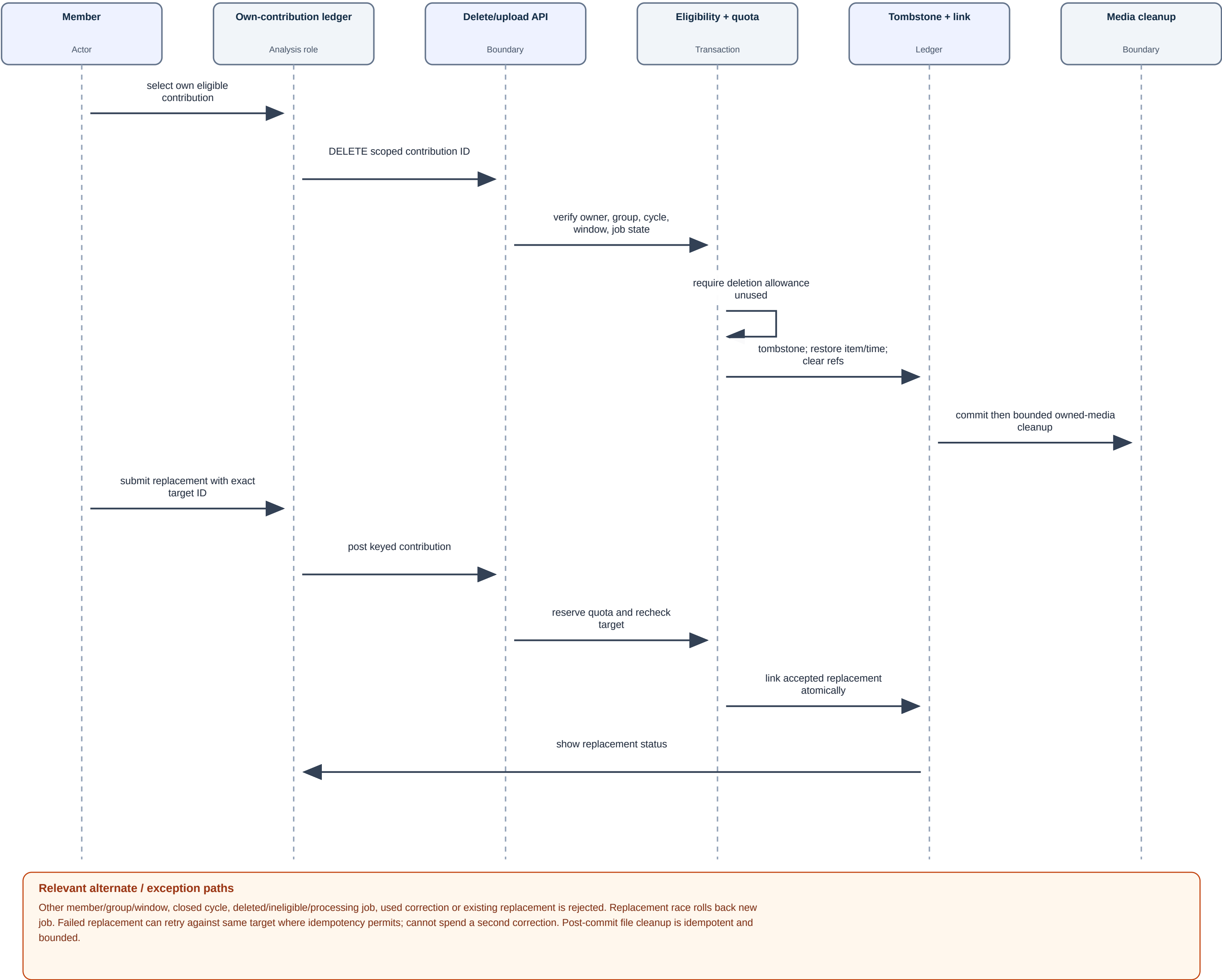
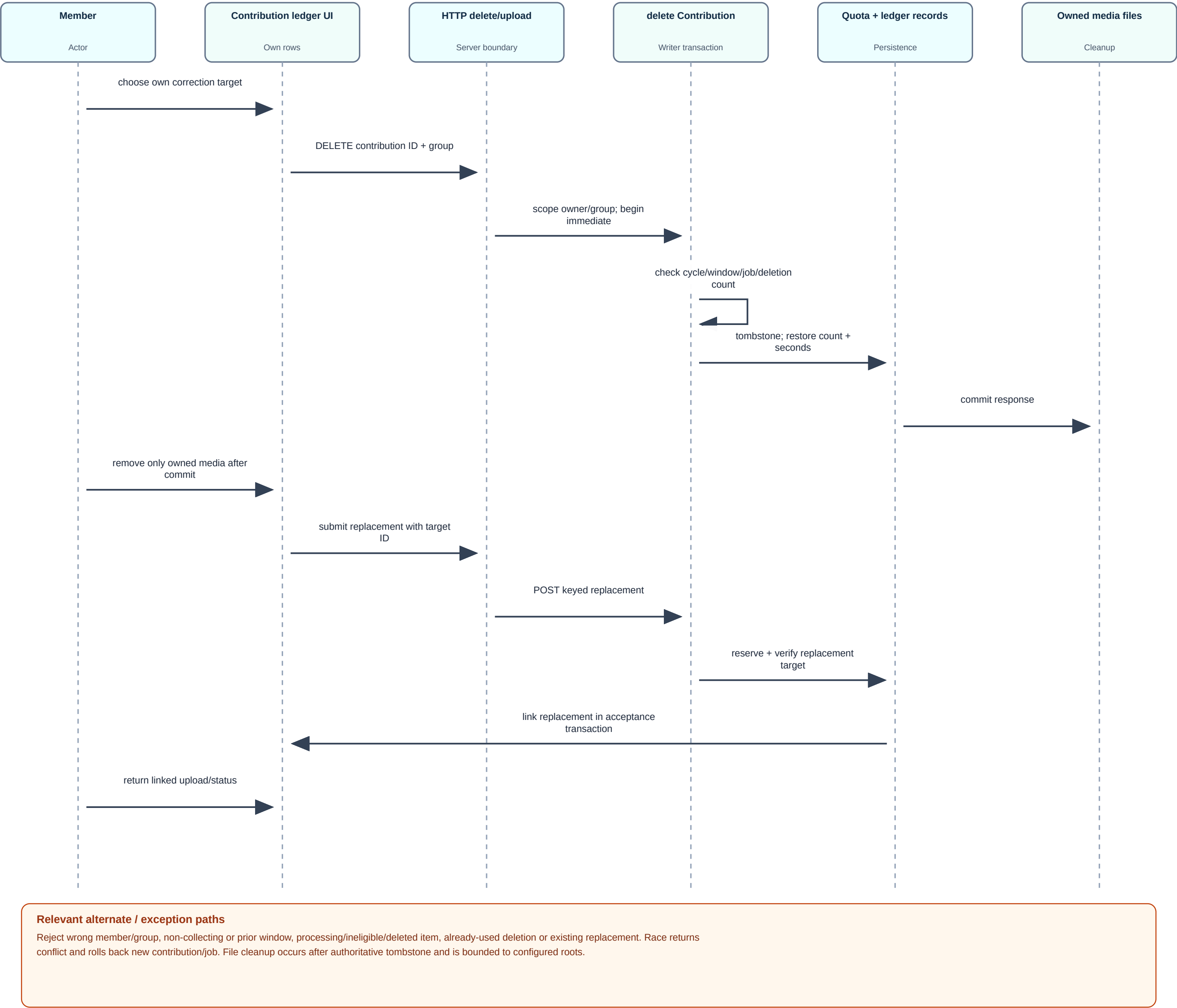


Figure M059 | CAPTURE | uc06 f2 design sequence

Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

UC06-F2 — Delete and replace once

Design sequence · same-window eligibility, allowance restoration and linked replacement



CYCLE model/source register | issue #359

Accepted model cut fc2b8c811e9c13b176113fa0bdb2c015798990b4; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

server/src/cycles/scheduler.ts:45

server/src/cycles/lifecycle.ts:200

server/src/cycles/lifecycle.ts:325

server/src/cycles/lifecycle.ts:276

server/src/cycles/lifecycle.ts:216

server/src/jobs/index.ts:502

server/src/db.ts:1071

server/src/jobs/index.ts:599

server/src/jobs/worker.ts:148

server/src/jobs/index.ts:131

server/src/jobs/index.ts:134

server/src/jobs/index.ts:380

server/src/jobs/index.ts:546

server/src/jobs/index.ts:785

server/src/ffmpeg.ts:463

server/src/jobs/index.ts:895

server/src/jobs/index.ts:963

server/src/db.ts:963

server/src/groups/real.ts:142

src/domain/premiere.ts:1

server/src/http.ts:2995

server/src/http.ts:3146

docs/domain/cycle-control-contract.md:1

docs/architecture/hosted-demo-persistence.md:21

server/src/cycles/lifecycle.ts:38

Figure M064 | CYCLE | uc07 analysis class

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

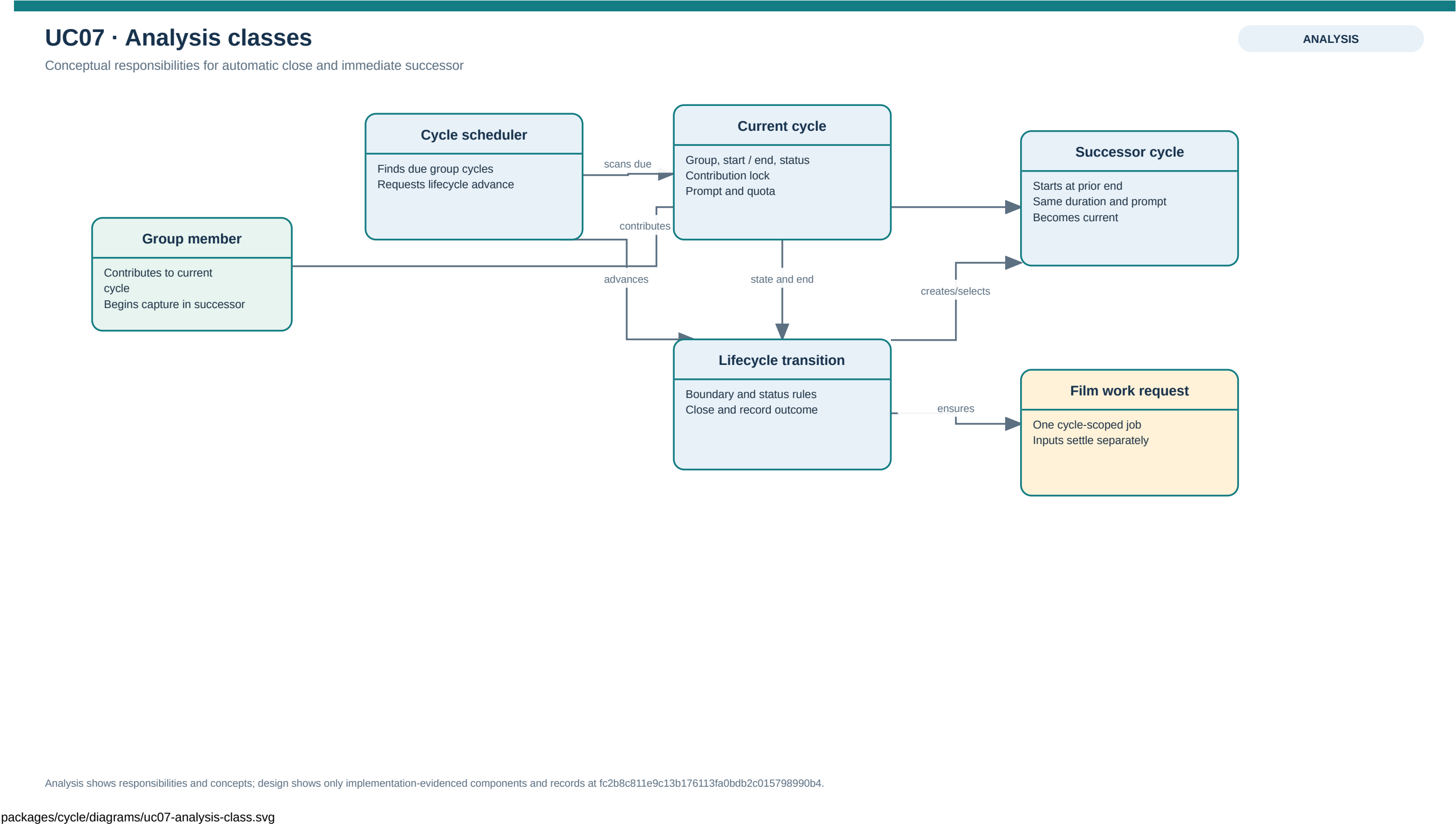


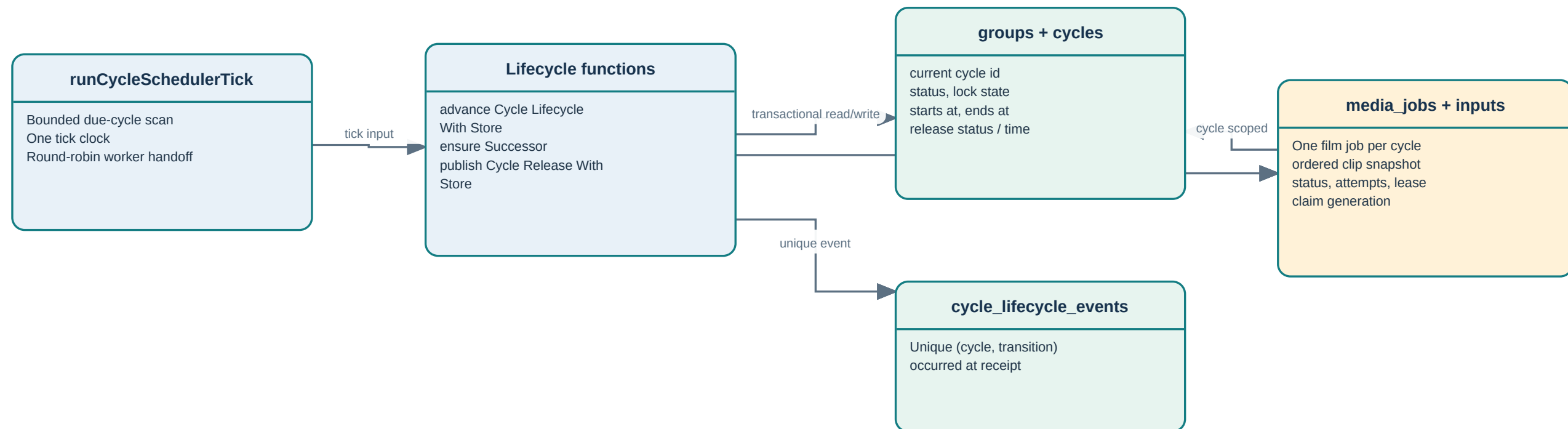
Figure M065 | CYCLE | uc07 design class

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC07 · Design classes and records

Pinned implementation responsibilities and persisted cycle transition

DESIGN



Analysis shows responsibilities and concepts; design shows only implementation-evidenced components and records at fc2b8c811e9c13b176113fa0bdb2c015798990b4.

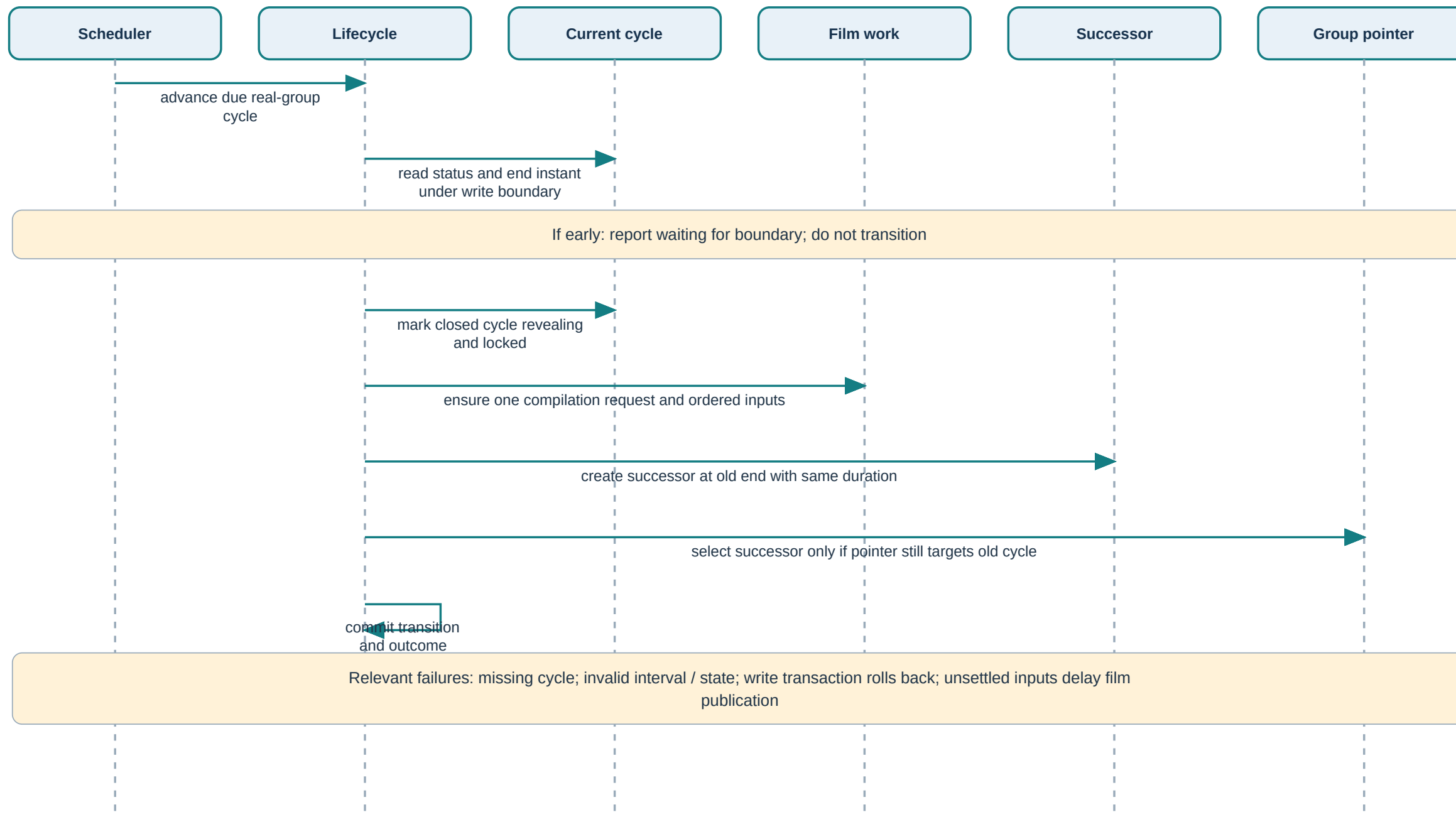
packages/cycle/diagrams/uc07-design-class.svg

Figure M066 | CYCLE | uc07 f1 analysis sequence
Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC07-F1 · Analysis sequence

ANALYSIS

Normal: close at boundary and open successor; exceptions are explicit

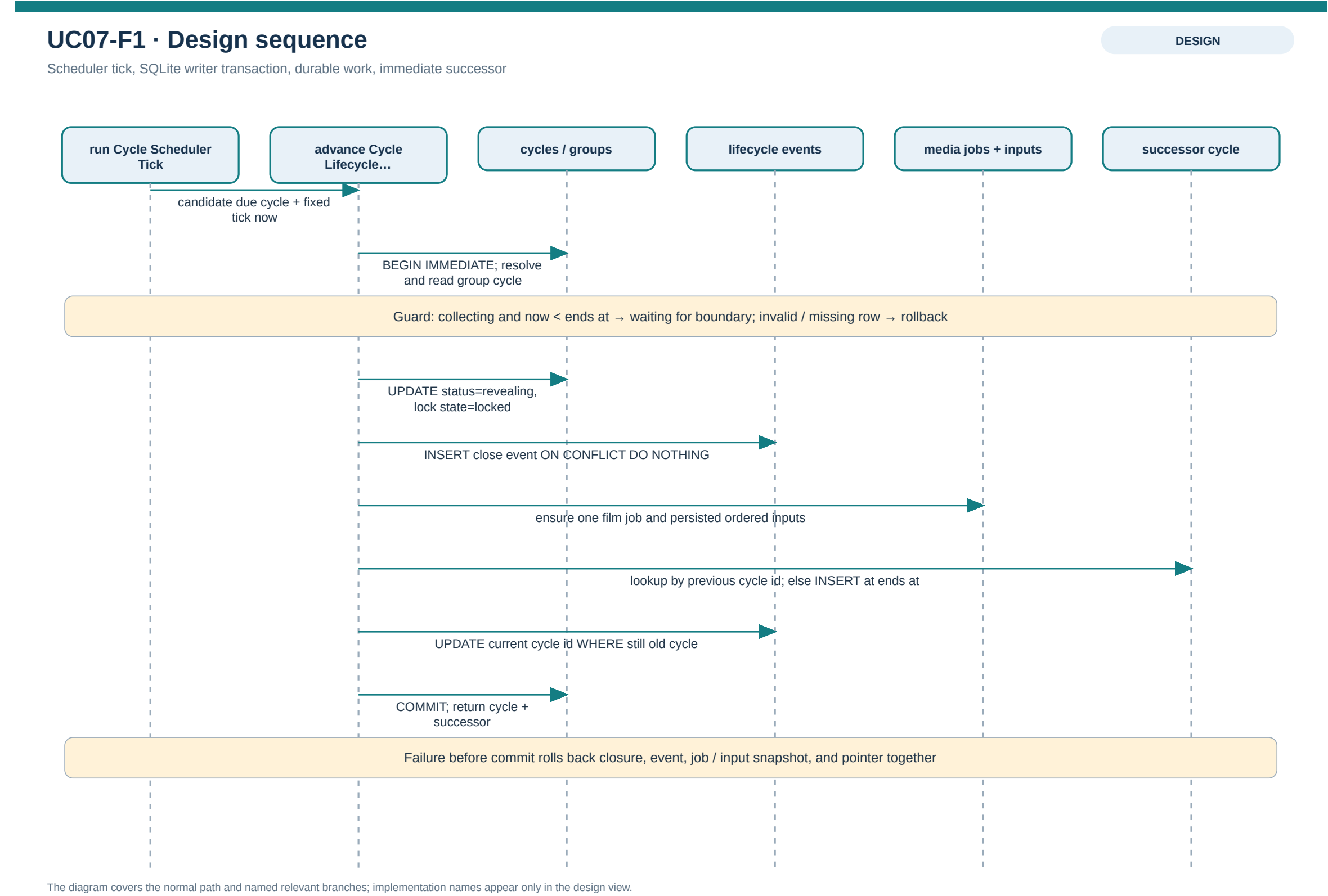


The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc07-f1-analysis-sequence.svg

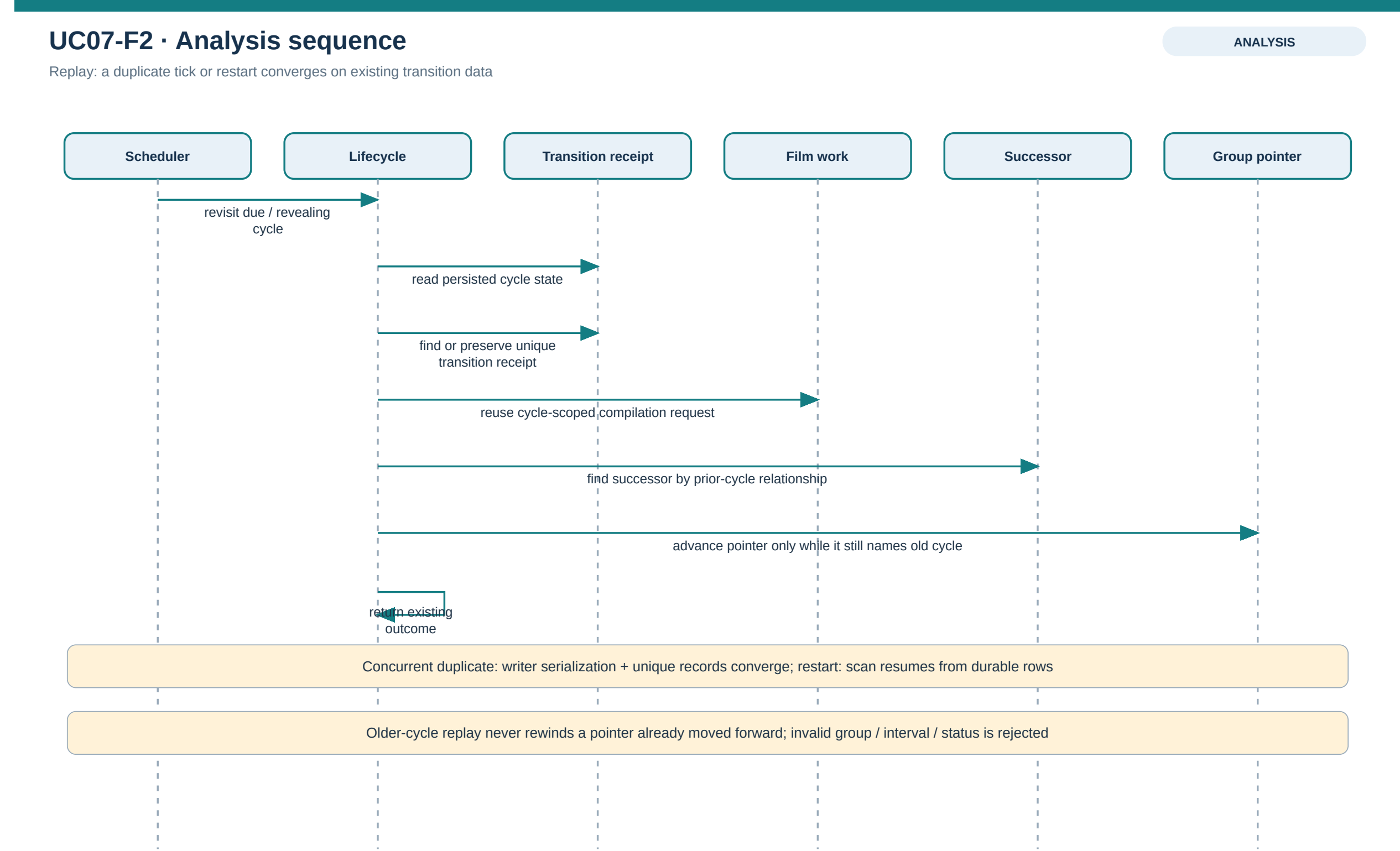
Figure M067 | CYCLE | uc07 f1 design sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model



packages/cycle/diagrams/uc07-f1-design-sequence.svg

Figure M068 | CYCLE | uc07 f2 analysis sequence
Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

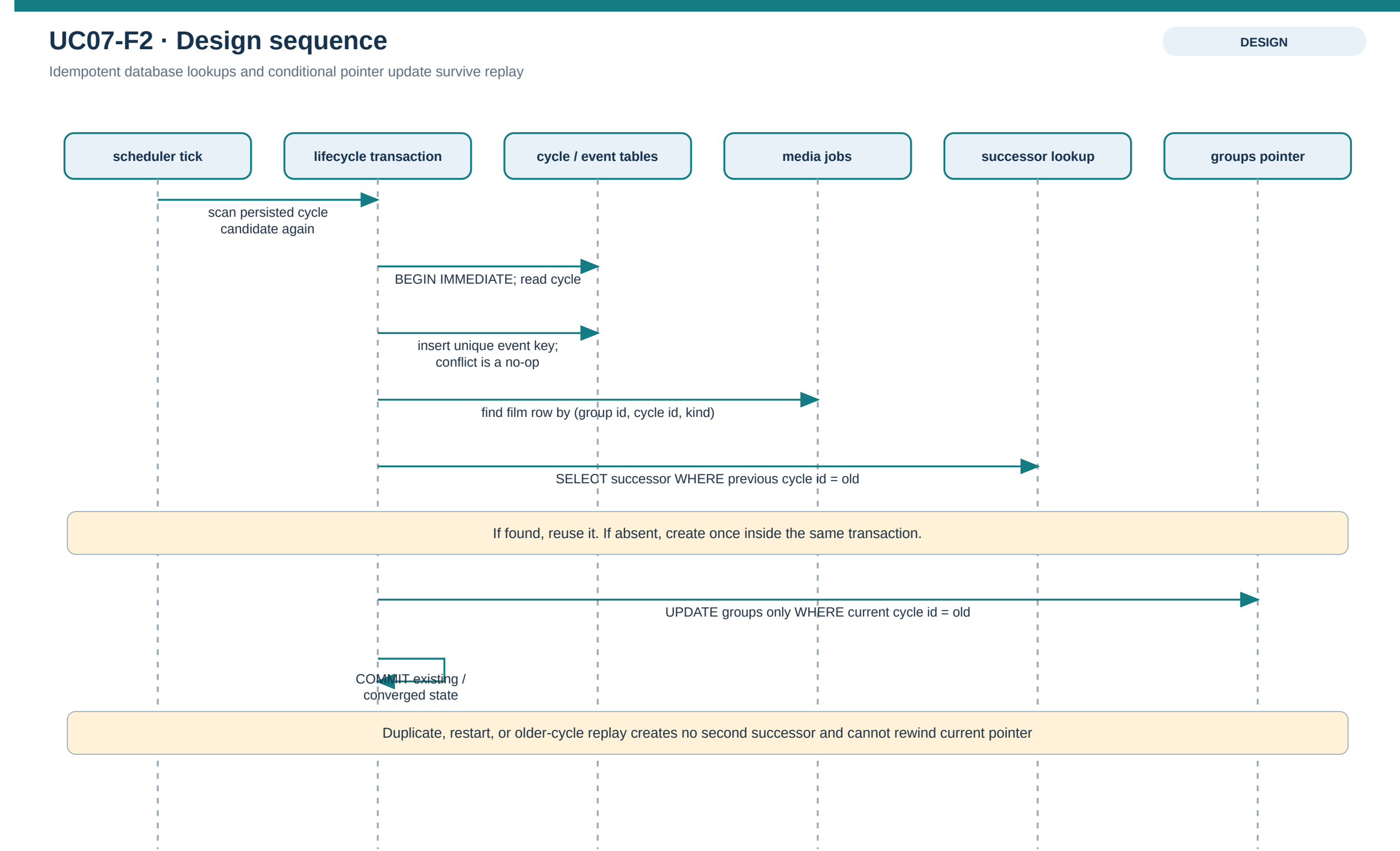


The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc07-f2-analysis-sequence.svg

Figure M069 | CYCLE | uc07 f2 design sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model



packages/cycle/diagrams/uc07-f2-design-sequence.svg

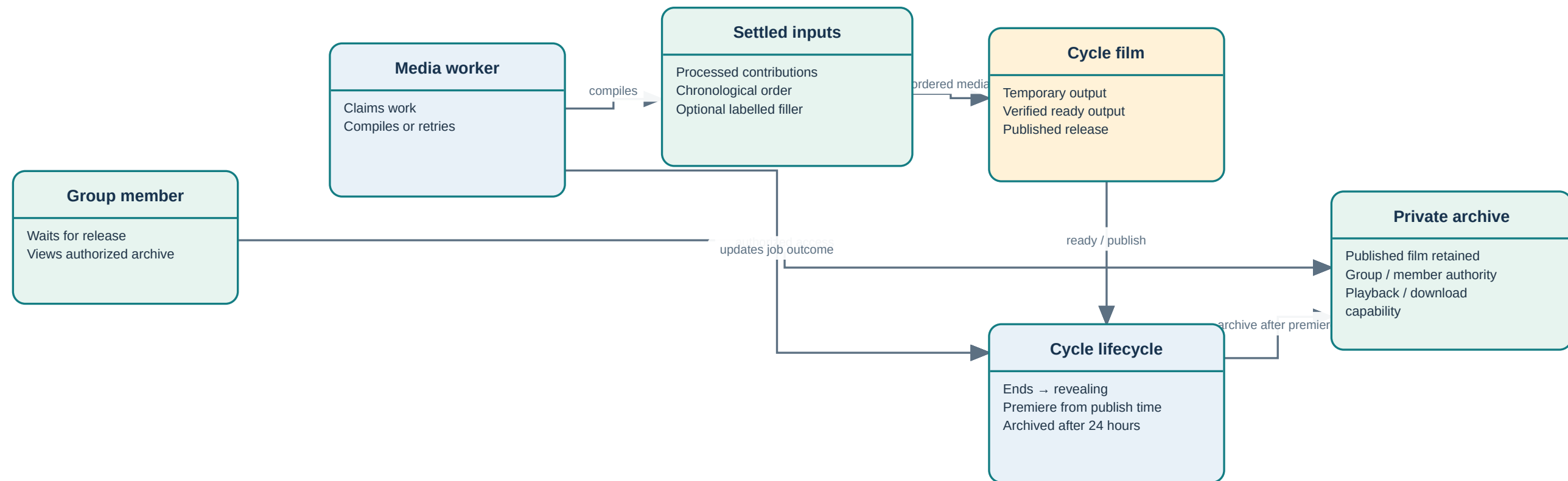
Figure M070 | CYCLE | uc08 analysis class

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08 · Analysis classes

Concepts for settled inputs, compilation, release, premiere, and archive

ANALYSIS



Analysis shows responsibilities and concepts; design shows only implementation-evidenced components and records at fc2b8c811e9c13b176113fa0bdb2c015798990b4.

packages/cycle/diagrams/uc08-analysis-class.svg

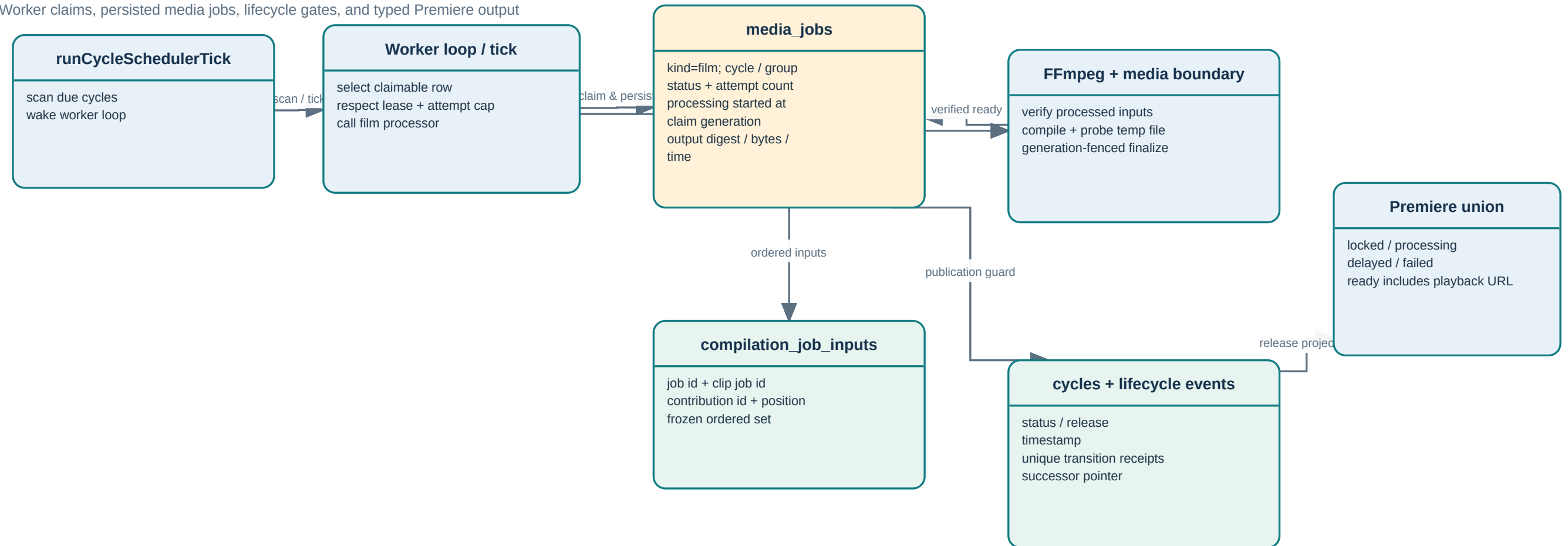
Figure M071 | CYCLE | uc08 design class

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08 · Design classes and records

DESIGN

Worker claims, persisted media jobs, lifecycle gates, and typed Premiere output



Analysis shows responsibilities and concepts; design shows only implementation-evidenced components and records at fc2b8c811e9c13b176113fa0bdb2c015798990b4.

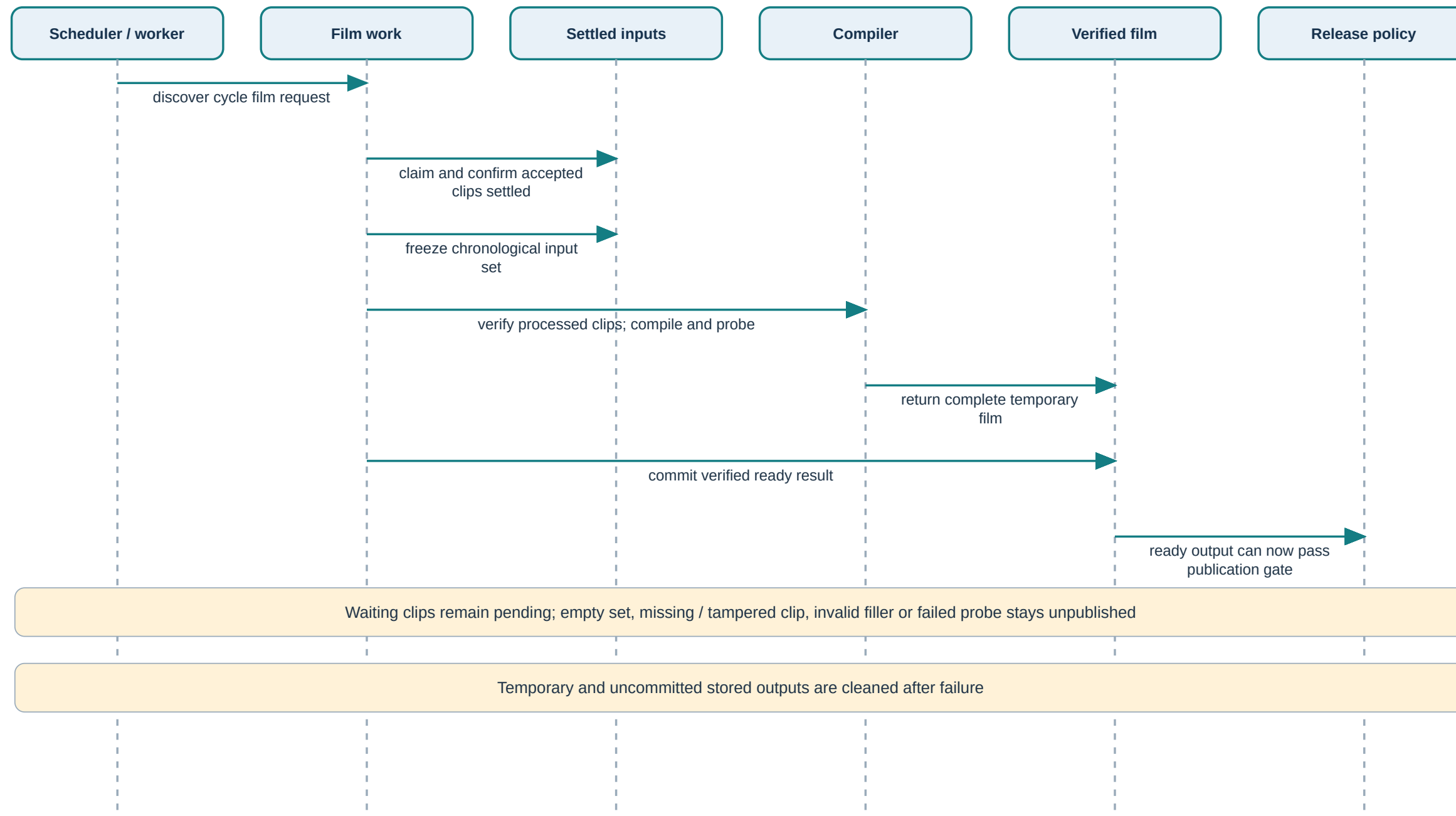
packages/cycle/diagrams/uc08-design-class.svg

Figure M072 | CYCLE | uc08 f1 analysis sequence
Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08-F1 · Analysis sequence

Normal compile from settled contributions to a verified film

ANALYSIS



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc08-f1-analysis-sequence.svg

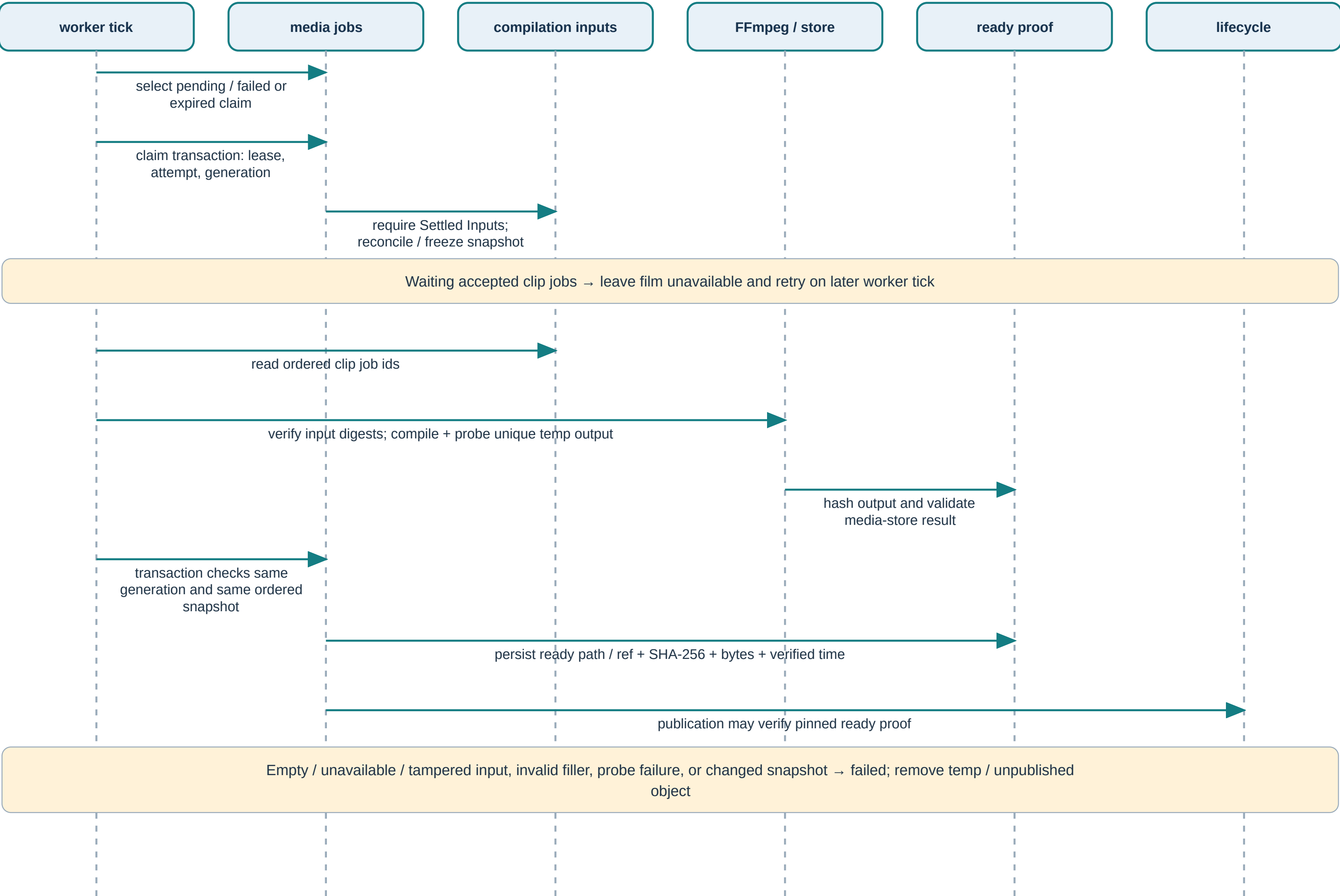
Figure M073 | CYCLE | uc08 f1 design sequence

Accepted source fc2b8c81 | issue #359 | Large landscape foldout | complete model

DESIGN

UC08-F1 · Design sequence

Persistent job, ordered inputs, worker processor, and fenced finalization



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

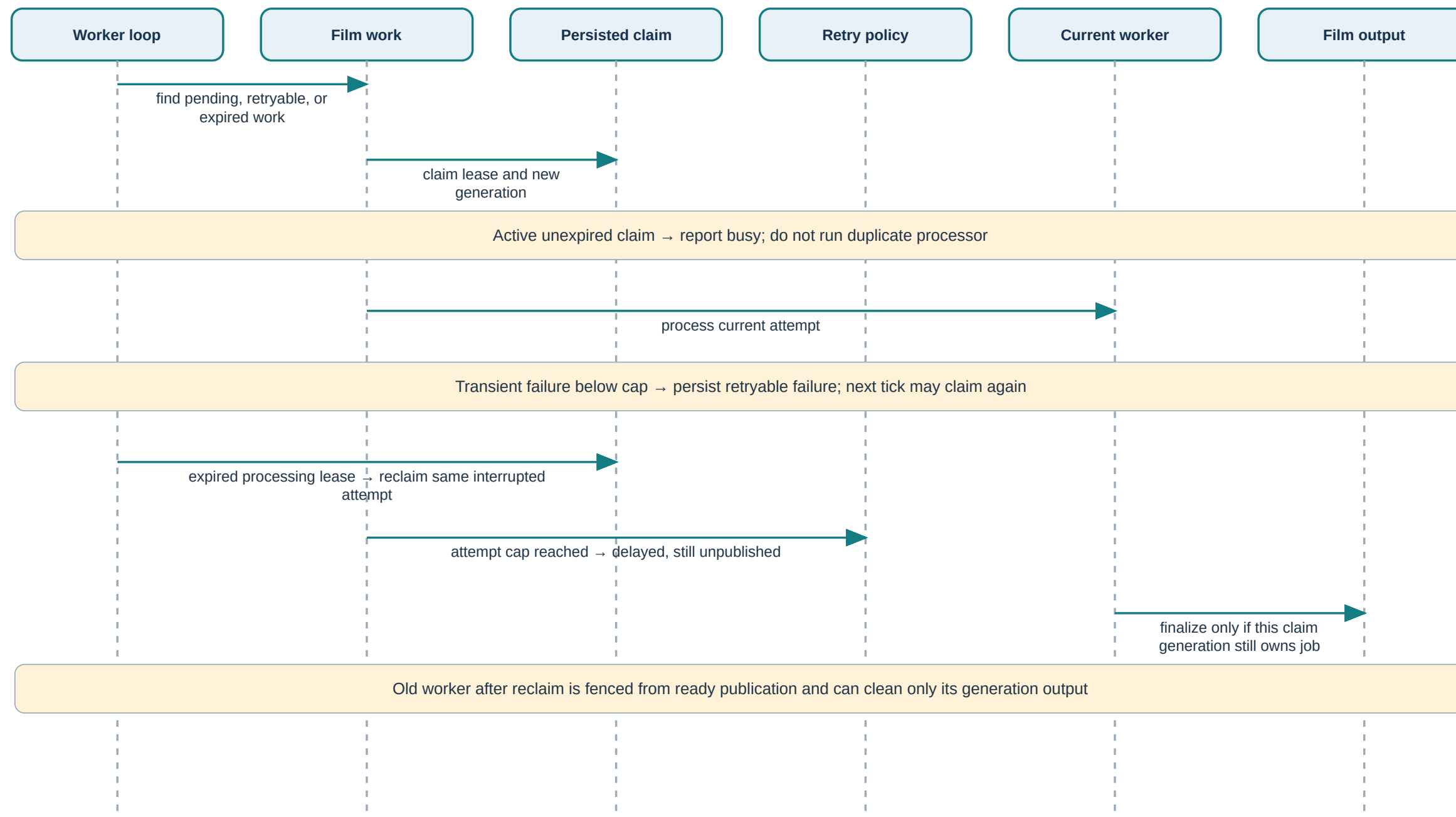
Figure M074 | CYCLE | uc08 f2 analysis sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08-F2 · Analysis sequence

ANALYSIS

Retry and lease recovery do not let an abandoned worker publish



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

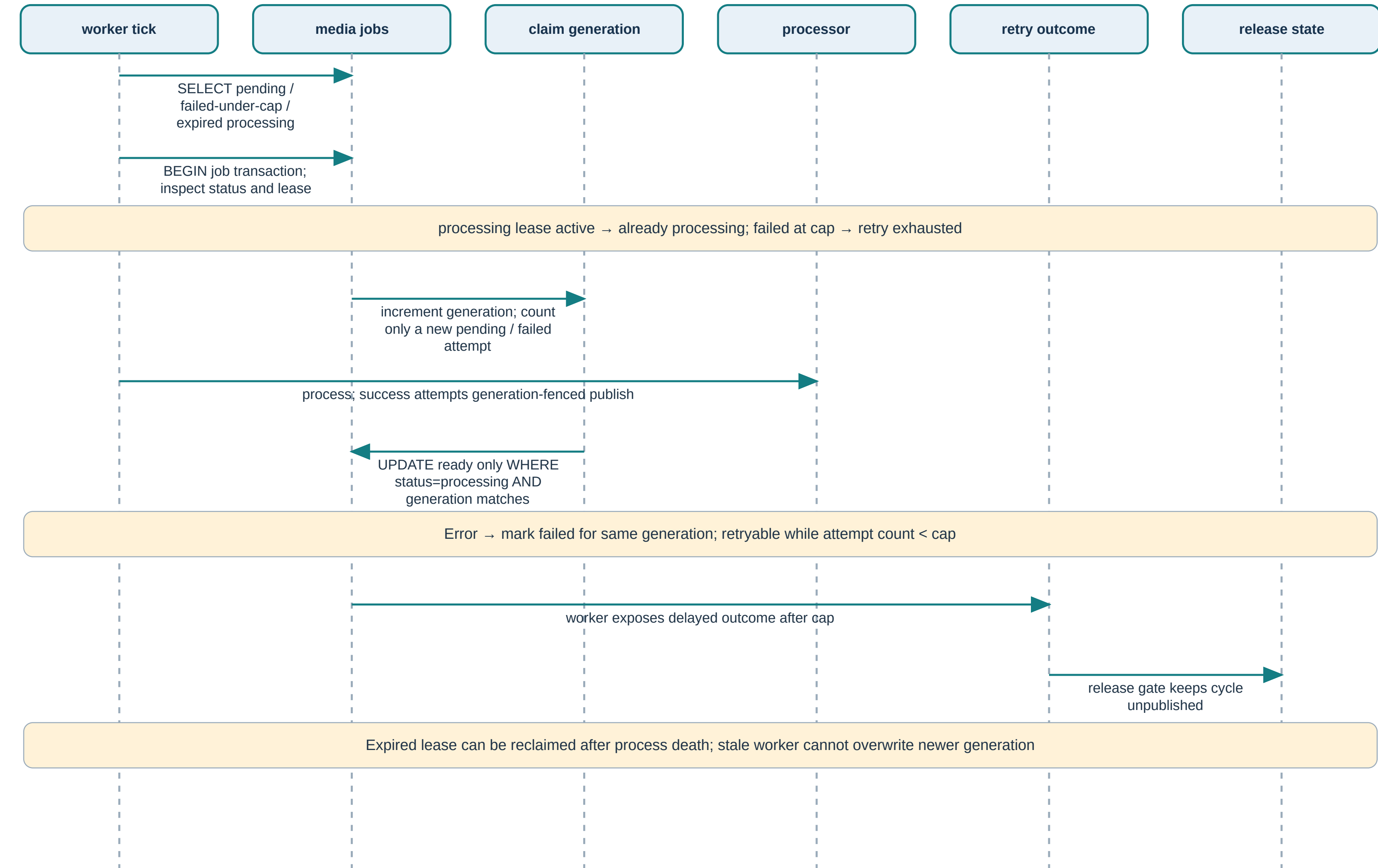
packages/cycle/diagrams/uc08-f2-analysis-sequence.svg

Figure M075 | CYCLE | uc08 f2 design sequence
 Accepted source fc2b8c81 | issue #359 | Large landscape foldout | complete model

UC08-F2 · Design sequence

DESIGN

Persisted attempt count, lease expiry, and claim-generation fence



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

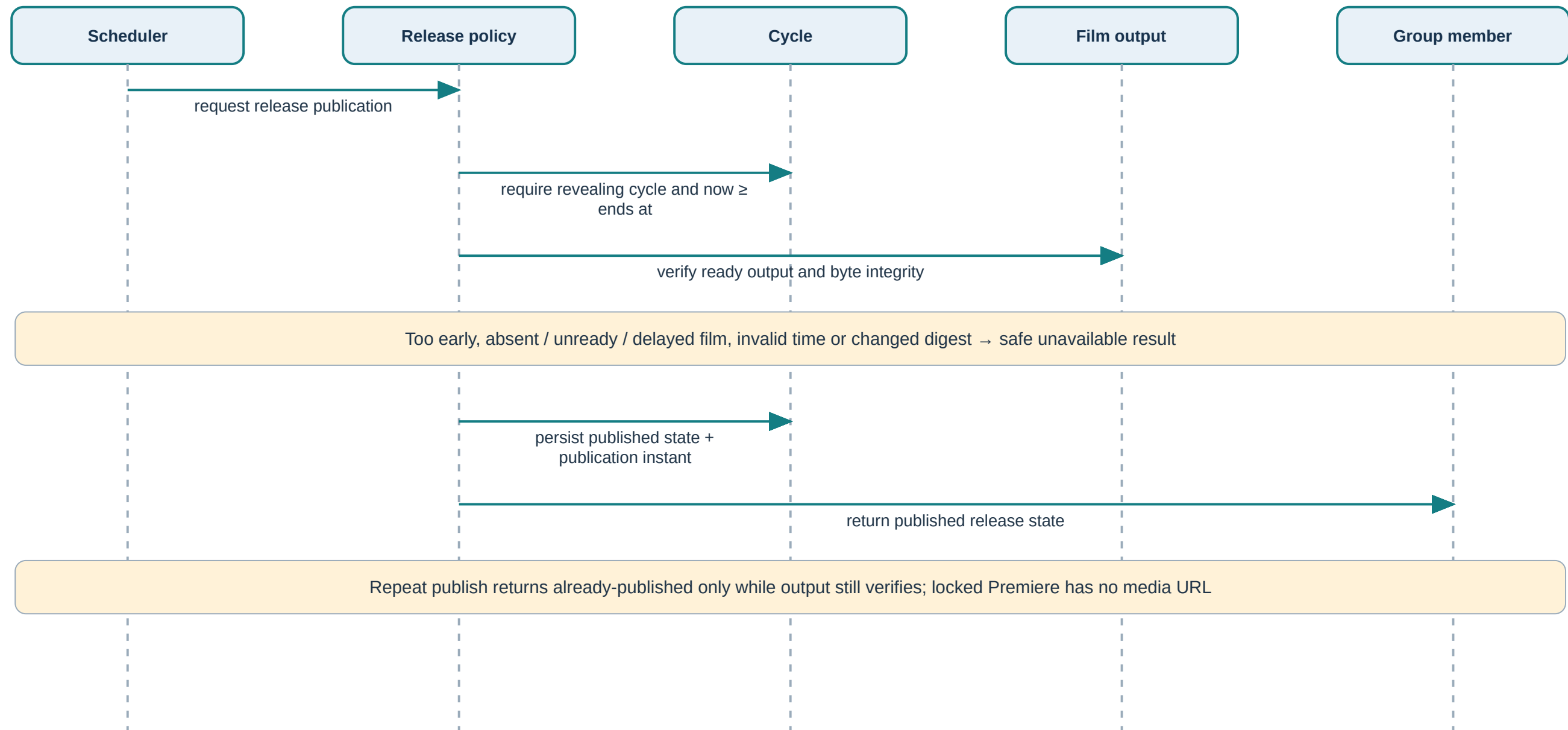
Figure M076 | CYCLE | uc08 f3 analysis sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08-F3 · Analysis sequence

ANALYSIS

Publish only a ready, verified film at/after the cycle boundary



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc08-f3-analysis-sequence.svg

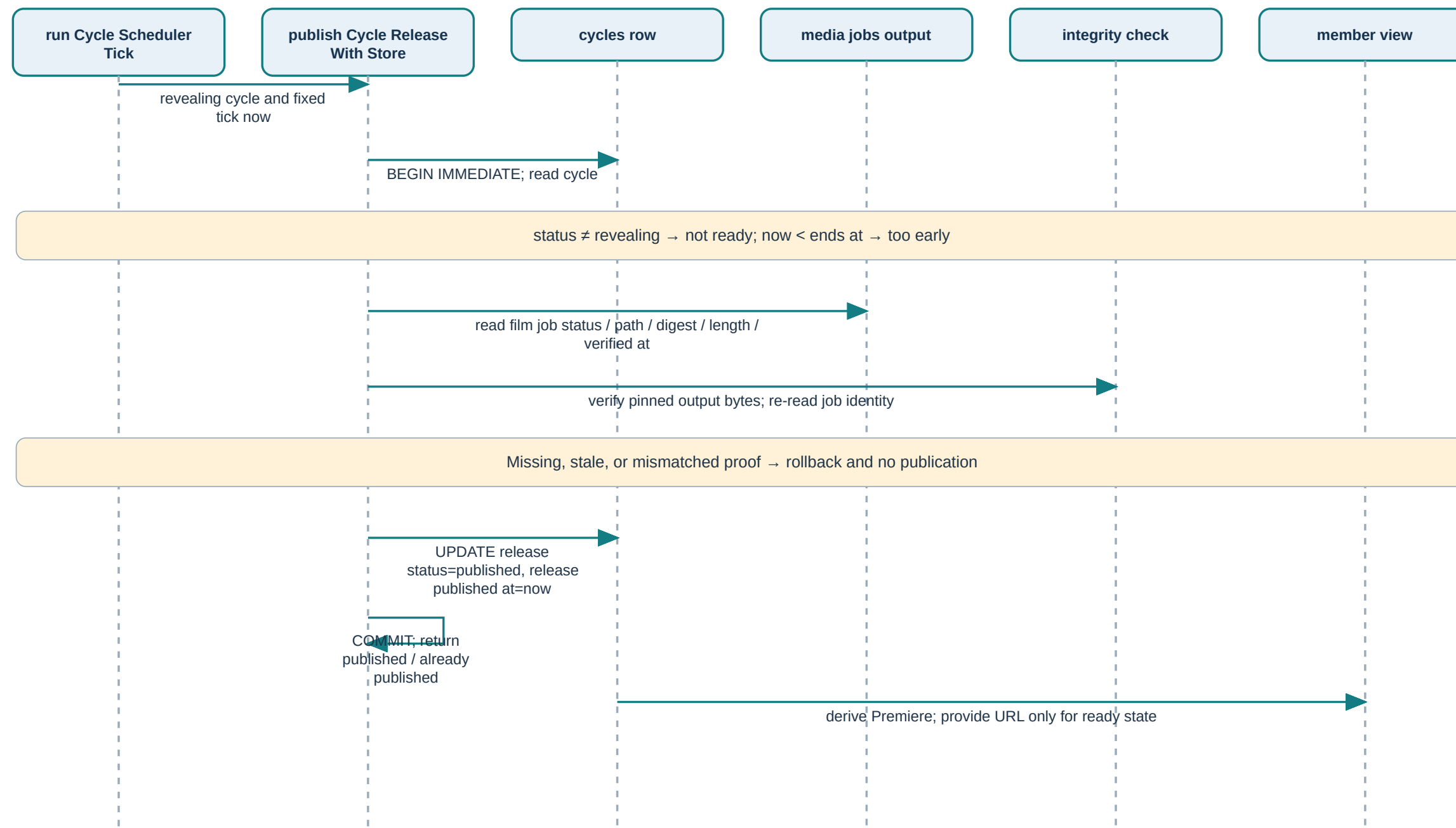
Figure M077 | CYCLE | uc08 f3 design sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08-F3 · Design sequence

DESIGN

End-time guard, integrity proof, and publication timestamp transaction

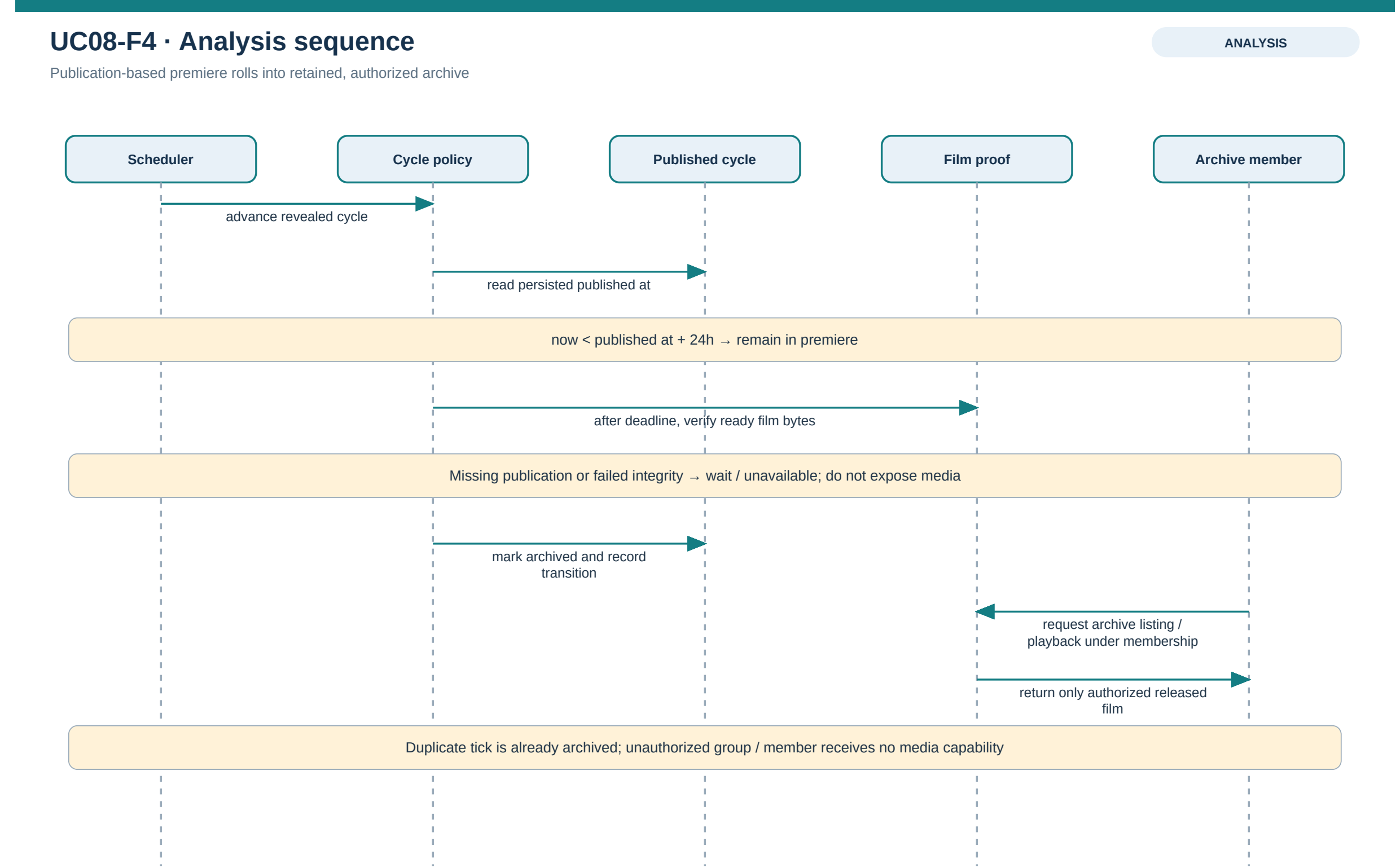


The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc08-f3-design-sequence.svg

Figure M078 | CYCLE | uc08 f4 analysis sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc08-f4-analysis-sequence.svg

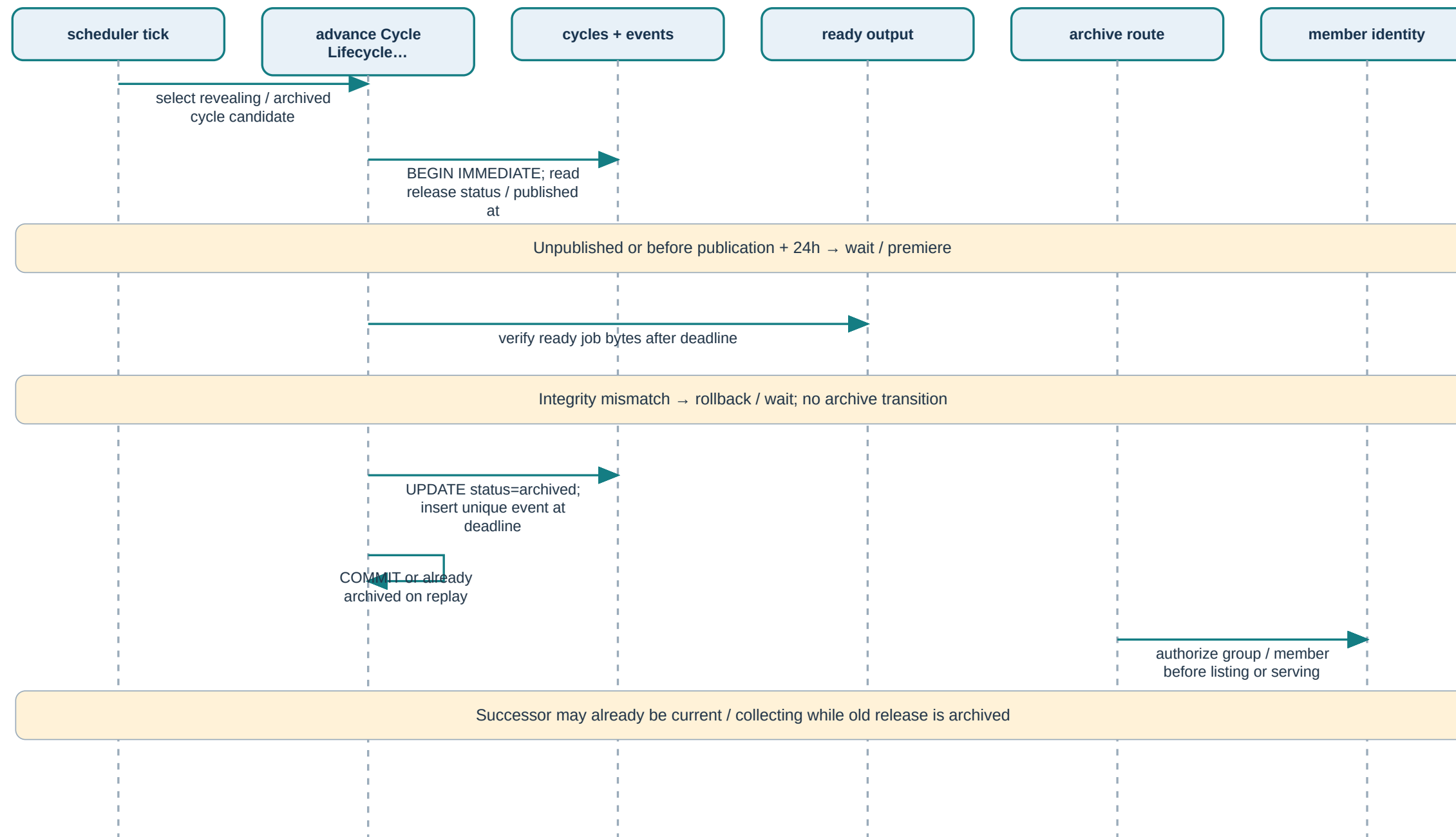
Figure M079 | CYCLE | uc08 f4 design sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

UC08-F4 · Design sequence

DESIGN

Archive after 24 hours from persisted publication; authorization remains server-side



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/uc08-f4-design-sequence.svg

PARTICIPATION model/source register | issue #360

Accepted model cut 1128b6a68985cf68215beb4a7a80fd00ec4242fa; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

src/reminders/RealGroupSettings.tsx:20

server/src/groups/settings.ts:5

server/src/http.ts:3783

server/src/reminders/schedule.ts:55

server/src/groups/settings.ts:44

server/src/groups/settings.ts:70

server/migrations/025-real-group-reminders.sql:3

src/reminders/private-reminder-client.ts:39

src/reminders/push-platform.ts:43

server/src/http.ts:3569

server/src/reminders/outbox.ts:158

server/src/reminders/outbox.ts:274

server/src/cli.ts:676

server/src/reminders/outbox.ts:303

server/src/reminders/outbox.ts:440

server/src/reminders/providers.ts:122

server/src/reminders/providers.ts:194

server/migrations/027-reminder-outbox.sql:3

src/chat/RealAccountChatScreen.tsx:50

src/chat/realtime-client.ts:171

src/chat/realtime-client.ts:208

src/chat/realtime-client.ts:329

src/chat/native-event-source.ts:229

server/src/chat/index.ts:156

server/src/chat/index.ts:206

server/src/chat/index.ts:282

server/src/chat/index.ts:426

server/src/http.ts:1626

server/src/http.ts:1667

server/src/http.ts:1811

server/src/http.ts:1868

server/src/realtime/index.ts:42

Figure M084 | PARTICIPATION | uc09 analysis class
 Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

uc09 — Owner prompt and timezone — analysis classes

Accepted source cut 1128b6a6 · domain responsibilities, no implementation class names

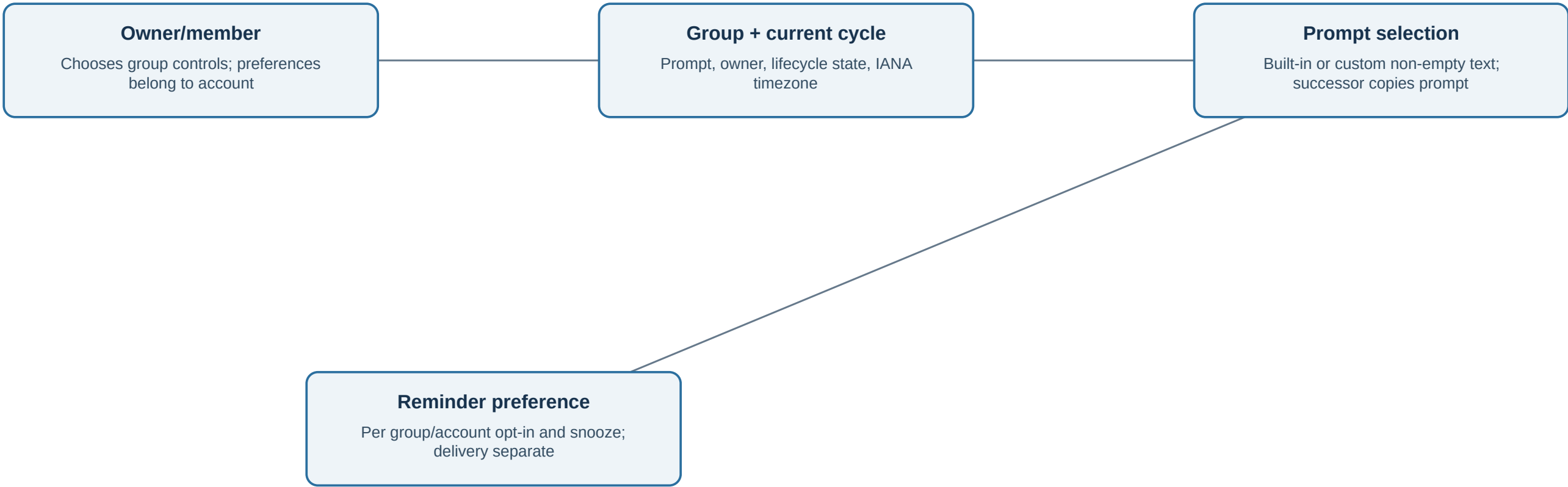


Figure M085 | PARTICIPATION | uc09 design class

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

uc09 — Owner prompt and timezone — design classes

Accepted source cut 1128b6a6 · source-backed modules, functions, transport and persisted records

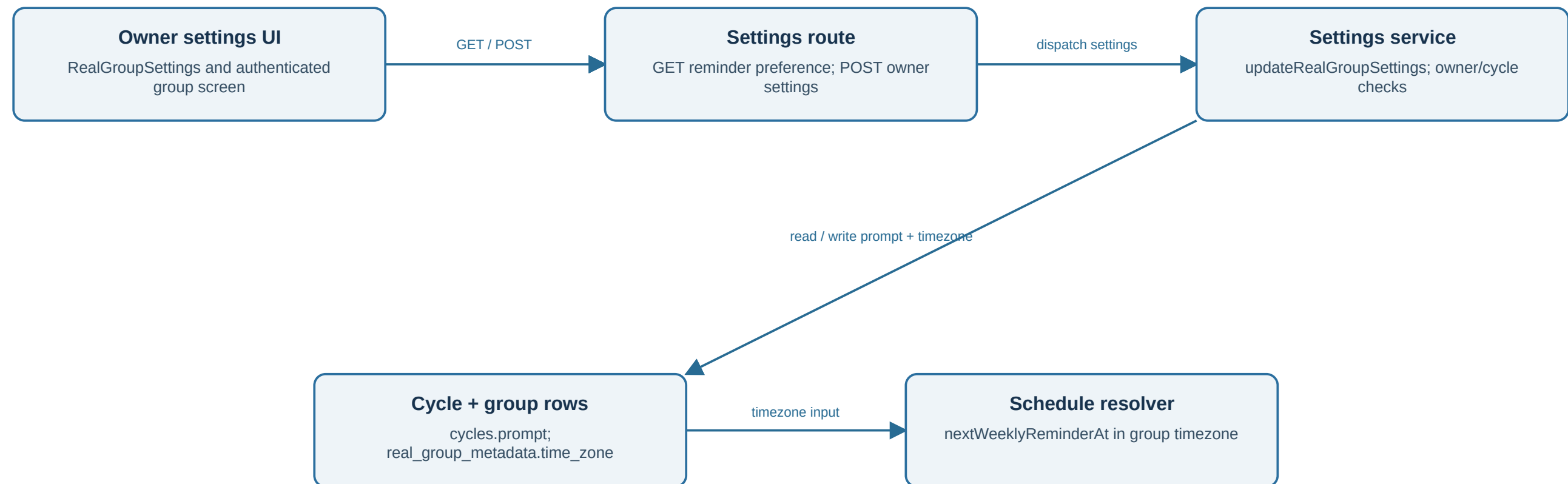
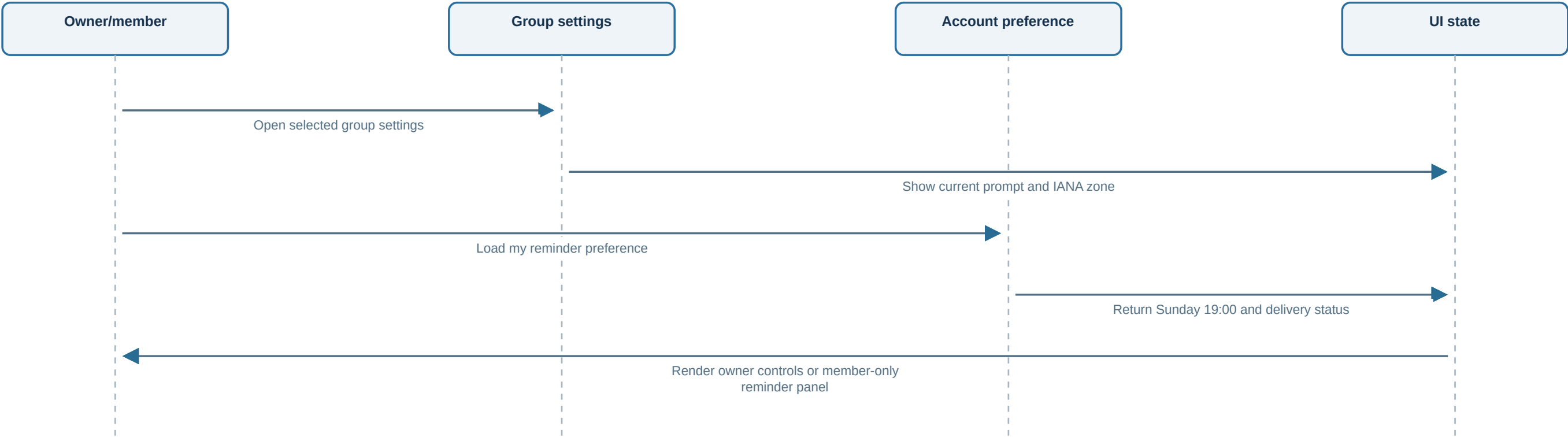


Figure M086 | PARTICIPATION | uc09 f1 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC09-F1 Load owner controls — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

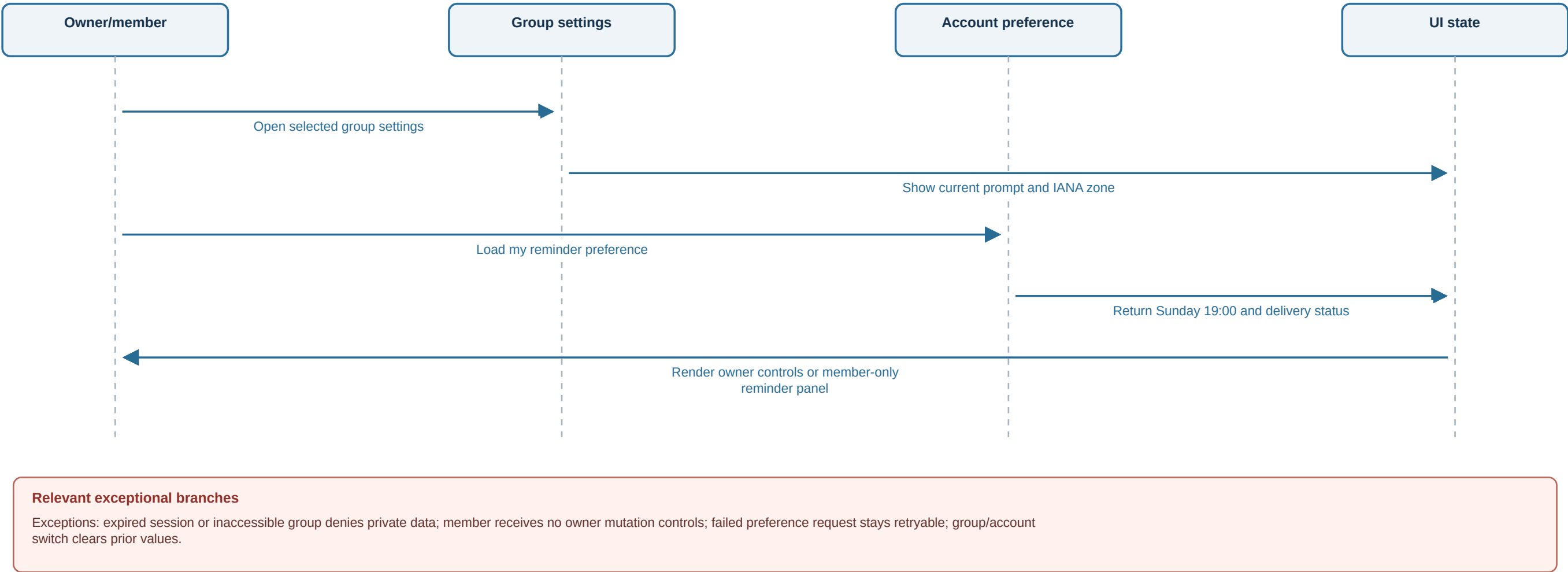
Exceptions: expired session or inaccessible group denies private data; member receives no owner mutation controls; failed preference request stays retryable; group/account switch clears prior values.

Figure M087 | PARTICIPATION | uc09 f1 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC09-F1 Load owner controls — design sequence

Accepted source cut 1128b6a6 · implemented components and records



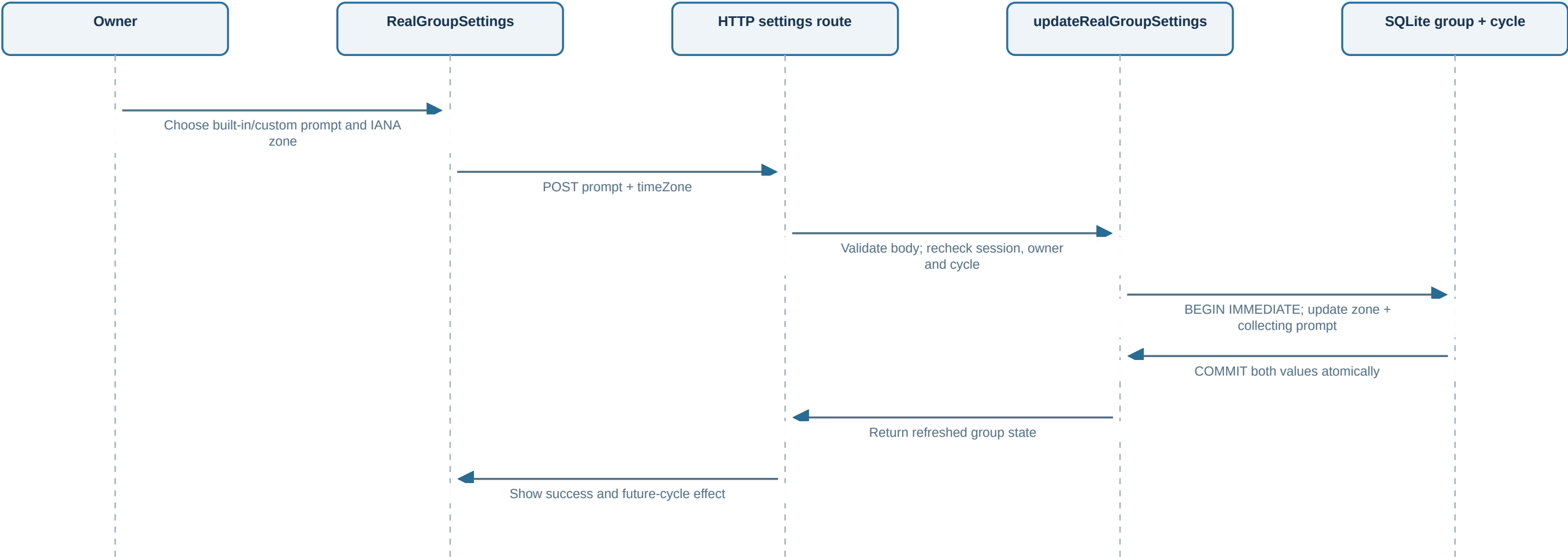
packages/participation/diagrams/uc09-f1-design-sequence.svg

Figure M088 | PARTICIPATION | uc09 f2 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC09-F2 Save prompt and timezone — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

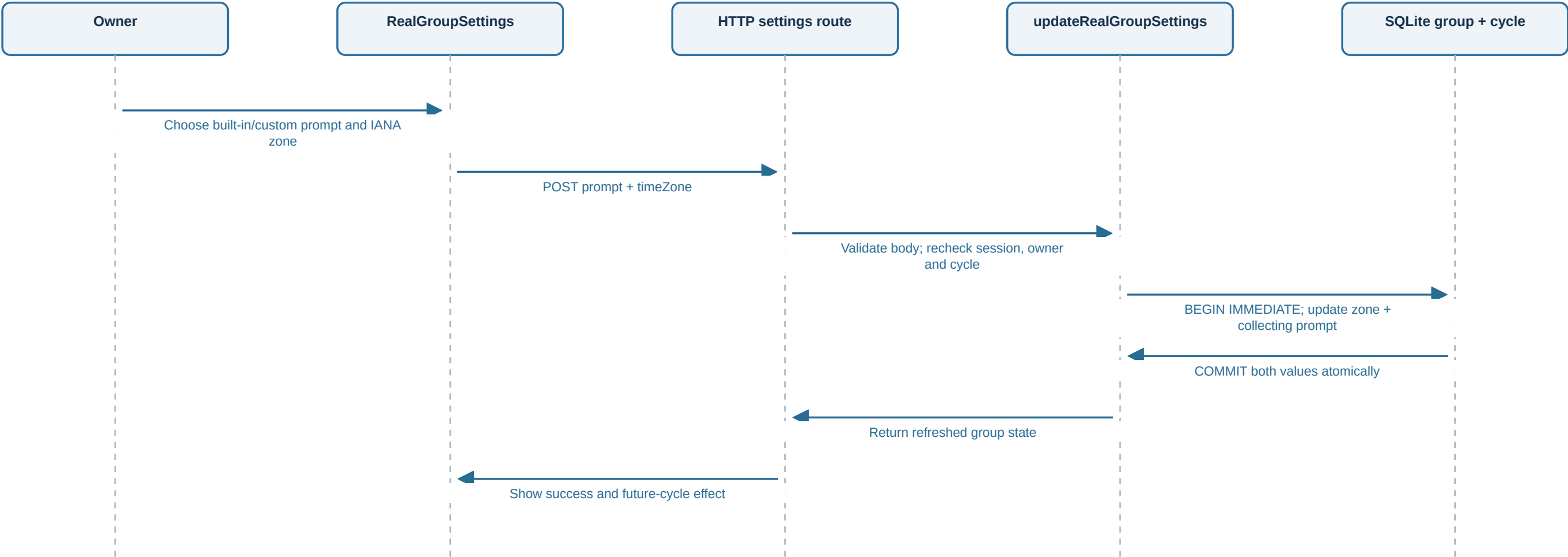
Exceptions: invalid prompt/zone returns 400; non-owner or lost membership returns 403; closed/elapsed cycle returns 409; transaction failure rolls back both updates.

Figure M089 | PARTICIPATION | uc09 f2 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC09-F2 Save prompt and timezone — design sequence

Accepted source cut 1128b6a6 · implemented components and records



Relevant exceptional branches

Exceptions: invalid prompt/zone returns 400; non-owner or lost membership returns 403; closed/elapsed cycle returns 409; transaction failure rolls back both updates.

Figure M090 | PARTICIPATION | uc10 analysis class
Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

uc10 — Reminder preference and delivery — analysis classes

Accepted source cut 1128b6a6 · domain responsibilities, no implementation class names

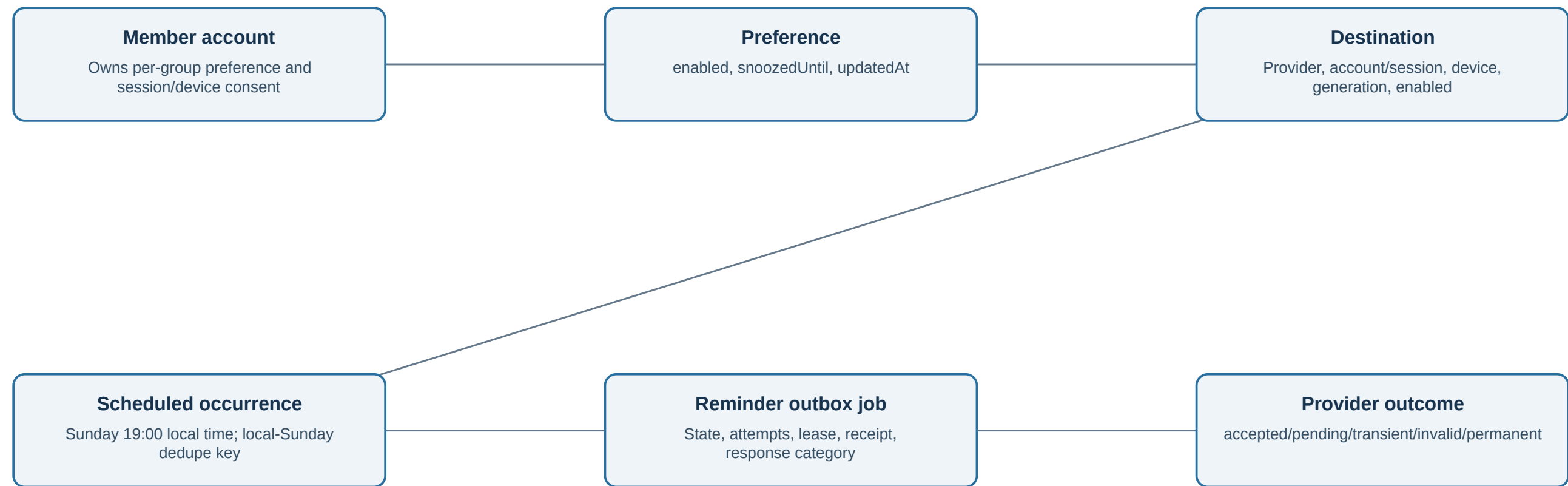


Figure M091 | PARTICIPATION | uc10 design class

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

uc10 — Reminder preference and delivery — design classes

Accepted source cut 1128b6a6 · source-backed modules, functions, transport and persisted records

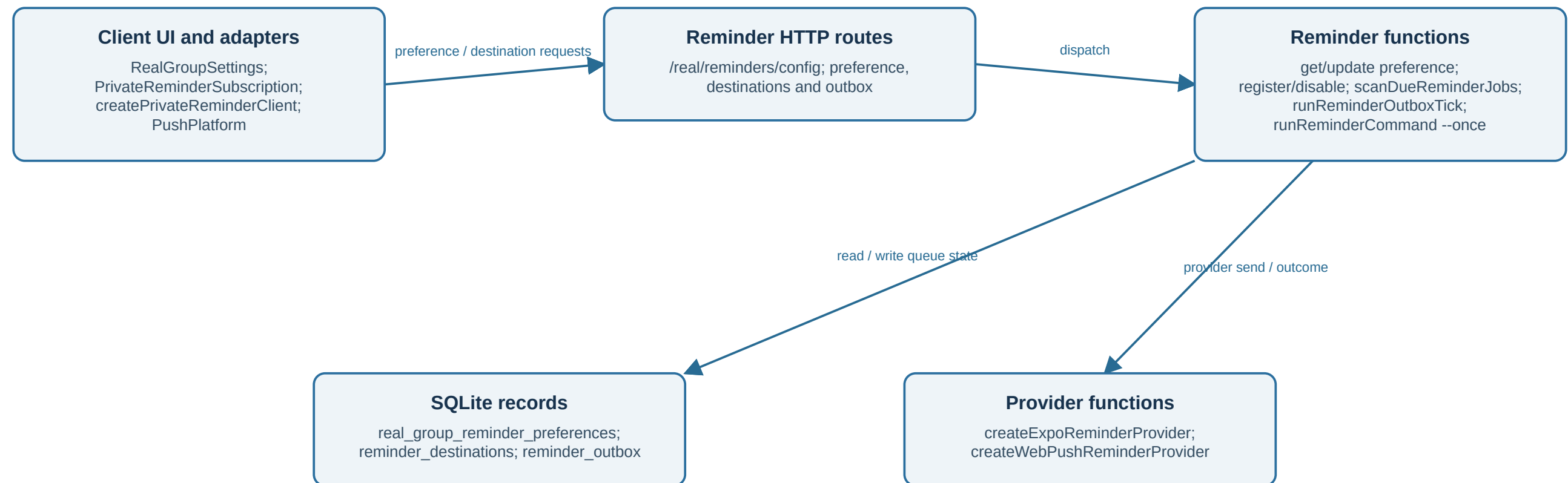
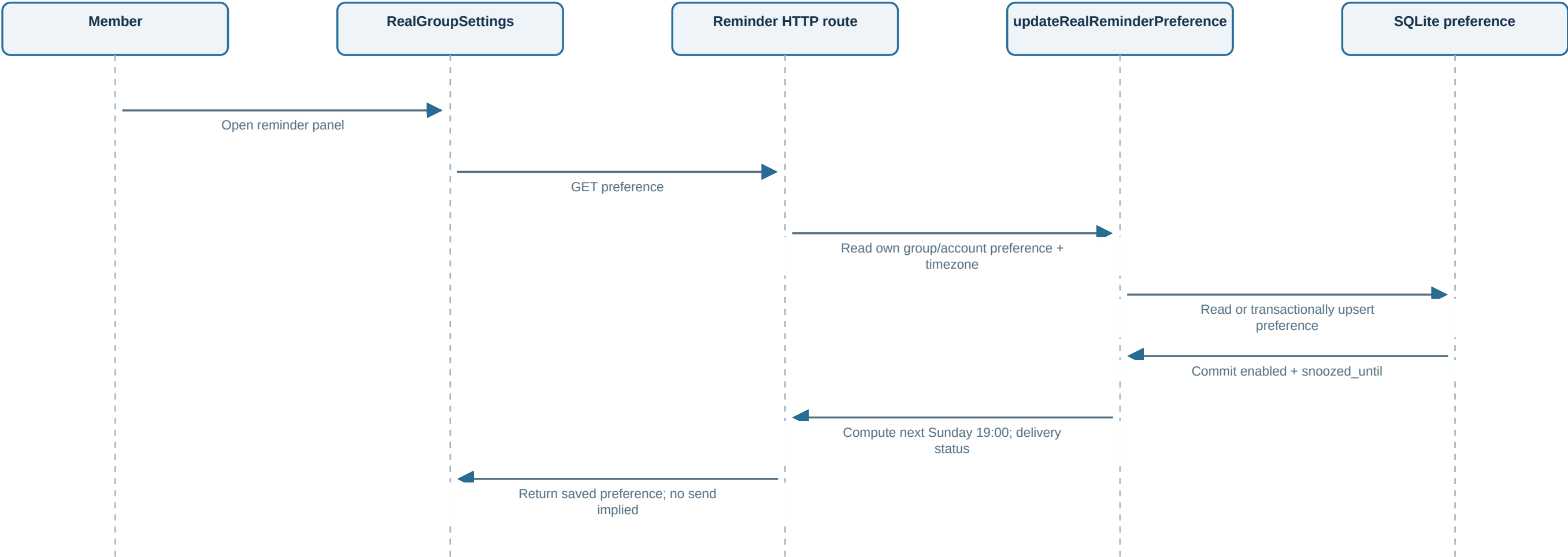


Figure M092 | PARTICIPATION | uc10 f1 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F1 Read and change my reminder preference — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

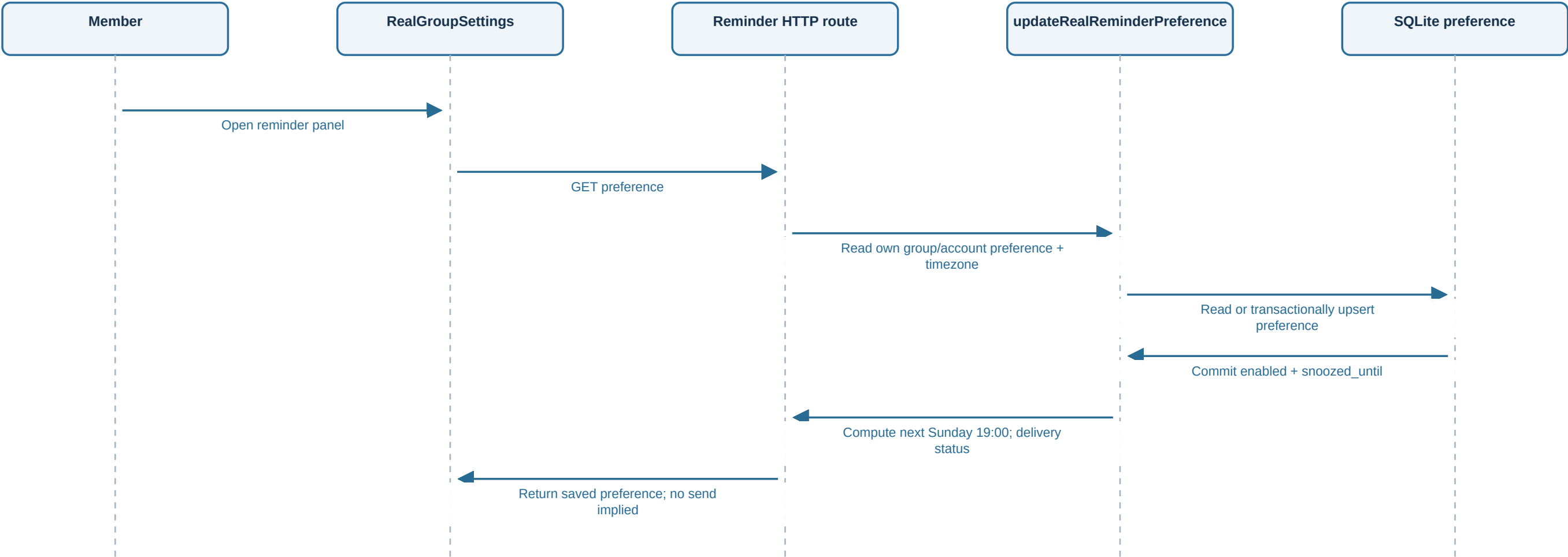
Exceptions: invalid date/boolean or snooze over 31 days is rejected; nonmember/expired session is denied; database failure preserves prior value; nonexistent local wall time is skipped.

Figure M093 | PARTICIPATION | uc10 f1 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F1 Read and change my reminder preference — design sequence

Accepted source cut 1128b6a6 · implemented components and records



Relevant exceptional branches

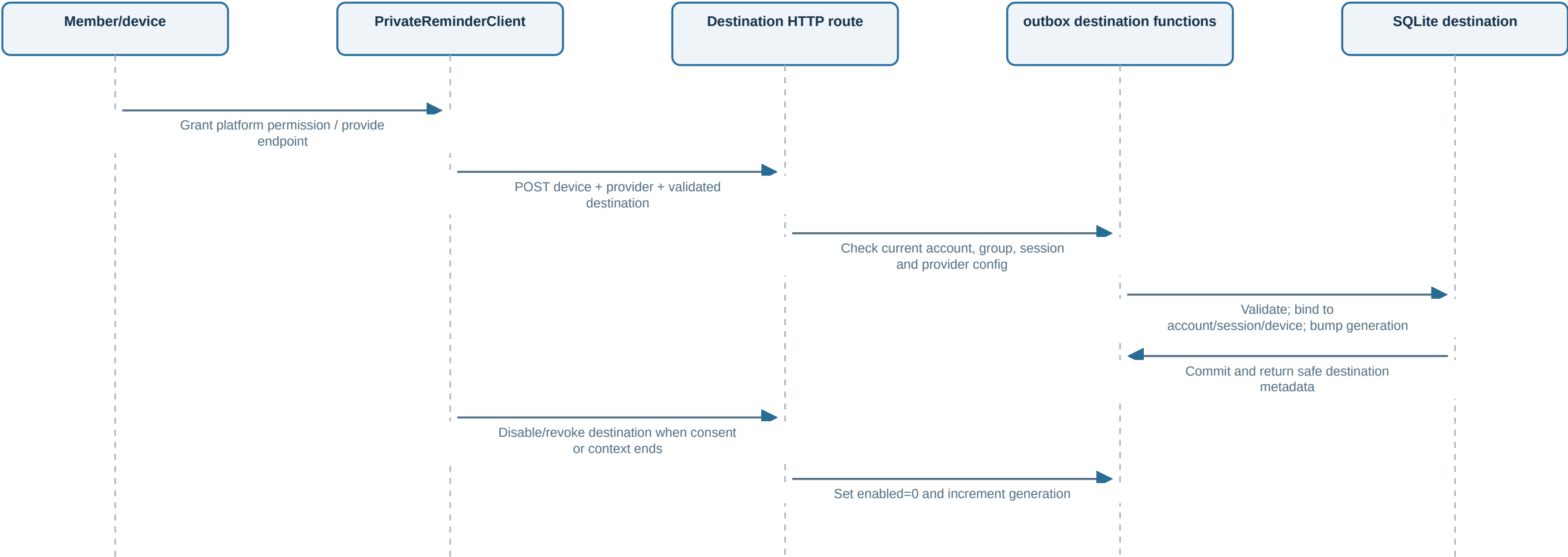
Exceptions: invalid date/boolean or snooze over 31 days is rejected; nonmember/expired session is denied; database failure preserves prior value; nonexistent local wall time is skipped.

Figure M094 | PARTICIPATION | uc10 f2 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F2 Register or disable device destination — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

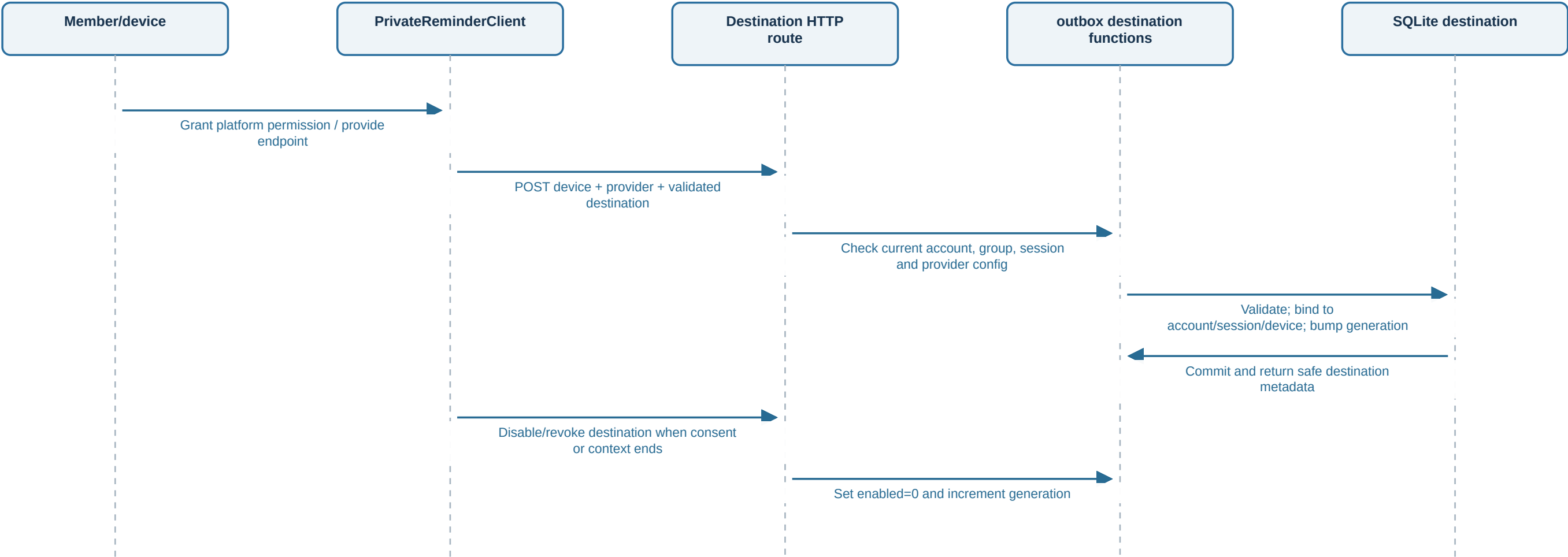
Exceptions: unsupported/denied permission, unconfigured provider, malformed endpoint/token, expired session, group mismatch or another live owner prevents registration; cleanup uncertainty is surfaced.

Figure M095 | PARTICIPATION | uc10 f2 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F2 Register or disable device destination — design sequence

Accepted source cut 1128b6a6 · implemented components and records



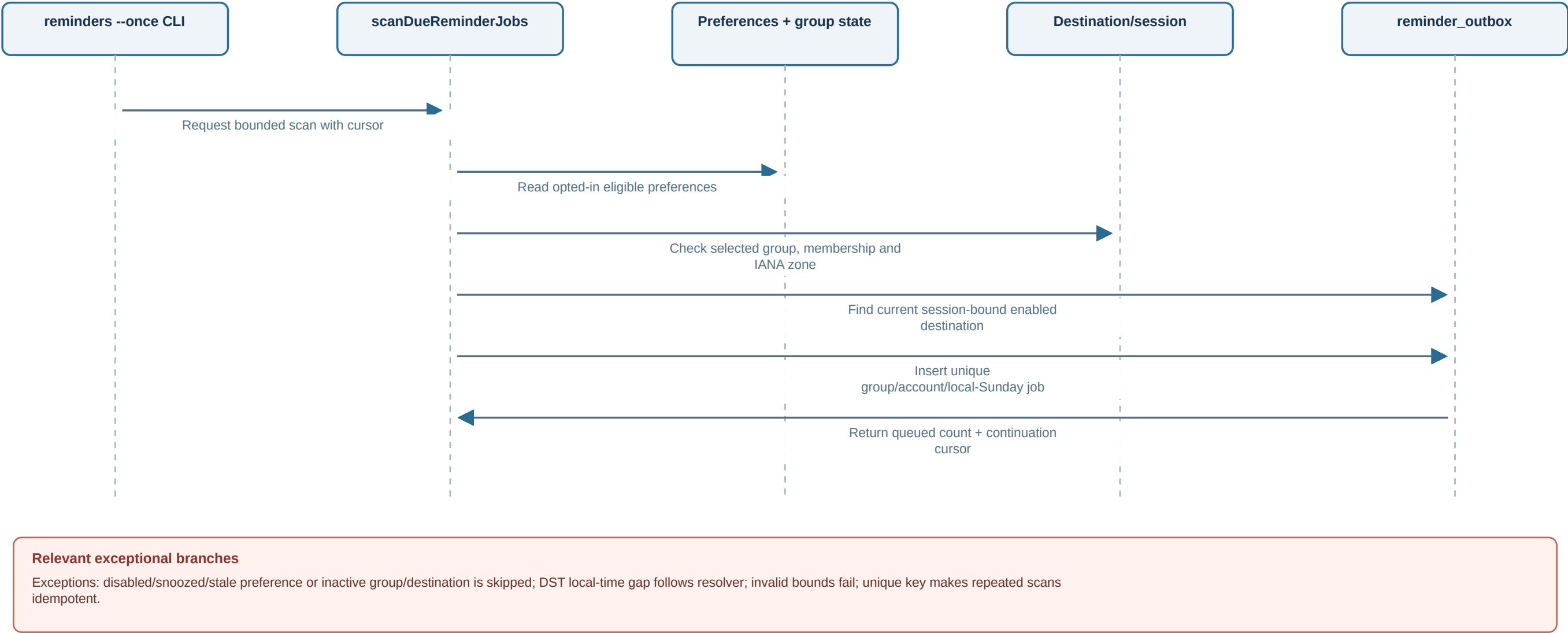
Relevant exceptional branches

Exceptions: unsupported/denied permission, unconfigured provider, malformed endpoint/token, expired session, group mismatch or another live owner prevents registration; cleanup uncertainty is surfaced.

Figure M096 | PARTICIPATION | uc10 f3 analysis sequence
Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F3 Scan and enqueue due reminder — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



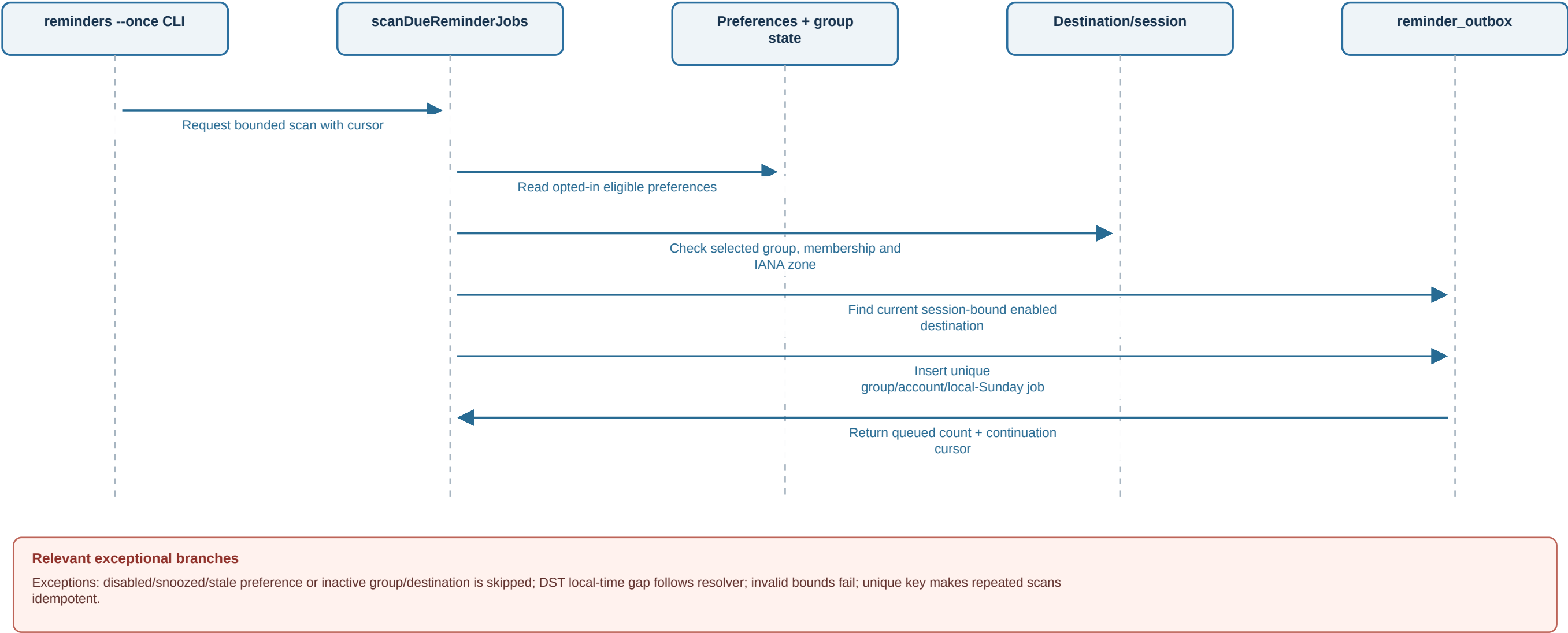
packages/participation/diagrams/uc10-f3-analysis-sequence.svg

Figure M097 | PARTICIPATION | uc10 f3 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F3 Scan and enqueue due reminder — design sequence

Accepted source cut 1128b6a6 · implemented components and records



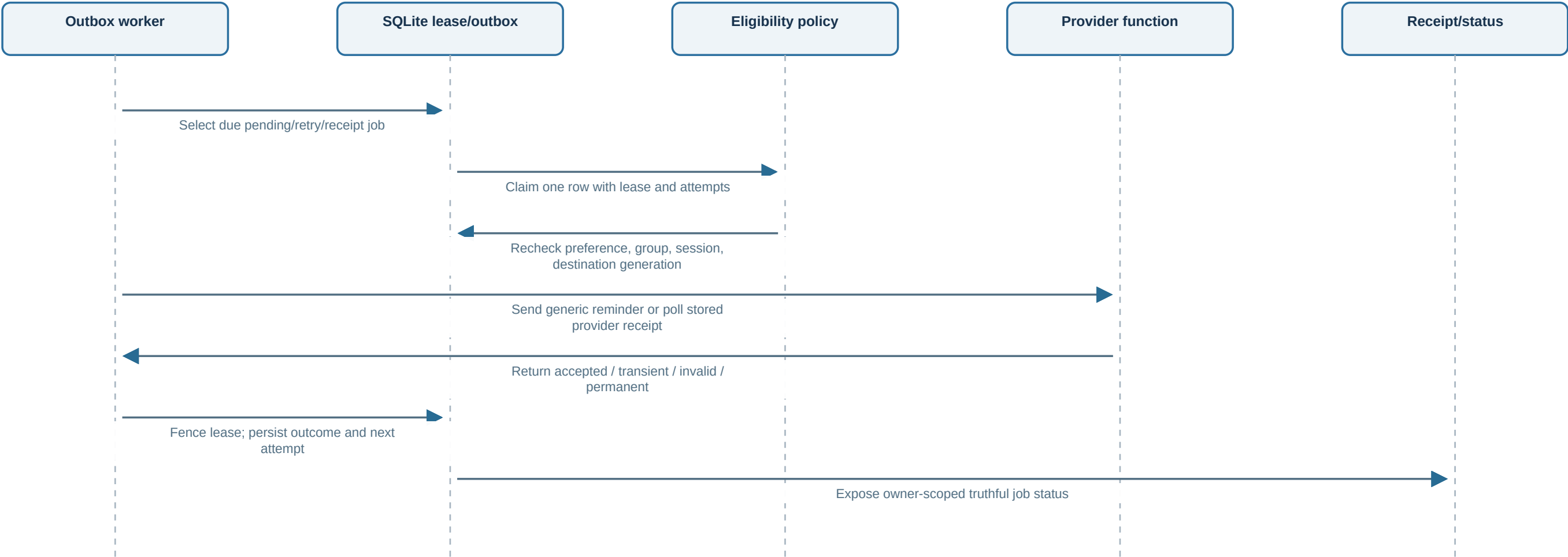
packages/participation/diagrams/uc10-f3-design-sequence.svg

Figure M098 | PARTICIPATION | uc10 f4 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F4 Send, receipt, retry or stop — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

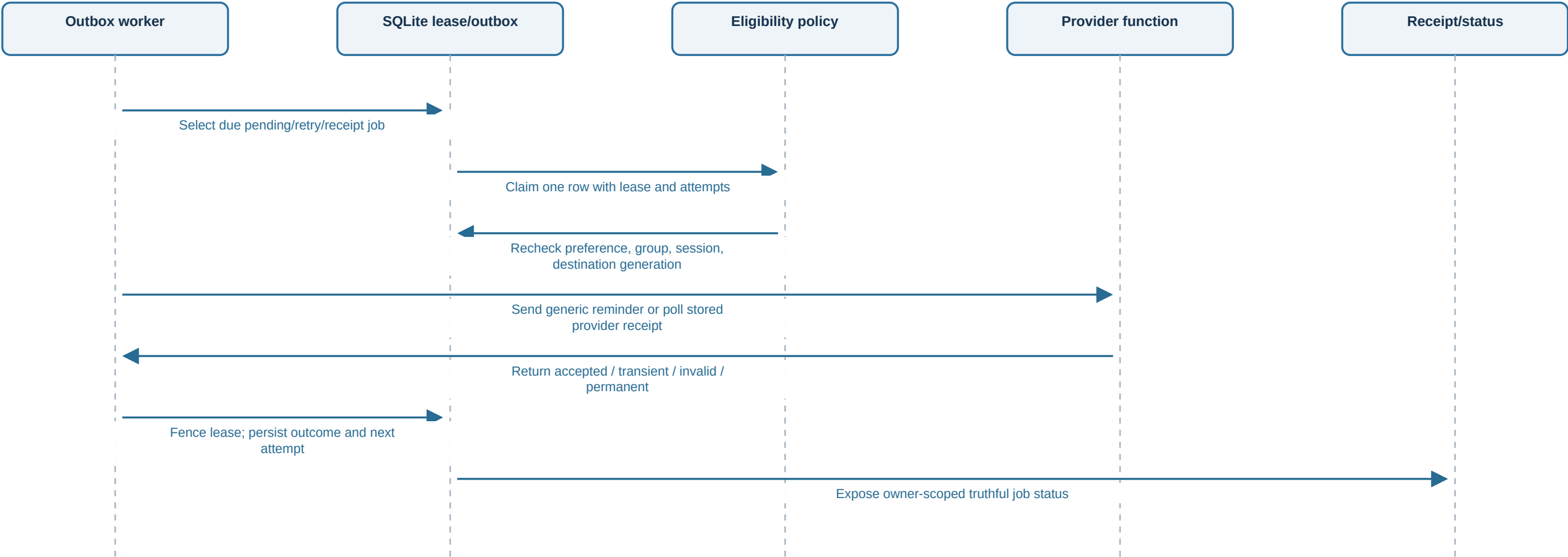
Exceptions: stale lease cannot overwrite; invalid destination is disabled; transient retries are bounded; permanent/exhausted send fails; consent loss cancels; receipt-unavailable is not device delivery.

Figure M099 | PARTICIPATION | uc10 f4 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC10-F4 Send, receipt, retry or stop — design sequence

Accepted source cut 1128b6a6 · implemented components and records



Relevant exceptional branches

Exceptions: stale lease cannot overwrite; invalid destination is disabled; transient retries are bounded; permanent/exhausted send fails; consent loss cancels; receipt-unavailable is not device delivery.

Figure M100 | PARTICIPATION | uc11 analysis class
 Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

uc11 — Private group chat — analysis classes

Accepted source cut 1128b6a6 · domain responsibilities, no implementation class names

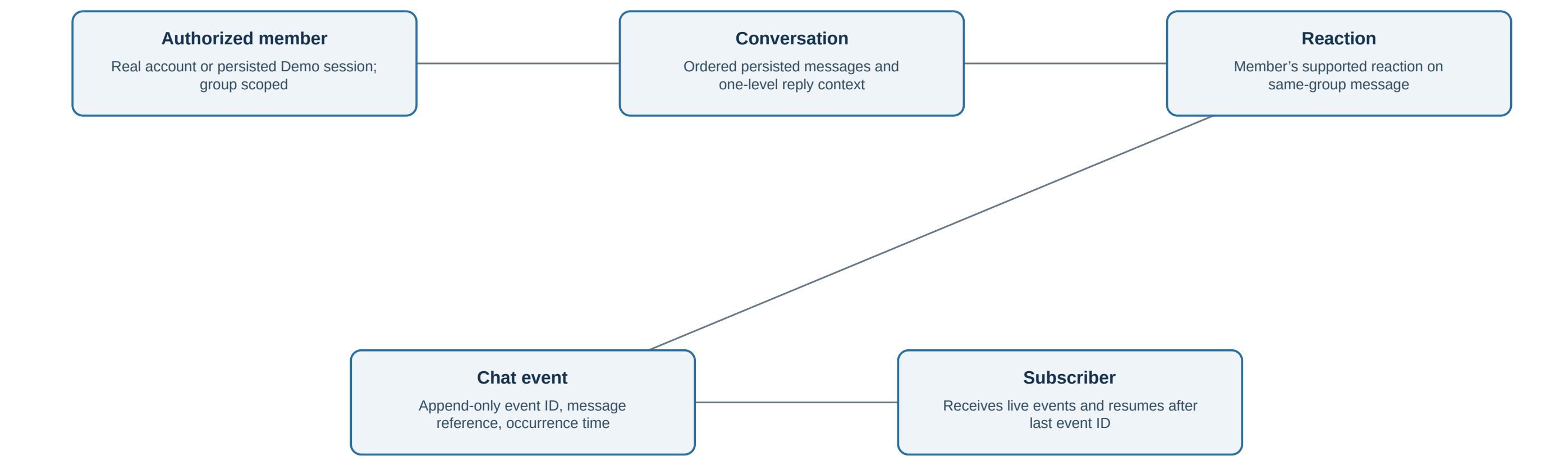


Figure M101 | PARTICIPATION | uc11 design class
Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

uc11 — Private group chat — design classes

Accepted source cut 1128b6a6 · source-backed modules, functions, transport and persisted records

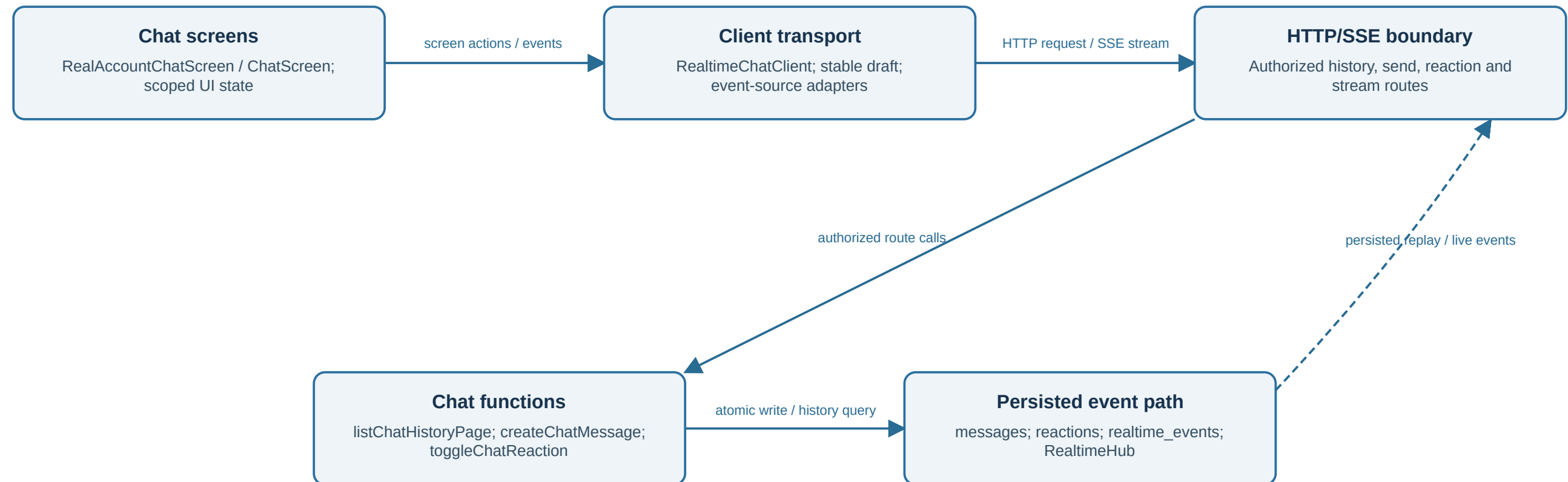


Figure M102 | PARTICIPATION | uc11 f1 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F1 Read bounded chat history — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities

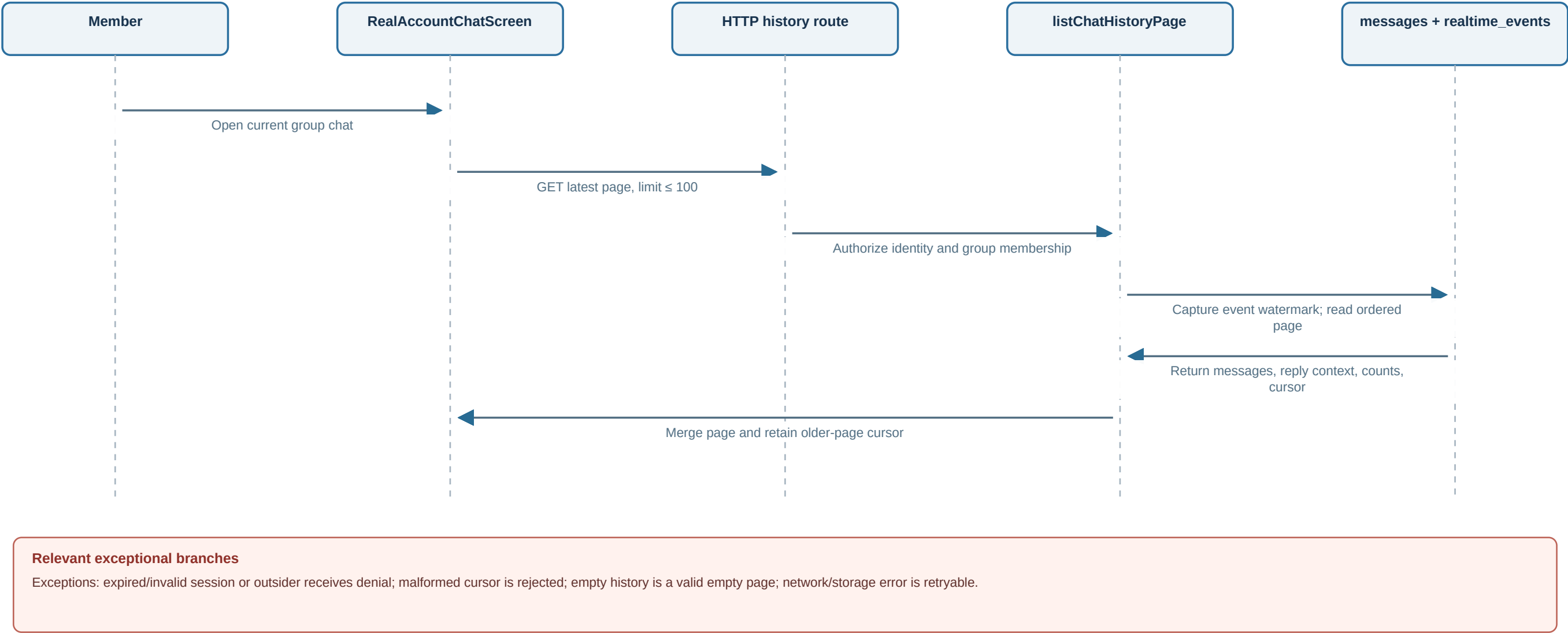


Figure M103 | PARTICIPATION | uc11 f1 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F1 Read bounded chat history — design sequence

Accepted source cut 1128b6a6 · implemented components and records

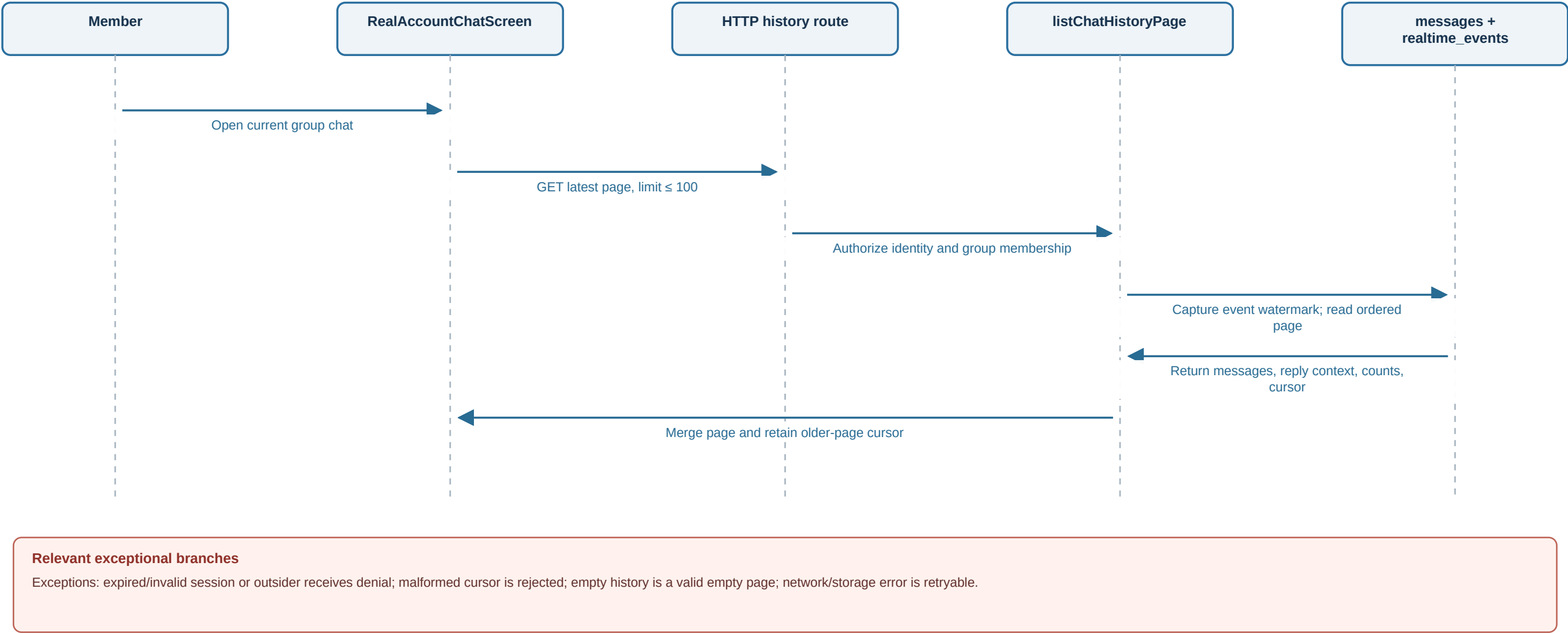
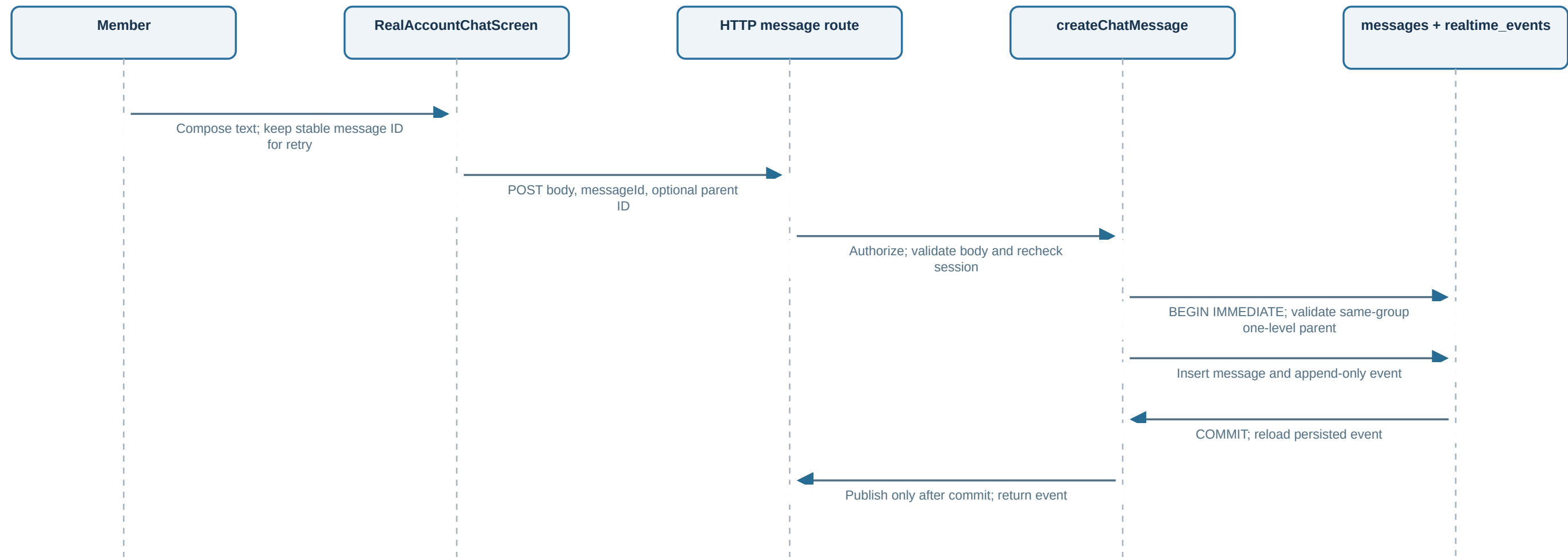


Figure M104 | PARTICIPATION | uc11 f2 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F2 Send text or one-level reply — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

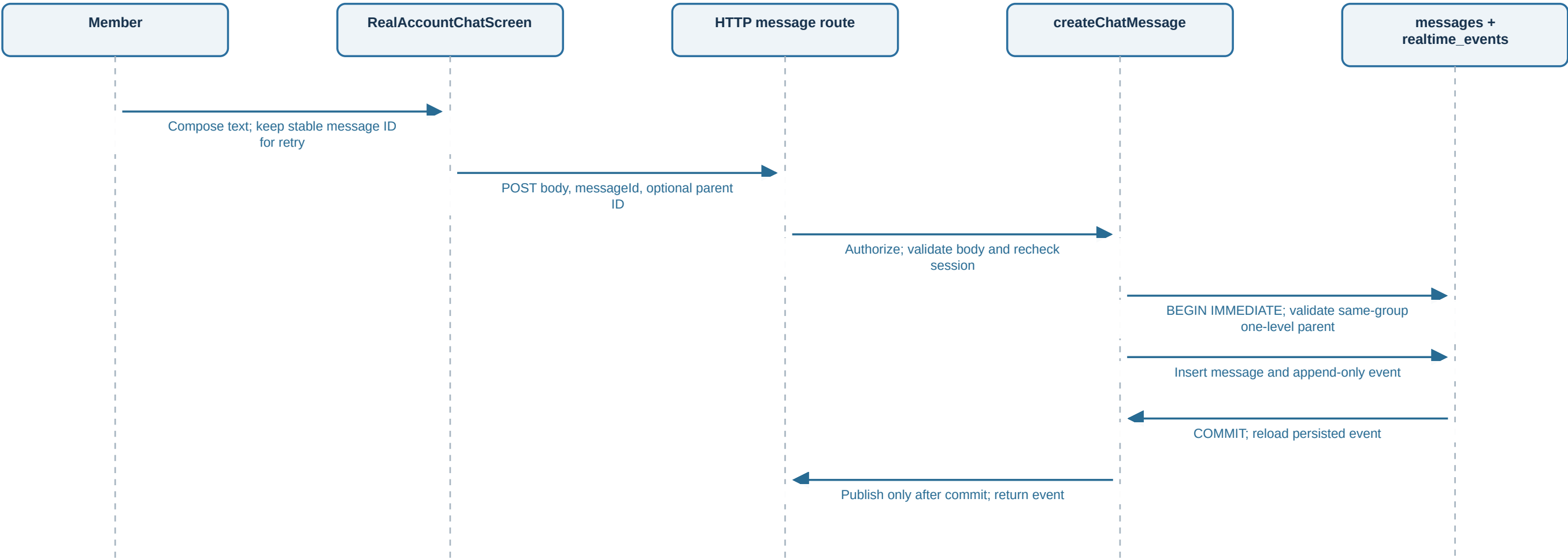
Exceptions: empty/long text, conflicting retry ID, missing/wrong-group parent or nested reply fails; denial is safe; transaction rollback leaves neither message nor event.

Figure M105 | PARTICIPATION | uc11 f2 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F2 Send text or one-level reply — design sequence

Accepted source cut 1128b6a6 · implemented components and records



Relevant exceptional branches

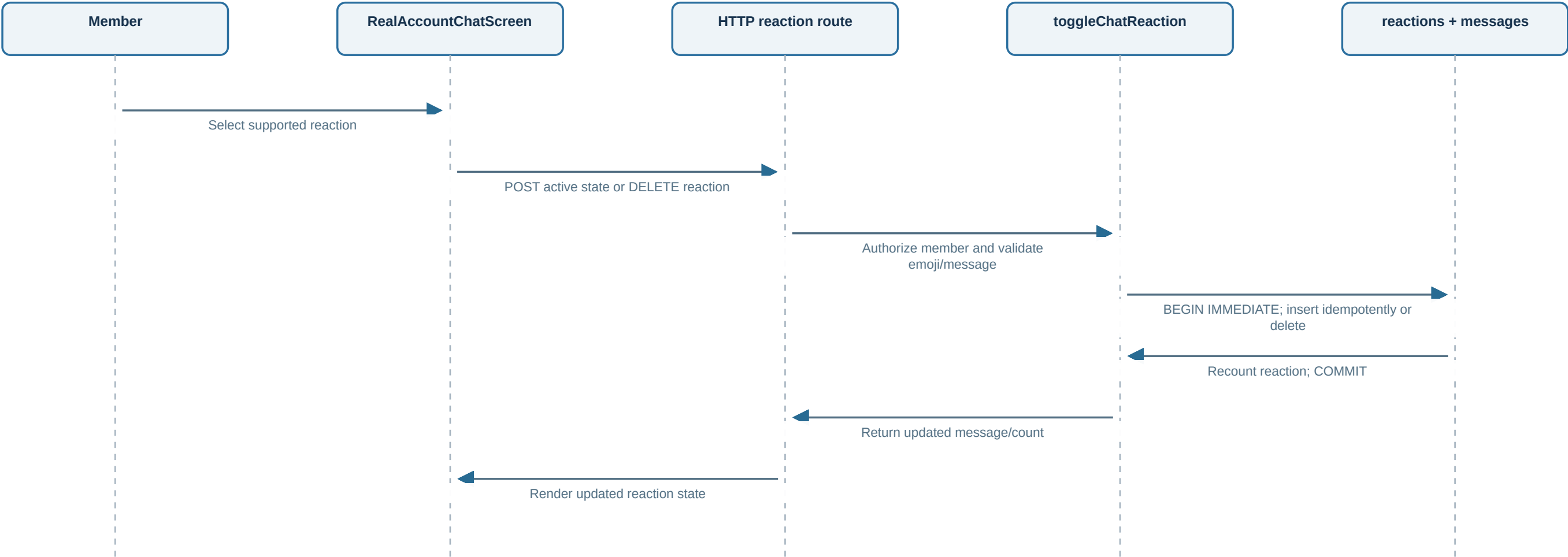
Exceptions: empty/long text, conflicting retry ID, missing/wrong-group parent or nested reply fails; denial is safe; transaction rollback leaves neither message nor event.

Figure M106 | PARTICIPATION | uc11 f3 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F3 Add, remove or toggle reaction — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

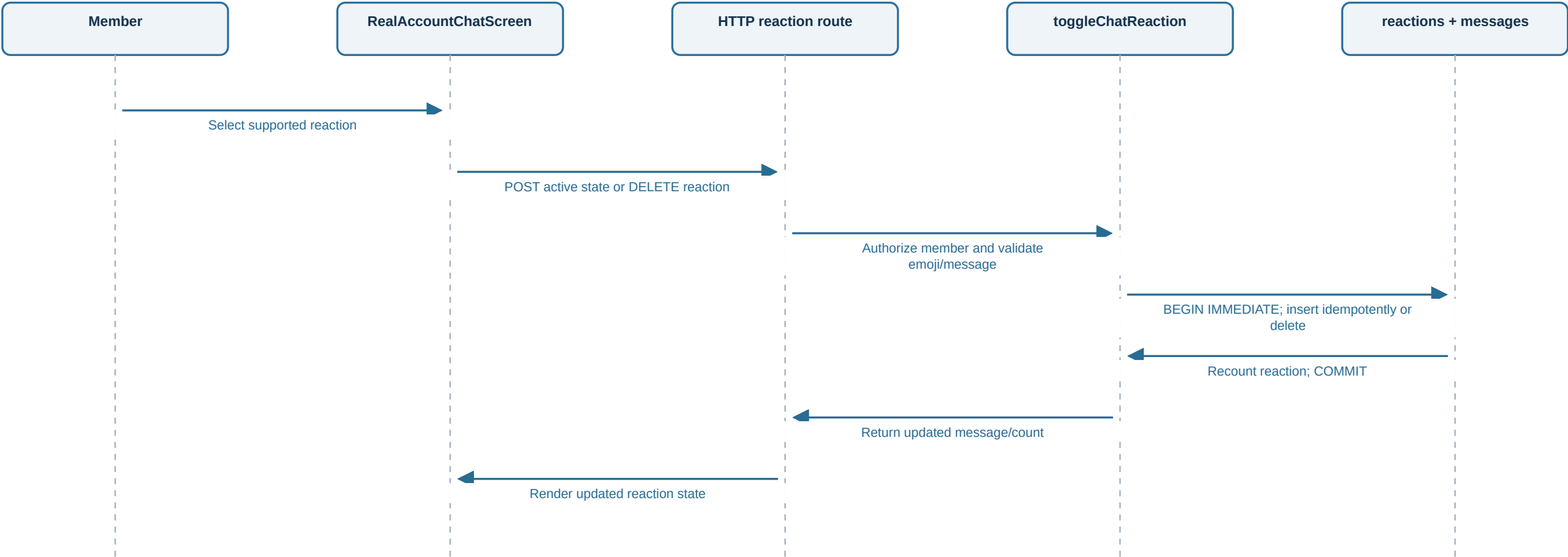
Exceptions: outsider/session expiry, unknown message, unsupported emoji or invalid timestamp is denied/rejected; reaction mutation does not create a message event.

Figure M107 | PARTICIPATION | uc11 f3 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F3 Add, remove or toggle reaction — design sequence

Accepted source cut 1128b6a6 · implemented components and records



Relevant exceptional branches

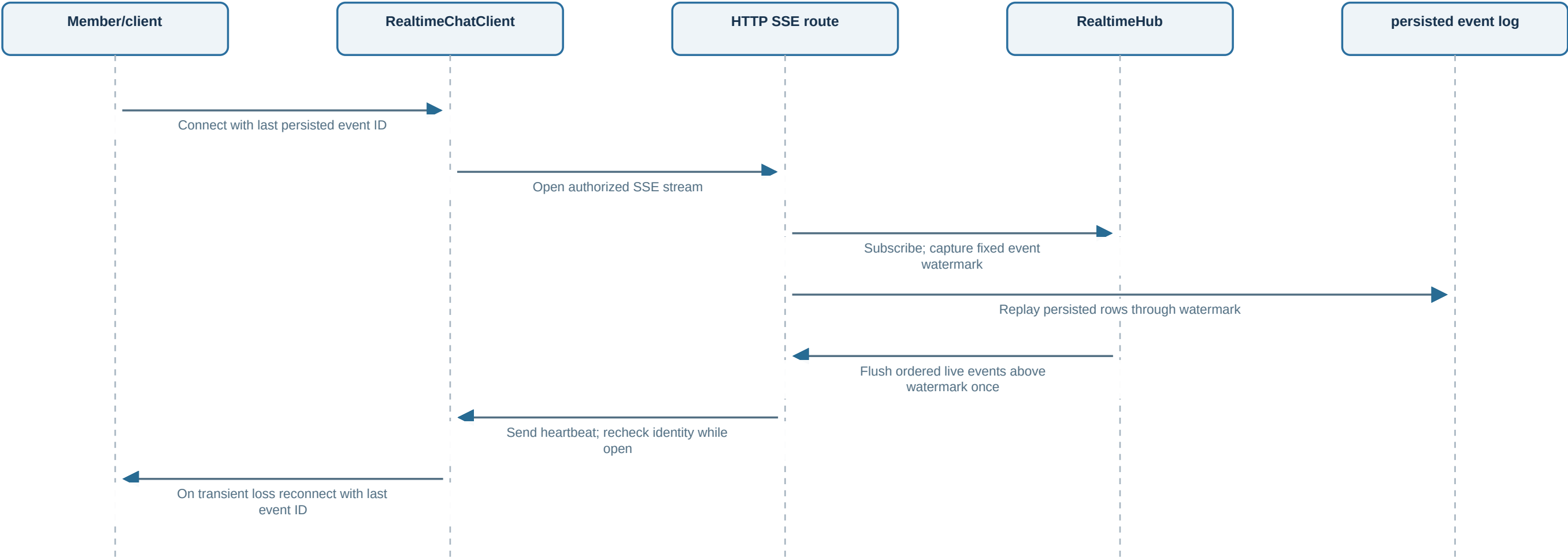
Exceptions: outsider/session expiry, unknown message, unsupported emoji or invalid timestamp is denied/rejected; reaction mutation does not create a message event.

Figure M108 | PARTICIPATION | uc11 f4 analysis sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F4 Reconnect and replay chat events — analysis sequence

Accepted source cut 1128b6a6 · conceptual responsibilities



Relevant exceptional branches

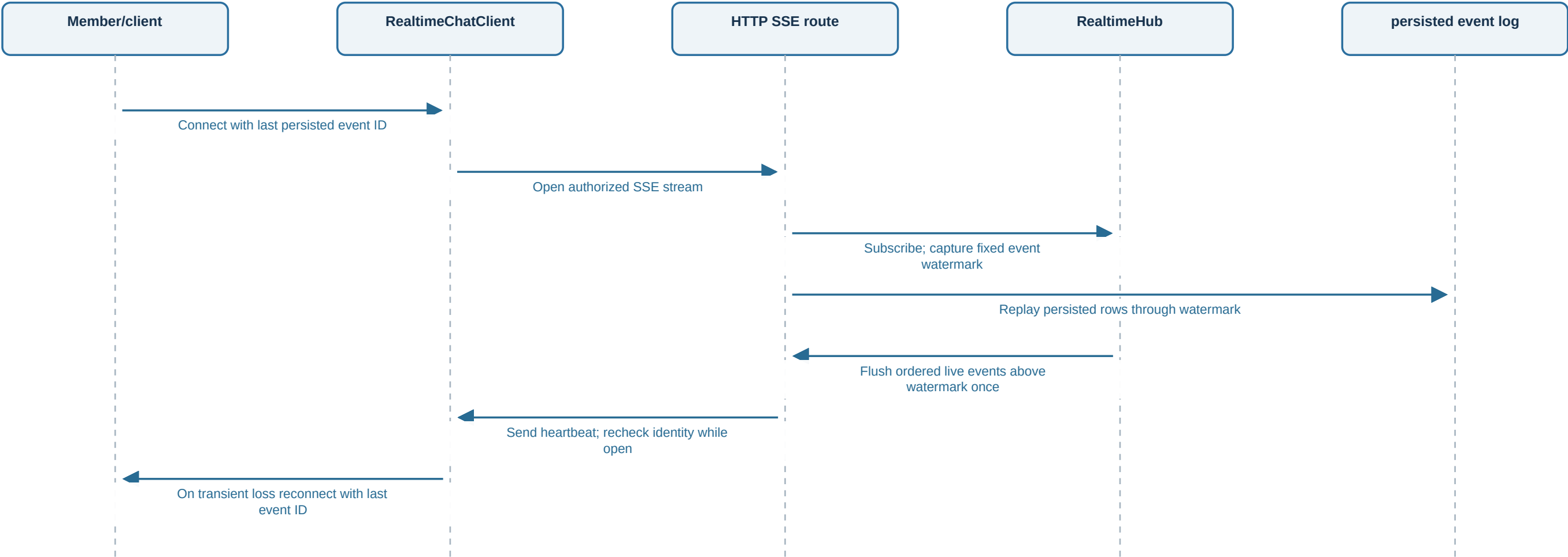
Exceptions: network loss marks reconnecting/disconnected; session expiry or membership loss ends stream and clears private UI state; replay suppresses duplicates across the watermark.

Figure M109 | PARTICIPATION | uc11 f4 design sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

UC11-F4 Reconnect and replay chat events — design sequence

Accepted source cut 1128b6a6 · implemented components and records



Relevant exceptional branches

Exceptions: network loss marks reconnecting/disconnected; session expiry or membership loss ends stream and clears private UI state; replay suppresses duplicates across the watermark.

ARCHIVE model/source register | issue #361

Accepted model cut 1128b6a68985cf68215beb4a7a80fd00ec4242fa; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

server/src/archive/capabilities.ts:21

server/src/http.ts:2995

server/src/archive/store-serving.ts:14

src/auth/real-account-client.ts:471

src/archive/ArchiveScreen.tsx:251

src/archive/archive-download.ts:54

src/invites/deep-links.ts:52

App.tsx:162

tests/e2e-real-account/group-switch-verification.mjs:3

public/sw.js:64

app.json:1

tests/real-account-archive-client.test.ts

tests/archive-download.test.ts

tests/deep-link-invites.test.ts

tests/pwa.test.mjs

server/src/http.ts:3263

Figure M114 | ARCHIVE | uc12 analysis class
Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12 — Analysis classes

Responsibilities in premiere state and archive access

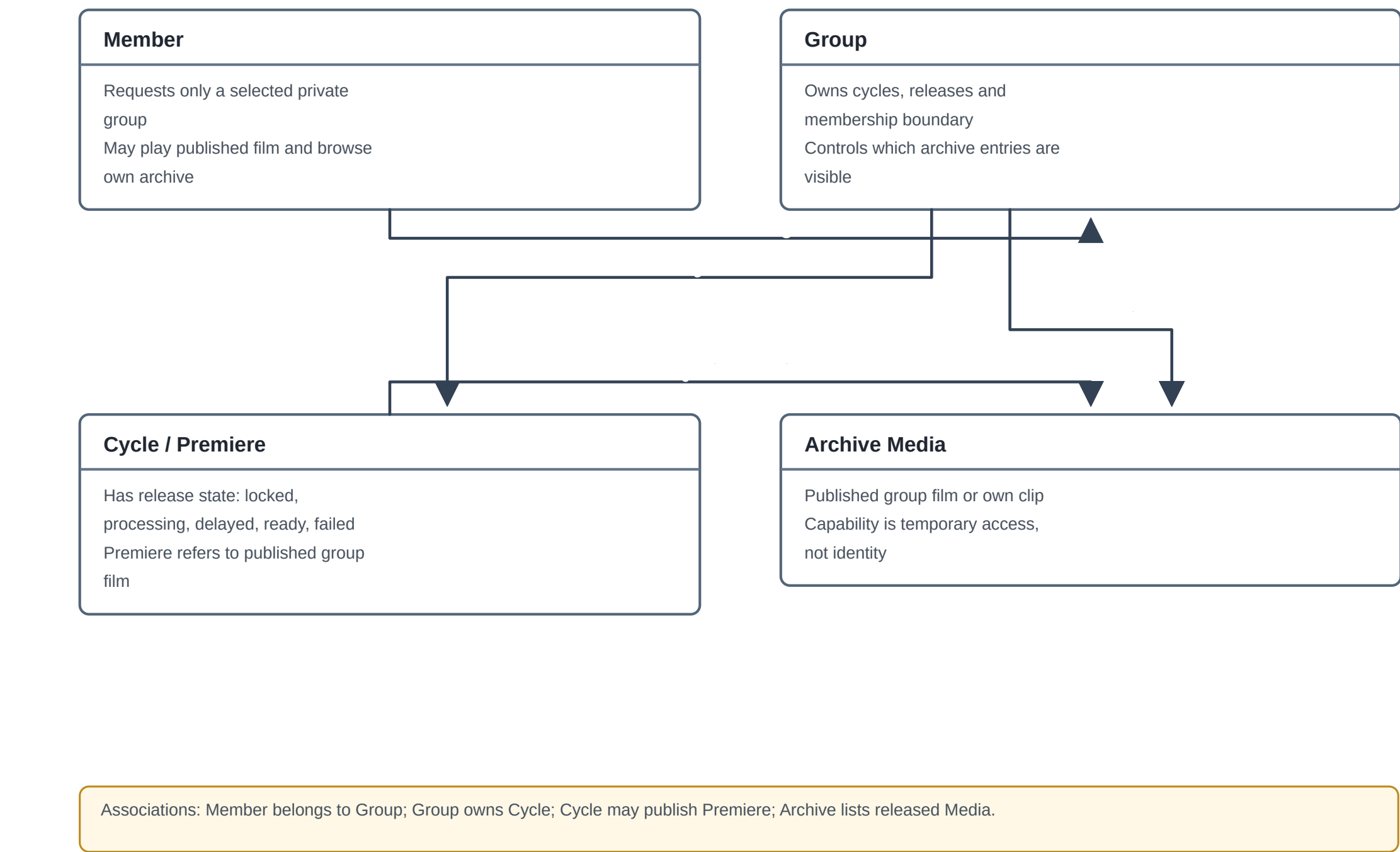


Figure M115 | ARCHIVE | uc12 design class
Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12 — Design classes

Code-backed server and client boundaries

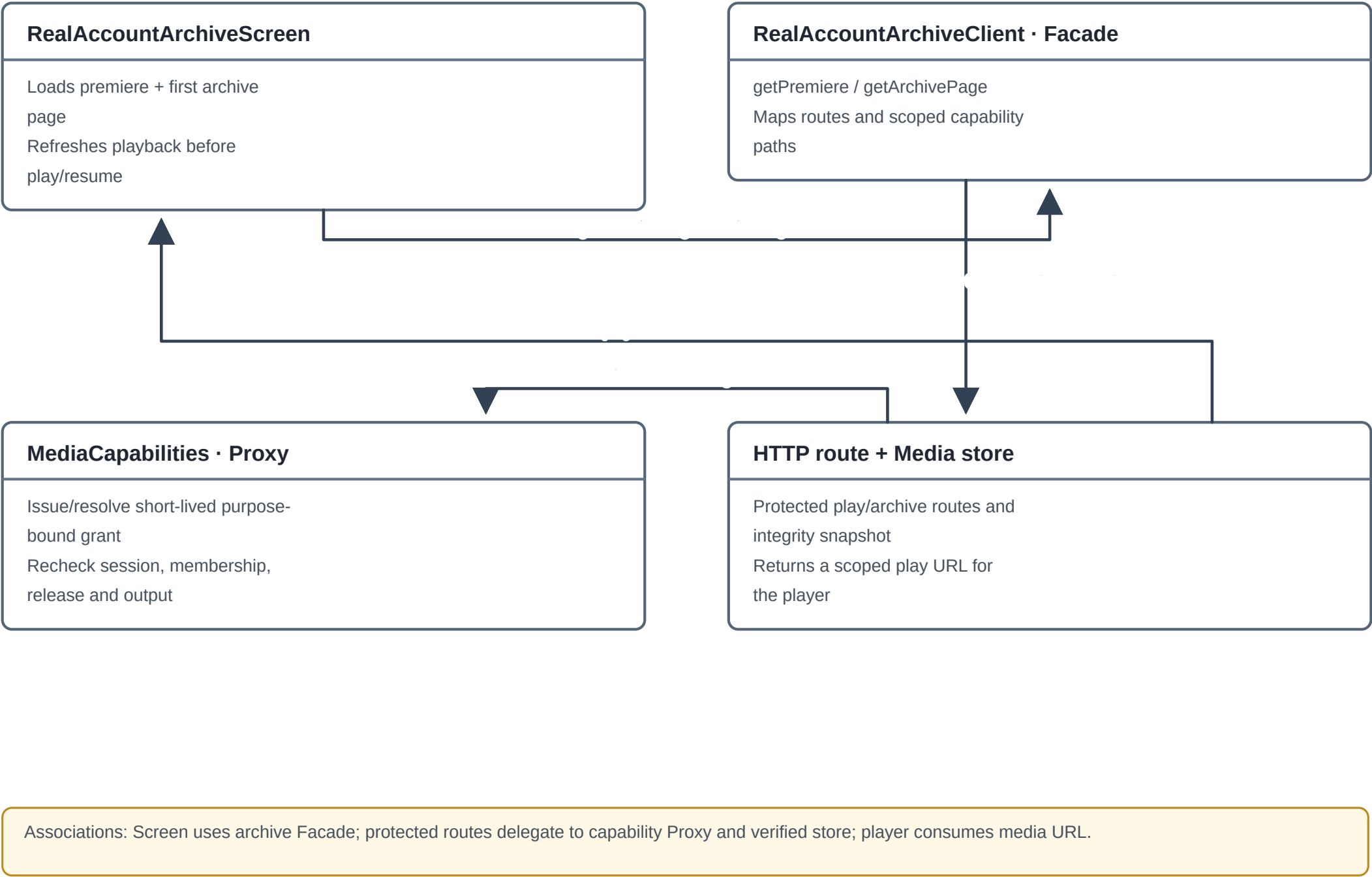


Figure M116 | ARCHIVE | uc12 f1 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12-F1 — Inspect premiere and archive state

Analysis responsibility model; platform-neutral concepts

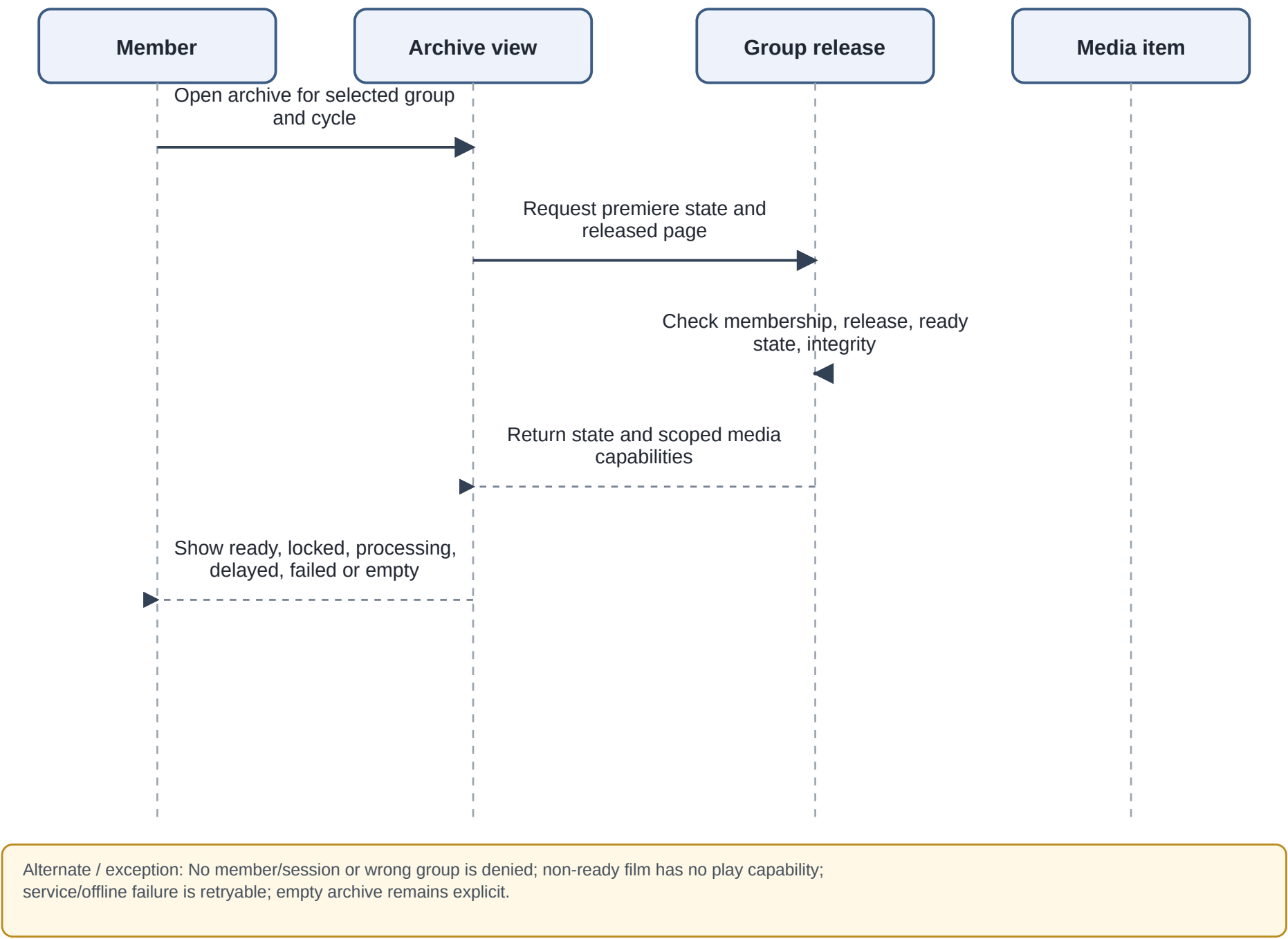


Figure M117 | ARCHIVE | uc12 f1 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12-F1 — Inspect premiere and archive state

Design interaction grounded in accepted code

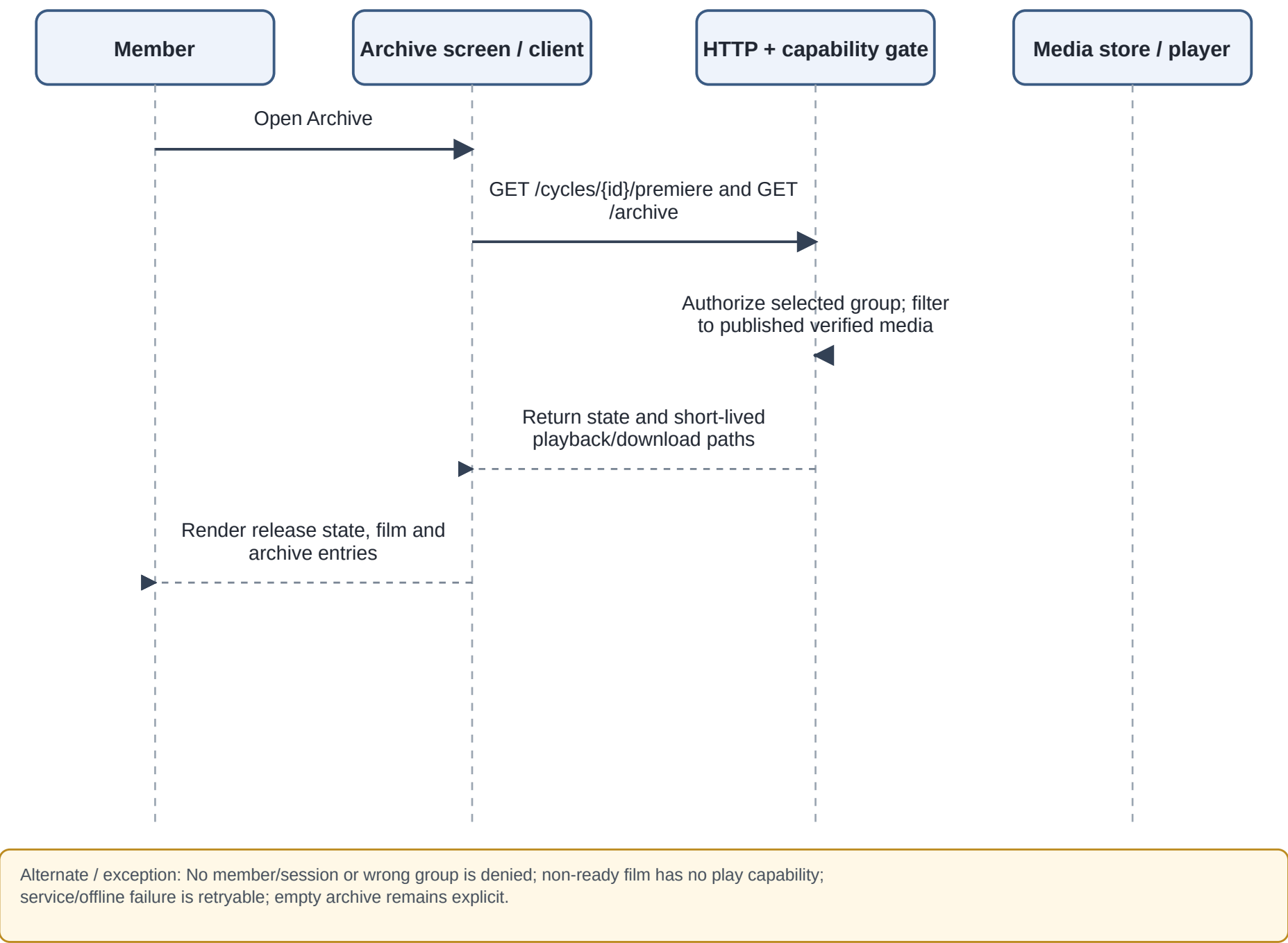


Figure M118 | ARCHIVE | uc12 f2 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12-F2 — Start or resume premiere playback

Analysis responsibility model; platform-neutral concepts

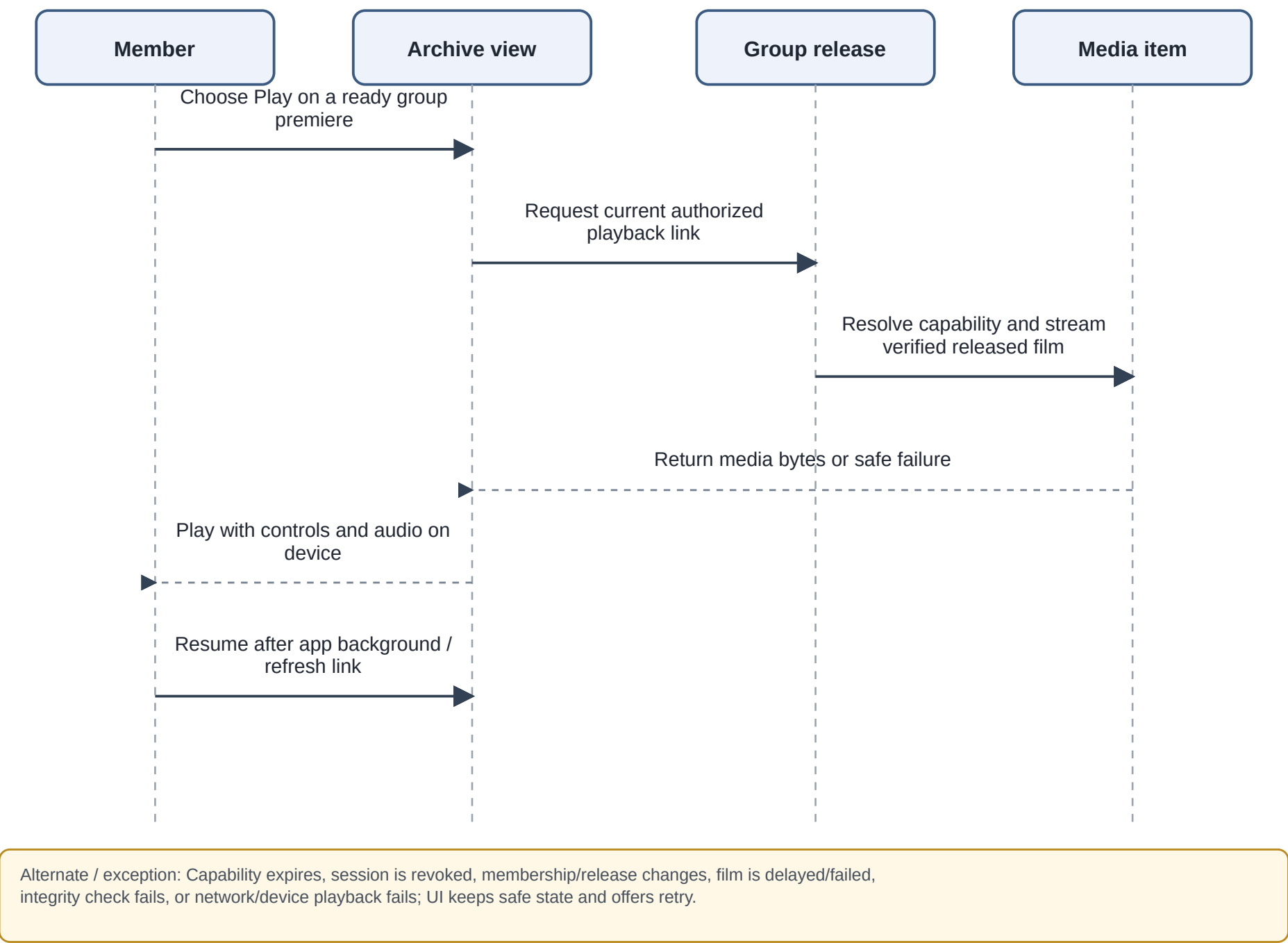


Figure M119 | ARCHIVE | uc12 f2 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12-F2 — Start or resume premiere playback

Design interaction grounded in accepted code

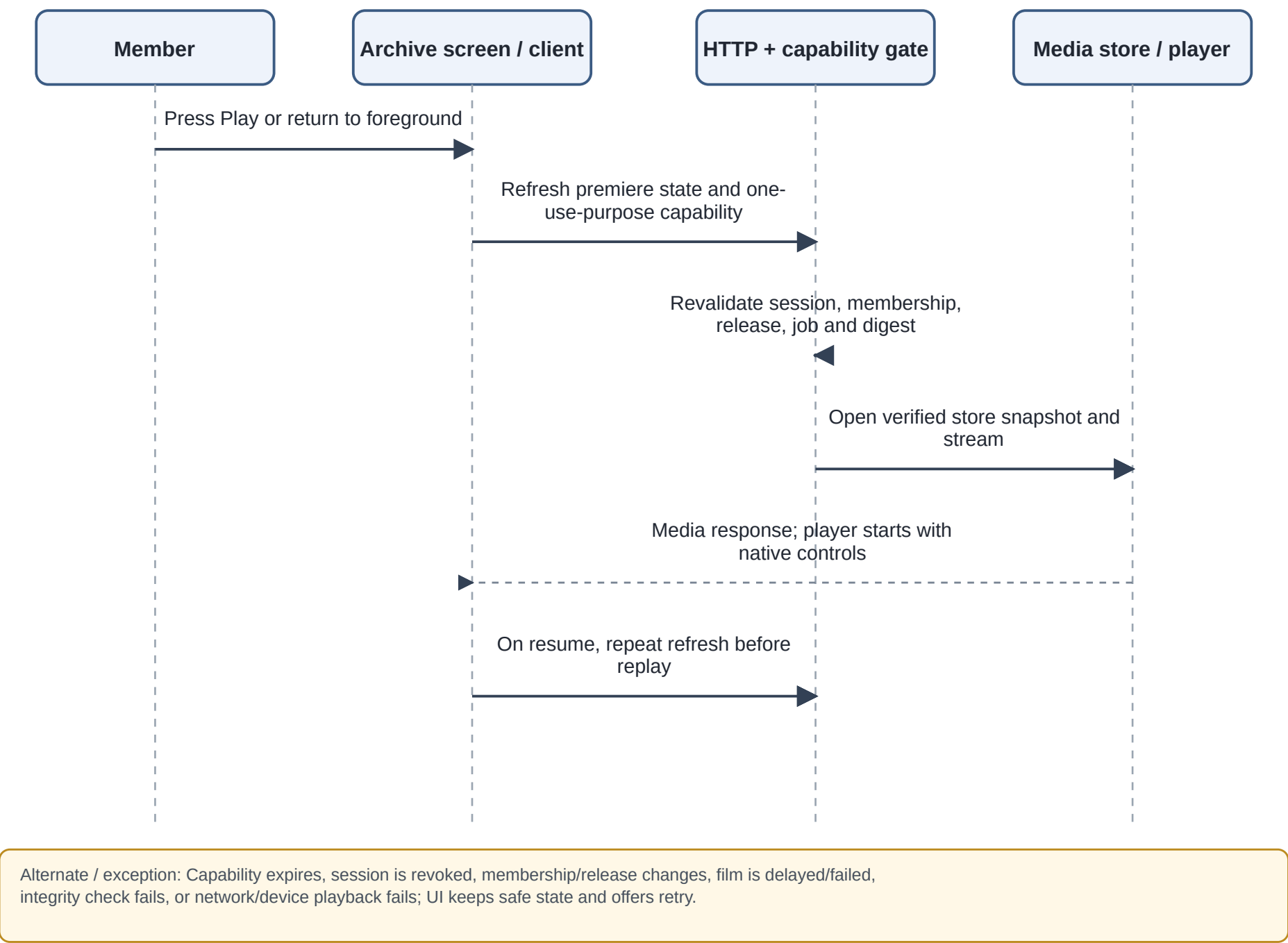


Figure M120 | ARCHIVE | uc12 f3 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12-F3 — Browse and page released archive

Analysis responsibility model; platform-neutral concepts

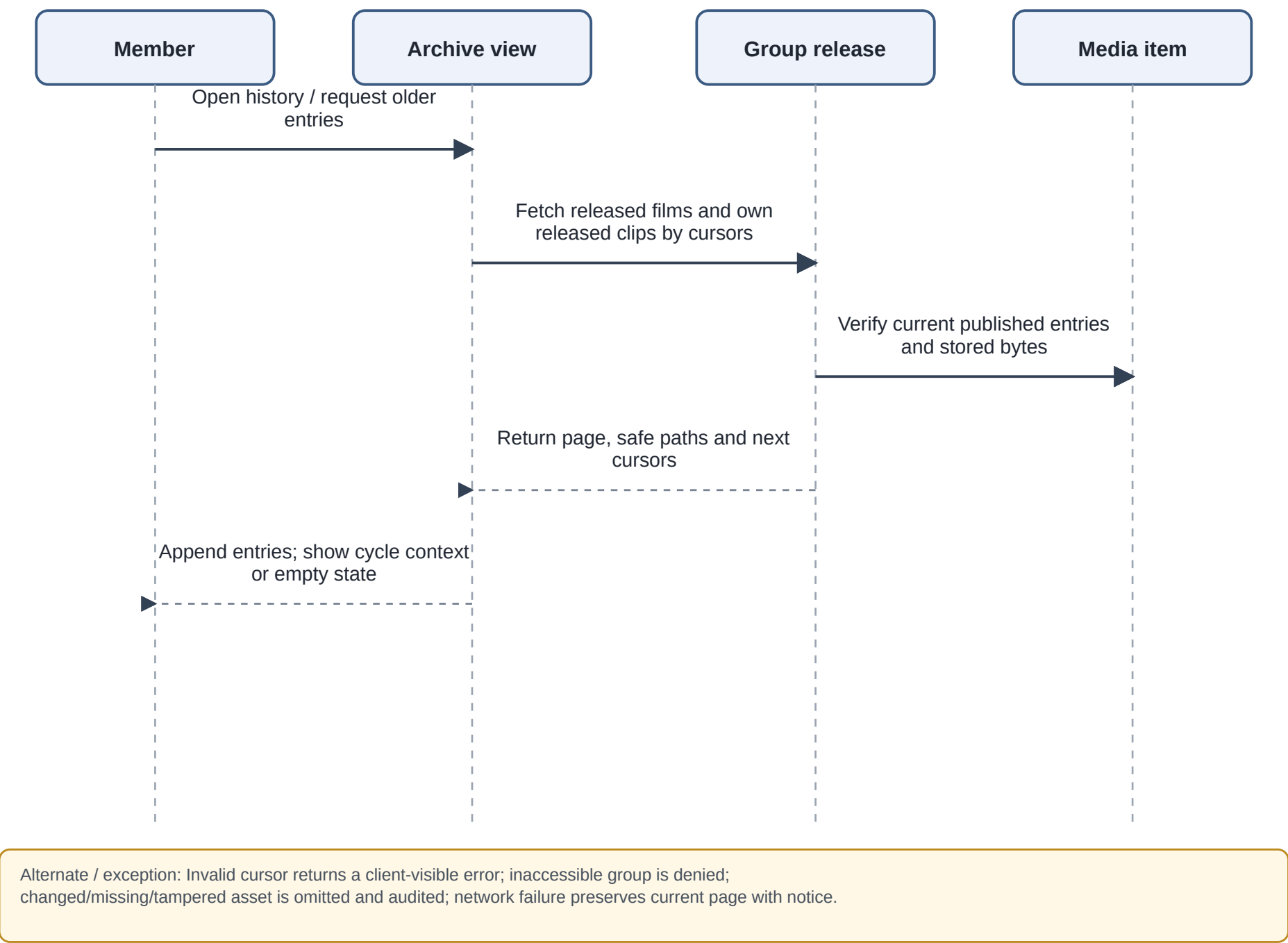


Figure M121 | ARCHIVE | uc12 f3 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC12-F3 — Browse and page released archive

Design interaction grounded in accepted code

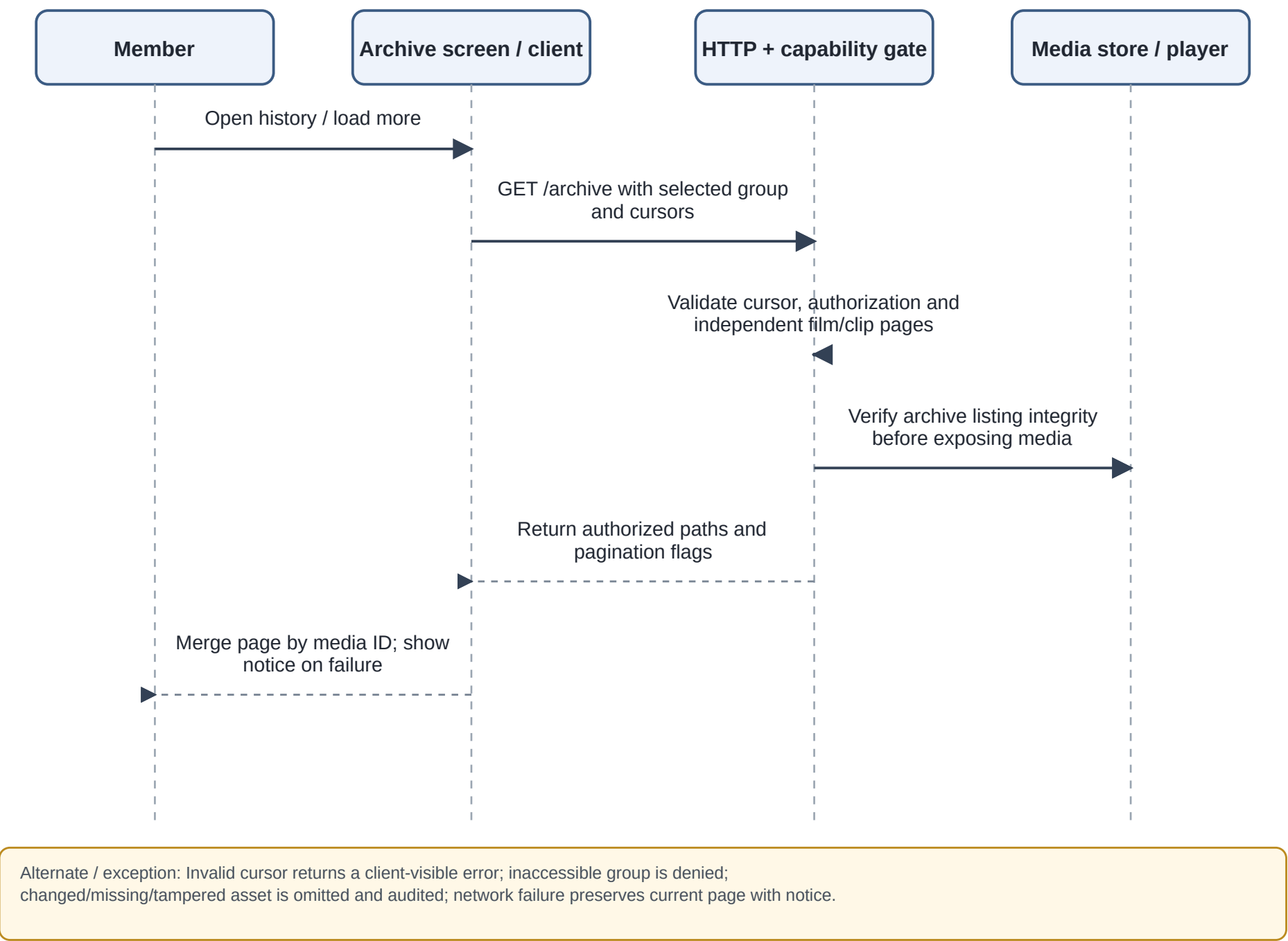


Figure M122 | ARCHIVE | uc13 analysis class

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC13 — Analysis classes

Own-clip and group-film delivery responsibilities

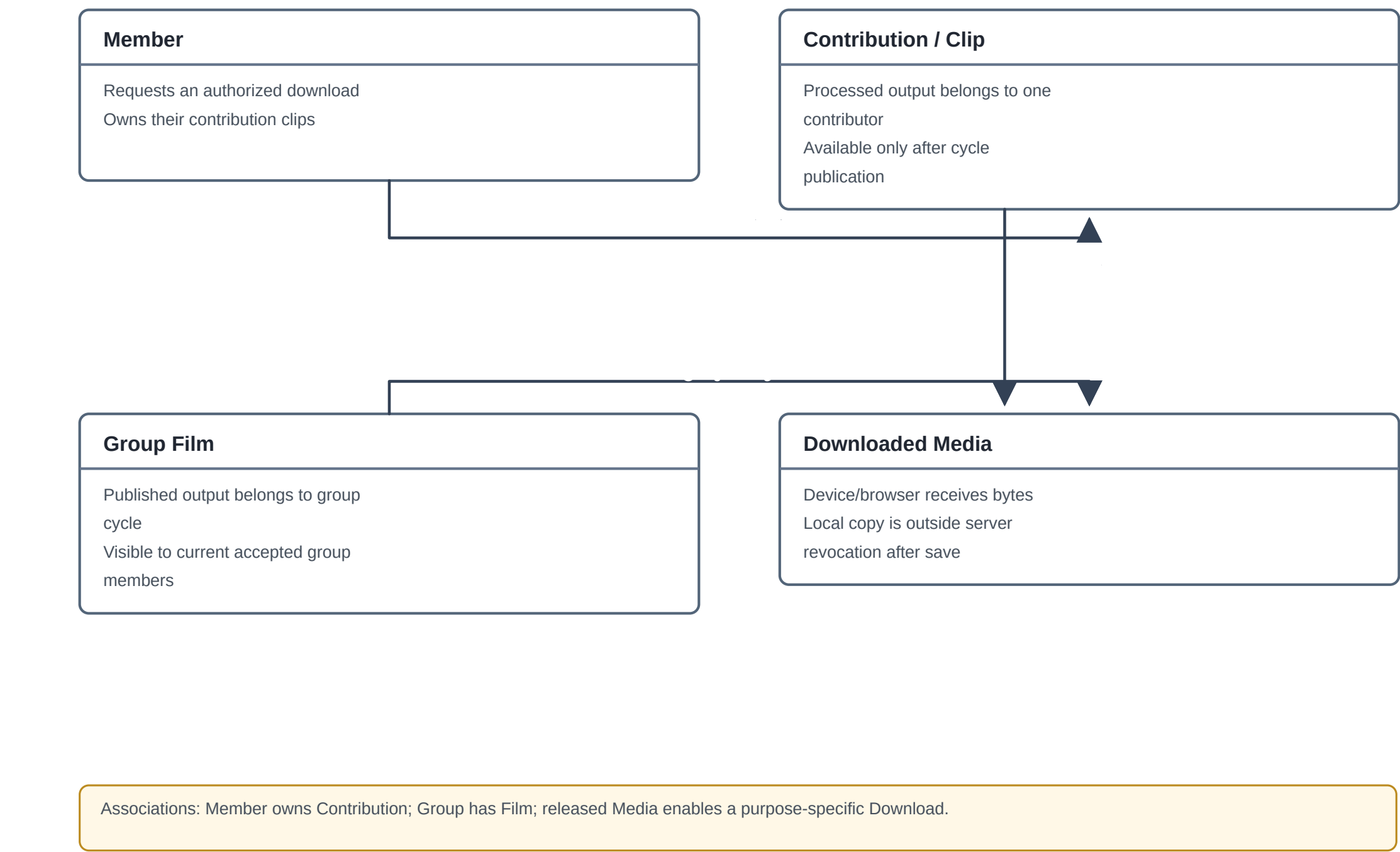


Figure M123 | ARCHIVE | uc13 design class
Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC13 — Design classes

Actual capability gate and platform delivery choices

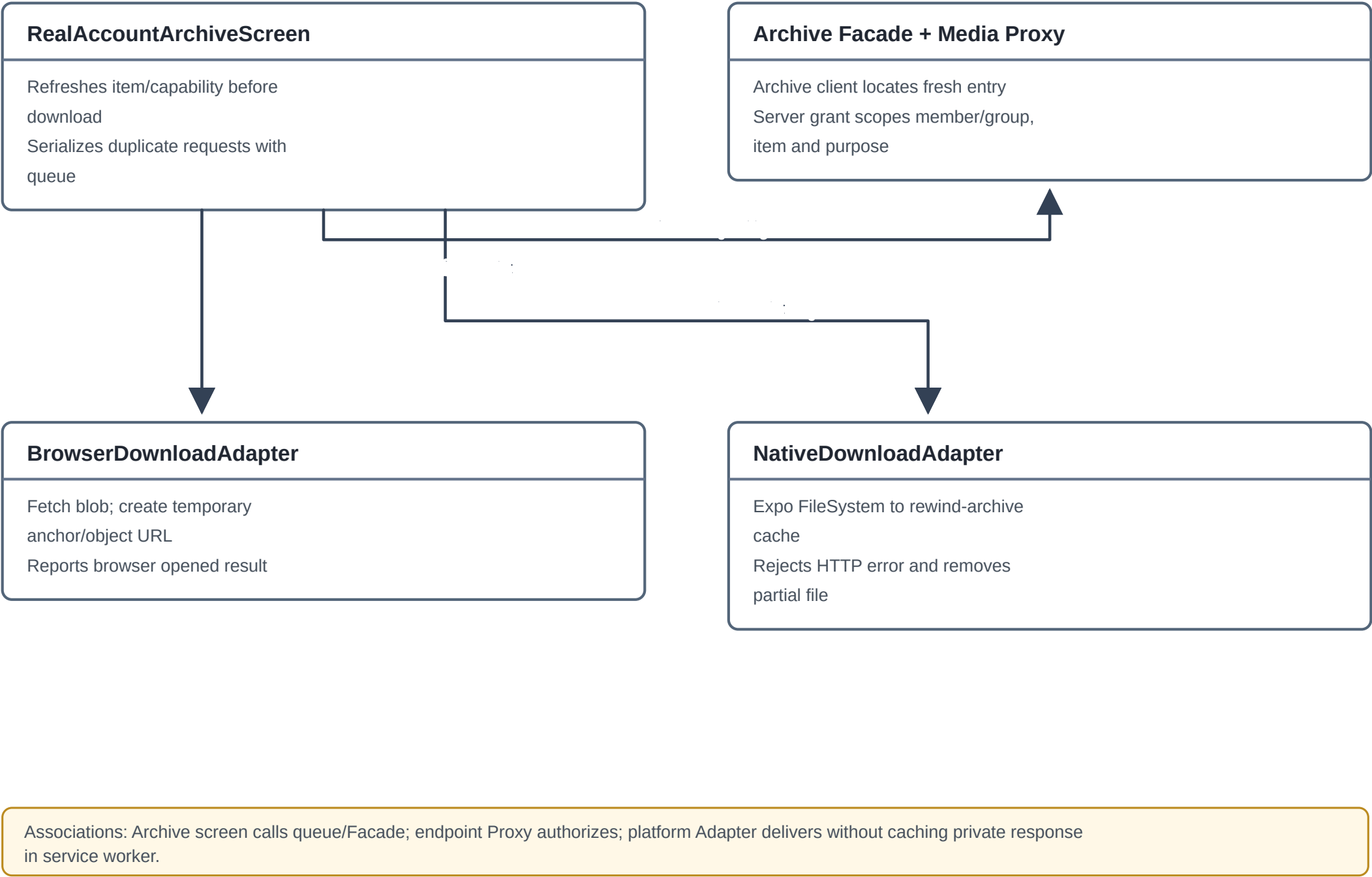


Figure M124 | ARCHIVE | uc13 f1 analysis sequence
 Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC13-F1 — Download my released clip

Analysis responsibility model; platform-neutral concepts

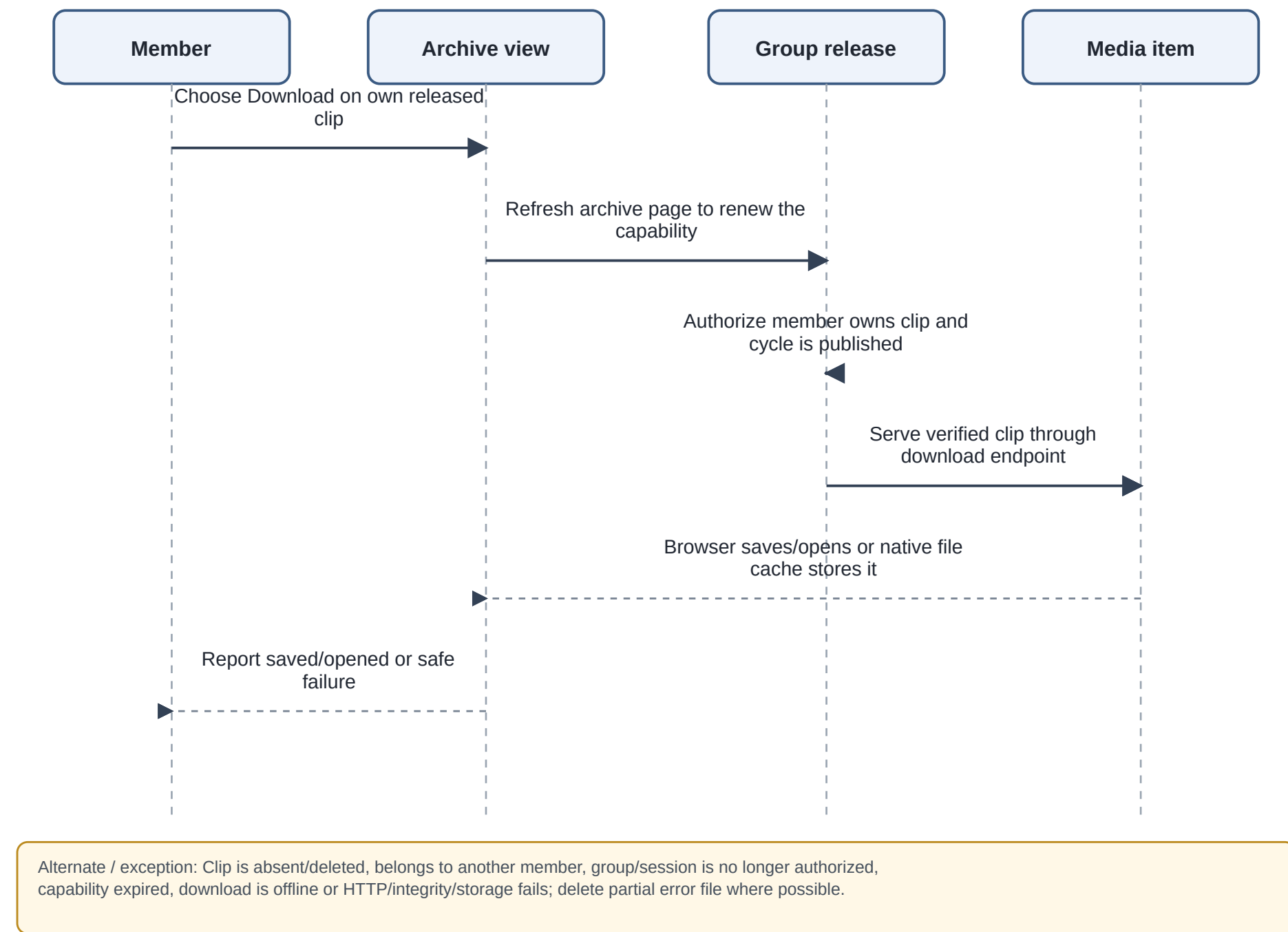


Figure M125 | ARCHIVE | uc13 f1 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC13-F1 — Download my released clip

Design interaction grounded in accepted code

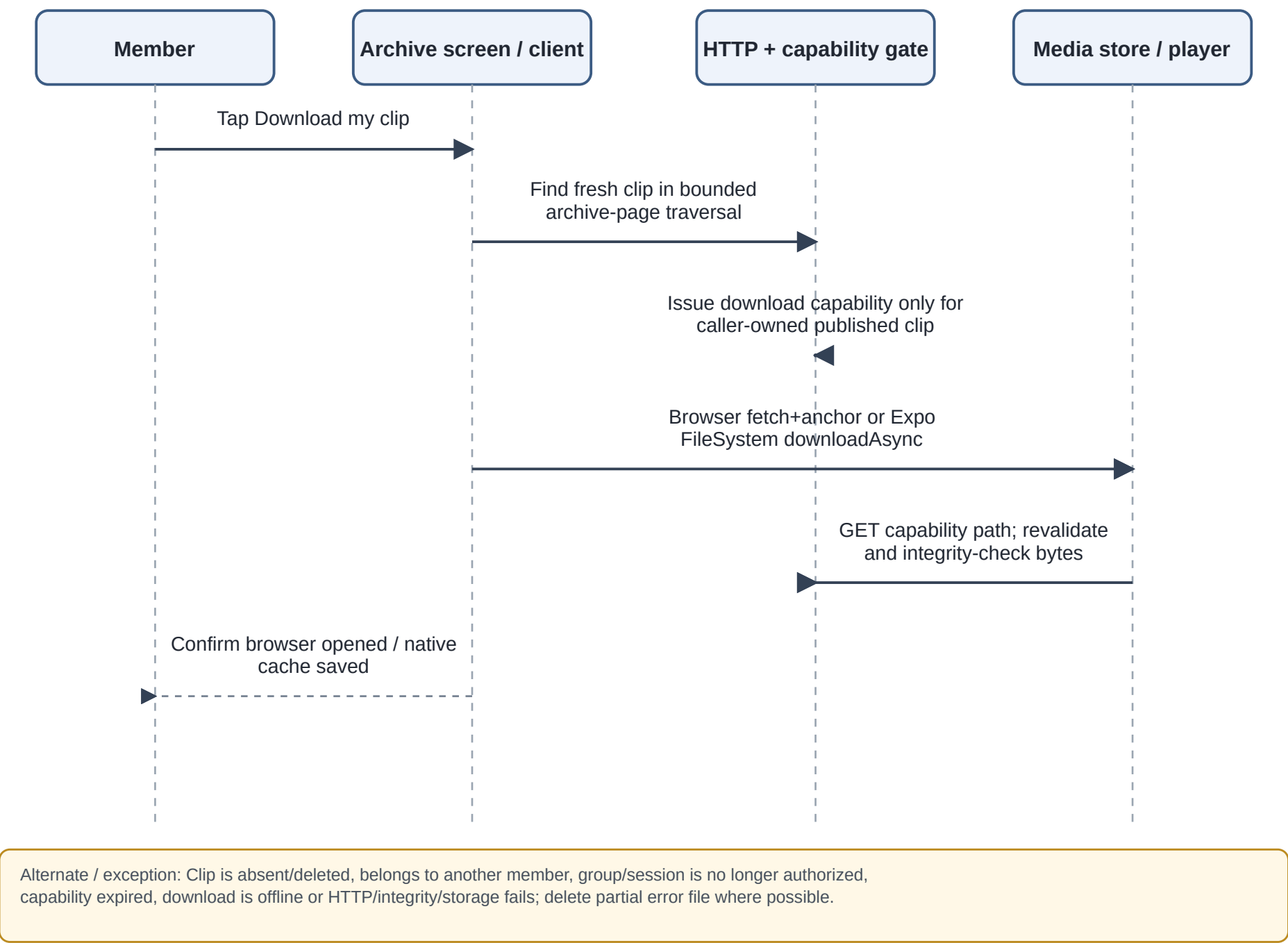


Figure M126 | ARCHIVE | uc13 f2 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC13-F2 — Download a released group film

Analysis responsibility model; platform-neutral concepts

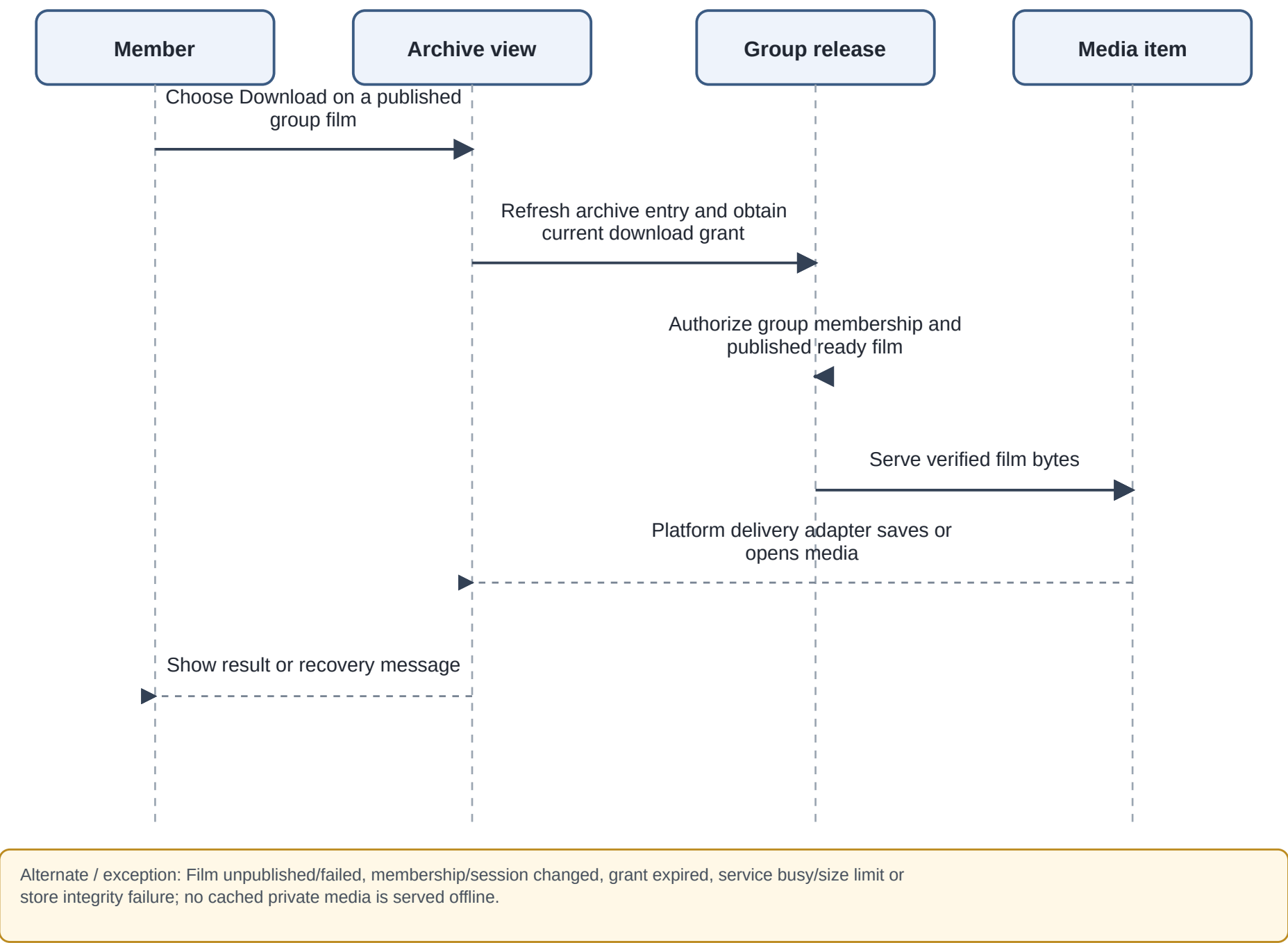


Figure M127 | ARCHIVE | uc13 f2 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC13-F2 — Download a released group film

Design interaction grounded in accepted code

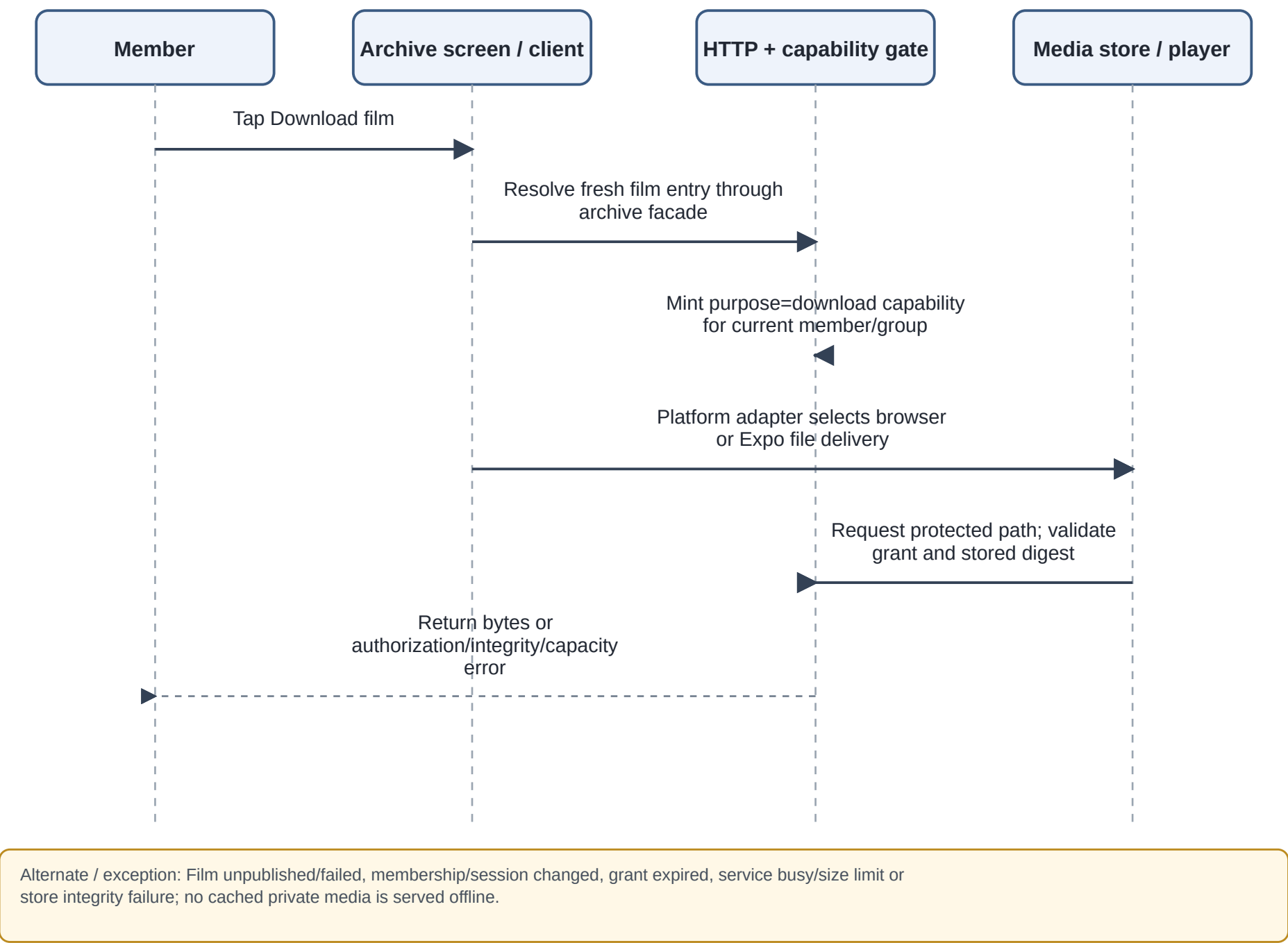


Figure M128 | ARCHIVE | uc14 analysis class
Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14 — Analysis classes

Installation, update and invite-opening responsibilities

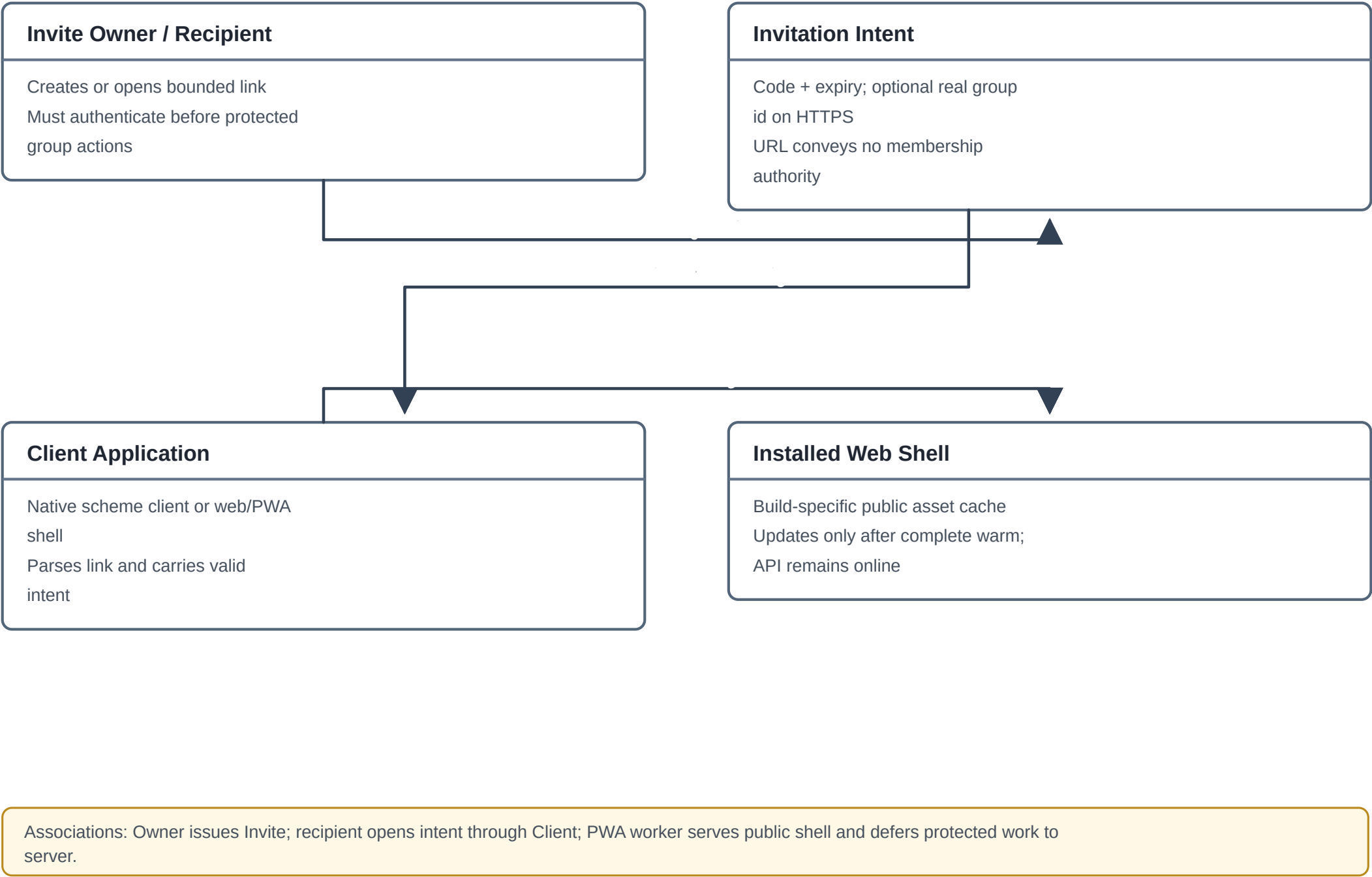
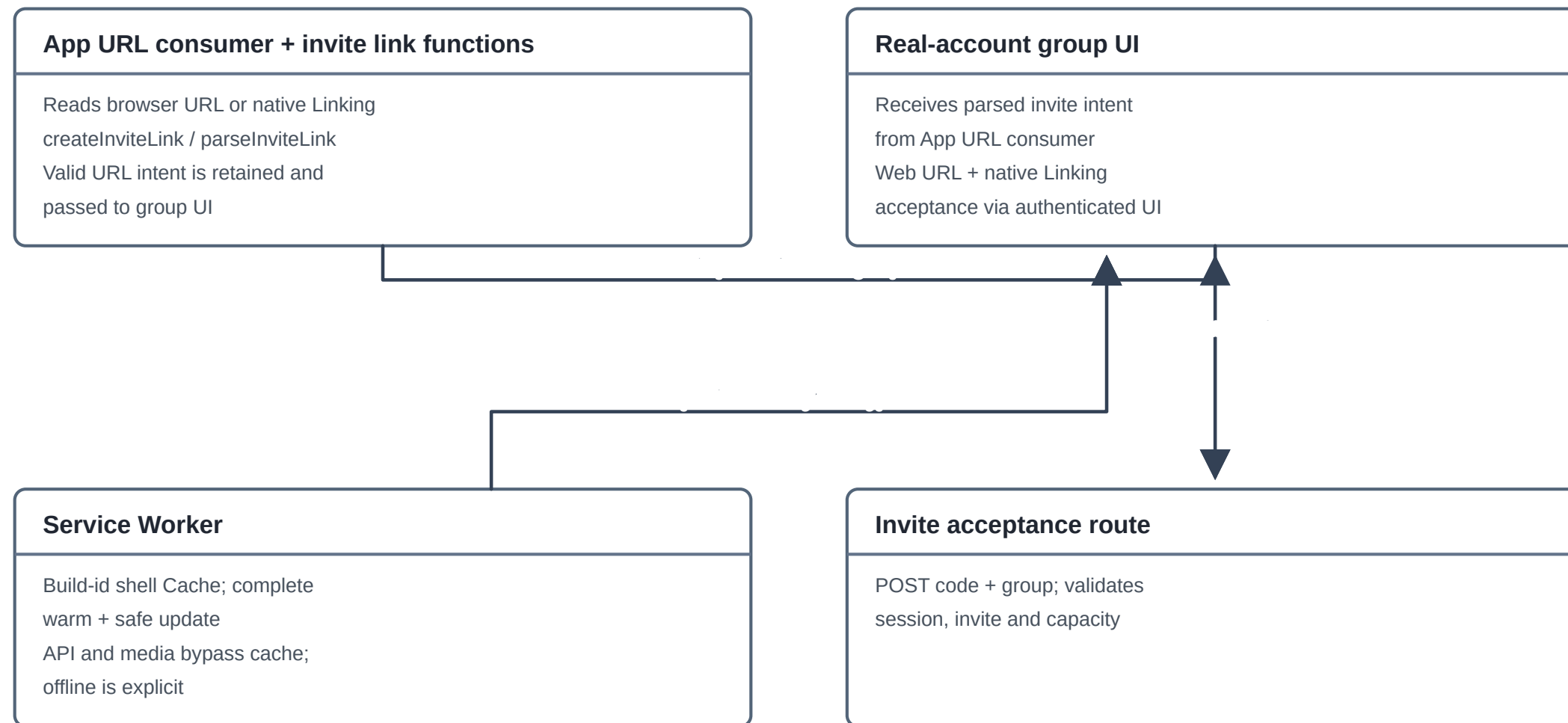


Figure M129 | ARCHIVE | uc14 design class

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14 — Design classes

Implemented URL consumption, invite parsing, and web lifecycle boundaries



Associations: App consumes browser/native URLs, parses invite intent, and passes valid intent to the group UI. The UI accepts through the authenticated route; physical-device URL association remains unverified.

Figure M130 | ARCHIVE | uc14 f1 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F1 — Install and open the web/PWA shell

Analysis responsibility model; platform-neutral concepts

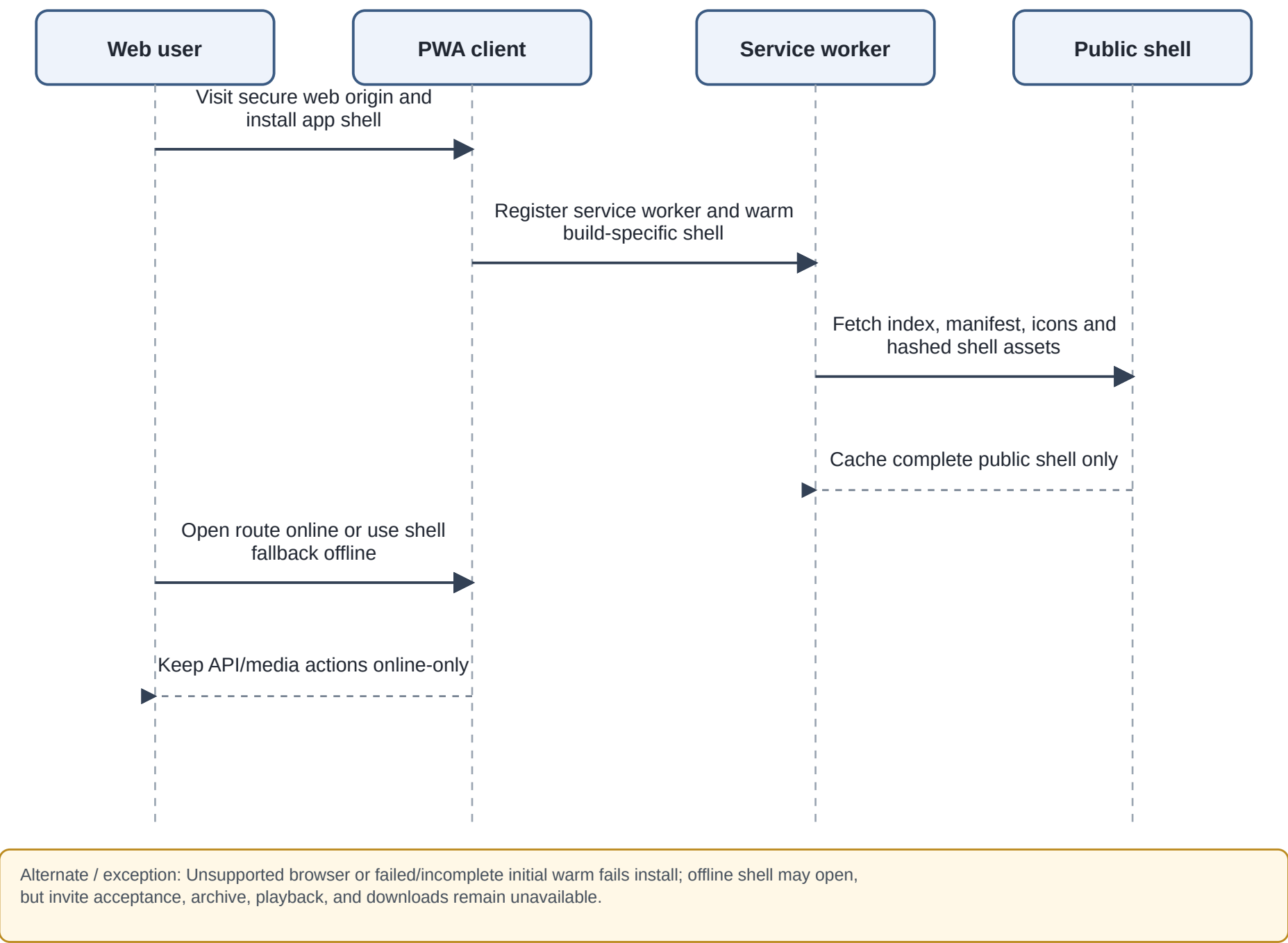


Figure M131 | ARCHIVE | uc14 f1 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F1 — Install and open the web/PWA shell

Design interaction grounded in accepted code

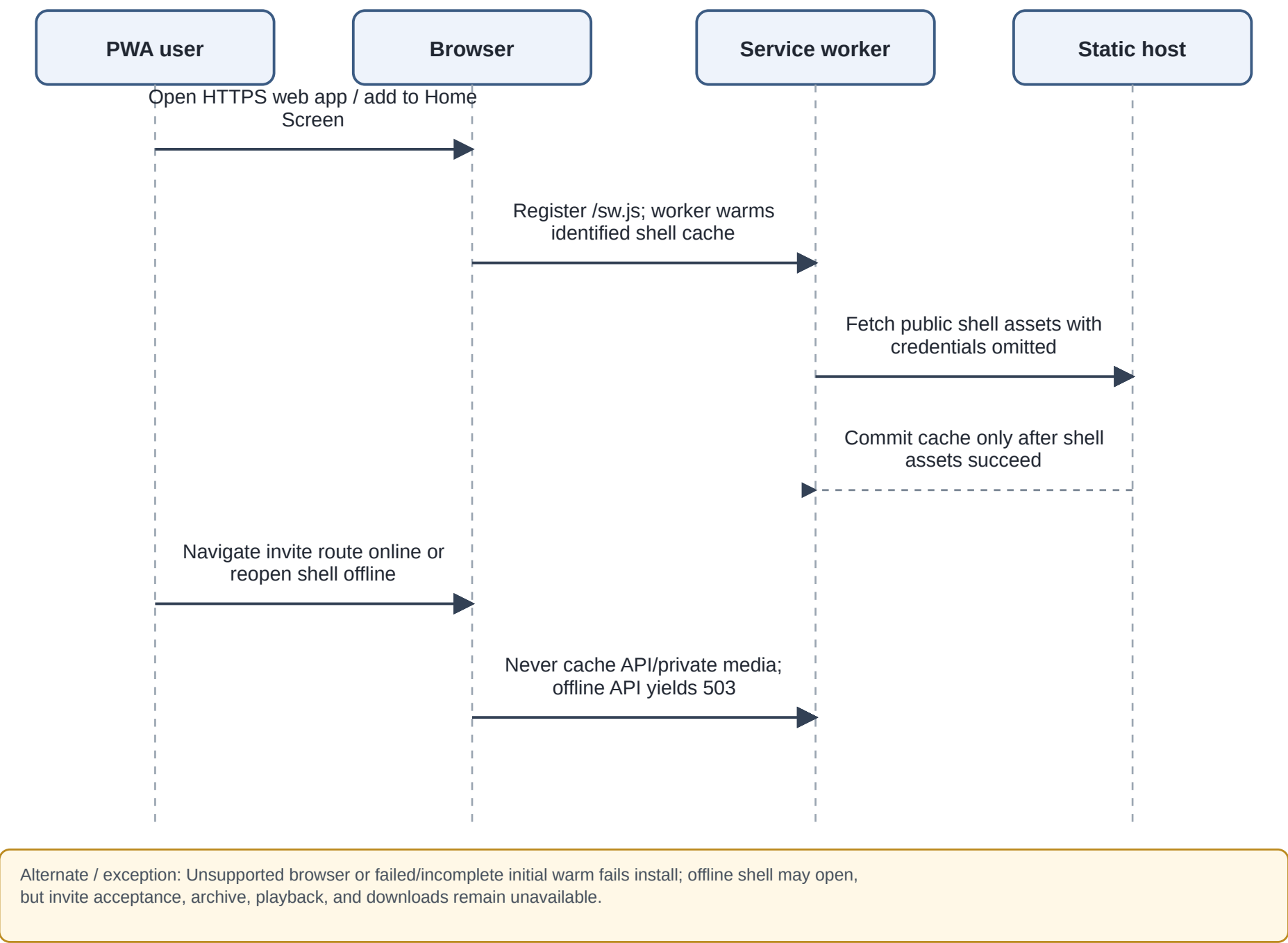


Figure M132 | ARCHIVE | uc14 f2 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F2 — Install an update without interrupting work

Analysis responsibility model; platform-neutral concepts

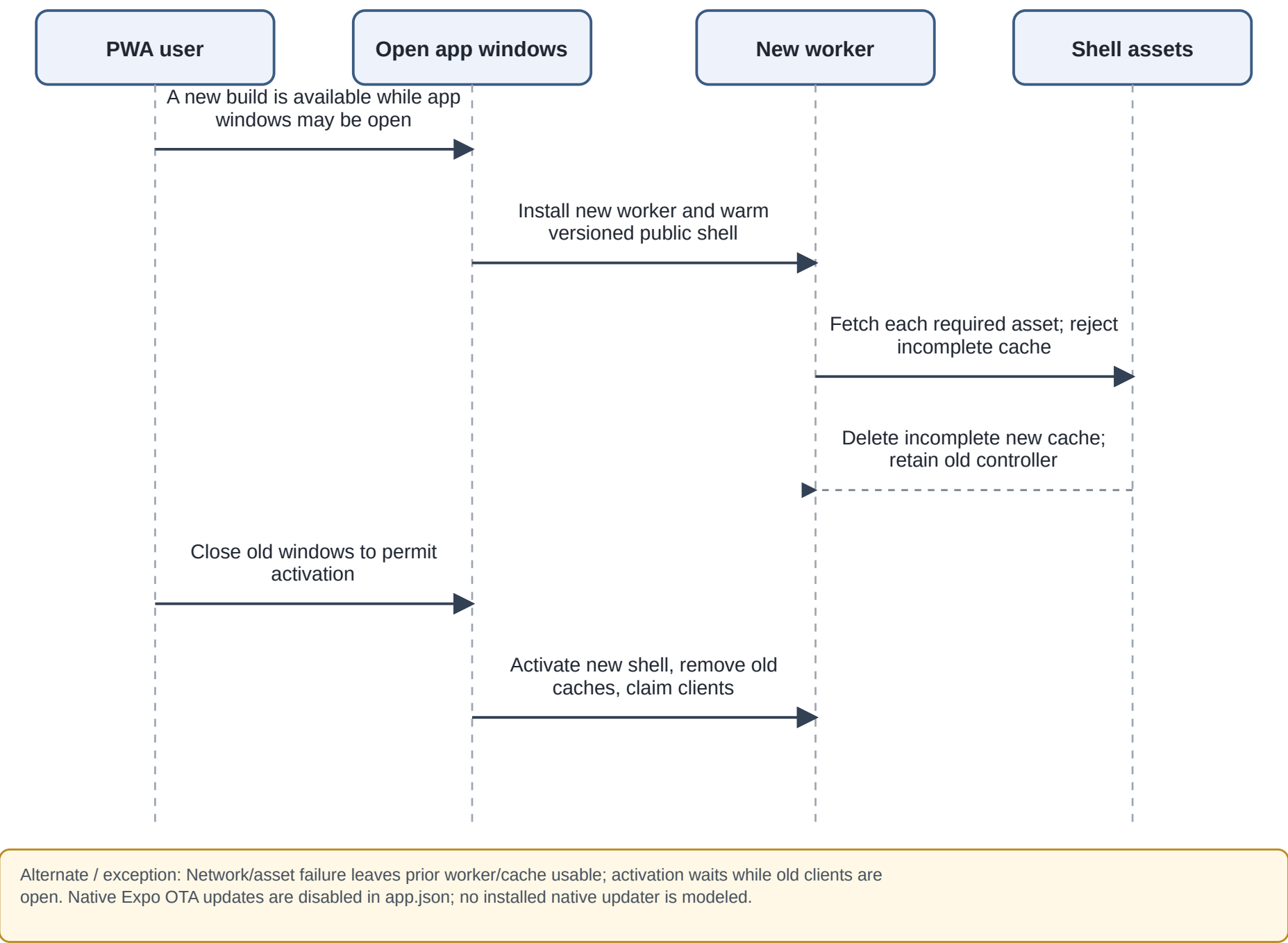


Figure M133 | ARCHIVE | uc14 f2 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F2 — Install an update without interrupting work

Design interaction grounded in accepted code

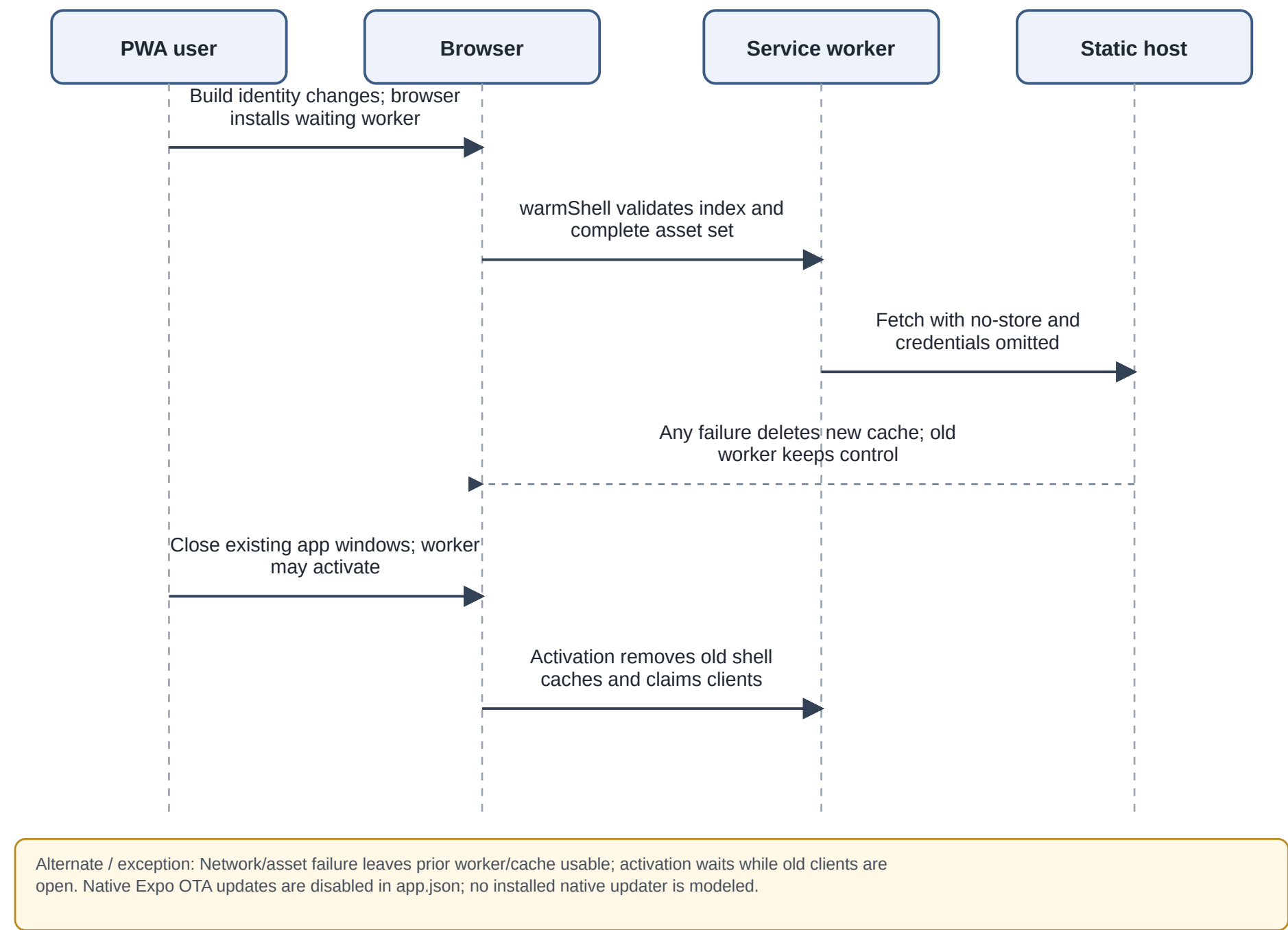


Figure M134 | ARCHIVE | uc14 f3 analysis sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F3 — Create and open a cross-client invite link

Analysis responsibility model; platform-neutral concepts

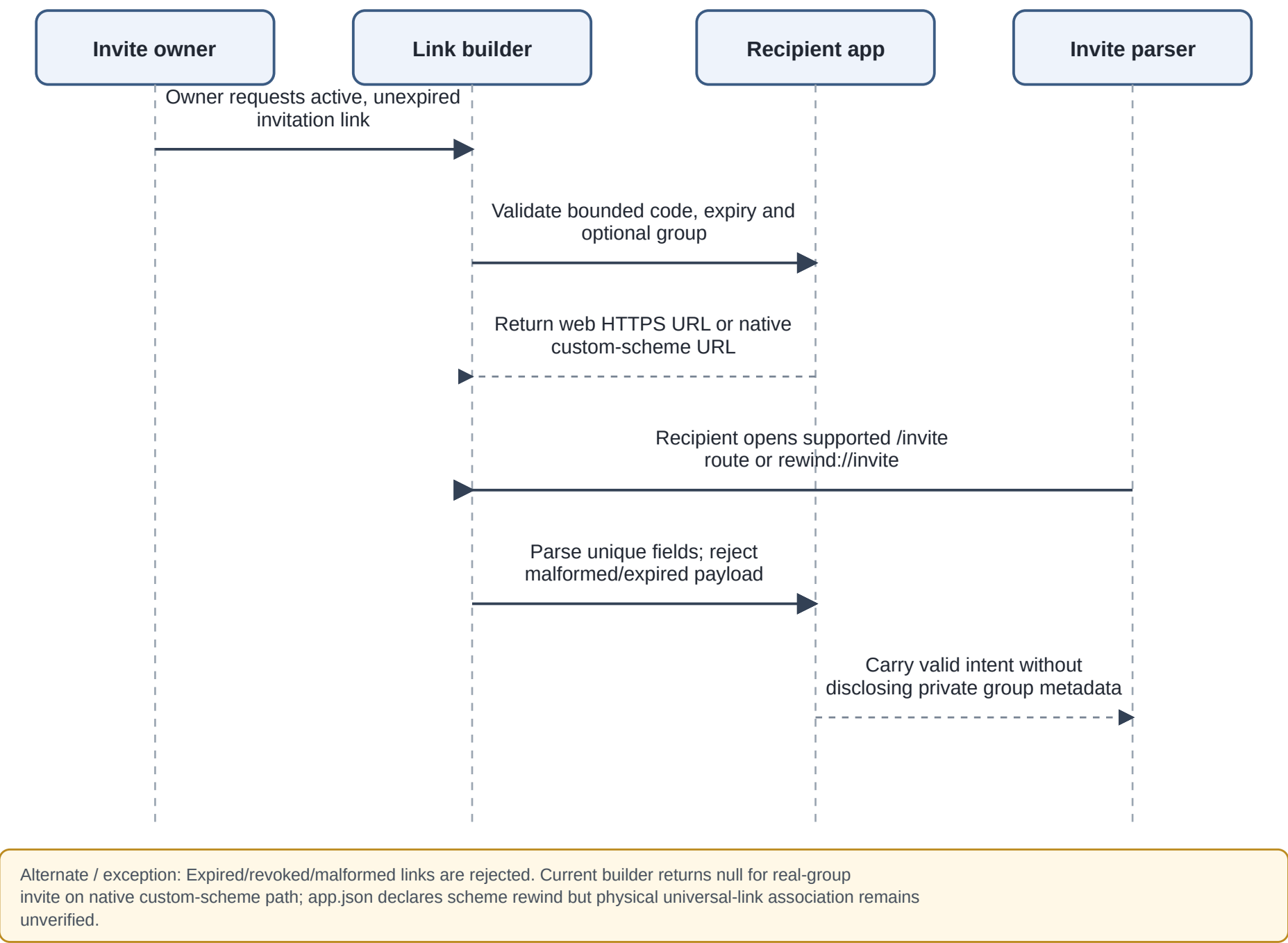


Figure M135 | ARCHIVE | uc14 f3 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F3 — Create and open a cross-client invite link

App URL consumer and invite parser are implemented at the source cut

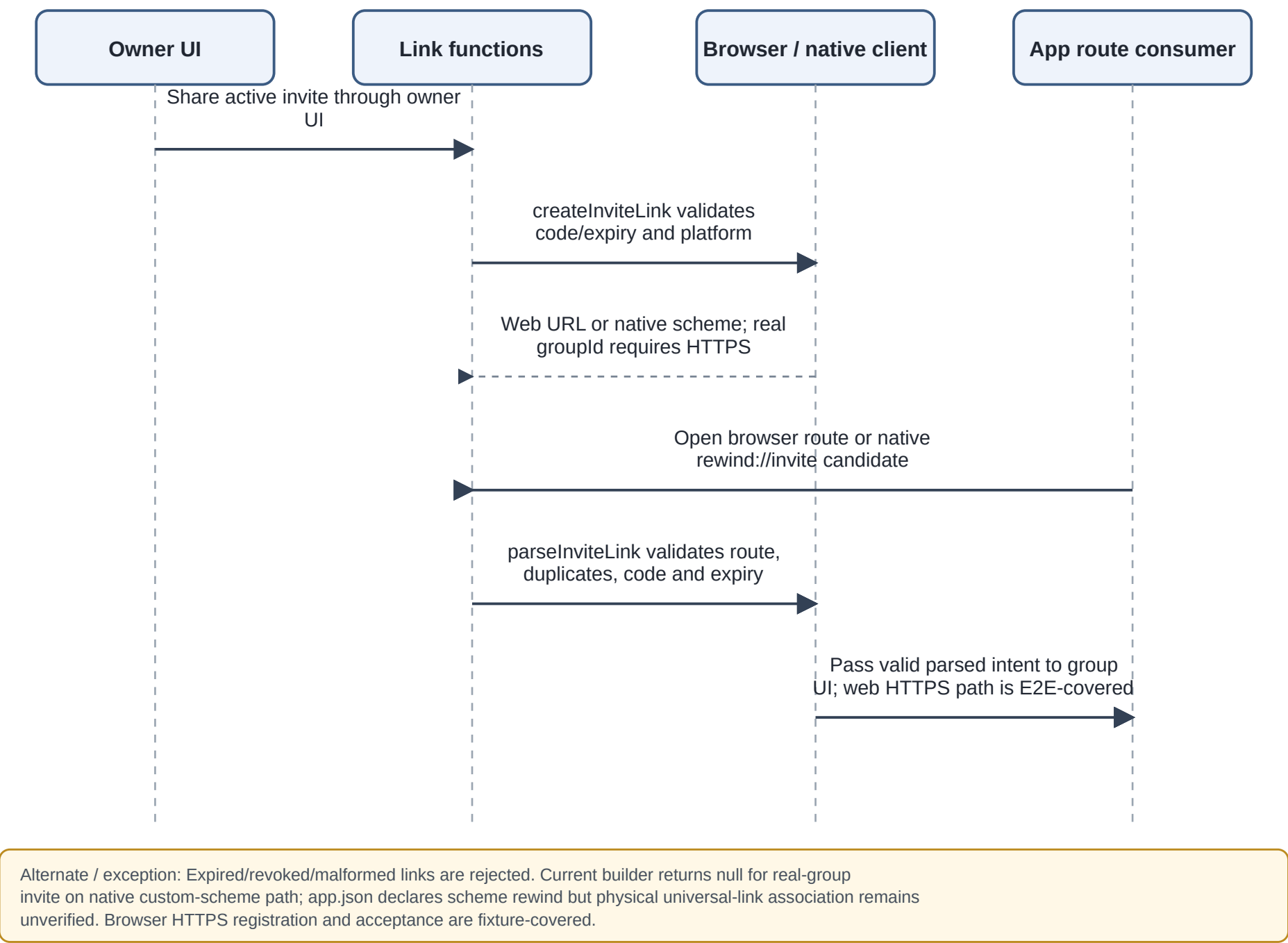


Figure M136 | ARCHIVE | uc14 f4 analysis sequence
Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F4 — Accept invite after client opens/authenticates

Analysis responsibility model; platform-neutral concepts

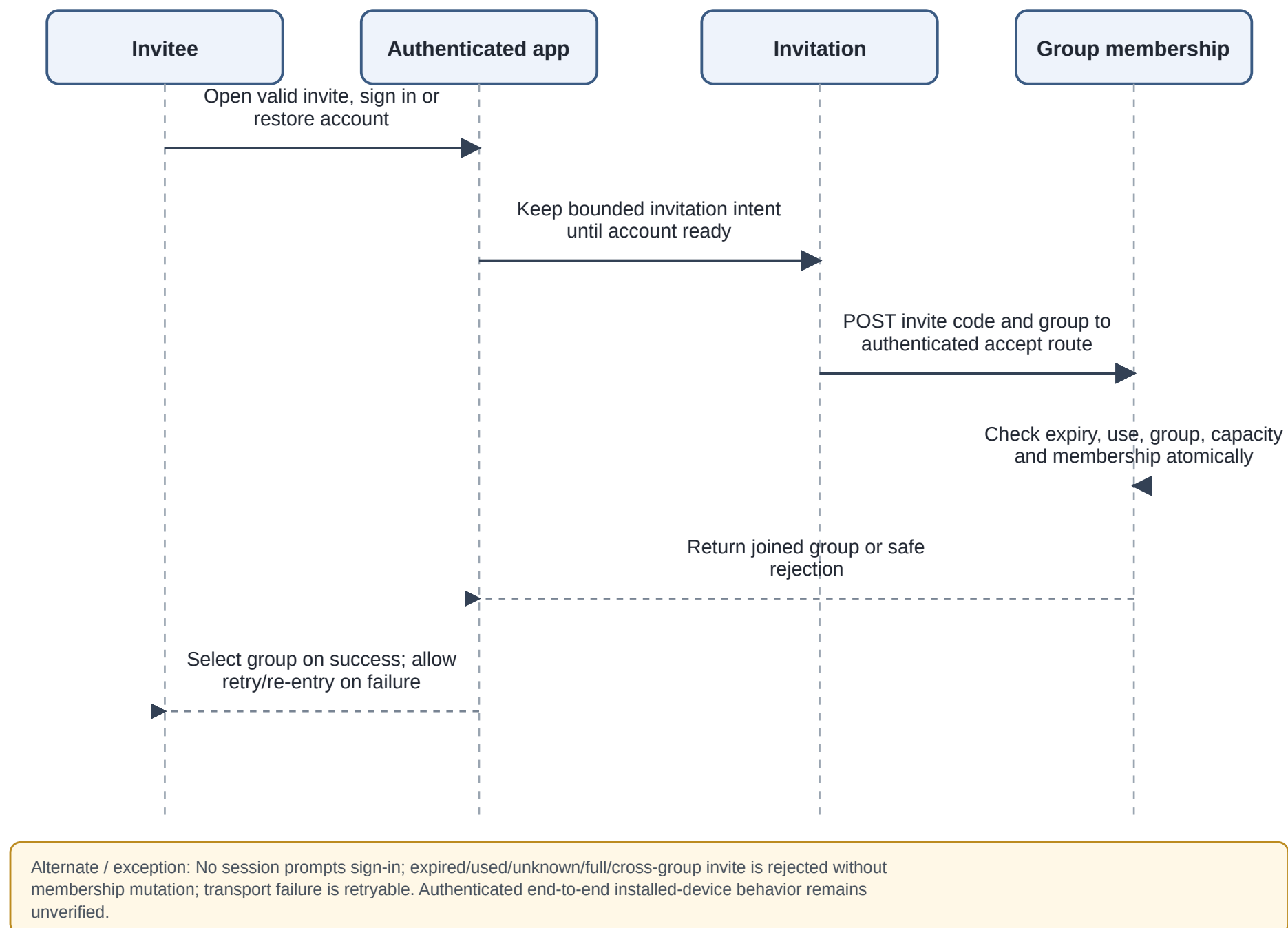
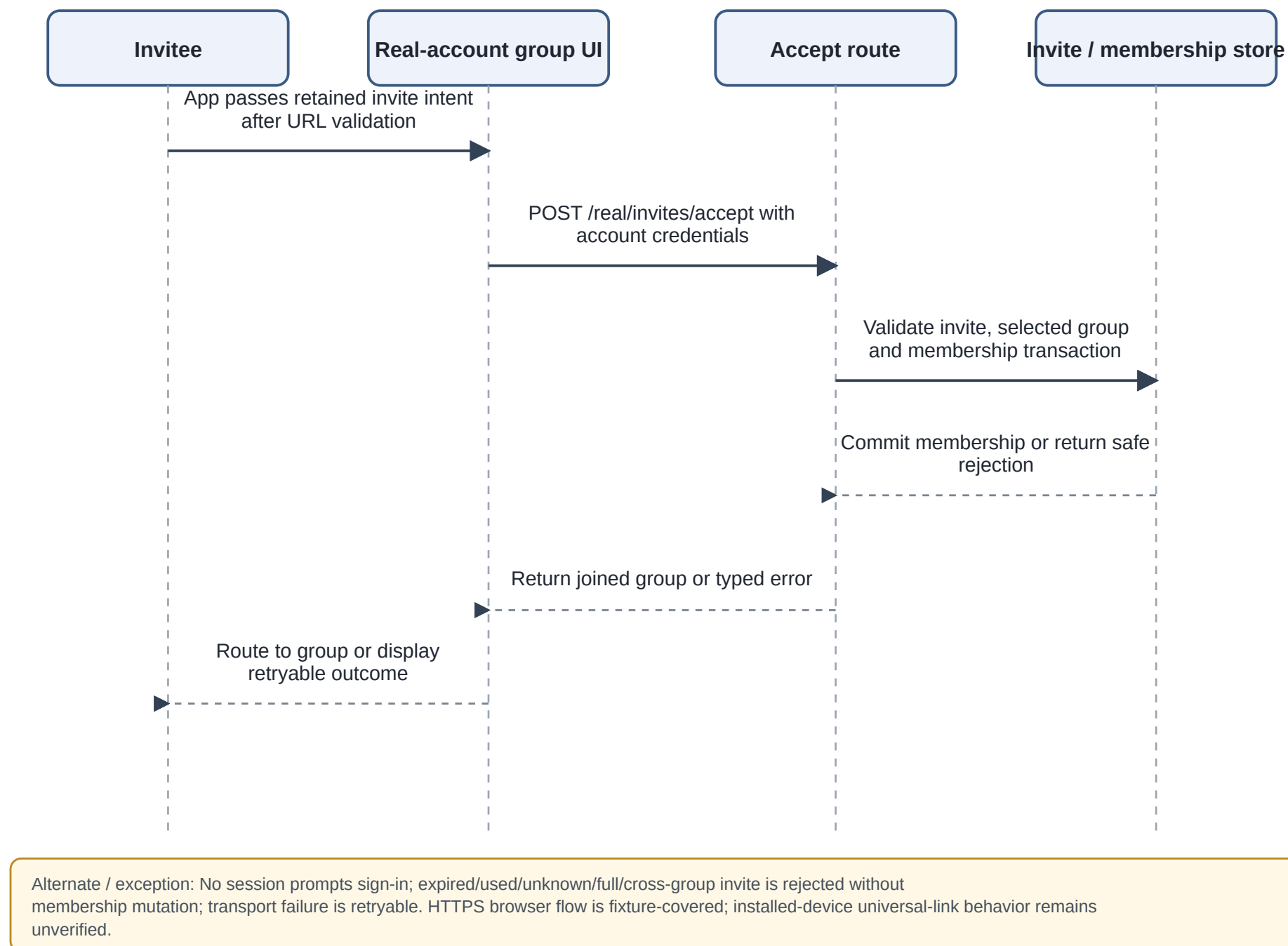


Figure M137 | ARCHIVE | uc14 f4 design sequence

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

UC14-F4 — Accept invite after client opens/authenticates

Design interaction grounded in accepted code



3.5 Design Problems and Patterns

3.5.1 Five code backed design problems

ACCESS compares Policy/Strategy with Chain of Responsibility for mixed identity and membership boundaries; actual implementation uses explicit adapters and function policy, with real-account predicates distinct from Demo. CAPTURE compares Adapter/Strategy/Template Method for platform/file/upload/processing variation. CYCLE compares State/Command/polling against functional persisted transitions and durable queue/leases. PARTICIPATION compares Observer/Adapter/Strategy for events, reconnect and provider/outbox coordination. ARCHIVE compares Proxy/Adapter/Facade for private capabilities and portable player/download delivery. Before figures are explicit design alternatives, not invented historical vulnerabilities. Current function-centered code must not be described as an unimplemented pattern class hierarchy. The package discussions supply candidate rationale, before/after class and sequence views and exact source pins; the incorporated before/after plates retain their accepted source identities, with baseline code links refreshed in the source crosswalk. Human pattern attribution remains pending.

Before models are conceptual alternatives, not historical vulnerabilities. After models retain actual functional policies, platform adapters and persisted transitions; no new class hierarchy or production refactor is claimed.

ACCESS model/source register | issue #357

Accepted model cut fc2b8c811e9c13b176113fa0bdb2c015798990b4; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

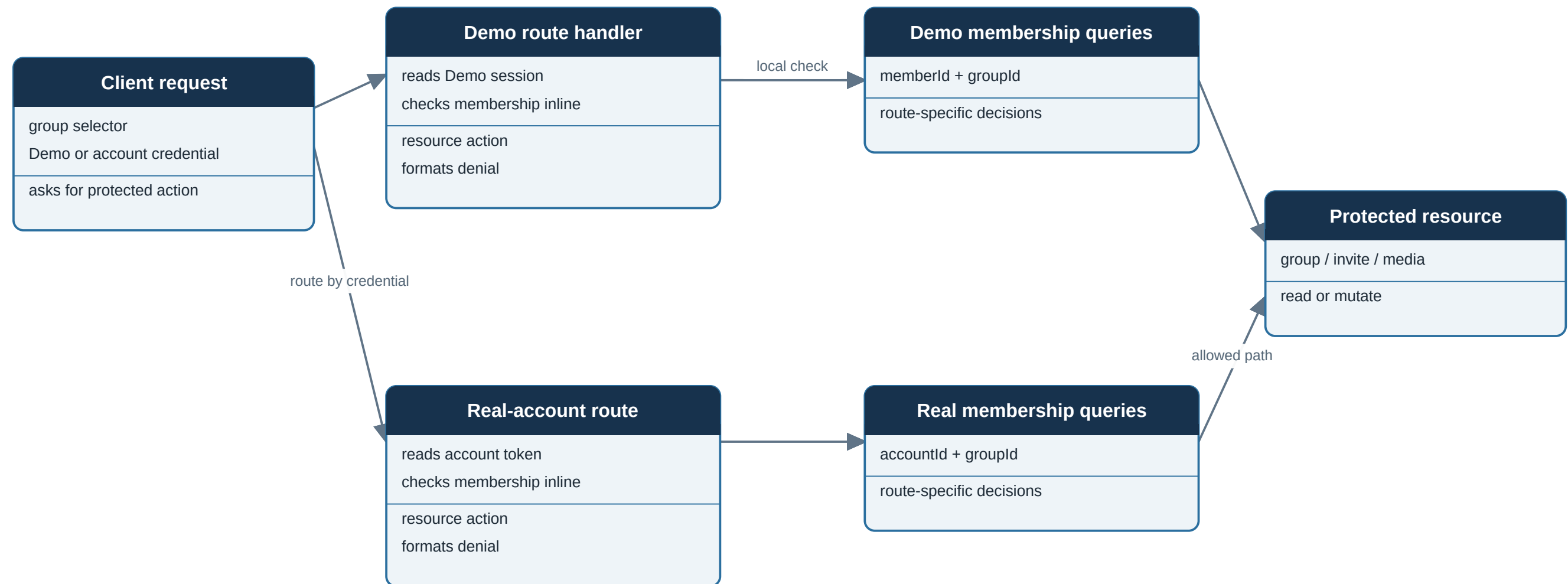
Package design-problem.md — requirements, flows, exceptions and code mapping

Figure M003 | ACCESS | problem before class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

Authorization problem — before candidate

Route-local brute-force alternative for comparison only; this is not asserted as historical deployed code.



Risk represented: repeated per-route checks can diverge. This diagram is an alternative under analysis, not a historical code claim.

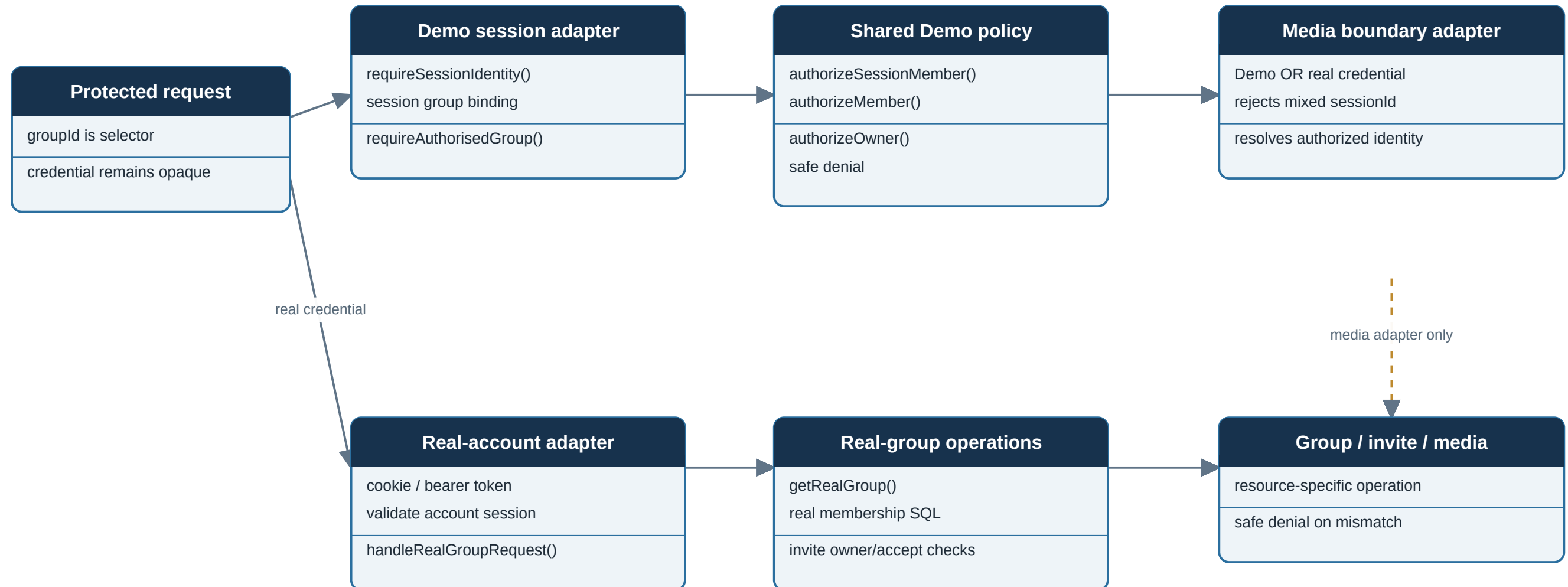
packages/access/diagrams/problem-before-class.svg

Figure M001 | ACCESS | problem after class

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

Authorization problem — implemented boundary design

Actual current structure: shared Demo policy plus explicit real-account session/group operations; the shared policy is not universal.

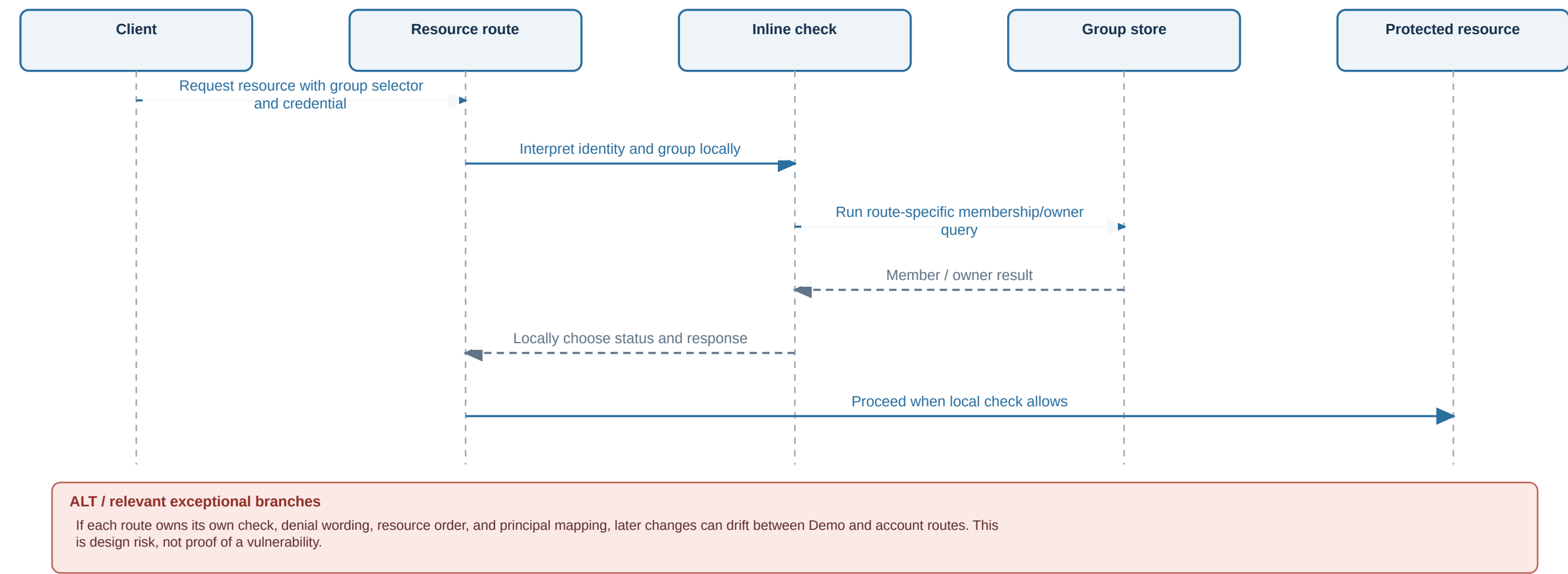


Actual gap: account-specific queries and Demo policy are distinct. No single Policy object spans both identity families at this SHA.

Figure M004 | ACCESS | problem before sequence
Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model

Authorization problem — before sequence

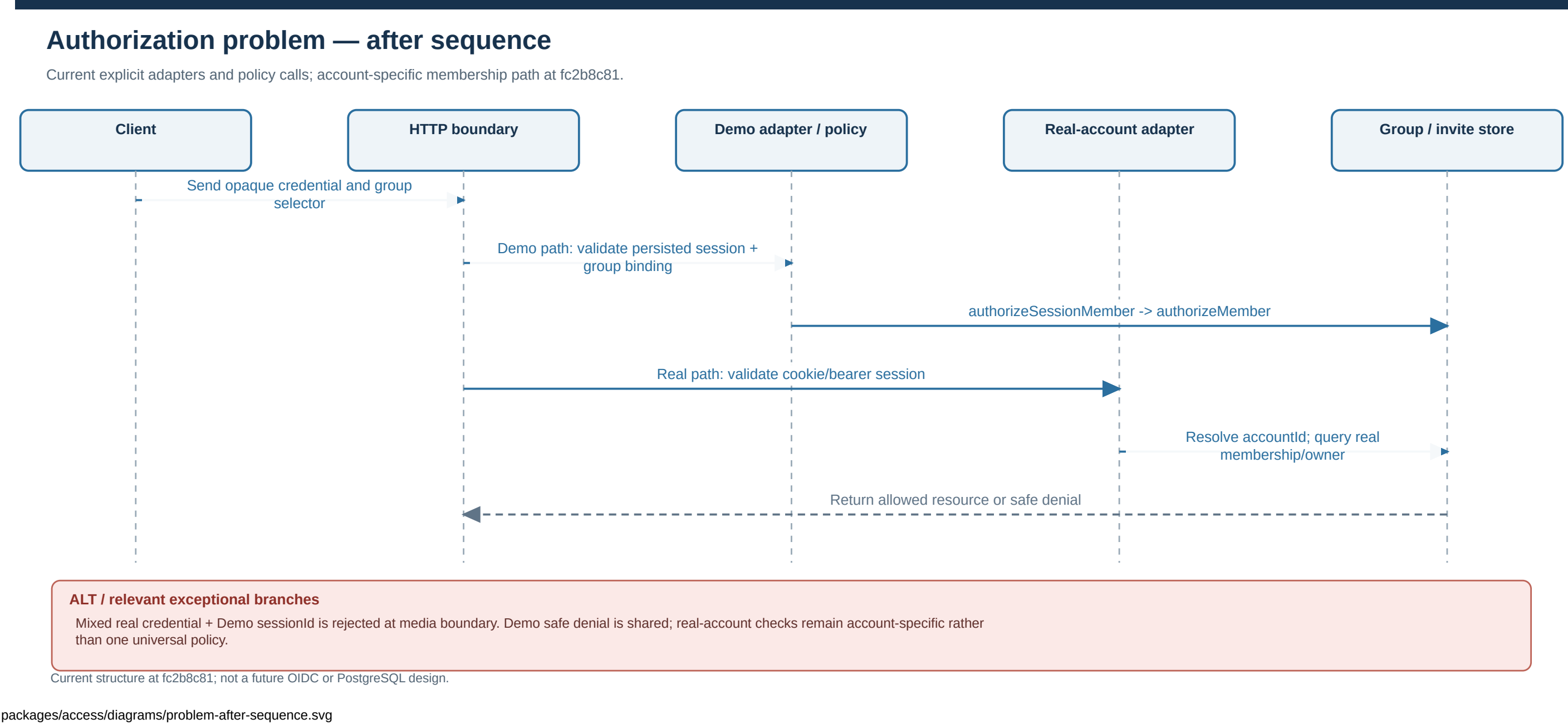
Illustrative route-local alternative; group and actor identifiers are not shown as authority in the current implementation.



packages/access/diagrams/problem-before-sequence.svg

Figure M002 | ACCESS | problem after sequence

Accepted source fc2b8c81 | issue #357 | A3 landscape | complete model



CAPTURE model/source register | issue #358

Accepted model cut fc2b8c811e9c13b176113fa0bdb2c015798990b4; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

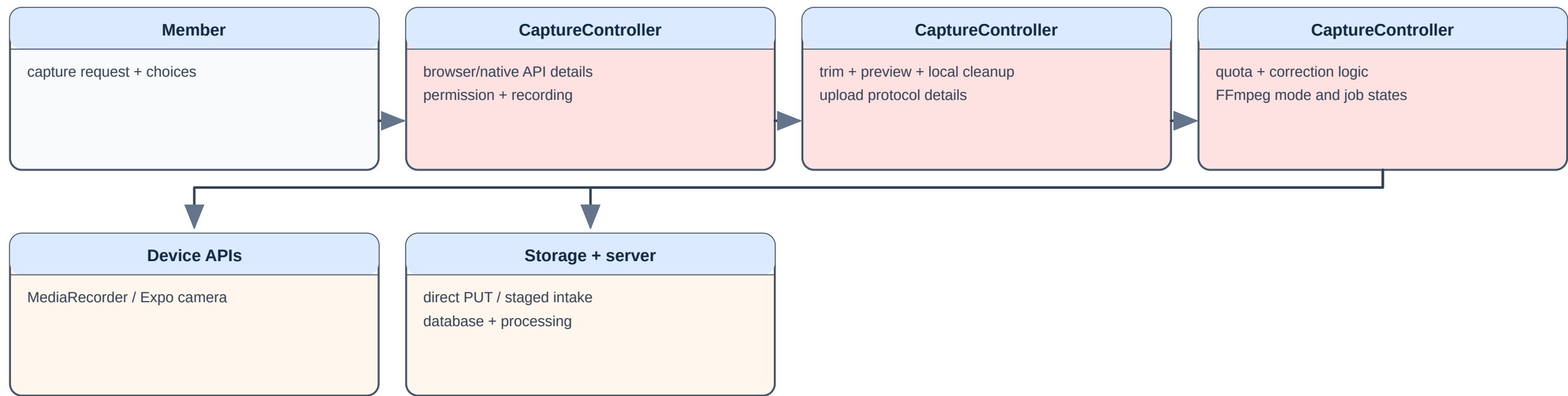
Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

Figure M038 | CAPTURE | problem before class
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

Before — variation pressure (conceptual)
Problem model only; it is not a historical class diagram from this repository.



Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

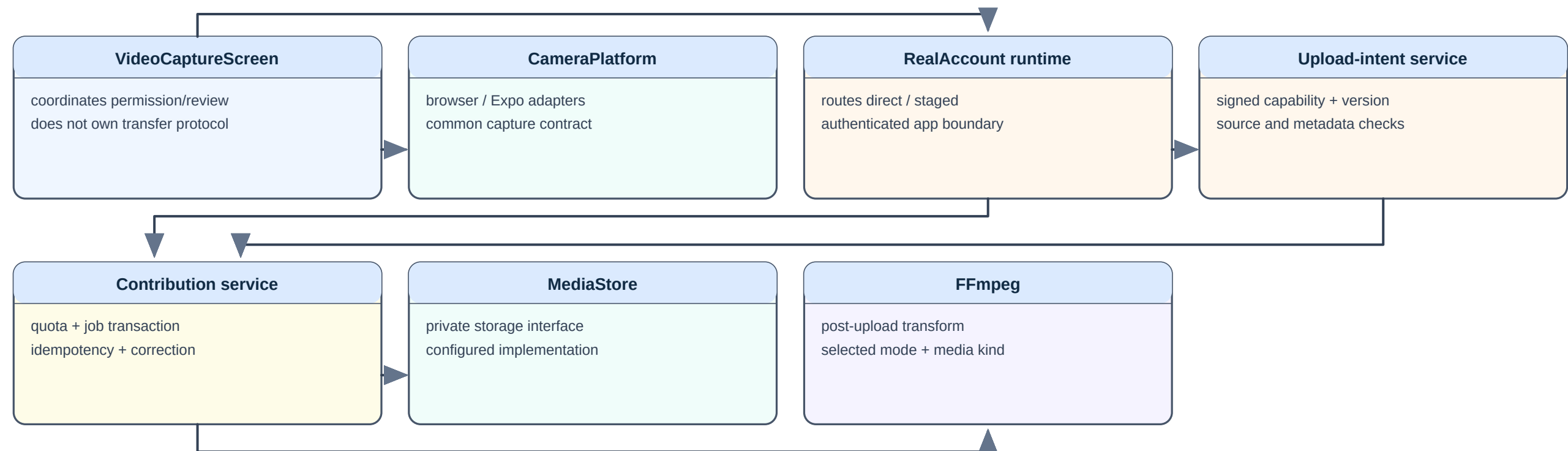
packages/capture/diagrams/problem-before-class.svg

Figure M036 | CAPTURE | problem after class

Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

After — accepted code boundaries

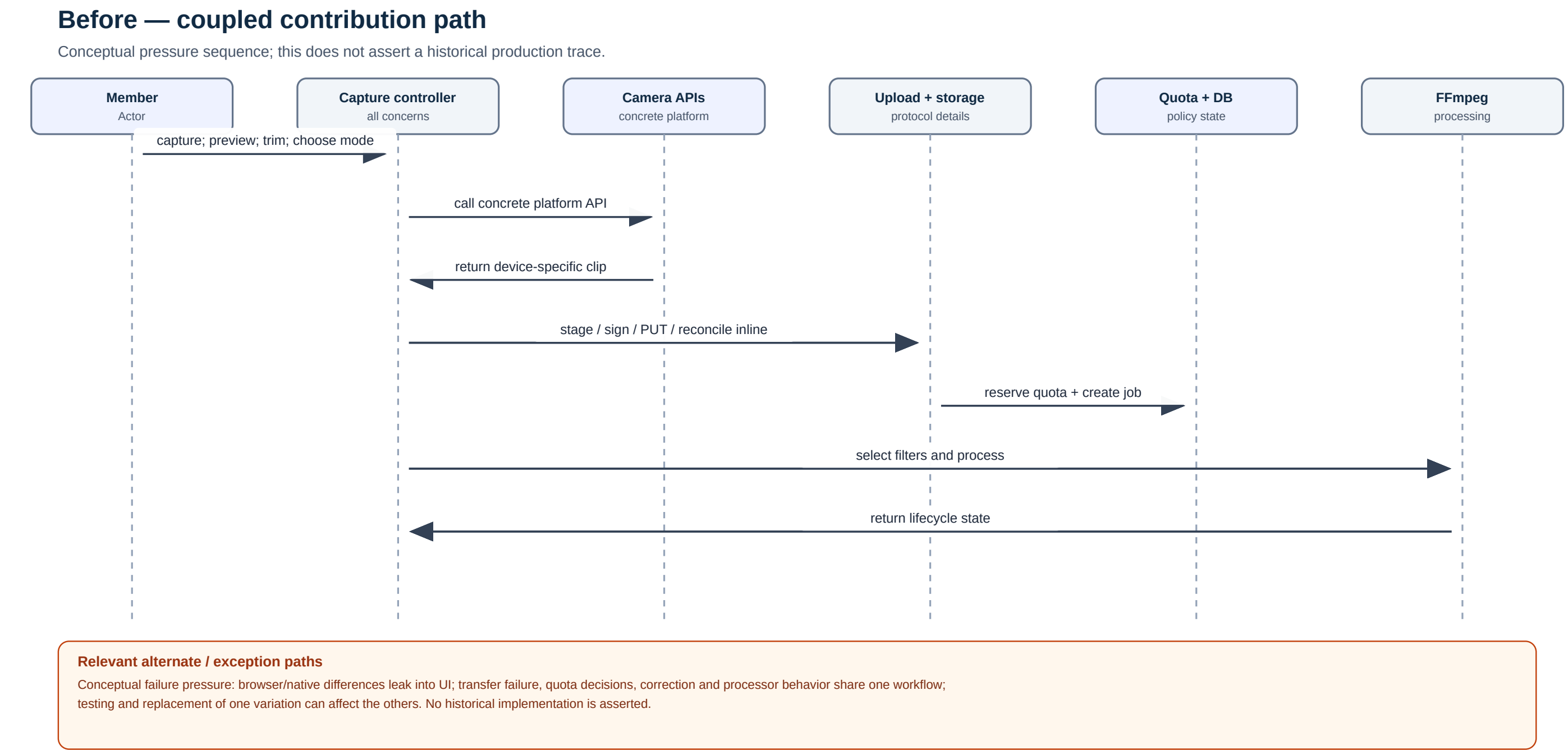
Snapshot already contains these seams; the diagram does not claim a measured refactor.



Solid arrows show a responsibility/use relationship. Analysis names are domain responsibilities; design names are present at the pinned code cut.

packages/capture/diagrams/problem-after-class.svg

Figure M039 | CAPTURE | problem before sequence
Accepted source fc2b8c81 | issue #358 | A3 landscape | complete model

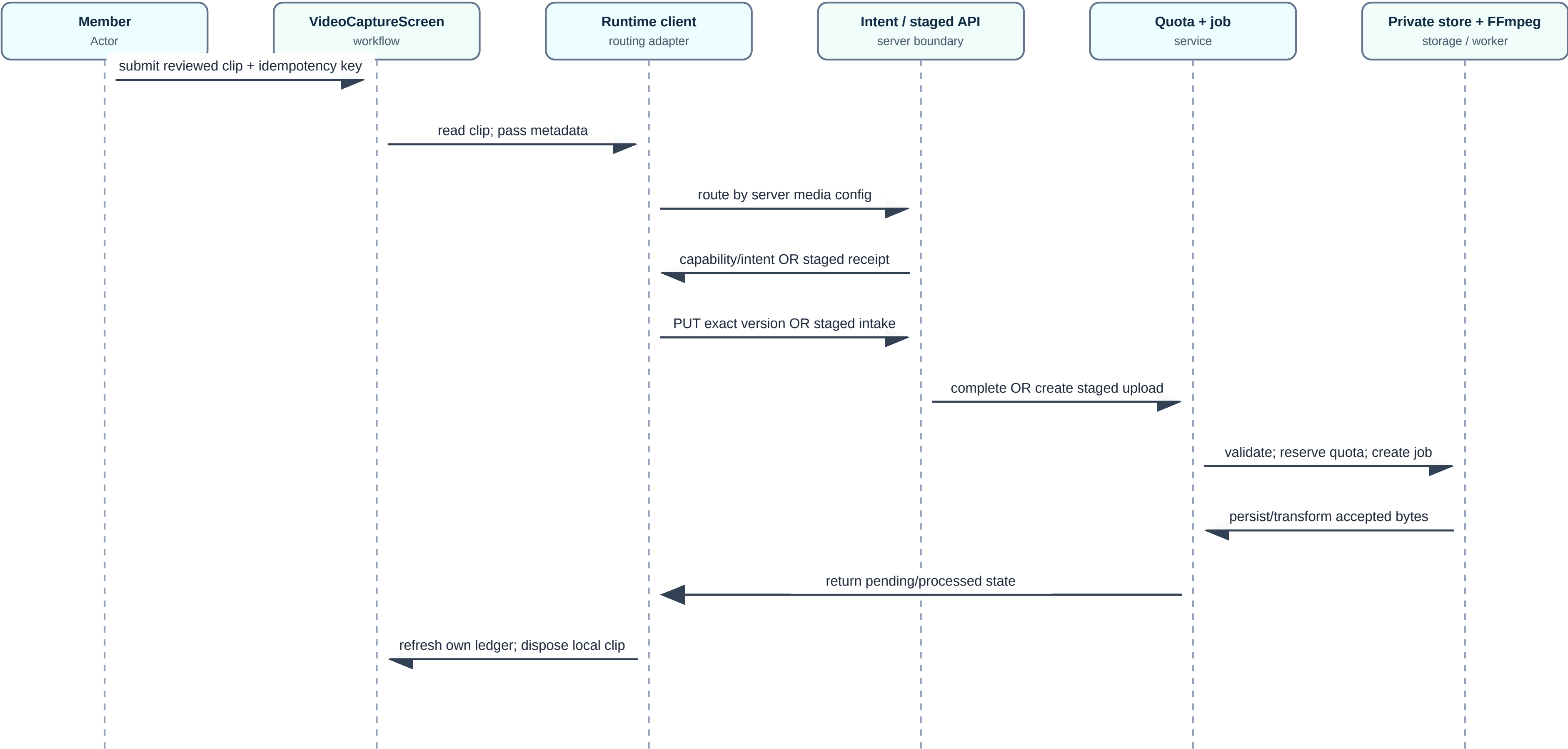


packages/capture/diagrams/problem-before-sequence.svg

Figure M037 | CAPTURE | problem after sequence
Accepted source fc2b8c81 | issue #358 | Large landscape foldout | complete model

After — configured transfer and processing

Code-backed sequence at accepted SHA; direct transfer is conditional on server configuration.



Relevant alternate / exception paths

Direct transfer reconciles intent/version before retry. Staged intake verifies its source binding. Both paths converge on server authority, quota and job creation. Processing errors remain job failures; pre-upload retro is not part of this implemented sequence.

CYCLE model/source register | issue #359

Accepted model cut fc2b8c811e9c13b176113fa0bdb2c015798990b4; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

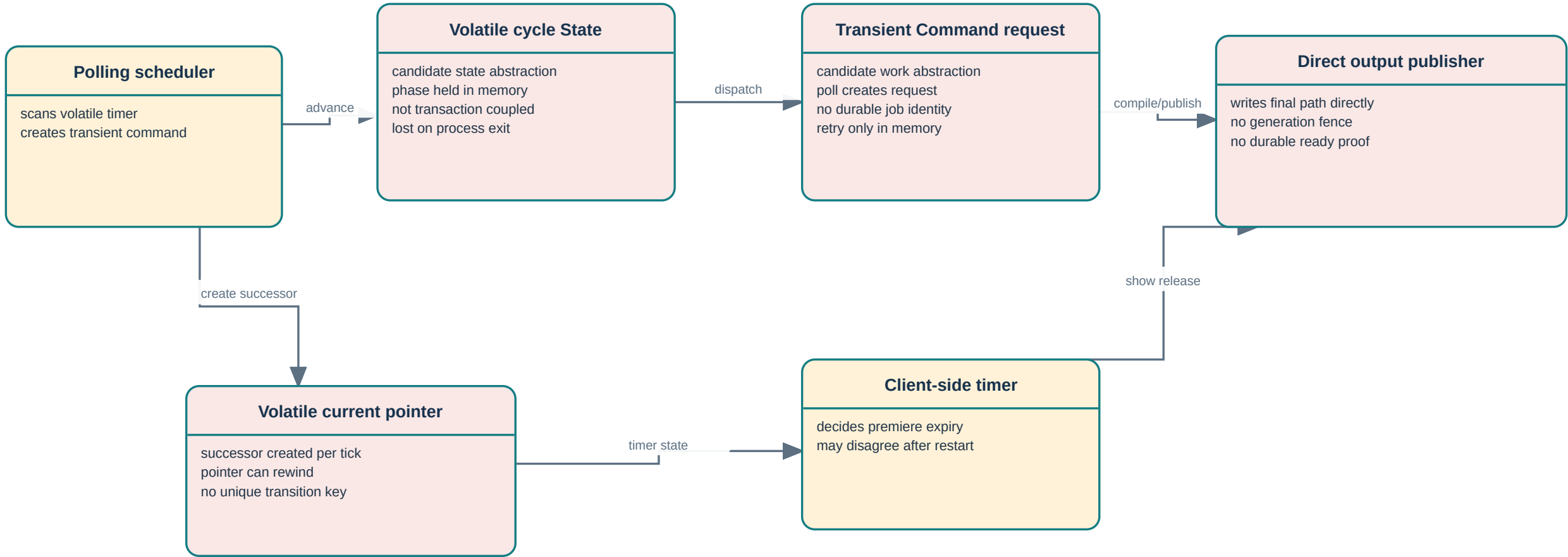
Figure M062 | CYCLE | problem before class

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

Before · Candidate volatile transition design

Hypothetical comparison only — not a historical Rewind implementation

BEFORE



Analysis shows responsibilities and concepts; design shows only implementation-evidenced components and records at fc2b8c811e9c13b176113fa0bdb2c015798990b4.

packages/cycle/diagrams/problem-before-class.svg

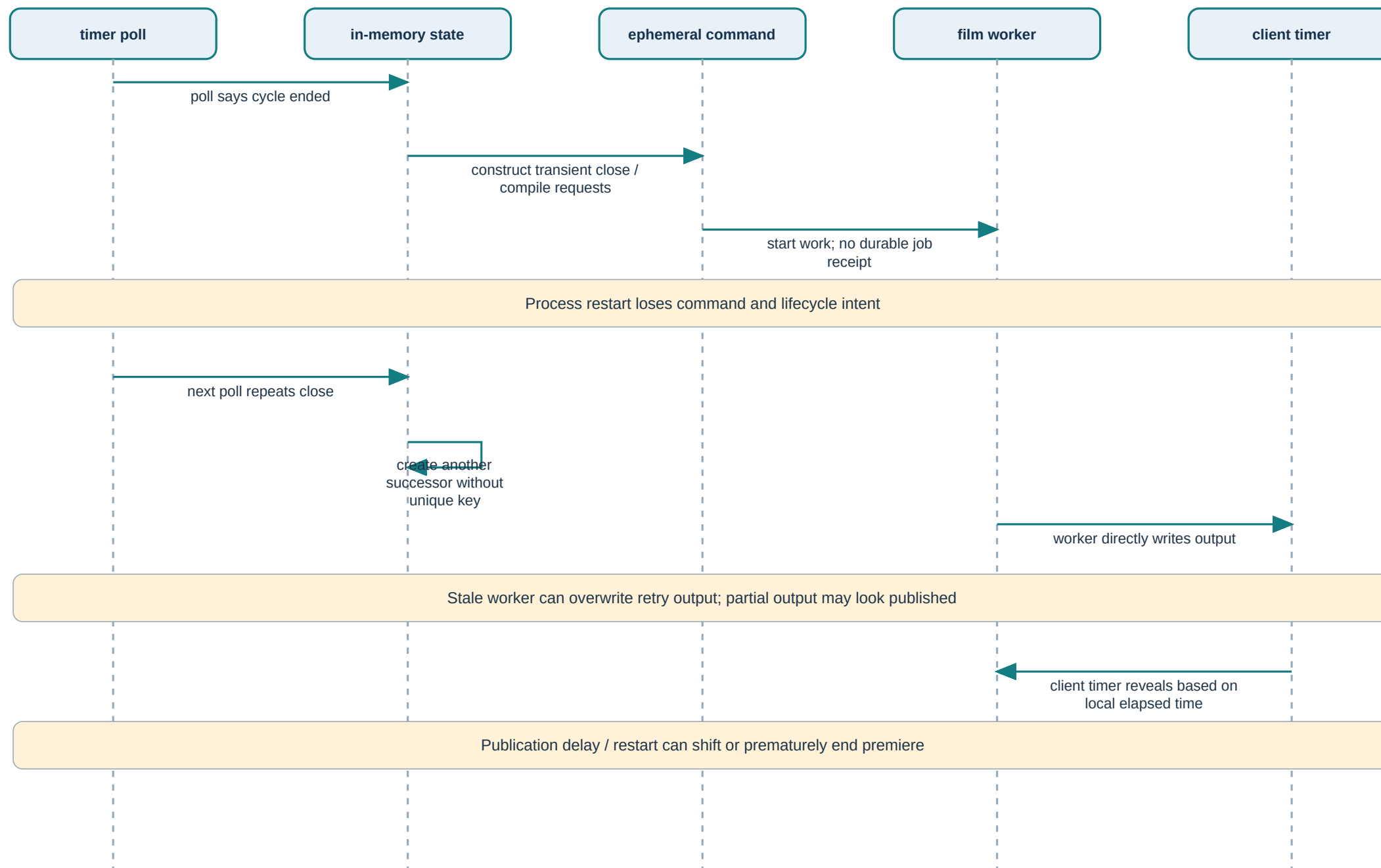
Figure M063 | CYCLE | problem before sequence

Accepted source fc2b8c81 | issue #359 | A3 landscape | complete model

Before · Candidate polling and ephemeral work

BEFORE

Failure comparison only — the sequence is not historical Rewind behavior



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

packages/cycle/diagrams/problem-before-sequence.svg

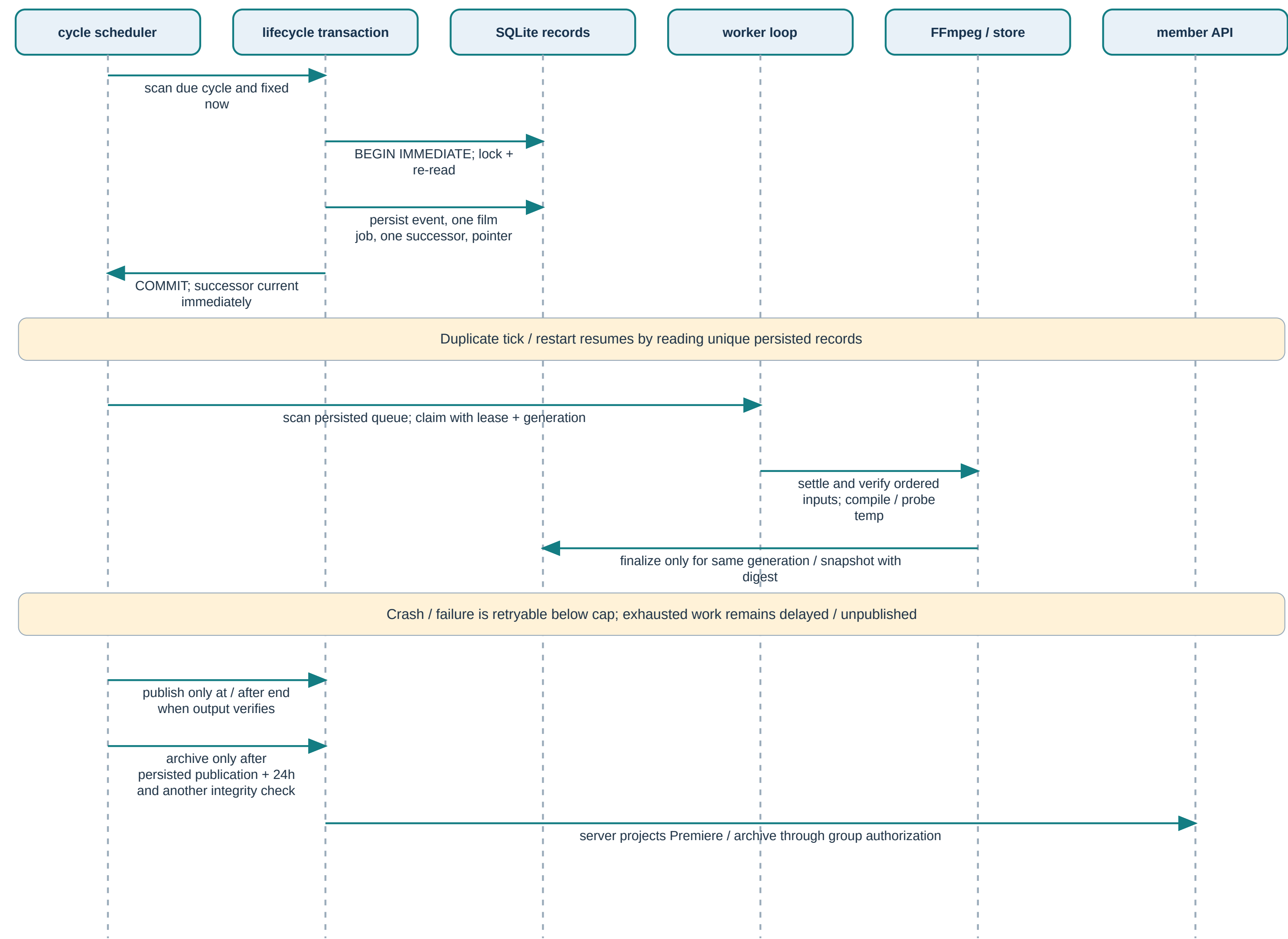
Figure M061 | CYCLE | problem after sequence

Accepted source fc2b8c81 | issue #359 | Large landscape foldout | complete model

After · Transactional lifecycle and durable compilation

AFTER

Restart-safe state transitions, claim-generation fencing, verified release



The diagram covers the normal path and named relevant branches; implementation names appear only in the design view.

PARTICIPATION model/source register | issue #360

Accepted model cut 1128b6a68985cf68215beb4a7a80fd00ec4242fa; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

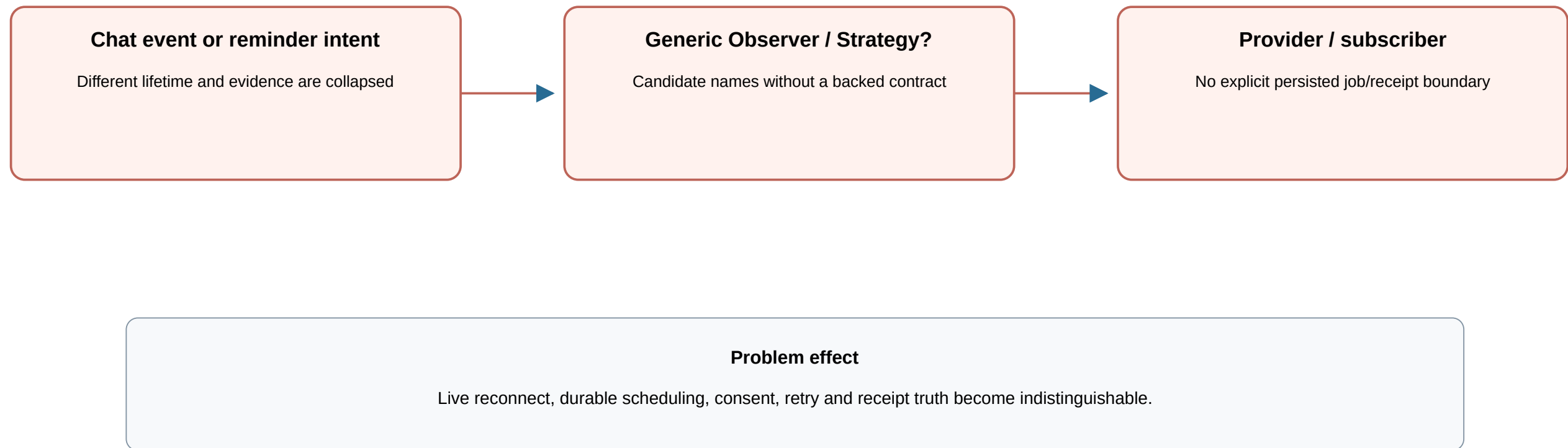
Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

Figure M082 | PARTICIPATION | problem before class
 Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

Before — pattern-only async composition (conceptual)

Alternative sketch, not an assertion about historical or current classes.



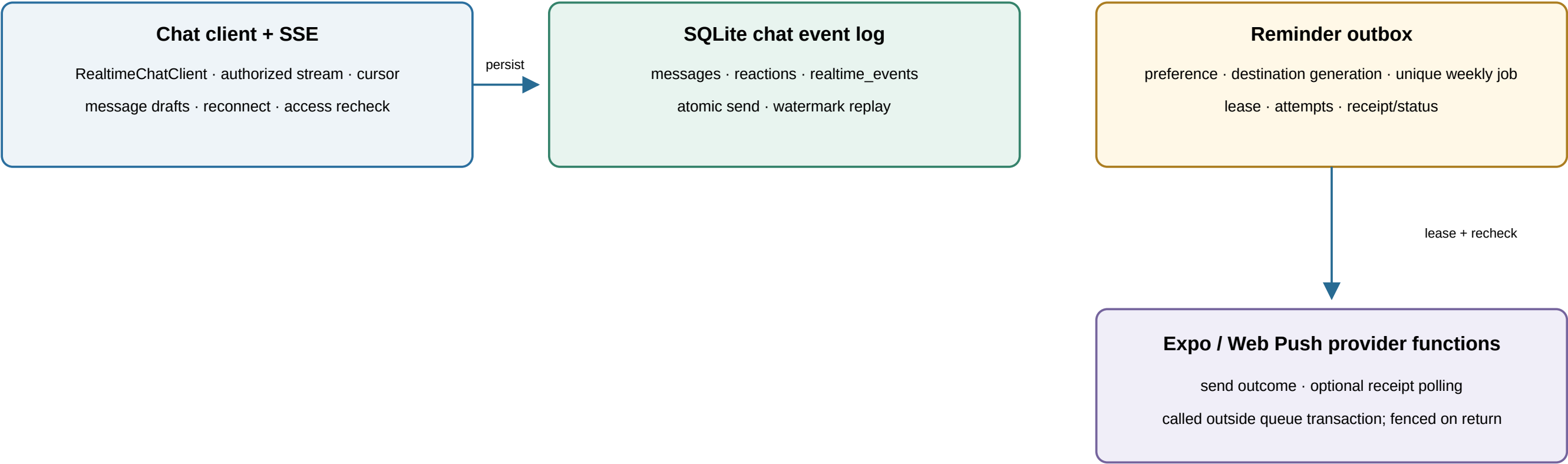
packages/participation/diagrams/problem-before-class.svg

Figure M080 | PARTICIPATION | problem after class

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

After — actual transport and persisted boundaries

Source-backed functions/modules; no fabricated pattern classes.



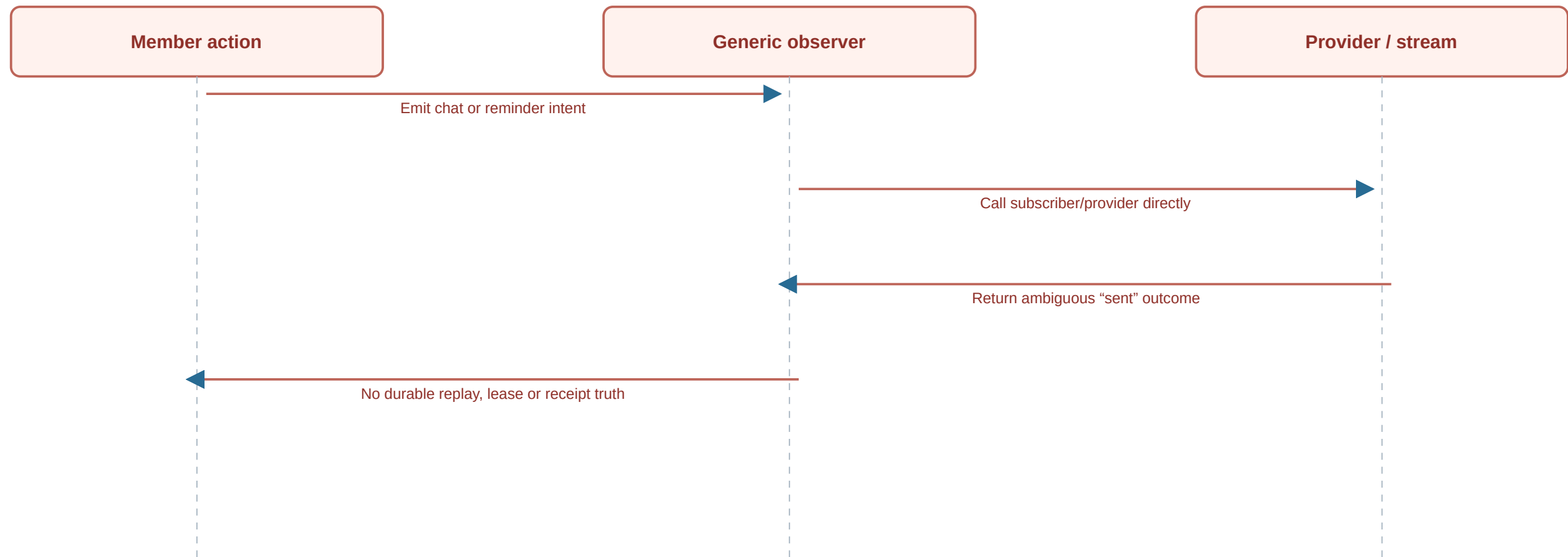
Evidence boundary (caption)

Chat: persisted message event + authorized replay. Reminder: consent-scoped job + provider outcome.
Provider acceptance/receipt does not prove OS presentation or human observation.

Figure M083 | PARTICIPATION | problem before sequence
Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

Before — pattern-only asynchronous request

Conceptual anti-pattern; no invented application class is asserted.



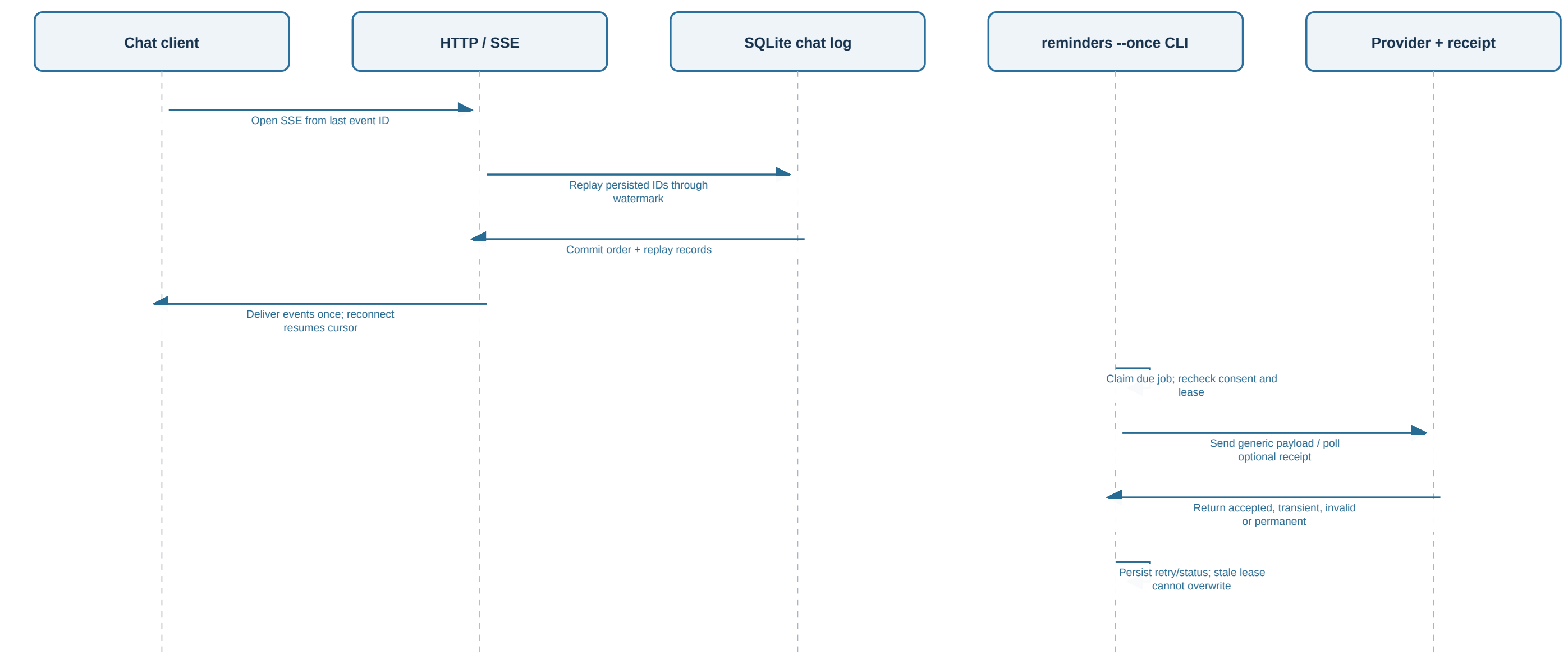
packages/participation/diagrams/problem-before-sequence.svg

Figure M081 | PARTICIPATION | problem after sequence

Accepted source 1128b6a6 | issue #360 | A3 landscape | complete model

After — actual chat replay and reminder delivery

Two independent paths; provider return is persisted as an outcome, not assumed device display.



packages/participation/diagrams/problem-after-sequence.svg

ARCHIVE model/source register | issue #361

Accepted model cut 1128b6a68985cf68215beb4a7a80fd00ec4242fa; refreshed code cut 25d5d83c3302491d5ea6796f31afd4a3593789a3.

Package use-cases.md — requirements, flows, exceptions and code mapping

Package models.md — requirements, flows, exceptions and code mapping

Package design-problem.md — requirements, flows, exceptions and code mapping

Figure M112 | ARCHIVE | problem before class

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

Pattern comparison — Before classes

Conceptual risk model only; not a historical implementation

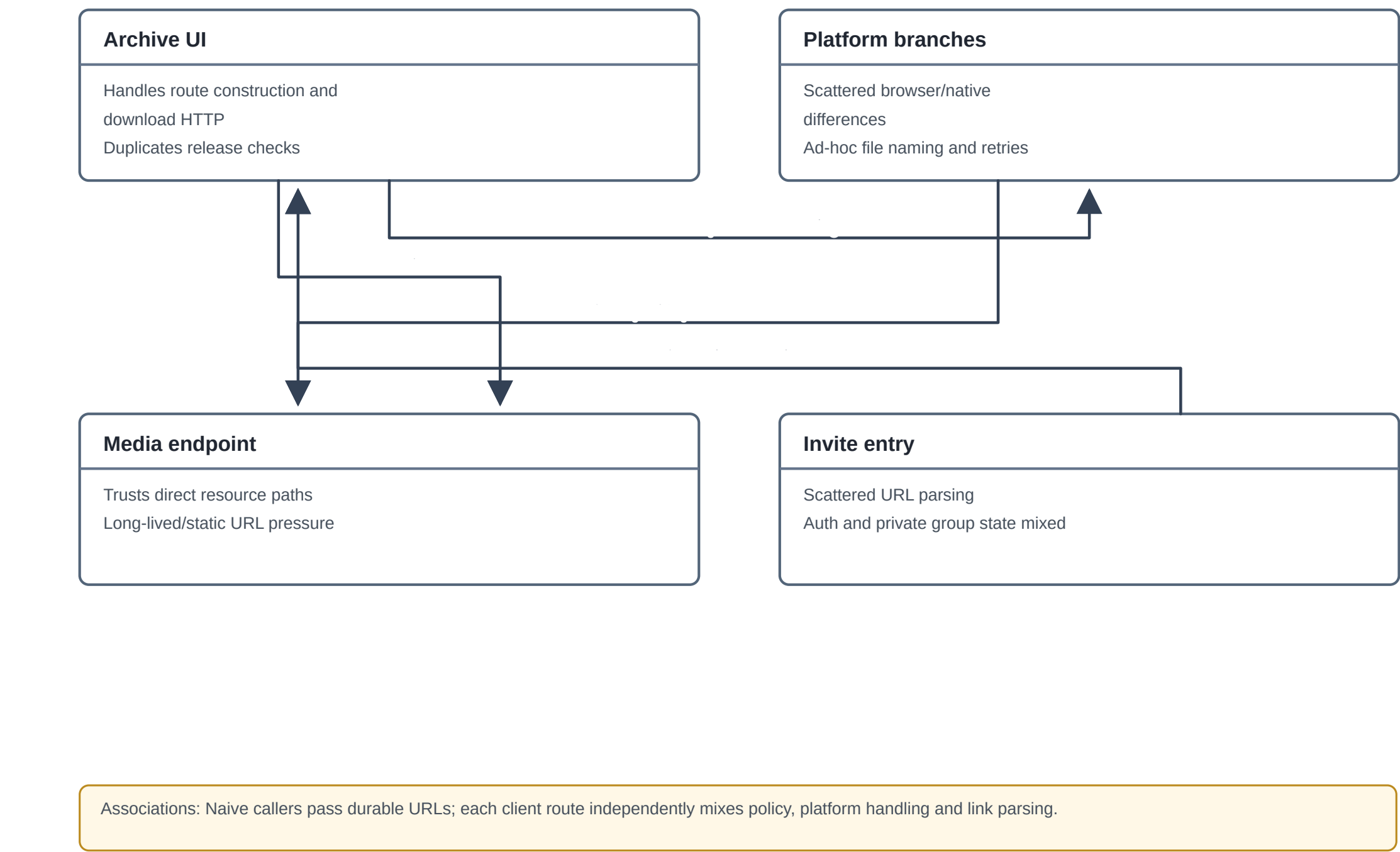
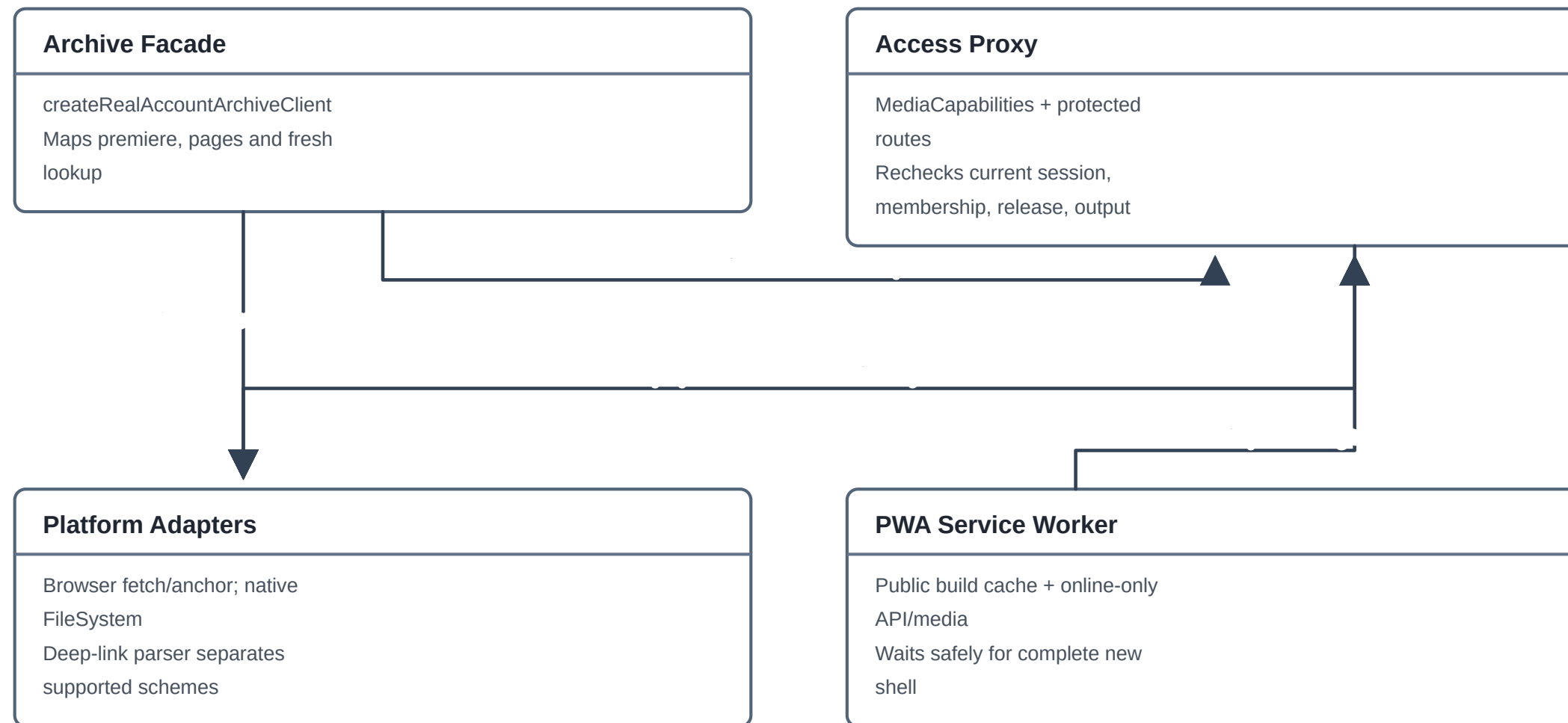


Figure M110 | ARCHIVE | problem after class

Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

Pattern comparison — After classes

Accepted implementation at pinned current dev cut



Associations: Facade coordinates API contract; Proxy guards media access; Adapter handles platform boundaries. These roles map to actual functions/modules.

Figure M113 | ARCHIVE | problem before sequence
Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

Pattern comparison — Before sequence
Analysis responsibility model; platform-neutral concepts

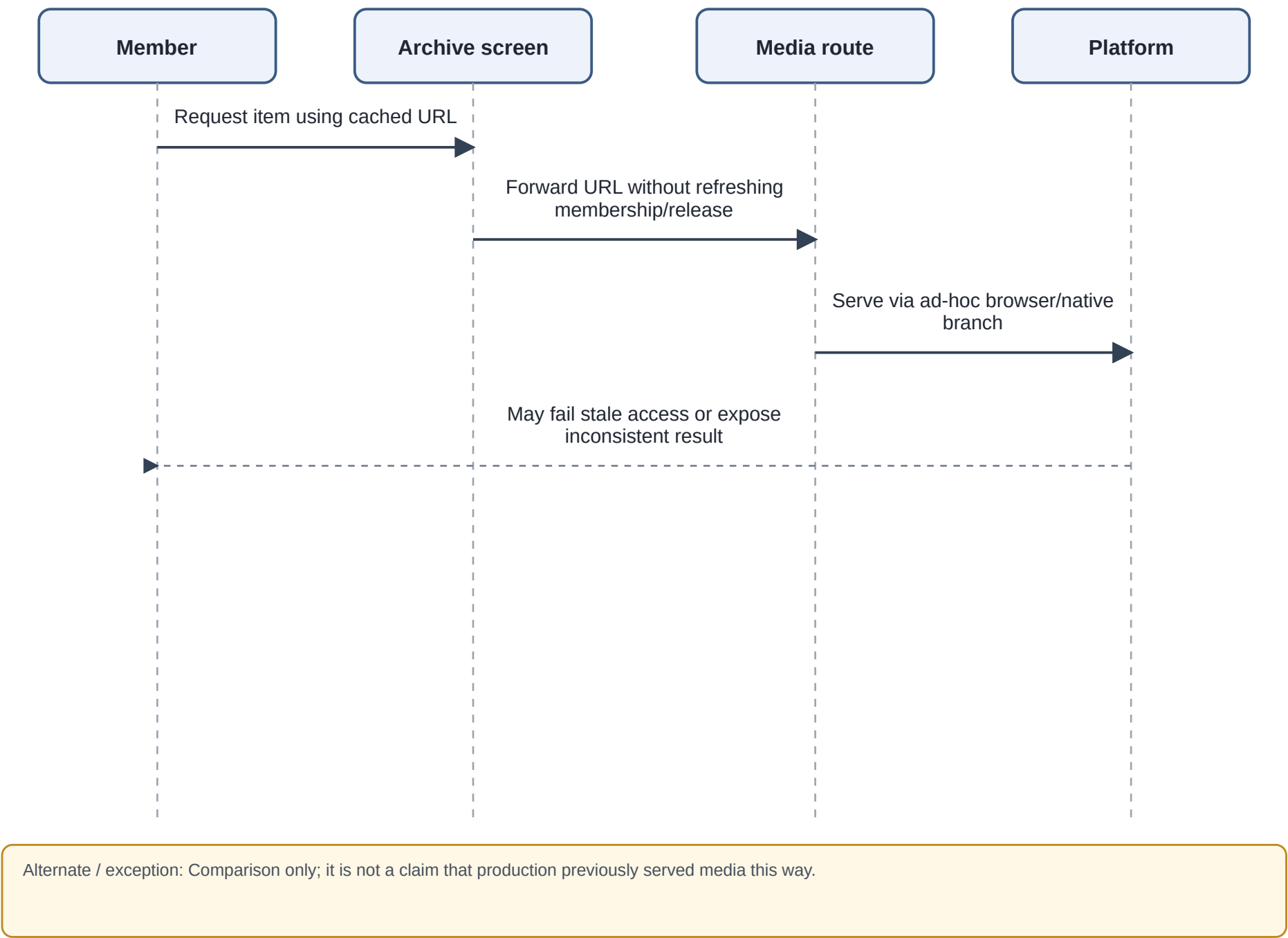
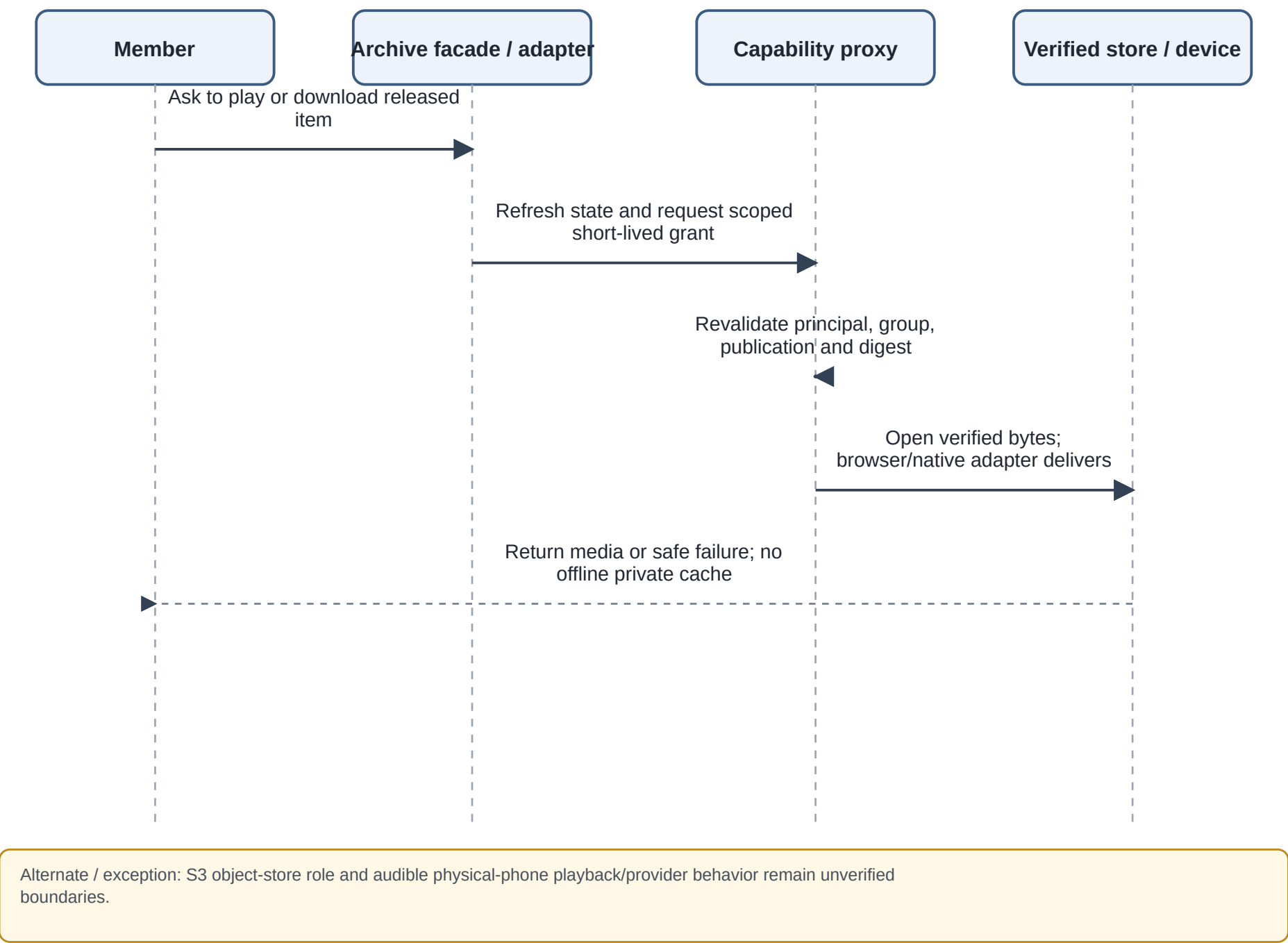


Figure M111 | ARCHIVE | problem after sequence
 Accepted source 1128b6a6 | issue #361 | A3 landscape | complete model

Pattern comparison — After sequence
 Design interaction grounded in accepted code

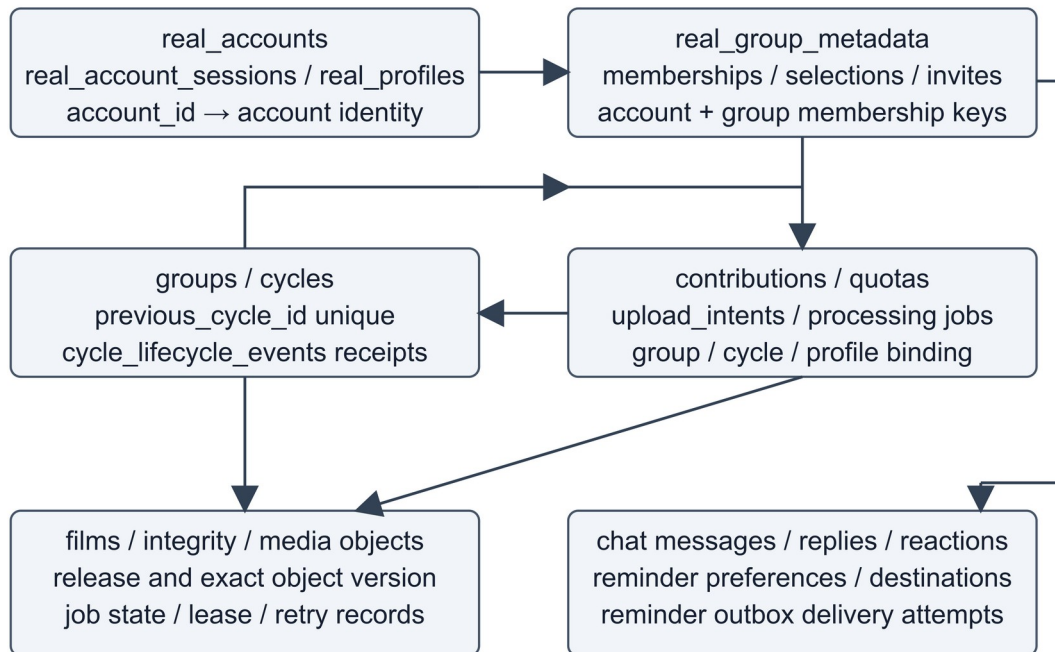


packages/archive/diagrams/problem-after-sequence.svg

3.6 Database Schemas

Current SQLite logical schema

Selected relation clusters; full DDL is in server/migrations



Demo identity records remain separate from real accounts.

PostgreSQL parity and managed restore are future obligations.

Figure 4 Selected logical SQLite relations. Full foreign keys and migration DDL remain the authoritative physical schema.

The current schema is SQLite through versioned migrations. Real accounts/sessions/profiles, group metadata/memberships/selections/invites stay separate from Demo identities. Group/cycle/contribution/quota/correction records connect to persisted media jobs, lifecycle events, films, chat/events and reminder preferences/destinations/outbox. Upload intents bind account/profile/group/cycle to exact object versions and final contribution/jobs. Foreign keys, unique retry identities and transaction boundaries enforce these relations. The current schema diagrams are selected logical views, with full DDL in server/migrations. PostgreSQL parity, data reconciliation, controlled cutover and managed restore are future obligations; no live PostgreSQL schema or restore is claimed.

4. DevSecOps and Development Lifecycle

4.1 Source Control Strategy

rewind-app is the authoritative repository for code, backlog and planning. Issue branches start from refreshed dev under guru/ and focused PRs target dev with Refs references. Routine dev integration requires up-to-date green aggregate Quality and resolved conversations; auth/private-media/migration/deployment/infra changes need human review. main remains the reviewed release branch. A merge is not deployment or user acceptance. Dirty primary Terraform/planning work and archived workspace boundaries remain protected. GitHub access in this run uses authenticated gh exclusively.

4.2 Continuous Integration

Quality classifies the exact diff and runs relevant format, lint, architecture, typecheck, production audit, root/infra, server/coverage, frontend/coverage, web/browser and cloud-free deployment lanes. The aggregate required check reports each lane truthfully; docs-only or unchanged browser lanes may be skipped by the accepted scope rules. Frontend statements have a 70 percent gate; server coverage is measured separately without a percentage gate. CodeQL, GitGuardian and release image/IaC scans supply distinct security evidence. Pending or absent scans are not a clean result. Local focused/fast checks precede relevant slower browser/media verification.

4.3 Continuous Delivery

Deploy dev requires successful Quality, verifies a release bundle and activates it after runtime/web health checks, with compatible rollback. The last successful run was 37142192598. Later runs 37143129969, 37143383024 and report-merge run 37149287273 failed with host ENOSPC during host-side bundle verification/staging, before activation. No manual retry or host cleanup was performed. Capacity recovery needs an authorized operator and subsequent deployment/health verification. Prepared dev/release roots and private-storage configuration still require reviewed live plans, inventory and activation. Main promotion, deployment and hosted acceptance remain distinct.

5. Individual members activity contribution summary

The proposal names the five members listed in §1.3. For each, the final report must record actual work, evidence, owned use case, owned design problem and rough effort supplied or confirmed by that person. These fields are pending. Jiayu and Long's research/review comments under #267 remain attributed observations. Issue #267 was independently accepted and closed through explicitly authorized substituted agent review after the source cut; this does not establish that the named researchers agreed or that feedback was timely. The closed unmerged #325 revision is not accepted code or document content. Agent-generated model drafts and automated test counts do not establish a member's hours, design judgment or lecturer approval.

Baseline refresh: accepted dev 25d5d83c3302491d5ea6796f31afd4a3593789a3 includes PR #398, a numeric filesystem-capacity signal. It does not recover capacity, establish inode/quota availability or prove a deployment. The accepted package cuts remain fc2b8c811e9c13b176113fa0bdb2c015798990b4 (ACCESS/CAPTURE/CYCLE) and 1128b6a68985cf68215beb4a7a80fd00ec4242fa (PARTICIPATION/ARCHIVE). Later photo-label (#387), chat API-prefix (#388) and portrait-guidance (#392) fixes change client presentation/transport without changing the accepted model responsibilities.

References and final draft gates

Issue #362 owns final assembly. Source packages are under packages/access, capture, cycle, participation and archive. The submitted proposal and report template are named retained sources; their SHA-256 identities are recorded in template-contract.md. Required final gates are factual member ownership/effort, actual Scrum/review records, final OIDC/PostgreSQL/backups/pre-upload retro, relevant hosted/device/provider acceptance, and final assessment sign-off. The draft preserves these outcomes without claiming completion.

Template fidelity: the original body and cover retain the supplied A4 geometry, styles, numbering, header logo and footer fields. Authorized landscape foldouts preserve complete models and restore A4 body geometry between blocks. Native contents/page fields retain their codes; page caches match the expanded render and update in Word. Factual contributions and final compliance remain pending.